Schema **SFERA_v3.00.xsd**

schema location: **..\..\Maintenance\GitHub\Rep\SFERA\IRS\Appendix_D-SFERA_XML_Schema_Definition\SFERA_v3.00.xsd**
attributeFormDefault: **unqualified**
elementFormDefault: **qualified**

Elements	Complex types	Simple types	Attributes
AbsoluteStop	AdvisedSpeed_ComplexType	adhesionCategoryType	adviceType
AdditionalInfo	ATPSystems_ComplexType	ClaimTypeType	aimEarlyArrival
AdditionalSpeedRestriction	BrakingForceCurve_ComplexType	currentType	ALSP_Front
AdditionalTrainNumber	CurrentLimitation_ComplexType	deviceType	alternativeItineraryProbability
AdhesionConditionsChange	Curve_ComplexType	distanceType	altitude
AdvisedSpeed	DAS_ModesComplexType	energyType	amountLevelCrossings
AlternativeItinerary	Error_ComplexType	forceType	applicationDistance
Applicable_OTN	GNSS_StaticPosition_ComplexType	gradientDirectionType	arrivalTime
ApplicableIfTrainRunsOnSP	Gradient_ComplexType	gradientValueType	arrivalWindow
ATOTS_ContactInfo	InfrastructureTractionType_ComplexType	LeavingEnteringStatusType	ascending
ATP_System_Identifier	LineAndTrackIdentifier_ComplexType	NC_CDTrainType	ASR_Front
ATP_Systems	MultilingualText_ComplexType	NC_TrainType	ASR_Speed
ATP_SystemsChange	NSPList_ComplexType	NID_C_Type	ATO_HandingOver
ATP_SystemsStart	OTN_ID_ComplexType	percentageType	ATO_State
AverageSpeed	PhysicalCharacteristics_ComplexType	powerTypekW	ATOTS_ID
AvoidSpeedRange	SegmentProfileIdentification_ComplexType	powerTypeW	automaticClosing
AxleLoadSpeedProfile	SegmentProfileInfo_ComplexType	relatedTrainRequestType	autonomy
B2G_Error	Signal_ComplexType	speedType	axleLoadCategory
B2G_EventPayload	SP_GenericArea_ComplexType	teltsi_CompanyCode	baliseGroupIdentifier
B2G_MessageResponse	SP_GenericPoint_ComplexType	teltsi_CountryIdentISO	baliseRelativePosition
B2G_ReplyPayload	SP_Zone_ComplexType	teltsi_DelayCode	boardingPassengers
B2G_Request	SpecificSSP_ComplexType	teltsi_DeltaTime	brakingForce
B2G_StatusReport	SSP_ComplexType	teltsi_FreeText	brakingReactionTime
Balise	TC_Features_ComplexType	teltsi_Numeric1-5	C_DAS_C_Able
BaliseGroup	TC_Identifier_ComplexType	teltsi_Numeric2-2	centralisedOpening
BatteryChargingArea	teltsi_CompositIdentifierOperationalType	teltsi_Numeric3-3	CI_InstanceNumber
BFC_NSPs	teltsi_LocationIdent	teltsi_Percentage	comfortableAcceleration
BFC_TractionType	TemporaryConstraints_ComplexType	teltsi_Speed	comfortableBraking
BrakingForceCurve	TimingPointIdentification_ComplexType	teltsi_String1-10	communicationNetworkChannel
BrakingForceCurveDataPoint	TimingPointZone_ComplexType	teltsi_String1-8	communicationNetworkType
C_DAS_C_Advice	TotalCurrent_ComplexType	teltsi_String4-4	connectionTrackType
C_DAS_C_AdviceRequest	TractionForceCurve_ComplexType	teltsi_WeightValueKilo	contactIdentifier
C_DAS_C_Shown	TractionType_ComplexType	tractionTypeType	contactNotes
CancelPreviousSignal	TrainIdentification_ComplexType	trainPartType	contactRole
CantDeficiencySSP	TrainType	UTCDateTimeType	contactType
ChangedStoppingLocationAlert	Voltage_ComplexType	UUIDv4_Type	correlation_ID
ChangeReason		voltageType	currentPassengers
ClaimType			curveDirection
CommunicationNetwork			curveRadius
ConnectionTrack			DAS_architecture
ConnectionTrackDescription			DAS_Behaviour
Contact			DAS_connectivity

ContactList
CurrentLimitation
CurrentLimitationChange
CurrentLimitationStart
CurveChange
Curves
CurveStart
DAS_DrivingModeRequest
DAS_Mode
DAS_ModesUnavailabilityArea
DAS_OperatingModeSelected
DAS_OperatingModesSupported
DateTimeMeasurement
DecisiveGradientArea
Delay
DelayNotice
DetailedDescriptionOfLocation
DynamicBrakeForceInhibitionArea
EarliestEndDateTime
EndSignalApplication
EnergyData
Engine
EngineTractionType
EstimatedVoltage
EstimatedVoltageChange
EstimatedVoltageConstraint
EstimatedVoltageStart
ExpectedCurrentData
ForceDrivingModeChange
ForceDrivingModeChangeRequest
G2B_Error
G2B_EventPayload
G2B_MessageResponse
G2B_ReplyPayload
G2B_Request
General_JP_Information
General_JP_Information_NSP
GeoLocalisationTimestamp
Gradient
GradientChange
GradientStart
GradientSteepest
GradientSteepestChange
GradientSteepestStart
GuardRails
DAS_drivingMode
dataFirstAvailable
dateEndApplication
datePublished
dateStartApplication
daylightSaving
Delay
DelayCategory
deltaSpeed
departureAuthorisationType
departureProcedureDuration
departureTime
description
destinationCompany
destinationDevice
directionOfApplicationOnSP
disembarkingPassengers
displayName
distance_EoA_Offset
distance_STP_Reached
distanceBeforeSignal
distanceToPowerConstraint
distanceToTractionTypeTransition
driverID
dynamicBrakeForceLimit
earliestDepartureAdviceTime
earliestDepartureTime
endKm
endKmAbs
endLine
endLineAbs
endLocation
endPointOfDownLineSpeedApplication
endSignal
endSignalAbs
endSignalApplicationMustBeClearedBy
endStationLong
endStationLongAbs
endStationShort
endStationShortAbs
endTime
endValidity
endValidityAbs
enginePower
engines

HandoverPoint
HandshakeAcknowledgement
HandshakeReject
HandshakeRejectReason
HandshakeRequest
IM_ID
JourneyProfile
JP_ChangeReason
JP_ContextInformation
JP_ContextInformation_NSP_Constraints
JP_InUse
JP_Request
KilometreReferencePoint
KM_Reference
LastAbsoluteOdometryPoint
LastForceStatusChangeMessage
LastNetworkSpecificEvent
LastPassedKmRef
LastPassedSignal
LastPlaintextMessage
LastStatusReportMessage
LatestEndDateTime
LeavingEnteringStatus
LevelCrossingArea
LevelCrossingDescription
LimitedDynamicBrakeForceArea
LineAndTrackIdentifier
LineAndTrackIdentifierChange
LineAndTrackIdentifierStart
LocationSubsidiaryReference
LocationText
LocoData
LocoWeight
LowAdhesion
MaxSpeed
MaxSpeedAnnouncement
MaxSpeedTrack
MessageHeader
MessageIdentifier
MessageReference
MessageText
MessageType
MessageTypeVersion
NetworkSpecificArea
NetworkSpecificCharacteristics
NetworkSpecificConstraint
errorCode
EuropeanVehicleNumber
expectedConsumptionCurrent
expectedRegenerativeCurrent
forecastedArrival
forecastedDeparture
frequency
gradient_PBD
gradientDirection
gradientDirection_PBD
gradientValue
groundContactIdentifier
highest_BFC_AdhesionRate
horizontalOffset
identifier
JP_SP_ConsistencyError
JP_Status
JP_Version
kmRef
language
latestDepartureAdviceTime
latitude
leaveASAP
levelCrossingGuarding
levelCrossingType
lineIdentifier
lineName
location
location_NegativeErrorMargin
location_PositiveErrorMargin
longitude
lowAdhesionCategory
mainContact
mandatoryStop
maxAdvisedSpeed
maxChargingCurrent
maxCurValue
maxDischargingCurrent
maximumCantDeficiency
maximumNumberOfCars
maximumNumberOfRSType
maximumRegenerativeBraking
maximumRegenerativeForce
maximumTrainLength
maxPower
measuredCurrent

NetworkSpecificEvent
NetworkSpecificEventRequest
NetworkSpecificParameter
NetworkSpecificPoint
NextTimingPointTrainInformation
NID_C
NoAdvicePresentedToDriver
NoStoppingArea
NSP_Change
NSP_GroupName
NSP_Start
OtherSpecificSSP
OTN_ID
OwnTrain
PassengerData
PassengerMovement
PassingPointLocation
PermittedBrakingDistanceArea
PhysicalCharacteristics_NSPs
PlaintextMessage
PlaintextMessageRequest
Platform
PlatformChange
PositionSpeed
PositionSpeedRequest
PowerAdvice
PowerConstraint
PreciseDelayCause
PreviousTimingPoint
PreviousTimingPointTrainInformation
RatedVoltage
RatedVoltageChange
RatedVoltageStart
ReasonText
Recipient
RecipientName
RegenerationTotalCurrent
RelatedTrainInformation
RelatedTrainInformationRequest
Reported_DAS_DrivingMode
RequestFromSegmentProfile
RestrictionReason
RollingStockApplicability
RT_TrainInformation
SameDirection
measuredVoltage
message_ID
minAdvisedSpeed
minimumDwellTime
minimumNumberOfCars
minimumNumberOfRSType
minimumPositionUpdateRequestPeriod
minimumTrainLength
name
NC_CDTrain
NC_Train
newAdhesionConditions
newSpeedLevel
nextStoppingPointSkip
NID_CTRACTION
NID_ENGINE
NID_OPERATIONAL
openingDoorSide
operationalConditionsFulfillment
optimalSpeed
originalLanguage
OTN
PBD_Brake
percentage
permittedBrakingDistance
plaintextMessageDirection
plannedDepartureTime
plannedStop
platformName
platformSide
platformUsage
positionOfLastPantograph
positionSpeedRequest
powerConstraintInstruction
powerConsumptionChange
preciseDelayReason
previousPlaintextMessageIdentifier
protectedLevelCrossing
rangeMaxSpeed
rangeMinSpeed
reasonCode
referenceIDAbs
relatedTrainRequest
relatedTrainSupported
relaxedCoupler

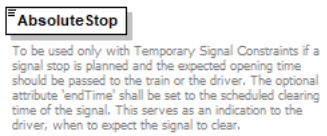
SegmentClaim
SegmentOccupationStatus
SegmentProfile
SegmentProfileInfo
SegmentProfileReference
Sender
SenderName
SessionTermination
SessionTerminationRequest
SFERA_B2G_EventMessage
SFERA_B2G_ReplyMessage
SFERA_B2G_RequestMessage
SFERA_G2B_EventMessage
SFERA_G2B_ReplyMessage
SFERA_G2B_RequestMessage
Signal
Signal_ATP_Systems
Signal_ID
Signal_NSPs
SignalApplication
SignalFunction
SignalInformation
SignalInformation_NSPs
SignalPhysicalCharacteristics
SignalTrainCategory
SignalTrainLength
SP_Areas
SP_Characteristics
SP_ContextInformation
SP_EndDeviation
SP_NextSP
SP_Points
SP_Request
SP_StartDeviation
SP_Zone
SpecificSSP
SSP_VisualReference
StartSignalApplication
StaticSpeedProfile
StaticSpeedProfileChange
StaticSpeedProfileStart
StationAheadSign
StatusReport_NSPs
StatusReportRequest
StopBeforePassingSignal
StoppingPointApplicability
remainingEngineCharge
remainingEnginePower
reportingTime
result
rollingStockType
rolloutCoefficientA
rolloutCoefficientB
rolloutCoefficientC
rotatingMassFactor
routingError
sessionTerminationReason
setValueTractionBraking
SFERA_version
signal_ID_Object
signal_ID_Physical
signalType
slipSlideReported
sourceCompany
sourceDevice
SP_Altitude
SP_BeforeOrAfter
SP_Direction
SP_Gauge
SP_ID
SP_Length
SP_Name
SP_Status
SP_VersionMajor
SP_VersionMinor
specific_SSP_Category
speed
SSP_Front
SSP_NC_CDDIFF
SSP_NC_DIFF
SSP_Speed
startEndQualifier
startKm
startLine
startLocation
startPointOfDownLineSpeedApplication
startSignal
startSignalApplicationMustBeClearedBy
startStationLong
startStationShort
startTime
startValidity

StoppingPointApplicableIfPrecedingSP
StoppingPointDepartureDetails
StoppingPointInformation
StoppingPointLocation
StopType
StopTypeDetails
SuperElevation
SuperElevationChange
SuperElevationStart
SwitchOffEddyCurrentBrakeArea
SwitchOffEddyCurrentEmergencyBrakeArea
SwitchOffMagneticShoeBrakeArea
SwitchOffRegenerativeBrakeArea
TAF_TAP_ID
TAF_TAP_Location
TAF_TAP_Location_NSP
TAF_TAP_LocationIdent
TAF_TAP_LocationName
TAF_TAP_LocationReference
TC_ChangeFeatures
TC_Features
TC_Features_NSPs
TC_Request
TC_RU_ID
teltsi_AgainstBooked
teltsi_AgainstReferenced
teltsi_AllocationCompany
teltsi_Altitude
teltsi_BookedLocationDateTime
teltsi_Company
teltsi_Core
teltsi_CountryCodeISO
teltsi_DelayCause
teltsi_GeographicCoordinates
teltsi_GeoLocalisation
teltsi_GeoLocalisationOnNetwork
teltsi_GNSS_DynamicPosition
teltsi_InternalReferenceIdentifier
teltsi_InterruptionDateTime
teltsi_InterruptionDescription
teltsi_InterruptionPoint
teltsi_InterruptionReason
teltsi_KeeperShortNameVKM
teltsi_Latitude
teltsi_LengthOverBuffers
stationTrack
statusReportsEnabled
stopLocationTolerance
stoppingPointOffset
stopTime
superElevationValue
TAF_TAP_location_abbreviation
TAF_TAP_location_type
TC_ID
TC_VersionMajor
TC_VersionMinor
teltsi_LocationSubsidiaryTypeCode
temporaryConstraintReference
temporaryConstraintType
temporarySignalConstraintType
text
TFC_estimatedVoltage
TFC_maxCurrentValue
timeout_ATOTS_Response
timestamp
TP_abbreviation
TP_Accurate_Stopping
TP_alignment
TP_ID
TP_Information
TP_latestArrivalTime
TP_Pass_Stop_Depart
TP_PlannedLatestArrivalTime
TP_priority
TP_StopSkipPass
TP_type
trackIdentifier
tractionForce
tractionReactionTime
trainCategoryCode
trainHold
trainLength
trainMaxSpeed
trainMoving
trainWeight
trp_ae
trp_aw
trp_be
trp_bw
tunnelCategory

teltsi_Location
teltsi_LocationPrimaryCode
teltsi_LocationSubsidiaryCode
teltsi_LocationSubsidiaryIdentification
teltsi_LocationSubsidiaryName
teltsi_Longitude
teltsi_NetworkProjectedLocation
teltsi_ObjectType
teltsi_OperationalTrainNumber
teltsi_PrimaryLocationName
teltsi_ReferencedLocationDateTime
teltsi_Remarks
teltsi_ScheduledDateTimeAtTransfer
teltsi_ScheduledTimeAtHandover
teltsi_SRID
teltsi_StartDate
teltsi_TimetableYear
teltsi_TotalLoadWeight
teltsi_TractionPositionInTrain
teltsi_TrainActivityType
teltsi_TrainDelay
teltsi_TransferPoint
teltsi_Variant
teltsi_WagonMaxSpeed
teltsi_WagonTrainPosition
teltsi_WagonWeightEmpty
TemporaryConstraintReason
TemporaryConstraints
TemporarySignalConstraints
TFC_NSPs
TFC_TractionType
TimingPoint
TimingPoint_NSPs
TimingPointConstraints
TimingPointConstraints_NSPs
TimingPointEstimation
TimingPointReference
TimingPointZone
TimingPointZoneReference
To_TAF_TAP_Location
TP_ID_Reference
TP_Name
TractionForceCurve
TractionForceCurveDataPoint
TractionTotalCurrent
tunnelFactor
unableToStop
UTC_Offset
V_DIFF
value
verticalOffset
virtualMarker
visualIdentifier
voltageValue
wagons
windHeading
windSpeed
XPath

TractionTypeTransition
TractionTypeTransitionBimodal
TractionTypeTransitionElectric
Train
Train_ATP_Systems
TrainCharacteristics
TrainCharacteristicsChange
TrainCharacteristicsRef
TrainCharacteristicsRef_NSPs
TrainIdentification
TrainLocation
TrainLocationInformation
TrainReady
TrainReadyStatus
TrainReadyTime
TrainsAfterOwn
TrainsBeforeOwn
Tunnel
Unavailable_DAS_OperatingModes
UnavailableTractionType
UnprotectedLevelCrossingStop
VehicleData
VirtualBalise
VirtualBalisePosition
WagonData
Wind

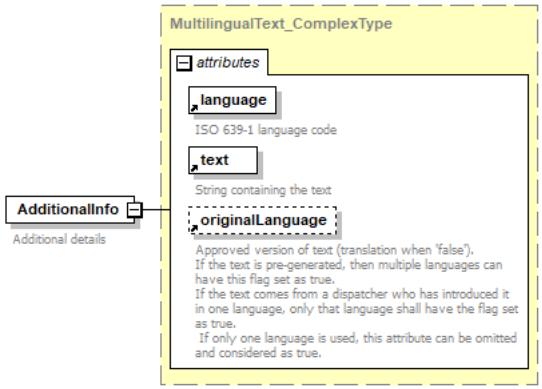
element **AbsoluteStop**

diagram	
type	**xs:boolean**
properties	content simple
used by	element **SignalInformation**
annotation	documentation To be used only with Temporary Signal Constraints if a signal stop is planned and the expected opening time should be passed to the train or the driver. The optional attribute 'endTime' shall be set to the scheduled clearing time of the signal. This serves as an indication to the driver, when to expect the signal to clear.
source	<xs:element name="AbsoluteStop" type="xs:boolean"> <xs:annotation> <xs:documentation>To be used only with Temporary Signal Constraints if a signal stop is planned and the expected opening time should be passed to the train or the driver. The optional attribute 'endTime' shall be set to the scheduled clearing time of the signal. This serves as an indication to the driver, when to expect the signal to clear.</xs:documentation>

```
  </xs:annotation>
</xs:element>
```

element **AdditionalInfo**

diagram



type **MultilingualText_ComplexType**

properties
- content complex

used by
- elements **HandshakeAcknowledgement** **HandshakeRequest** **KM_Reference** **SessionTerminationRequest**
- complexType **Error_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

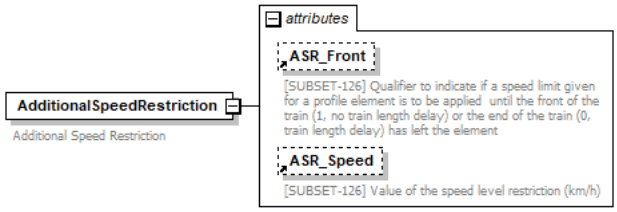
annotation
- documentation
  Additional details

source

```
<xs:element name="AdditionalInfo" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Additional details</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **AdditionalSpeedRestriction**

diagram



properties

content complex

used by

element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
ASR_Front	**xs:boolean**				documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (1, no train length delay) or the end of the train (0, train length delay) has left the element
ASR_Speed	**speedType**				documentation [SUBSET-126] Value of the speed level restriction (km/h)

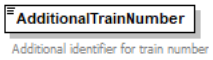
annotation

documentation
Additional Speed Restriction

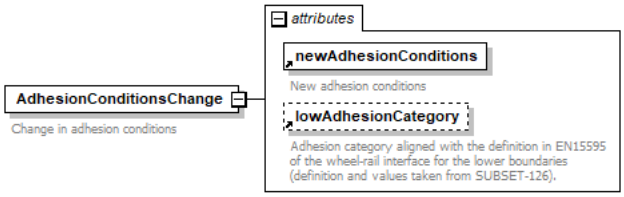
source

```
<xs:element name="AdditionalSpeedRestriction">
  <xs:annotation>
    <xs:documentation>Additional Speed Restriction</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="ASR_Front"/>
    <xs:attribute ref="ASR_Speed"/>
  </xs:complexType>
</xs:element>
```

element **AdditionalTrainNumber**

diagram	
type	**xs:string**
properties	content simple
used by	element **RT_TrainInformation** complexType **OTN_ID_ComplexType**
annotation	documentation Additional identifier for train number
source	<xs:element name="AdditionalTrainNumber" type="xs:string"> <xs:annotation> <xs:documentation>Additional identifier for train number</xs:documentation> </xs:annotation> </xs:element>

element **AdhesionConditionsChange**

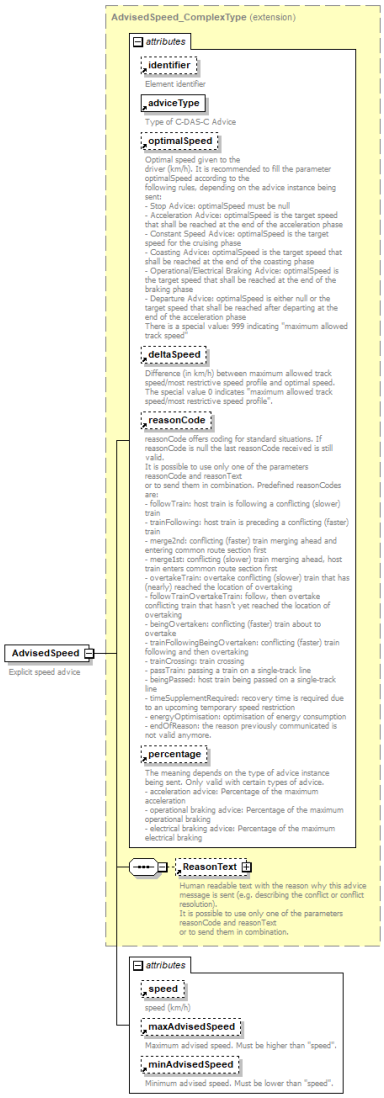
diagram	
properties	content complex
used by	element **B2G_StatusReport**

attributes

Name	Type	Use	Default	Fixed	Annotation
newAdhesionConditions	**derived by: xs:string**	required			documentation New adhesion conditions
lowAdhesionCategory	**adhesionCategoryType**				documentation Adhesion category aligned with the definition in EN15595 of the wheel-rail interface for the lower boundaries (definition and values taken from SUBSET-126).

annotation	documentation Change in adhesion conditions
source	<xs:element name="AdhesionConditionsChange"> <xs:annotation> <xs:documentation>Change in adhesion conditions</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="newAdhesionConditions" use="required"/> <xs:attribute ref="lowAdhesionCategory"/> </xs:complexType> </xs:element>

element **AdvisedSpeed**

diagram	
type	extension of **AdvisedSpeed_ComplexType**
properties	content complex
children	**ReasonText**
used by	element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
adviceType	**derived by: xs:string**	required			documentation Type of C-DAS-C Advice
optimalSpeed	**speedType**				documentation Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on

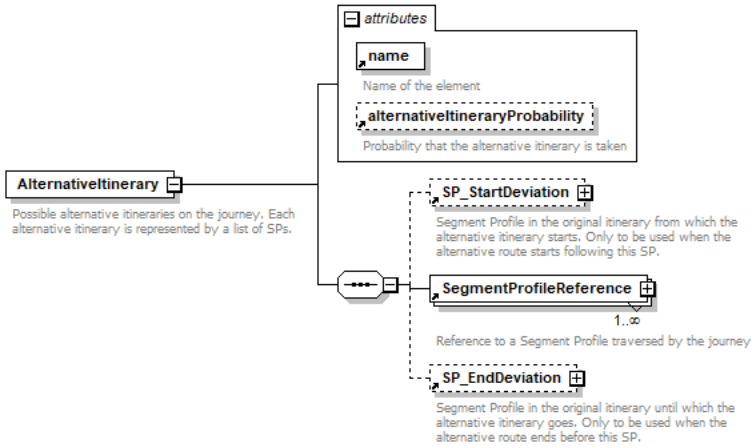
			the advice instance being sent: - Stop Advice: optimalSpeed must be null - Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase - Constant Speed Advice: optimalSpeed is the target speed for the cruising phase - Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase - Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase - Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase There is a special value: 999 indicating "maximum allowed track speed"
	deltaSpeed	**speedType**	documentation Difference (in km/h) between maximum allowed track speed/most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed/most restrictive speed profile".
	reasonCode	**derived by:** **xs:string**	documentation reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid. It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are: - followTrain: host train is following a conflicting (slower) train - trainFollowing: host train is preceding a conflicting (faster) train - merge2nd: conflicting (faster) train merging ahead and entering common route section first - merge1st: conflicting (slower) train merging ahead, host train enters common route section first - overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking - followTrainOvertakeTrain: follow, then overtake

			conflicting train that hasn't yet reached the location of overtaking - beingOvertaken: conflicting (faster) train about to overtake - trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking - trainCrossing: train crossing - passTrain: passing a train on a single-track line - beingPassed: host train being passed on a single-track line - timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction - energyOptimisation: optimisation of energy consumption - endOfReason: the reason previously communicated is not valid anymore.
	percentage	**percentageType**	documentation The meaning depends on the type of advice instance being sent. Only valid with certain types of advice. - acceleration advice: Percentage of the maximum acceleration - operational braking advice: Percentage of the maximum operational braking - electrical braking advice: Percentage of the maximum electrical braking
	speed	**speedType**	documentation speed (km/h)
	maxAdvisedSpeed	**speedType**	documentation Maximum advised speed. Must be higher than "speed".
	minAdvisedSpeed	**speedType**	documentation Minimum advised speed. Must be lower than "speed".
annotation	documentation Explicit speed advice		
source			

```
<xs:element name="AdvisedSpeed">
 <xs:annotation>
  <xs:documentation>Explicit speed advice</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="AdvisedSpeed_ComplexType">
    <xs:attribute ref="speed"/>
    <xs:attribute ref="maxAdvisedSpeed"/>
    <xs:attribute ref="minAdvisedSpeed"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **AlternativeItinerary**

diagram



properties

content complex

children **SP_StartDeviation** **SegmentProfileReference** **SP_EndDeviation**

used by

element **JourneyProfile**

attributes

Name	Type	Use	Default	Fixed	Annotation
name	**xs:string**	required			documentation Name of the element
alternativeItineraryProbability	**percentageType**				documentation Probability that the alternative itinerary is taken

annotation

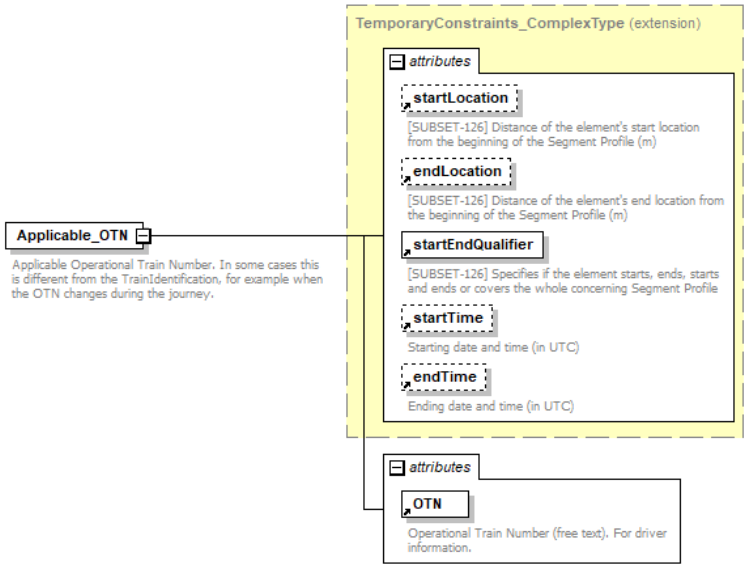
documentation
Possible alternative itineraries on the journey. Each alternative itinerary is represented by a list of SPs.

source

```
<xs:element name="AlternativeItinerary">
  <xs:annotation>
    <xs:documentation>Possible alternative itineraries on the journey. Each alternative itinerary is represented by a list of SPs.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="SP_StartDeviation" minOccurs="0"/>
      <xs:element ref="SegmentProfileReference" maxOccurs="unbounded"/>
      <xs:element ref="SP_EndDeviation" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute ref="name" use="required"/>
    <xs:attribute ref="alternativeItineraryProbability"/>
  </xs:complexType>
</xs:element>
```

element **Applicable_OTN**

diagram



type: extension of **TemporaryConstraints_ComplexType**

properties: content complex

used by: element **JP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
startTime	**UTCDateTimeType**				documentation Starting date and time (in UTC)
endTime	**UTCDateTimeType**				documentation Ending date and time (in UTC)

	OTN	**xs:string**	required	documentation Operational Train Number (free text). For driver information.

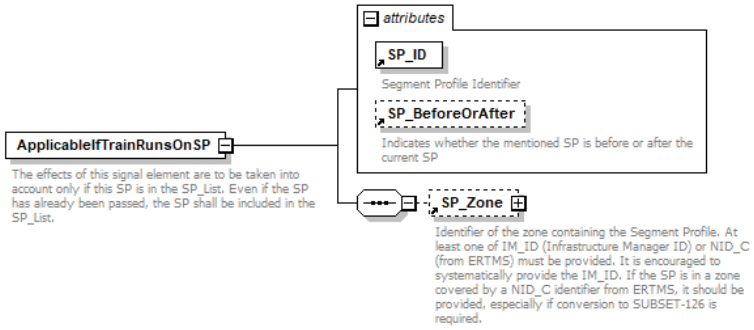
annotation	documentation Applicable Operational Train Number. In some cases this is different from the TrainIdentification, for example when the OTN changes during the journey.

source

```
<xs:element name="Applicable_OTN">
 <xs:annotation>
  <xs:documentation>Applicable Operational Train Number. In some cases this is different from the TrainIdentification, for example when the OTN changes during the journey.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="TemporaryConstraints_ComplexType">
    <xs:attribute ref="OTN" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **ApplicableIfTrainRunsOnSP**

diagram



properties	content complex
children	**SP_Zone**
used by	element **SignalApplication**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_BeforeOrAfter	**derived by: xs:string**				documentation Indicates whether the mentioned SP is before or after the current SP

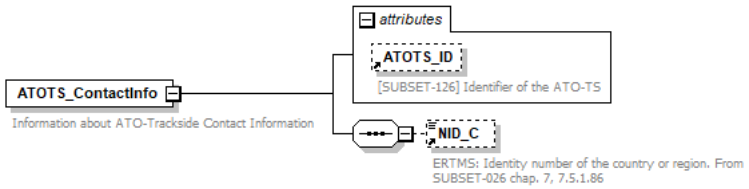
annotation	documentation The effects of this signal element are to be taken into account only if this SP is in the SP_List. Even if the SP has already been passed, the SP shall be included in the SP_List.

source

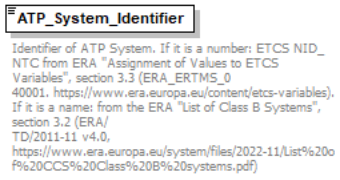
```
<xs:element name="ApplicableIfTrainRunsOnSP">
 <xs:annotation>
  <xs:documentation>The effects of this signal element are to be taken into account only if this SP
```

	is in the SP_List. Even if the SP has already been passed, the SP shall be included in the SP_List.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="SP_Zone" minOccurs="0"/> </xs:sequence> <xs:attribute ref="SP_ID" use="required"/> <xs:attribute ref="SP_BeforeOrAfter"/> </xs:complexType> </xs:element>

element **ATOTS_ContactInfo**

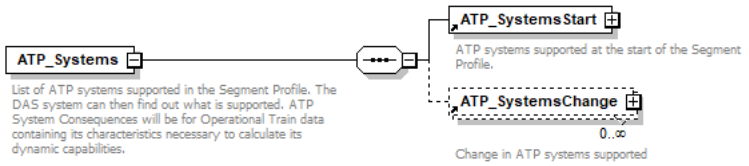
diagram	
properties	content complex
children	**NID_C**
used by	element **SP_NextSP**
attributes	Name: ATOTS_ID; Type: **xs:string**; Use: ; Default: ; Fixed: ; Annotation documentation: [SUBSET-126] Identifier of the ATO-TS
annotation	documentation Information about ATO-Trackside Contact Information
source	<xs:element name="ATOTS_ContactInfo"> <xs:annotation> <xs:documentation>Information about ATO-Trackside Contact Information</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="NID_C" minOccurs="0"/> </xs:sequence> <xs:attribute ref="ATOTS_ID"/> </xs:complexType> </xs:element>

element **ATP_System_Identifier**

diagram	
type	extension of **xs:string**

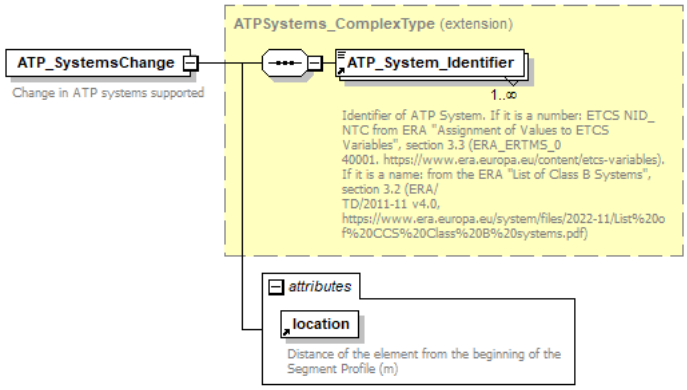
properties	content complex
used by	elements **RT_TrainInformation** **StaticSpeedProfile** complexTypes **ATPSystems_ComplexType** **BrakingForceCurve_ComplexType**
annotation	documentation Identifier of ATP System. If it is a number: ETCS NID_ NTC from ERA "Assignment of Values to ETCS Variables", section 3.3 (ERA_ERTMS_0 40001. https://www.era.europa.eu/content/etcs-variables). If it is a name: from the ERA "List of Class B Systems", section 3.2 (ERA/ TD/2011-11 v4.0, https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf)
source	<xs:element name="ATP_System_Identifier"> <xs:annotation> <xs:documentation>Identifier of ATP System. If it is a number: ETCS NID_ NTC from ERA "Assignment of Values to ETCS Variables", section 3.3 (ERA_ERTMS_0 40001. https://www.era.europa.eu/content/etcs-variables). If it is a name: from the ERA "List of Class B Systems", section 3.2 (ERA/ TD/2011-11 v4.0, https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf)</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="xs:string"/> </xs:simpleContent> </xs:complexType> </xs:element>

element **ATP_Systems**

diagram	
properties	content complex
children	**ATP_SystemsStart** **ATP_SystemsChange**
used by	element **SP_Characteristics**
annotation	documentation List of ATP systems supported in the Segment Profile. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities.
source	<xs:element name="ATP_Systems"> <xs:annotation> <xs:documentation>List of ATP systems supported in the Segment Profile. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="ATP_SystemsStart"/> <xs:element ref="ATP_SystemsChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **ATP_SystemsChange**

diagram



type: extension of **ATPSystems_ComplexType**

properties: content complex

children: **ATP_System_Identifier**

used by: element **ATP_Systems**

attributes

Name	Type	Use	Default	Fixed	Annotation
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

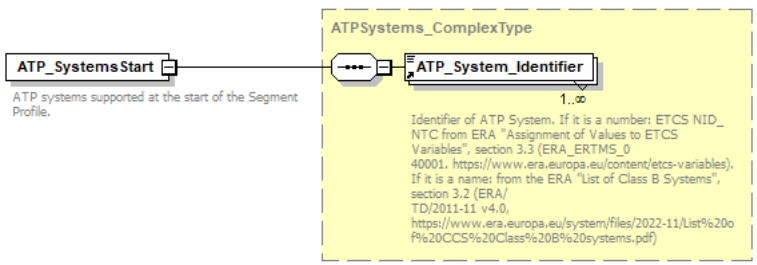
documentation
Change in ATP systems supported

source

```
<xs:element name="ATP_SystemsChange">
 <xs:annotation>
  <xs:documentation>Change in ATP systems supported</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="ATPSystems_ComplexType">
    <xs:attribute ref="location" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

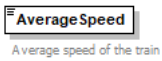
element **ATP_SystemsStart**

diagram

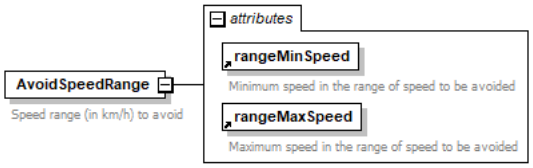


type	**ATPSystems_ComplexType**
properties	content complex
children	**ATP_System_Identifier**
used by	element **ATP_Systems**
annotation	documentation ATP systems supported at the start of the Segment Profile.
source	<xs:element name="ATP_SystemsStart" type="ATPSystems_ComplexType"> <xs:annotation> <xs:documentation>ATP systems supported at the start of the Segment Profile.</xs:documentation> </xs:annotation> </xs:element>

element **AverageSpeed**

diagram	
used by	element **TrainLocationInformation**
annotation	documentation Average speed of the train
source	<xs:element name="AverageSpeed"> <xs:annotation> <xs:documentation>Average speed of the train</xs:documentation> </xs:annotation> </xs:element>

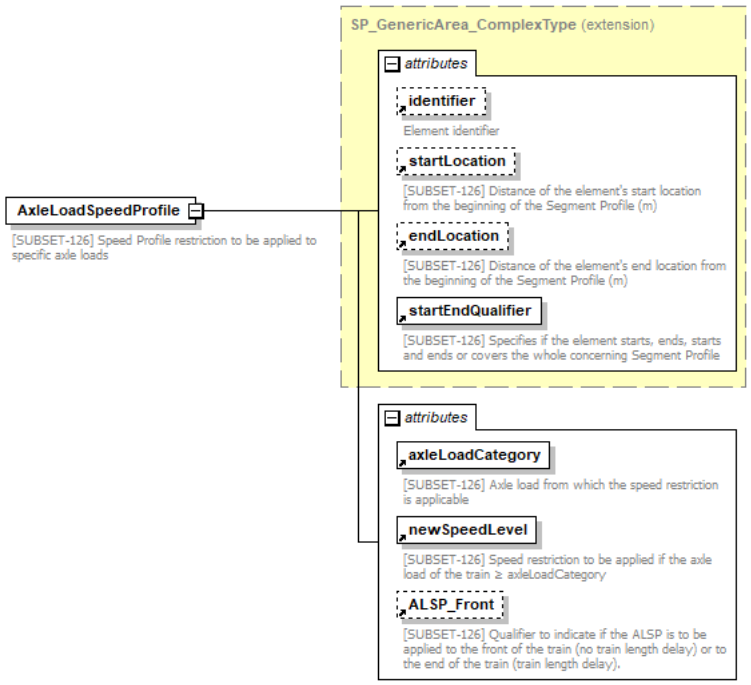
element **AvoidSpeedRange**

diagram	
properties	content complex
used by	element **SignalInformation**
attributes	Name / Type / Use / Default / Fixed / Annotation rangeMinSpeed — **speedType** — required — documentation: Minimum speed in the range of speed to be avoided rangeMaxSpeed — **speedType** — required — documentation: Maximum speed in the range of speed to be avoided
annotation	documentation Speed range (in km/h) to avoid
source	<xs:element name="AvoidSpeedRange">

```
  <xs:annotation>
    <xs:documentation>Speed range (in km/h) to avoid</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="rangeMinSpeed" use="required"/>
    <xs:attribute ref="rangeMaxSpeed" use="required"/>
  </xs:complexType>
</xs:element>
```

element **AxleLoadSpeedProfile**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

used by: element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)

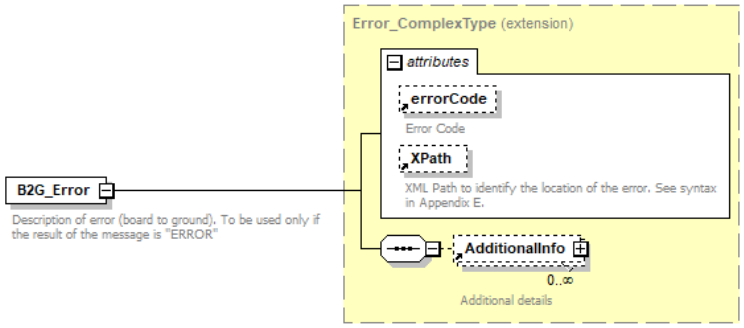
	startEndQualifier	**derived by: xs:string**	required	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
	axleLoadCategory	**derived by: xs:string**	required	documentation [SUBSET-126] Axle load from which the speed restriction is applicable
	newSpeedLevel	**speedType**	required	documentation [SUBSET-126] Speed restriction to be applied if the axle load of the train ≥ axleLoadCategory
	ALSP_Front	**xs:boolean**		documentation [SUBSET-126] Qualifier to indicate if the ALSP is to be applied to the front of the train (no train length delay) or to the end of the train (train length delay).
annotation	documentation [SUBSET-126] Speed Profile restriction to be applied to specific axle loads			

source

```
<xs:element name="AxleLoadSpeedProfile">
  <xs:annotation>
    <xs:documentation>[SUBSET-126] Speed Profile restriction to be applied to specific axle loads</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SP_GenericArea_ComplexType">
        <xs:attribute ref="axleLoadCategory" use="required"/>
        <xs:attribute ref="newSpeedLevel" use="required"/>
        <xs:attribute ref="ALSP_Front"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **B2G_Error**

diagram



type: extension of **Error_ComplexType**

properties: content complex

children: **AdditionalInfo**

used by: element **B2G_MessageResponse**

attributes

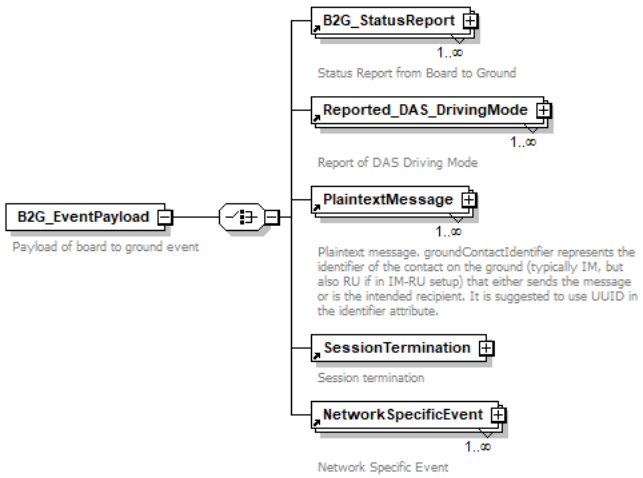
Name	Type	Use	Default	Fixed	Annotation
errorCode	**xs:string**				documentation Error Code
XPath	**xs:string**				documentation XML Path to identify the location of the error. See syntax in Appendix E.

annotation: documentation
Description of error (board to ground). To be used only if the result of the message is "ERROR"

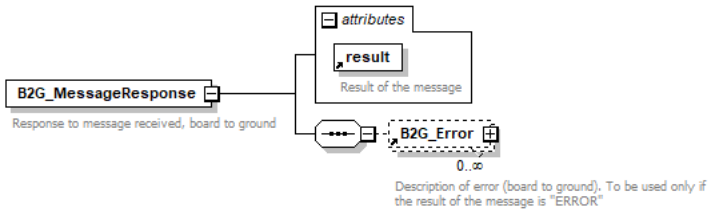
source

```
<xs:element name="B2G_Error">
  <xs:annotation>
    <xs:documentation>Description of error (board to ground). To be used only if the result of the message is "ERROR"</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="Error_ComplexType"/>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **B2G_EventPayload**

diagram	
properties	content complex
children	B2G_StatusReport Reported_DAS_DrivingMode PlaintextMessage SessionTermination NetworkSpecificEvent
used by	element SFERA_B2G_EventMessage
annotation	documentation Payload of board to ground event
source	<xs:element name="B2G_EventPayload"> <xs:annotation> <xs:documentation>Payload of board to ground event</xs:documentation> </xs:annotation> <xs:complexType> <xs:choice> <xs:element ref="B2G_StatusReport" maxOccurs="unbounded"/> <xs:element ref="Reported_DAS_DrivingMode" maxOccurs="unbounded"/> <xs:element ref="PlaintextMessage" maxOccurs="unbounded"/> <xs:element ref="SessionTermination"/> <xs:element ref="NetworkSpecificEvent" maxOccurs="unbounded"/> </xs:choice> </xs:complexType> </xs:element>

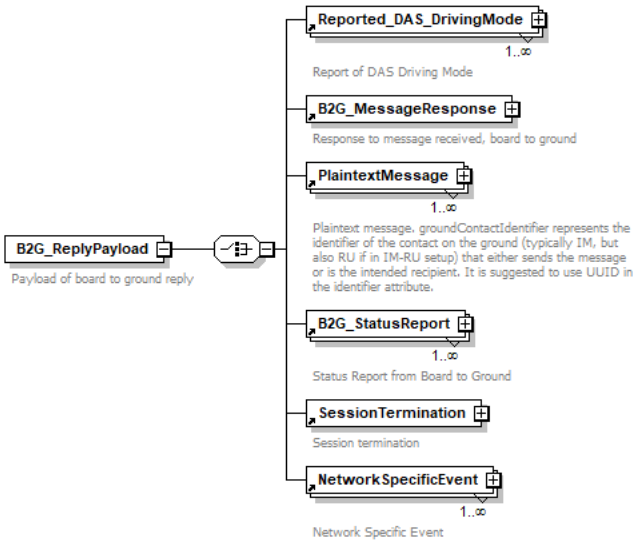
element **B2G_MessageResponse**

diagram	
properties	content complex
children	B2G_Error
used by	element B2G_ReplyPayload

attributes	Name	Type	Use	Default	Fixed	Annotation
	result	**derived by: xs:string**	required			documentation Result of the

	message
annotation	documentation Response to message received, board to ground
source	<xs:element name="B2G_MessageResponse"> <xs:annotation> <xs:documentation>Response to message received, board to ground</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="B2G_Error" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="result" use="required"/> </xs:complexType> </xs:element>

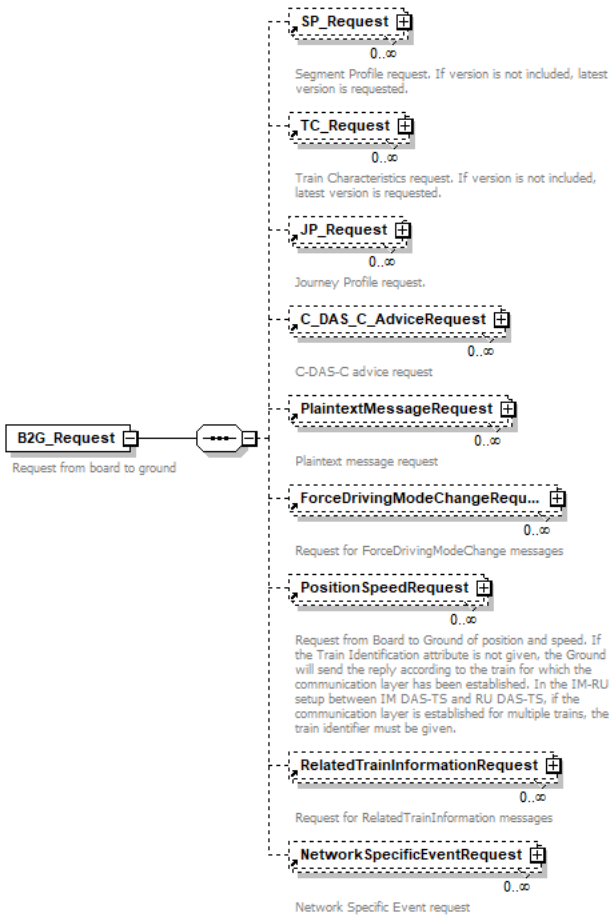
element **B2G_ReplyPayload**

diagram	
properties	content complex
children	Reported_DAS_DrivingMode B2G_MessageResponse PlaintextMessage B2G_StatusReport SessionTermination NetworkSpecificEvent
used by	element SFERA_B2G_ReplyMessage
annotation	documentation Payload of board to ground reply
source	<xs:element name="B2G_ReplyPayload"> <xs:annotation> <xs:documentation>Payload of board to ground reply</xs:documentation> </xs:annotation> <xs:complexType> <xs:choice> <xs:element ref="Reported_DAS_DrivingMode" maxOccurs="unbounded"/> <xs:element ref="B2G_MessageResponse"/> <xs:element ref="PlaintextMessage" maxOccurs="unbounded"/> <xs:element ref="B2G_StatusReport" maxOccurs="unbounded"/> <xs:element ref="SessionTermination"/> <xs:element ref="NetworkSpecificEvent" maxOccurs="unbounded"/>

```
    </xs:choice>
  </xs:complexType>
</xs:element>
```

element **B2G_Request**

diagram



properties

content complex

children

SP_Request TC_Request JP_Request C_DAS_C_AdviceRequest PlaintextMessageRequest ForceDrivingModeChangeRequest PositionSpeedRequest RelatedTrainInformationRequest NetworkSpecificEventRequest

used by

element SFERA_B2G_RequestMessage

annotation

documentation
Request from board to ground

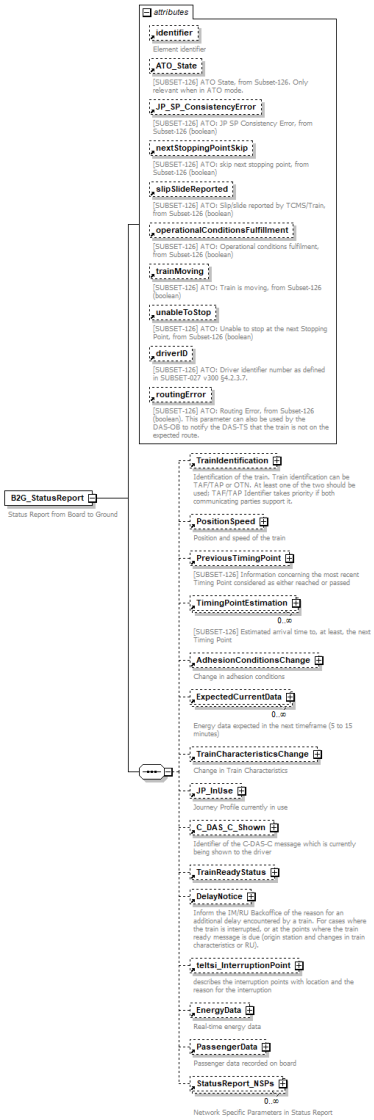
source

```
<xs:element name="B2G_Request">
  <xs:annotation>
    <xs:documentation>Request from board to ground</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="SP_Request" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="TC_Request" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="JP_Request" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="C_DAS_C_AdviceRequest" minOccurs="0" maxOccurs="unbounded"/>
```

```
      <xs:element ref="PlaintextMessageRequest" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="ForceDrivingModeChangeRequest" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="PositionSpeedRequest" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="RelatedTrainInformationRequest" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="NetworkSpecificEventRequest" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **B2G_StatusReport**

diagram



properties	content complex
children	**TrainIdentification PositionSpeed PreviousTimingPoint TimingPointEstimation AdhesionConditionsChange ExpectedCurrentData TrainCharacteristicsChange JP_InUse C_DAS_C_Shown TrainReadyStatus DelayNotice teltsi_InterruptionPoint EnergyData PassengerData StatusReport_NSPs**
used by	elements **B2G_EventPayload B2G_ReplyPayload**

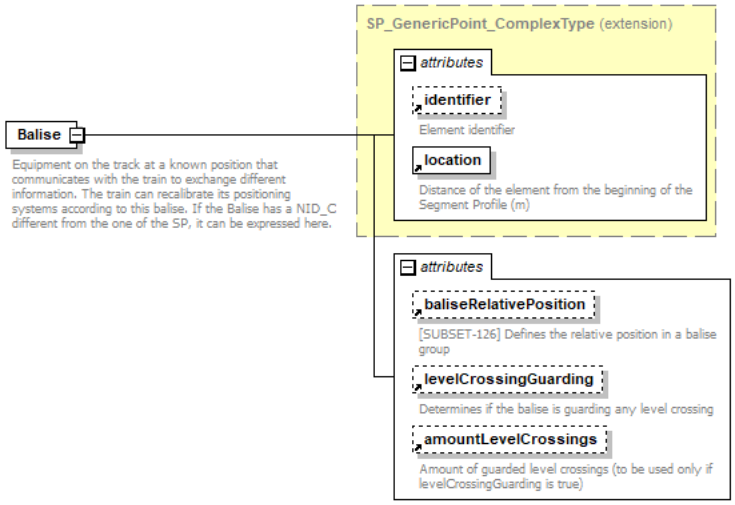
attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
ATO_State	**derived by: xs:string**				documentation [SUBSET-126] ATO State, from Subset-126. Only relevant when in ATO mode.
JP_SP_ConsistencyError	**xs:boolean**				documentation [SUBSET-126] ATO: JP SP Consistency Error, from Subset-126 (boolean)
nextStoppingPointSkip	**xs:boolean**				documentation [SUBSET-126] ATO: skip next stopping point, from Subset-126 (boolean)
slipSlideReported	**xs:boolean**				documentation [SUBSET-126] ATO: Slip/slide reported by TCMS/Train, from Subset-126 (boolean)
operationalConditionsFulfillment	**xs:boolean**				documentation [SUBSET-126] ATO: Operational conditions fulfilment, from Subset-126 (boolean)
trainMoving	**xs:boolean**				documentation [SUBSET-126] ATO: Train is moving, from Subset-126 (boolean)
unableToStop	**xs:boolean**				documentation [SUBSET-126] ATO: Unable to stop at the next Stopping Point, from Subset-126 (boolean)
driverID	**xs:string**				documentation [SUBSET-126] ATO: Driver identifier number as defined in SUBSET-027 v300 §4.2.3.7.
routingError	**xs:boolean**				documentation [SUBSET-126] ATO: Routing Error, from Subset-126 (boolean). This parameter can

	also be used by the DAS-OB to notify the DAS-TS that the train is not on the expected route.
annotation	documentation Status Report from Board to Ground
source	<xs:element name="B2G_StatusReport"> <xs:annotation> <xs:documentation>Status Report from Board to Ground</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> <xs:element ref="PositionSpeed" minOccurs="0"/> <xs:element ref="PreviousTimingPoint" minOccurs="0"/> <xs:element ref="TimingPointEstimation" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="AdhesionConditionsChange" minOccurs="0"/> <xs:element ref="ExpectedCurrentData" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TrainCharacteristicsChange" minOccurs="0"/> <xs:element ref="JP_InUse" minOccurs="0"/> <xs:element ref="C_DAS_C_Shown" minOccurs="0"/> <xs:element ref="TrainReadyStatus" minOccurs="0"/> <xs:element ref="DelayNotice" minOccurs="0"/> <xs:element ref="teltsi_InterruptionPoint" minOccurs="0"/> <xs:element ref="EnergyData" minOccurs="0"/> <xs:element ref="PassengerData" minOccurs="0"/> <xs:element ref="StatusReport_NSPs" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="ATO_State"/> <xs:attribute ref="JP_SP_ConsistencyError"/> <xs:attribute ref="nextStoppingPointSkip"/> <xs:attribute ref="slipSlideReported"/> <xs:attribute ref="operationalConditionsFulfillment"/> <xs:attribute ref="trainMoving"/> <xs:attribute ref="unableToStop"/> <xs:attribute ref="driverID"/> <xs:attribute ref="routingError"/> </xs:complexType> </xs:element>

element **Balise**

diagram



type: extension of **SP_GenericPoint_ComplexType**

properties: content complex

used by: element **BaliseGroup**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation: Element identifier
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)
baliseRelativePosition	**xs:unsignedByte**				documentation: [SUBSET-126] Defines the relative position in a balise group
levelCrossingGuarding	**xs:boolean**				documentation: Determines if the balise is guarding any level crossing
amountLevelCrossings	**xs:unsignedByte**				documentation: Amount of guarded level crossings (to be used only if levelCrossingGuarding is true)

annotation

documentation
Equipment on the track at a known position that communicates with the train to exchange different information. The train can recalibrate its positioning systems according to this balise. If the Balise has a NID_C different from the one of the SP, it can be expressed here.

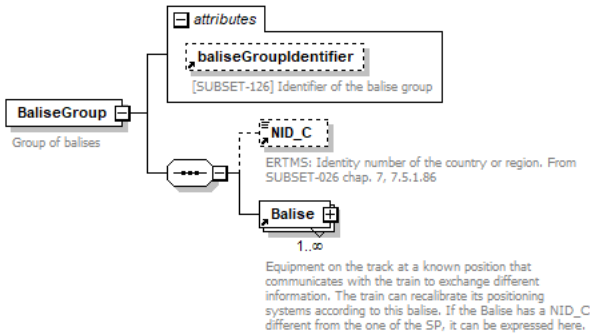
source

```
<xs:element name="Balise">
  <xs:annotation>
    <xs:documentation>Equipment on the track at a known position that communicates with the train to exchange different information. The train can recalibrate its positioning systems according to this balise. If the Balise has a NID_C different from the one of the SP, it can be expressed here.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SP_GenericPoint_ComplexType">
```

```
      <xs:attribute ref="baliseRelativePosition"/>
      <xs:attribute ref="levelCrossingGuarding"/>
      <xs:attribute ref="amountLevelCrossings"/>
    </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **BaliseGroup**

diagram



properties: content complex

children: **NID_C** **Balise**

used by: element **SP_Points**

attributes

Name	Type	Use	Default	Fixed	Annotation
baliseGroupIdentifier	**xs:unsignedShort**				documentation [SUBSET-126] Identifier of the balise group

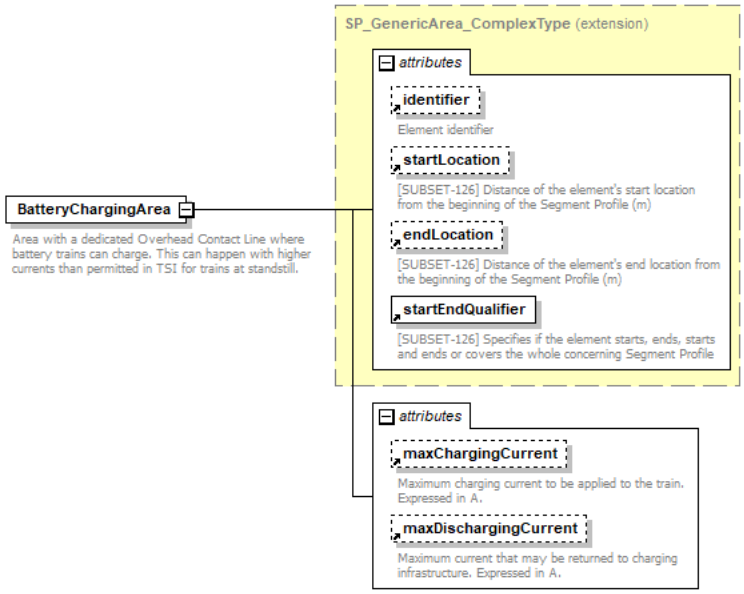
annotation: documentation
Group of balises

source

```
<xs:element name="BaliseGroup">
 <xs:annotation>
  <xs:documentation>Group of balises</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="NID_C" minOccurs="0"/>
   <xs:element ref="Balise" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="baliseGroupIdentifier"/>
 </xs:complexType>
</xs:element>
```

element **BatteryChargingArea**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

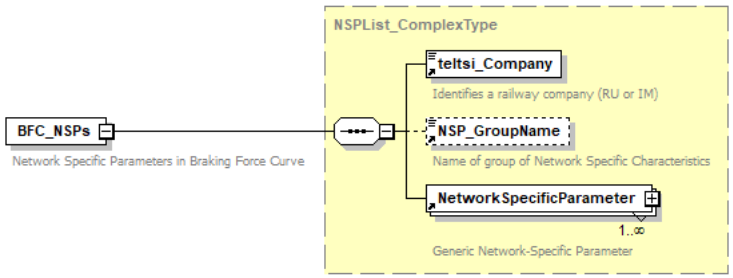
used by: element **SP_Areas**

attributes

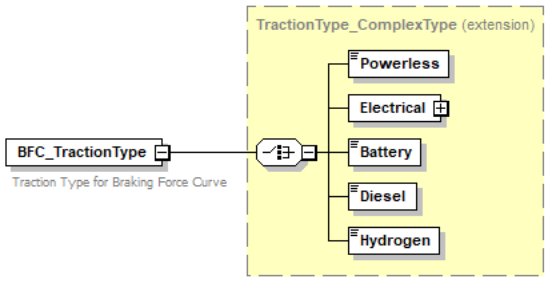
Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
maxChargingCurrent	**currentType**				documentation Maximum charging current to be

	maxDischargingCurrent **currentType** applied to the train. Expressed in A. documentation Maximum current that may be returned to charging infrastructure. Expressed in A.
annotation	documentation Area with a dedicated Overhead Contact Line where battery trains can charge. This can happen with higher currents than permitted in TSI for trains at standstill.
source	<xs:element name="BatteryChargingArea"> <xs:annotation> <xs:documentation>Area with a dedicated Overhead Contact Line where battery trains can charge. This can happen with higher currents than permitted in TSI for trains at standstill.</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericArea_ComplexType"> <xs:attribute ref="maxChargingCurrent"/> <xs:attribute ref="maxDischargingCurrent"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

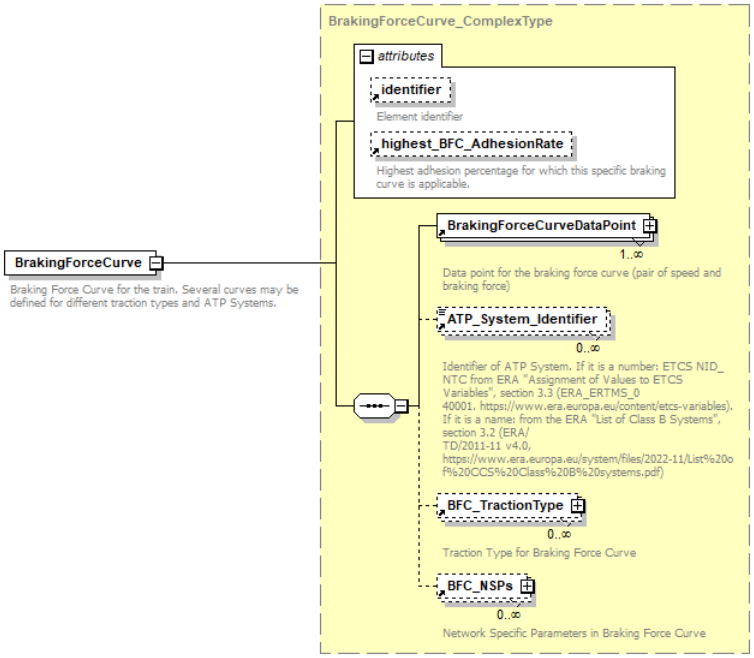
element **BFC_NSPs**

diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	complexType **BrakingForceCurve_ComplexType**
annotation	documentation Network Specific Parameters in Braking Force Curve
source	<xs:element name="BFC_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Braking Force Curve</xs:documentation> </xs:annotation> </xs:element>

element **BFC_TractionType**

diagram	
type	extension of **TractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Battery** **Diesel** **Hydrogen**
used by	complexType **BrakingForceCurve_ComplexType**
annotation	documentation Traction Type for Braking Force Curve
source	<xs:element name="BFC_TractionType"> <xs:annotation> <xs:documentation>Traction Type for Braking Force Curve</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="TractionType_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

element **BrakingForceCurve**

diagram	
type	**BrakingForceCurve_ComplexType**
properties	content complex
children	**BrakingForceCurveDataPoint** **ATP_System_Identifier** **BFC_TractionType** **BFC_NSPs**
used by	complexType **TC_Features_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
highest_BFC_AdhesionRate	**percentageType**				documentation Highest adhesion percentage for which this specific braking curve is applicable.

annotation

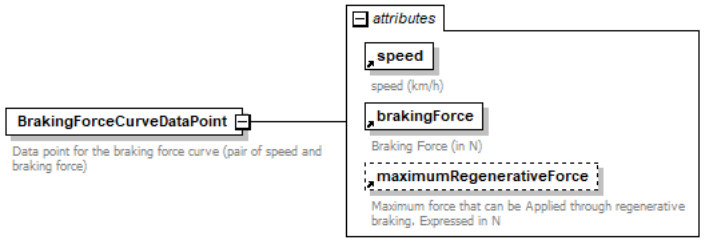
documentation
Braking Force Curve for the train. Several curves may be defined for different traction types and ATP Systems.

source

```
<xs:element name="BrakingForceCurve" type="BrakingForceCurve_ComplexType">
 <xs:annotation>
  <xs:documentation>Braking Force Curve for the train. Several curves may be defined for different traction types and ATP Systems.</xs:documentation>
 </xs:annotation>
</xs:element>
```

element **BrakingForceCurveDataPoint**

diagram



properties

content complex

used by

complexType **BrakingForceCurve_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
speed	**speedType**	required			documentation speed (km/h)
brakingForce	**forceType**	required			documentation Braking Force (in N)
maximumRegenerativeForce	**xs:double**				documentation Maximum force that can be Applied through regenerative braking. Expressed in N

annotation

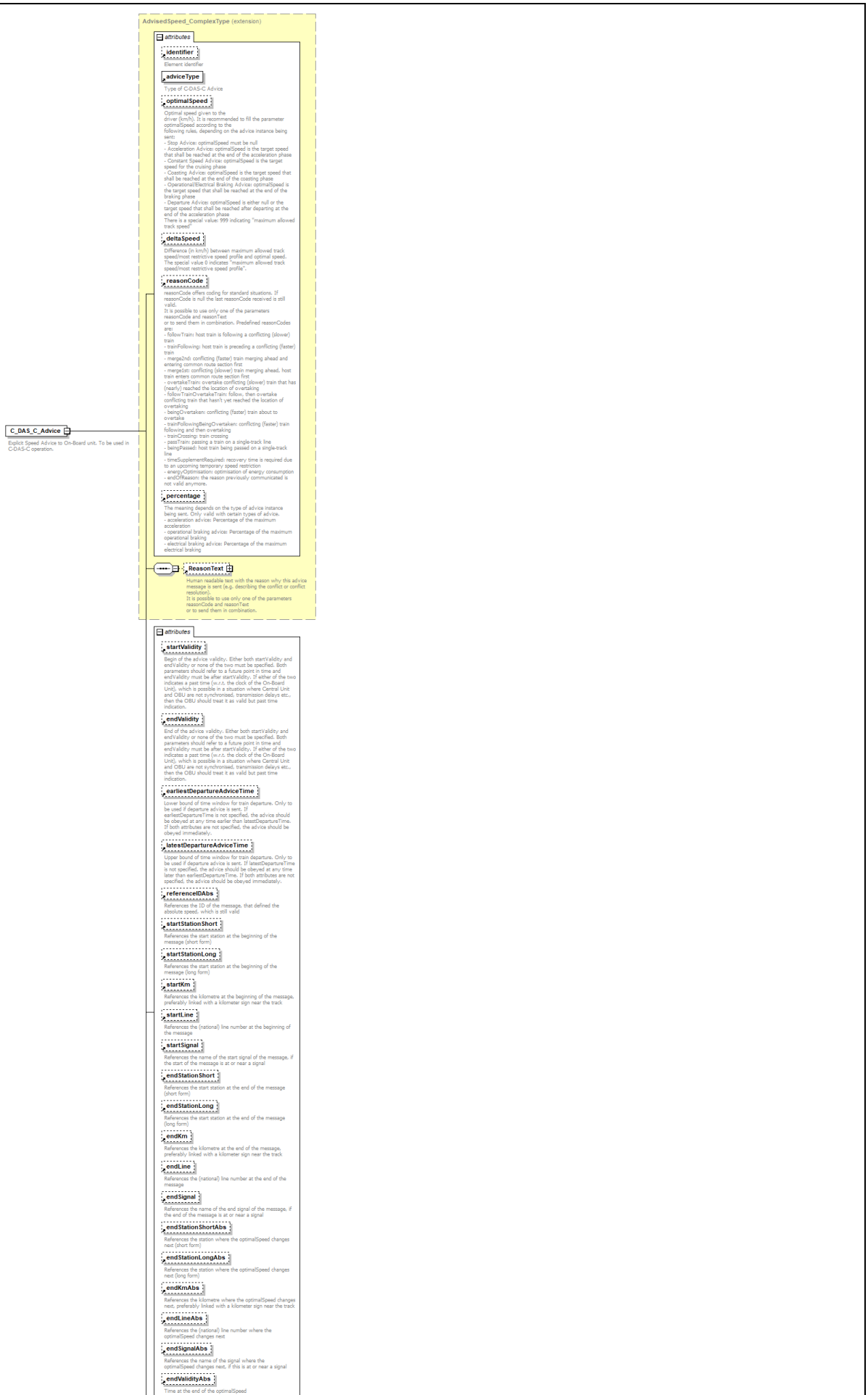
documentation
Data point for the braking force curve (pair of speed and braking force)

source

```
<xs:element name="BrakingForceCurveDataPoint">
  <xs:annotation>
    <xs:documentation>Data point for the braking force curve (pair of speed and braking force)</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="speed" use="required"/>
    <xs:attribute ref="brakingForce" use="required"/>
    <xs:attribute ref="maximumRegenerativeForce"/>
  </xs:complexType>
</xs:element>
```

element **C_DAS_C_Advice**

diagram


AdvisedSpeed_ComplexType (extension)
attributes
identifier
Element identifier
adviceType
Type of C-DAS-C Advice
optimalSpeed
Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on the advice instance being sent:
- Stop Advice: optimalSpeed must be null
- Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase
- Constant Speed Advice: optimalSpeed is the target speed for the cruising phase
- Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase
- Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase
- Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase
There is a special value: 999 indicating "maximum allowed track speed"
deltaSpeed
Difference (in km/h) between maximum allowed track speed(most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed(most restrictive speed profile".
reasonCode
reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid.
It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are:
- followTrain: host train is following a conflicting (slower) train
- trainFollowing: host train is preceding a conflicting (faster) train
- merge2nd: conflicting (faster) train merging ahead and entering common route section first
- merge1st: conflicting (slower) train merging ahead, host train enters common route section first
- overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking
- followTrainOvertakeTrain: follow, then overtake conflicting train that hasn't yet reached the location of overtaking
- beingOvertaken: conflicting (faster) train about to overtake
- trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking
- trainCrossing: train crossing
- passTrain: passing a train on a single-track line
- beingPassed: host train being passed on a single-track line
- timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction
- energyOptimisation: optimisation of energy consumption
- endOfReason: the reason previously communicated is not valid anymore.
percentage
The meaning depends on the type of advice instance being sent. Only valid with certain types of advice.
- acceleration advice: Percentage of the maximum acceleration
- operational braking advice: Percentage of the maximum operational braking
- electrical braking advice: Percentage of the maximum electrical braking
ReasonText
Human readable text with the reason why this advice message is sent (e.g. describing the conflict or conflict resolution).
It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination.
C_DAS_C_Advice
Explicit Speed Advice to On-Board unit. To be used in C-DAS-C operation.
attributes
startValidity
Begin of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
endValidity
End of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
earliestDepartureAdviceTime
Lower bound of time window for train departure. Only to be used if departure advice is sent. If earliestDepartureTime is not specified, the advice should be obeyed at any time earlier than latestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
latestDepartureAdviceTime
Upper bound of time window for train departure. Only to be used if departure advice is sent. If latestDepartureTime is not specified, the advice should be obeyed at any time later than earliestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
referenceIDAbs
References the ID of the message, that defined the absolute speed, which is still valid
startStationShort
References the start station at the beginning of the message (short form)
startStationLong
References the start station at the beginning of the message (long form)
startKm
References the kilometre at the beginning of the message, preferably linked with a kilometer sign near the track
startLine
References the (national) line number at the beginning of the message
startSignal
References the name of the start signal of the message, if the start of the message is at or near a signal
endStationShort
References the start station at the end of the message (short form)
endStationLong
References the start station at the end of the message (long form)
endKm
References the kilometre at the end of the message, preferably linked with a kilometer sign near the track
endLine
References the (national) line number at the end of the message
endSignal
References the name of the end signal of the message, if the end of the message is at or near a signal
endStationShortAbs
References the station where the optimalSpeed changes next (short form)
endStationLongAbs
References the station where the optimalSpeed changes next (long form)
endKmAbs
References the kilometre where the optimalSpeed changes next, preferably linked with a kilometer sign near the track
endLineAbs
References the (national) line number where the optimalSpeed changes next
endSignalAbs
References the name of the signal where the optimalSpeed changes next, if this is at or near a signal
endValidityAbs
Time at the end of the optimalSpeed

type	extension of **AdvisedSpeed_ComplexType**
properties	content complex
children	**ReasonText** **TrainIdentification** **SegmentProfileInfo** **LocationText**
used by	elements **G2B_EventPayload** **G2B_ReplyPayload**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
adviceType	**derived by: xs:string**	required			documentation Type of C-DAS-C Advice
optimalSpeed	**speedType**				documentation Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on the advice instance being sent: - Stop Advice: optimalSpeed must be null - Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase - Constant Speed Advice: optimalSpeed is the target speed for the cruising phase - Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase - Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase - Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase There is a special value: 999 indicating "maximum allowed track speed"
deltaSpeed	**speedType**				documentation Difference (in km/h) between maximum allowed track speed/most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed/most restrictive speed profile".
reasonCode	**derived by: xs:string**				documentation reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid. It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are:

			- followTrain: host train is following a conflicting (slower) train - trainFollowing: host train is preceding a conflicting (faster) train - merge2nd: conflicting (faster) train merging ahead and entering common route section first - merge1st: conflicting (slower) train merging ahead, host train enters common route section first - overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking - followTrainOvertakeTrain: follow, then overtake conflicting train that hasn't yet reached the location of overtaking - beingOvertaken: conflicting (faster) train about to overtake - trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking - trainCrossing: train crossing - passTrain: passing a train on a single-track line - beingPassed: host train being passed on a single-track line - timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction - energyOptimisation: optimisation of energy consumption - endOfReason: the reason previously communicated is not valid anymore.
	percentage	**percentageType**	documentation The meaning depends on the type of advice instance being sent. Only valid with certain types of advice. - acceleration advice: Percentage of the maximum acceleration - operational braking advice: Percentage of the maximum operational braking - electrical braking advice: Percentage of the maximum electrical braking
	startValidity	**UTCDateTimeType**	documentation Begin of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-

			Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
	endValidity	**UTCDateTimeType**	documentation End of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
	earliestDepartureAdviceTime	**UTCDateTimeType**	documentation Lower bound of time window for train departure. Only to be used if departure advice is sent. If earliestDepartureTime is not specified, the advice should be obeyed at any time earlier than latestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
	latestDepartureAdviceTime	**UTCDateTimeType**	documentation Upper bound of time window for train departure. Only to be used if departure advice is sent. If latestDepartureTime is not specified, the advice should be obeyed at any time later than earliestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
	referenceIDAbs	**UUIDv4_Type**	documentation References the ID of the message, that defined the absolute speed, which is still valid
	startStationShort	**xs:string**	documentation References the start station at the beginning of the message (short form)
	startStationLong	**xs:string**	documentation References the start station at the beginning of the message (long form)
	startKm	**xs:double**	documentation References the kilometre at the beginning of the message, preferably linked with a kilometer sign near the track
	startLine	**xs:int**	documentation References the (national) line number at the beginning of the

			message
	startSignal	**xs:string**	documentation References the name of the start signal of the message, if the start of the message is at or near a signal
	endStationShort	**xs:string**	documentation References the start station at the end of the message (short form)
	endStationLong	**xs:string**	documentation References the start station at the end of the message (long form)
	endKm	**xs:double**	documentation References the kilometre at the end of the message, preferably linked with a kilometer sign near the track
	endLine	**xs:int**	documentation References the (national) line number at the end of the message
	endSignal	**xs:string**	documentation References the name of the end signal of the message, if the end of the message is at or near a signal
	endStationShortAbs	**xs:string**	documentation References the station where the optimalSpeed changes next (short form)
	endStationLongAbs	**xs:string**	documentation References the station where the optimalSpeed changes next (long form)
	endKmAbs	**xs:double**	documentation References the kilometre where the optimalSpeed changes next, preferably linked with a kilometer sign near the track
	endLineAbs	**xs:int**	documentation References the (national) line number where the optimalSpeed changes next
	endSignalAbs	**xs:string**	documentation References the name of the signal where the optimalSpeed changes next, if this is at or near a signal
	endValidityAbs	**UTCDateTimeType**	documentation Time at the end of the optimalSpeed
annotation	documentation Explicit Speed Advice to On-Board unit. To be used in C-DAS-C operation.		

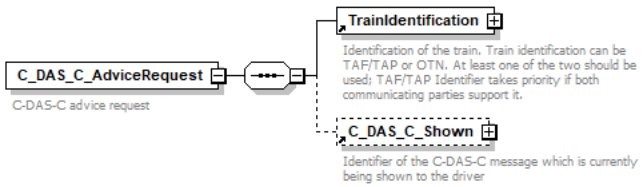
source

```
<xs:element name="C_DAS_C_Advice">
 <xs:annotation>
  <xs:documentation>Explicit Speed Advice to On-Board unit. To be used in C-DAS-C operation.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="AdvisedSpeed_ComplexType">
    <xs:sequence>
     <xs:element ref="TrainIdentification"/>
     <xs:element ref="SegmentProfileInfo" minOccurs="0" maxOccurs="unbounded"/>
```

```
        <xs:element ref="LocationText" minOccurs="0"/>
      </xs:sequence>
      <xs:attribute ref="startValidity"/>
      <xs:attribute ref="endValidity"/>
      <xs:attribute ref="earliestDepartureAdviceTime"/>
      <xs:attribute ref="latestDepartureAdviceTime"/>
      <xs:attribute ref="referenceIDAbs"/>
      <xs:attribute ref="startStationShort"/>
      <xs:attribute ref="startStationLong"/>
      <xs:attribute ref="startKm"/>
      <xs:attribute ref="startLine"/>
      <xs:attribute ref="startSignal"/>
      <xs:attribute ref="endStationShort"/>
      <xs:attribute ref="endStationLong"/>
      <xs:attribute ref="endKm"/>
      <xs:attribute ref="endLine"/>
      <xs:attribute ref="endSignal"/>
      <xs:attribute ref="endStationShortAbs"/>
      <xs:attribute ref="endStationLongAbs"/>
      <xs:attribute ref="endKmAbs"/>
      <xs:attribute ref="endLineAbs"/>
      <xs:attribute ref="endSignalAbs"/>
      <xs:attribute ref="endValidityAbs"/>
    </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **C_DAS_C_AdviceRequest**

diagram



properties

content complex

children **TrainIdentification C_DAS_C_Shown**

used by

element **B2G_Request**

annotation

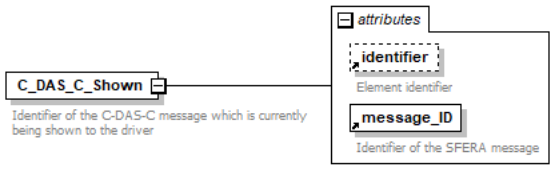
documentation
C-DAS-C advice request

source

```
<xs:element name="C_DAS_C_AdviceRequest">
 <xs:annotation>
  <xs:documentation>C-DAS-C advice request</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="TrainIdentification"/>
   <xs:element ref="C_DAS_C_Shown" minOccurs="0"/>
  </xs:sequence>
 </xs:complexType>
</xs:element>
```

element **C_DAS_C_Shown**

diagram



properties: content complex

used by: elements **B2G_StatusReport** **C_DAS_C_AdviceRequest**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
message_ID	**UUIDv4_Type**	required			documentation Identifier of the SFERA message

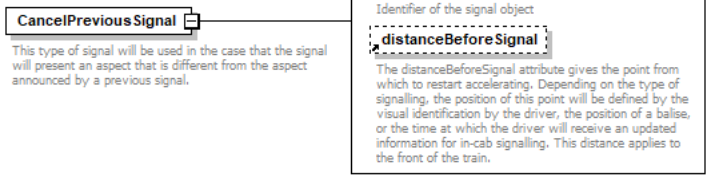
annotation: documentation
Identifier of the C-DAS-C message which is currently being shown to the driver

source

```
<xs:element name="C_DAS_C_Shown">
  <xs:annotation>
    <xs:documentation>Identifier of the C-DAS-C message which is currently being shown to the driver</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="identifier"/>
    <xs:attribute ref="message_ID" use="required"/>
  </xs:complexType>
</xs:element>
```

element **CancelPreviousSignal**

diagram



properties: content complex

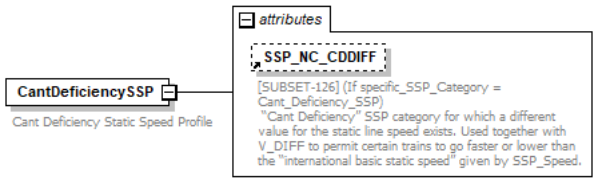
used by: element **SignalInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
signal_ID_Physical	**xs:string**	required			documentation Identifier of the physical signal on the track
signal_ID_Object	**xs:string**				documentation Identifier of the signal object

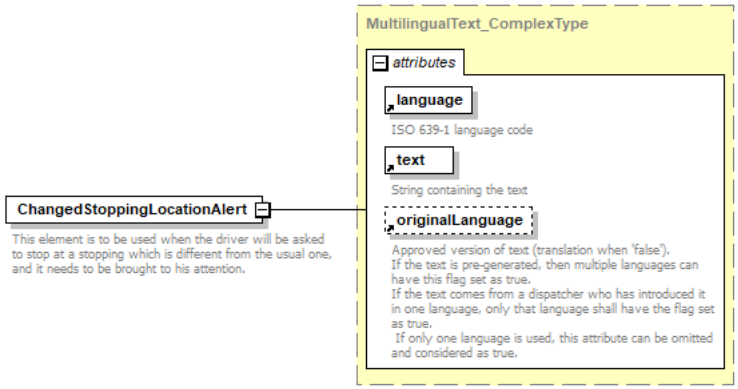
	distanceBeforeSignal **distanceType**	documentation The distanceBeforeSignal attribute gives the point from which to restart accelerating. Depending on the type of signalling, the position of this point will be defined by the visual identification by the driver, the position of a balise, or the time at which the driver will receive an updated information for in-cab signalling. This distance applies to the front of the train.
annotation	documentation This type of signal will be used in the case that the signal will present an aspect that is different from the aspect announced by a previous signal.	
source	<xs:element name="CancelPreviousSignal"> <xs:annotation> <xs:documentation>This type of signal will be used in the case that the signal will present an aspect that is different from the aspect announced by a previous signal. </xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="signal_ID_Physical" use="required"/> <xs:attribute ref="signal_ID_Object"/> <xs:attribute ref="distanceBeforeSignal"/> </xs:complexType> </xs:element>	

element **CantDeficiencySSP**

diagram					
properties	content complex				
used by	complexType **SpecificSSP_ComplexType**				
attributes	Name SSP_NC_CDDIFF	Type **NC_CDTrainType**	Use	Default	Fixed	Annotation documentation [SUBSET-126] (If specific_SSP_Category = Cant_Deficiency_SSP) "Cant Deficiency" SSP category for which a different value for the static line speed exists. Used together with V_DIFF to permit certain trains to go faster or lower than the "international basic static speed" given by

	SSP_Speed.
annotation	documentation Cant Deficiency Static Speed Profile
source	<xs:element name="CantDeficiencySSP"> <xs:annotation> <xs:documentation>Cant Deficiency Static Speed Profile</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="SSP_NC_CDDIFF"/> </xs:complexType> </xs:element>

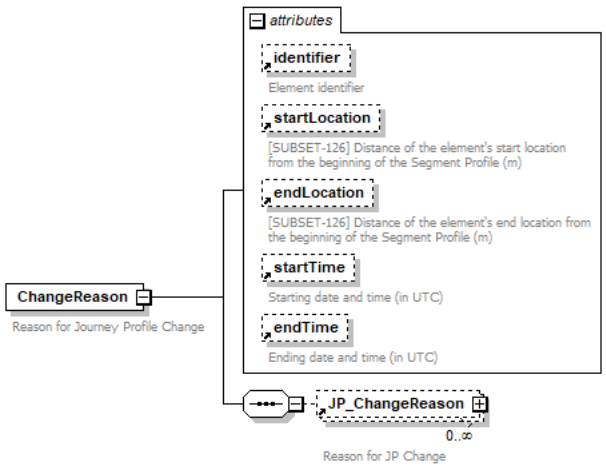
element **ChangedStoppingLocationAlert**

diagram	
type	**MultilingualText_ComplexType**
properties	content complex
used by	element **StoppingPointInformation**

attributes	Name	Type	Use	Default	Fixed	Annotation
	language	**derived by: xs:string**	required			documentation ISO 639-1 language code
	text	**xs:string**	required			documentation String containing the text
	originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag

	set as true. If only one language is used, this attribute can be omitted and considered as true.
annotation	documentation This element is to be used when the driver will be asked to stop at a stopping which is different from the usual one, and it needs to be brought to his attention.
source	<xs:element name="ChangedStoppingLocationAlert" type="MultilingualText_ComplexType"> <xs:annotation> <xs:documentation>This element is to be used when the driver will be asked to stop at a stopping which is different from the usual one, and it needs to be brought to his attention.</xs:documentation> </xs:annotation> </xs:element>

element **ChangeReason**

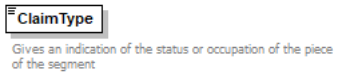
diagram	
properties	content complex
children	**JP_ChangeReason**
used by	element **SegmentProfileReference**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location

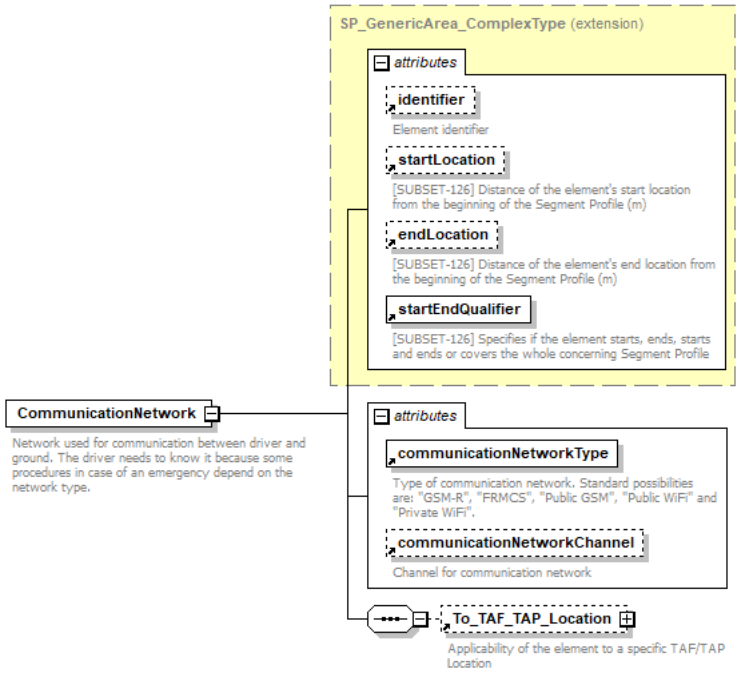
	startTime **UTCDateTimeType** from the beginning of the Segment Profile (m) documentation Starting date and time (in UTC) endTime **UTCDateTimeType** documentation Ending date and time (in UTC)
annotation	documentation Reason for Journey Profile Change
source	<xs:element name="ChangeReason"> <xs:annotation> <xs:documentation>Reason for Journey Profile Change</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="JP_ChangeReason" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="startLocation"/> <xs:attribute ref="endLocation"/> <xs:attribute ref="startTime"/> <xs:attribute ref="endTime"/> </xs:complexType> </xs:element>

element **ClaimType**

diagram	
type	**ClaimTypeType**
properties	content simple
used by	element **SegmentOccupationStatus**
facets	Kind Value Annotation enumeration Planned enumeration Reserved enumeration ReservedImplicit enumeration Occupied enumeration Unknown
annotation	documentation Gives an indication of the status or occupation of the piece of the segment
source	<xs:element name="ClaimType" type="ClaimTypeType"> <xs:annotation> <xs:documentation>Gives an indication of the status or occupation of the piece of the segment</xs:documentation> </xs:annotation> </xs:element>

element **CommunicationNetwork**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

children: **To_TAF_TAP_Location**

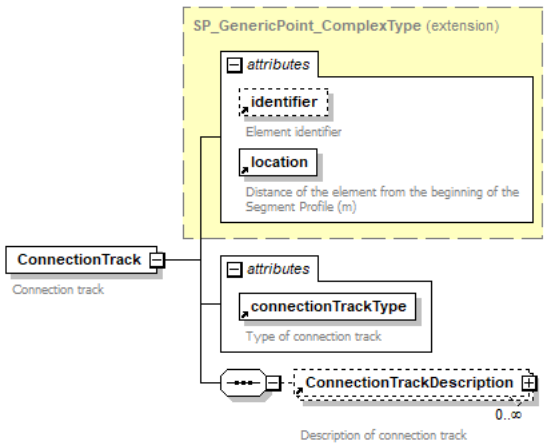
used by: element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

	communicationNetworkType	**xs:string**	required	documentation Type of communication network. Standard possibilities are: "GSM-R", "FRMCS", "Public GSM", "Public WiFi" and "Private WiFi".
	communicationNetworkChannel	**xs:string**		documentation Channel for communication network
annotation	documentation Network used for communication between driver and ground. The driver needs to know it because some procedures in case of an emergency depend on the network type.			
source	<xs:element name="CommunicationNetwork"> <xs:annotation> <xs:documentation>Network used for communication between driver and ground. The driver needs to know it because some procedures in case of an emergency depend on the network type.</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericArea_ComplexType"> <xs:sequence> <xs:element ref="To_TAF_TAP_Location" minOccurs="0"/> </xs:sequence> <xs:attribute ref="communicationNetworkType" use="required"/> <xs:attribute ref="communicationNetworkChannel"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>			

element **ConnectionTrack**

diagram	
type	extension of **SP_GenericPoint_ComplexType**
properties	content complex

children	**ConnectionTrackDescription**
used by	element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
connectionTrackType	**derived by: xs:string**	required			documentation Type of connection track

annotation

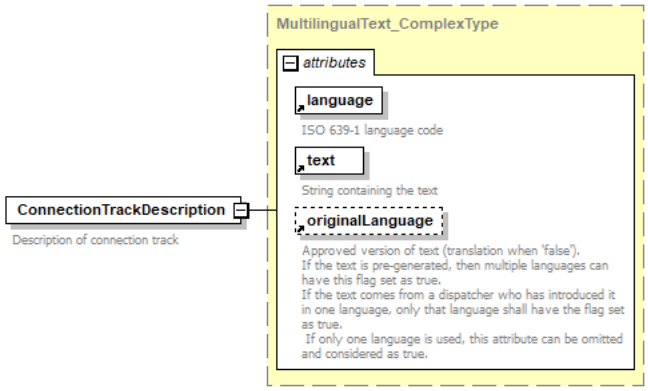
documentation
Connection track

source

```
<xs:element name="ConnectionTrack">
 <xs:annotation>
  <xs:documentation>Connection track</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericPoint_ComplexType">
    <xs:sequence>
     <xs:element ref="ConnectionTrackDescription" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="connectionTrackType" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **ConnectionTrackDescription**

diagram



type	**MultilingualText_ComplexType**
properties	content complex
used by	element **ConnectionTrack**

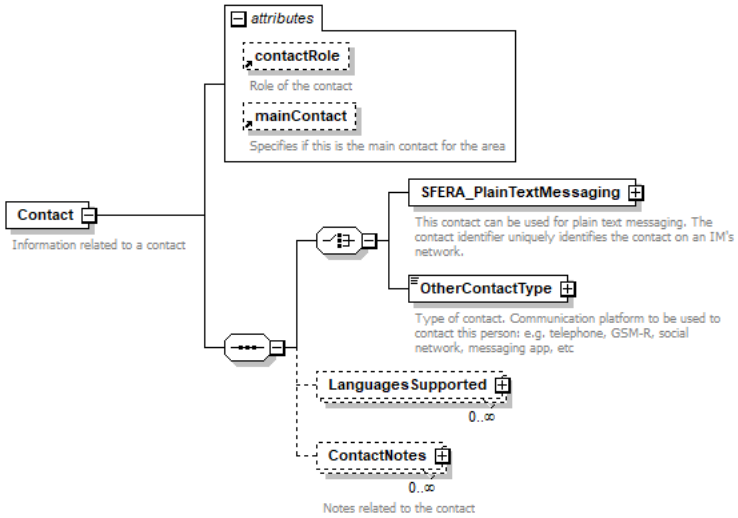
attributes

Name	Type	Use	Default	Fixed	Annotation

	language	**derived by: xs:string**	required	documentation ISO 639-1 language code
	text	**xs:string**	required	documentation String containing the text
	originalLanguage	**xs:boolean**		documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
annotation	documentation Description of connection track			
source	<xs:element name="ConnectionTrackDescription" type="MultilingualText_ComplexType"> <xs:annotation> <xs:documentation>Description of connection track</xs:documentation> </xs:annotation> </xs:element>			

element **Contact**

diagram



properties: content complex

children: **SFERA_PlainTextMessaging** **OtherContactType** **LanguagesSupported** **ContactNotes**

used by: element **ContactList**

attributes

Name	Type	Use	Default	Fixed	Annotation
contactRole	**xs:string**				documentation Role of the contact
mainContact	**xs:boolean**				documentation Specifies if this is the main contact for the area

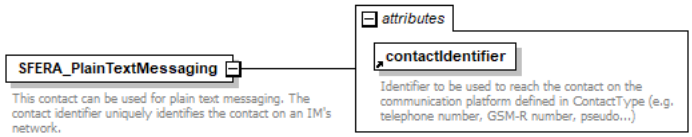
annotation: documentation
Information related to a contact

source

```
<xs:element name="Contact">
 <xs:annotation>
  <xs:documentation>Information related to a contact</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:choice>
    <xs:element name="SFERA_PlainTextMessaging">
     <xs:annotation>
      <xs:documentation>This contact can be used for plain text messaging. The contact identifier uniquely identifies the contact on an IM's network.</xs:documentation>
     </xs:annotation>
     <xs:complexType>
      <xs:attribute ref="contactIdentifier" use="required"/>
     </xs:complexType>
    </xs:element>
    <xs:element name="OtherContactType">
     <xs:annotation>
      <xs:documentation>Type of contact. Communication platform to be used to contact this person: e.g. telephone, GSM-R, social network, messaging app, etc</xs:documentation>
     </xs:annotation>
     <xs:complexType>
      <xs:simpleContent>
```

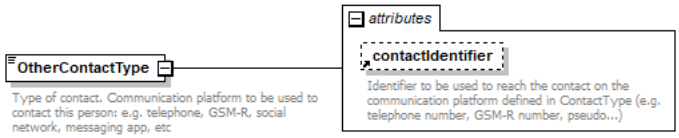
```
            <xs:extension base="xs:string">
              <xs:attribute ref="contactIdentifier"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:choice>
    <xs:element name="LanguagesSupported" minOccurs="0" maxOccurs="unbounded">
      <xs:complexType>
        <xs:attribute ref="language" use="required"/>
        <xs:attribute ref="originalLanguage"/>
      </xs:complexType>
    </xs:element>
    <xs:element name="ContactNotes" type="MultilingualText_ComplexType" minOccurs="0"
maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation>Notes related to the contact</xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
  <xs:attribute ref="contactRole"/>
  <xs:attribute ref="mainContact"/>
 </xs:complexType>
</xs:element>
```

element **Contact/SFERA_PlainTextMessaging**

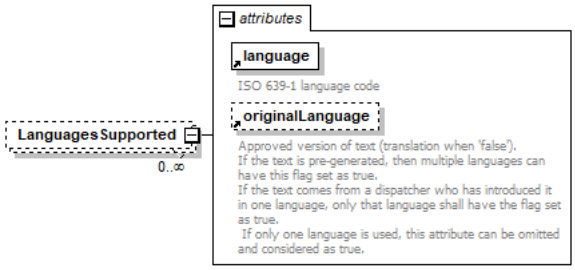
diagram	
properties	content complex
attributes	Name: contactIdentifier; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation: Identifier to be used to reach the contact on the communication platform defined in ContactType (e.g. telephone number, GSM-R number, pseudo...)
annotation	documentation This contact can be used for plain text messaging. The contact identifier uniquely identifies the contact on an IM's network.
source	<xs:element name="SFERA_PlainTextMessaging"> <xs:annotation> <xs:documentation>This contact can be used for plain text messaging. The contact identifier uniquely identifies the contact on an IM's network.</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="contactIdentifier" use="required"/>

```
  </xs:complexType>
</xs:element>
```

element **Contact/OtherContactType**

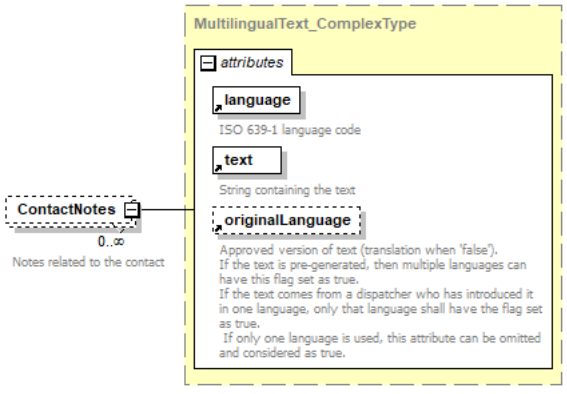
diagram	
type	extension of **xs:string**
properties	content complex
attributes	Name: contactIdentifier; Type: **xs:string**; Use: ; Default: ; Fixed: ; Annotation: documentation: Identifier to be used to reach the contact on the communication platform defined in ContactType (e.g. telephone number, GSM-R number, pseudo...)
annotation	documentation Type of contact. Communication platform to be used to contact this person: e.g. telephone, GSM-R, social network, messaging app, etc
source	<xs:element name="OtherContactType"> <xs:annotation> <xs:documentation>Type of contact. Communication platform to be used to contact this person: e.g. telephone, GSM-R, social network, messaging app, etc</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="xs:string"> <xs:attribute ref="contactIdentifier"/> </xs:extension> </xs:simpleContent> </xs:complexType> </xs:element>

element **Contact/LanguagesSupported**

diagram	
properties	minOcc 0 maxOcc unbounded

	content complex
attributes	Name: language — Type: derived by: xs:string — Use: required — Default: — Fixed: — Annotation: documentation ISO 639-1 language code Name: originalLanguage — Type: xs:boolean — Use: — Default: — Fixed: — Annotation: documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
source	<xs:element name="LanguagesSupported" minOccurs="0" maxOccurs="unbounded"> <xs:complexType> <xs:attribute ref="language" use="required"/> <xs:attribute ref="originalLanguage"/> </xs:complexType> </xs:element>

element **Contact/ContactNotes**

diagram	
type	**MultilingualText_ComplexType**
properties	minOcc 0 maxOcc unbounded content complex

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

annotation

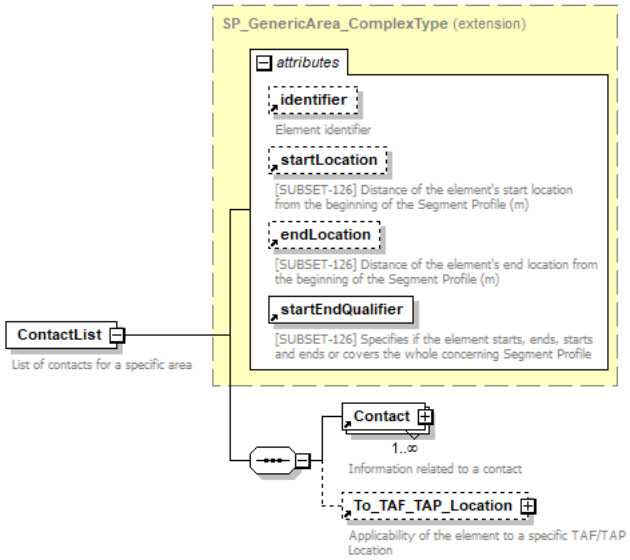
documentation
Notes related to the contact

source

```
<xs:element name="ContactNotes" type="MultilingualText_ComplexType" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>Notes related to the contact</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **ContactList**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

children: **Contact** **To_TAF_TAP_Location**

used by: element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

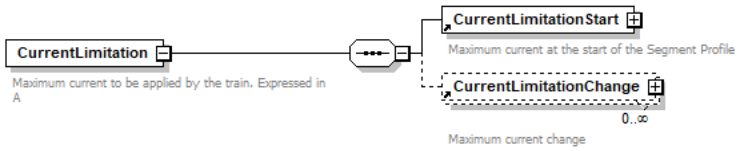
annotation: documentation
List of contacts for a specific area

source

```
<xs:element name="ContactList">
```

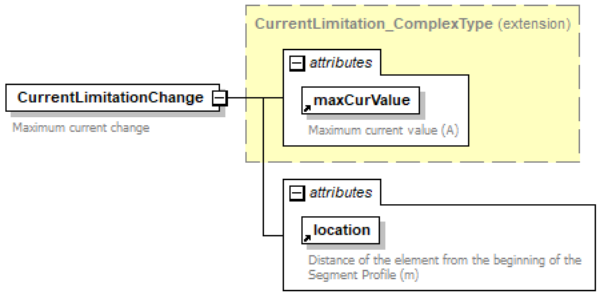
```
<xs:annotation>
  <xs:documentation>List of contacts for a specific area</xs:documentation>
</xs:annotation>
<xs:complexType>
  <xs:complexContent>
    <xs:extension base="SP_GenericArea_ComplexType">
      <xs:sequence>
        <xs:element ref="Contact" maxOccurs="unbounded"/>
        <xs:element ref="To_TAF_TAP_Location" minOccurs="0"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
</xs:element>
```

element **CurrentLimitation**

diagram	
properties	content complex
children	**CurrentLimitationStart** **CurrentLimitationChange**
used by	element **SP_Characteristics**
annotation	documentation Maximum current to be applied by the train. Expressed in A
source	<xs:element name="CurrentLimitation"> <xs:annotation> <xs:documentation>Maximum current to be applied by the train. Expressed in A</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="CurrentLimitationStart"/> <xs:element ref="CurrentLimitationChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **CurrentLimitationChange**

diagram



type: extension of **CurrentLimitation_ComplexType**

properties: content complex

used by: element **CurrentLimitation**

attributes

Name	Type	Use	Default	Fixed	Annotation
maxCurValue	**currentType**	required			documentation: Maximum current value (A)
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

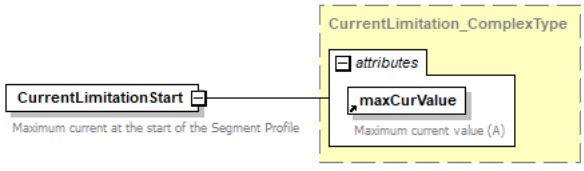
annotation: documentation
Maximum current change

source

```
<xs:element name="CurrentLimitationChange">
 <xs:annotation>
  <xs:documentation>Maximum current change</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="CurrentLimitation_ComplexType">
    <xs:attribute ref="location" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **CurrentLimitationStart**

diagram



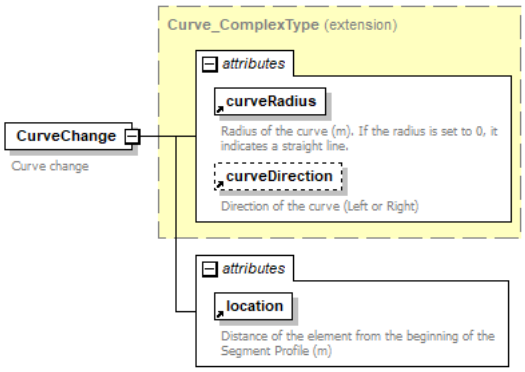
type: **CurrentLimitation_ComplexType**

properties: content complex

used by: element **CurrentLimitation**

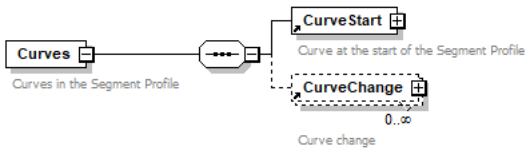
attributes	Name	Type	Use	Default	Fixed	Annotation
	maxCurValue	**currentType**	required			documentation Maximum current value (A)
annotation	documentation Maximum current at the start of the Segment Profile					
source	<xs:element name="CurrentLimitationStart" type="CurrentLimitation_ComplexType"> <xs:annotation> <xs:documentation>Maximum current at the start of the Segment Profile</xs:documentation> </xs:annotation> </xs:element>					

element **CurveChange**

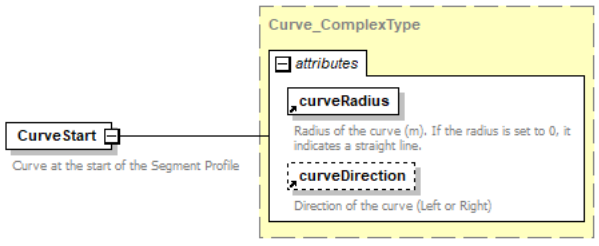
diagram						
type	extension of **Curve_ComplexType**					
properties	content complex					
used by	element **Curves**					
attributes	Name	Type	Use	Default	Fixed	Annotation
	curveRadius	**xs:unsignedShort**	required			documentation Radius of the curve (m). If the radius is set to 0, it indicates a straight line.
	curveDirection	**derived by: xs:string**				documentation Direction of the curve (Left or Right)
	location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
annotation	documentation Curve change					
source	<xs:element name="CurveChange"> <xs:annotation> <xs:documentation>Curve change</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent>					

```
      <xs:extension base="Curve_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **Curves**

diagram	
properties	content complex
children	**CurveStart** **CurveChange**
used by	element **SP_Characteristics**
annotation	documentation Curves in the Segment Profile
source	<xs:element name="Curves"> <xs:annotation> <xs:documentation>Curves in the Segment Profile</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="CurveStart"/> <xs:element ref="CurveChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

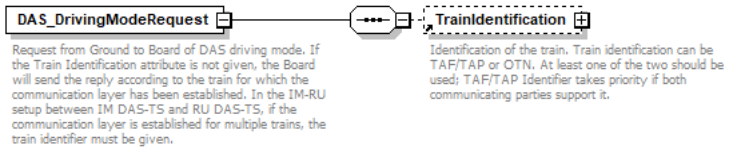
element **CurveStart**

diagram	
type	**Curve_ComplexType**
properties	content complex
used by	element **Curves**

attributes	Name	Type	Use	Default	Fixed	Annotation
	curveRadius	**xs:unsignedShort**	required			documentation Radius of the curve (m). If the radius is set to 0, it indicates a

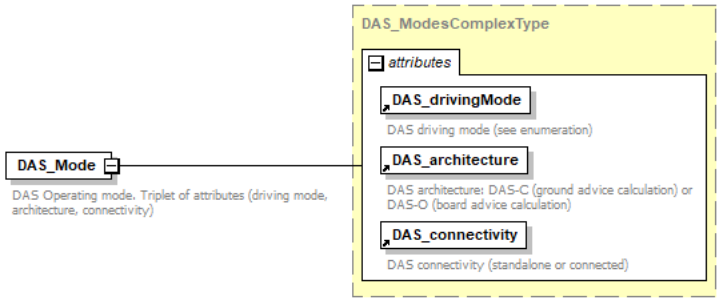
	curveDirection **derived by: xs:string** straight line. documentation Direction of the curve (Left or Right)
annotation	documentation Curve at the start of the Segment Profile
source	<xs:element name="CurveStart" type="Curve_ComplexType"> <xs:annotation> <xs:documentation>Curve at the start of the Segment Profile</xs:documentation> </xs:annotation> </xs:element>

element **DAS_DrivingModeRequest**

diagram	
properties	content complex
children	**TrainIdentification**
used by	element **G2B_Request**
annotation	documentation Request from Ground to Board of DAS driving mode. If the Train Identification attribute is not given, the Board will send the reply according to the train for which the communication layer has been established. In the IM-RU setup between IM DAS-TS and RU DAS-TS, if the communication layer is established for multiple trains, the train identifier must be given.
source	<xs:element name="DAS_DrivingModeRequest"> <xs:annotation> <xs:documentation>Request from Ground to Board of DAS driving mode. If the Train Identification attribute is not given, the Board will send the reply according to the train for which the communication layer has been established. In the IM-RU setup between IM DAS-TS and RU DAS-TS, if the communication layer is established for multiple trains, the train identifier must be given.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **DAS_Mode**

diagram



type **DAS_ModesComplexType**

properties
content complex

attributes

Name	Type	Use	Default	Fixed	Annotation
DAS_drivingMode	**derived by: xs:string**	required			documentation DAS driving mode (see enumeration)
DAS_architecture	**derived by: xs:string**	required			documentation DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation)
DAS_connectivity	**derived by: xs:string**	required			documentation DAS connectivity (standalone or connected)

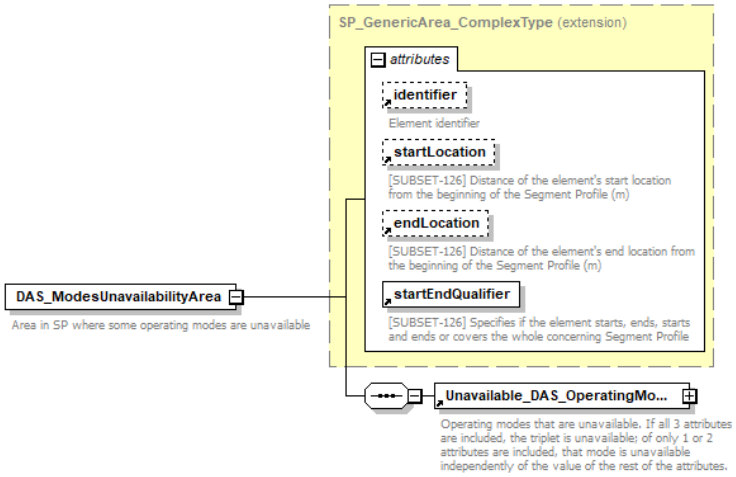
annotation
documentation
DAS Operating mode. Triplet of attributes (driving mode, architecture, connectivity)

source

```
<xs:element name="DAS_Mode" type="DAS_ModesComplexType">
  <xs:annotation>
    <xs:documentation>DAS Operating mode. Triplet of attributes (driving mode, architecture, connectivity)</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **DAS_ModesUnavailabilityArea**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

children: **Unavailable_DAS_OperatingModes**

used by: element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation: Element identifier
startLocation	**distanceType**				documentation: [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation: [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation: [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

annotation

documentation
Area in SP where some operating modes are unavailable

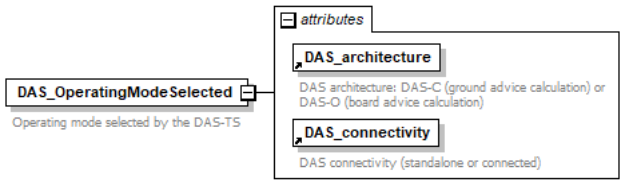
source

```
<xs:element name="DAS_ModesUnavailabilityArea">
  <xs:annotation>
    <xs:documentation>Area in SP where some operating modes are unavailable</xs:documentation>
```

```
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SP_GenericArea_ComplexType">
        <xs:sequence>
          <xs:element ref="Unavailable_DAS_OperatingModes"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **DAS_OperatingModeSelected**

diagram



properties

content complex

used by

element **HandshakeAcknowledgement**

attributes

Name	Type	Use	Default	Fixed	Annotation
DAS_architecture	**derived by: xs:string**	required			documentation DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation)
DAS_connectivity	**derived by: xs:string**	required			documentation DAS connectivity (standalone or connected)

annotation

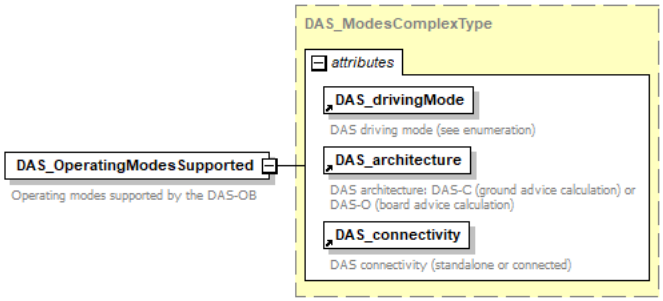
documentation
Operating mode selected by the DAS-TS

source

```
<xs:element name="DAS_OperatingModeSelected">
  <xs:annotation>
    <xs:documentation>Operating mode selected by the DAS-TS</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="DAS_architecture" use="required"/>
    <xs:attribute ref="DAS_connectivity" use="required"/>
  </xs:complexType>
</xs:element>
```

element **DAS_OperatingModesSupported**

diagram



type **DAS_ModesComplexType**

properties
- content complex

used by
- element **HandshakeRequest**

attributes

Name	Type	Use	Default	Fixed	Annotation
DAS_drivingMode	**derived by: xs:string**	required			documentation DAS driving mode (see enumeration)
DAS_architecture	**derived by: xs:string**	required			documentation DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation)
DAS_connectivity	**derived by: xs:string**	required			documentation DAS connectivity (standalone or connected)

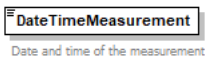
annotation
documentation
Operating modes supported by the DAS-OB

source

```
<xs:element name="DAS_OperatingModesSupported" type="DAS_ModesComplexType">
  <xs:annotation>
    <xs:documentation>Operating modes supported by the DAS-OB</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **DateTimeMeasurement**

diagram



type **UTCDateTimeType**

properties
- content simple

used by
- element **TrainLocationInformation**

facets

Kind	Value	Annotation
pattern	\d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z	

annotation
documentation
Date and time of the measurement

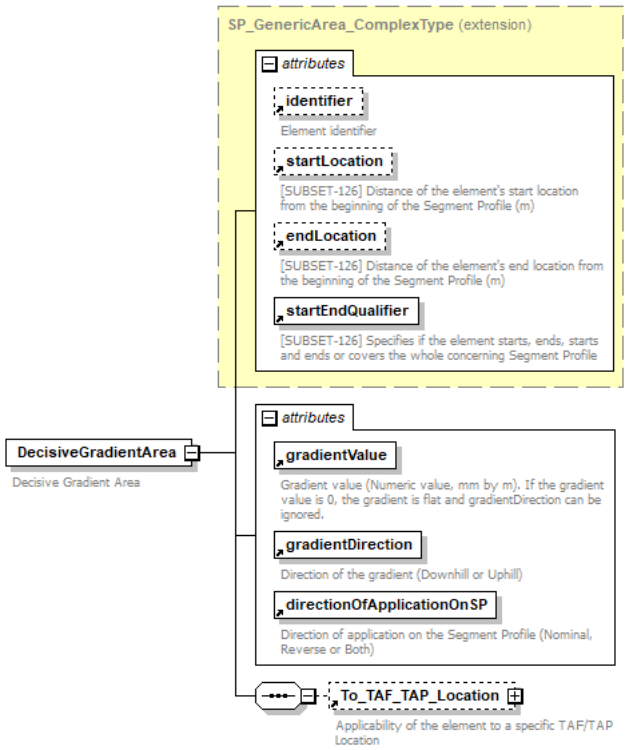
source

```
<xs:element name="DateTimeMeasurement" type="UTCDateTimeType">
```

```
  <xs:annotation>
    <xs:documentation>Date and time of the measurement</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **DecisiveGradientArea**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

children: **To_TAF_TAP_Location**

used by: element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation

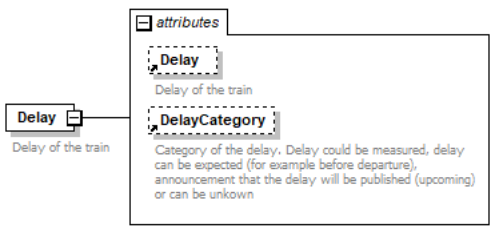
	gradientValue	**gradientValueType**	required	[SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile documentation Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
	gradientDirection	**gradientDirectionType**	required	documentation Direction of the gradient (Downhill or Uphill)
	directionOfApplicationOnSP	**derived by: xs:string**	required	documentation Direction of application on the Segment Profile (Nominal, Reverse or Both)
annotation	documentation Decisive Gradient Area			

source

```
<xs:element name="DecisiveGradientArea">
 <xs:annotation>
  <xs:documentation>Decisive Gradient Area</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericArea_ComplexType">
    <xs:sequence>
     <xs:element ref="To_TAF_TAP_Location" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute ref="gradientValue" use="required"/>
    <xs:attribute ref="gradientDirection" use="required"/>
    <xs:attribute ref="directionOfApplicationOnSP" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **Delay**

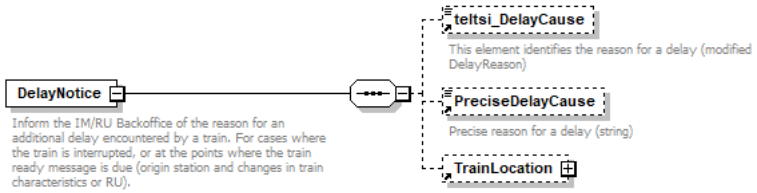
diagram



properties	content complex

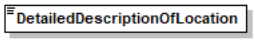
used by	element **TrainLocationInformation**
attributes	Name: Delay — Type: **xs:duration** — Use: — Default: — Fixed: — Annotation: documentation: Delay of the train Name: DelayCategory — Type: **derived by: xs:string** — Use: — Default: — Fixed: — Annotation: documentation: Category of the delay. Delay could be measured, delay can be expected (for example before departure), announcement that the delay will be published (upcoming) or can be unkown
annotation	documentation Delay of the train
source	<xs:element name="Delay"> <xs:annotation> <xs:documentation>Delay of the train</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="Delay"/> <xs:attribute ref="DelayCategory"/> </xs:complexType> </xs:element>

element **DelayNotice**

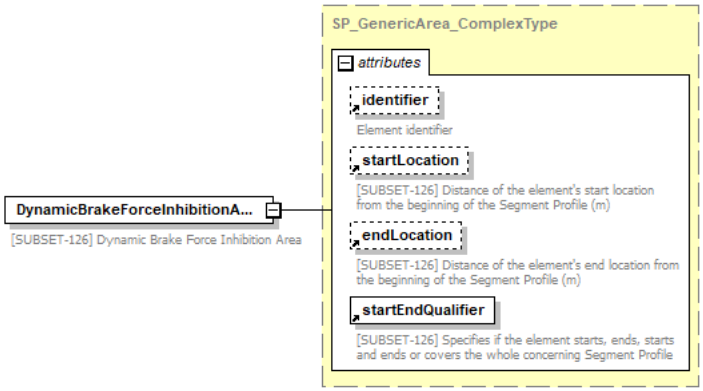
diagram	
properties	content complex
children	**teltsi_DelayCause PreciseDelayCause TrainLocation**
used by	element **B2G_StatusReport**
annotation	documentation Inform the IM/RU Backoffice of the reason for an additional delay encountered by a train. For cases where the train is interrupted, or at the points where the train ready message is due (origin station and changes in train characteristics or RU).
source	<xs:element name="DelayNotice"> <xs:annotation> <xs:documentation>Inform the IM/RU Backoffice of the reason for an additional delay encountered by a train. For cases where the train is interrupted, or at the points where the train ready message is due (origin station and changes in train characteristics or

```
RU).</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="teltsi_DelayCause" minOccurs="0"/>
      <xs:element ref="PreciseDelayCause" minOccurs="0"/>
      <xs:element ref="TrainLocation" minOccurs="0"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **DetailedDescriptionOfLocation**

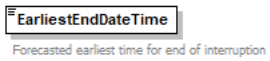
diagram	
type	**teltsi_FreeText**
properties	content simple
facets	Kind Value Annotation minLength 1 maxLength 255
source	<xs:element name="DetailedDescriptionOfLocation" type="teltsi_FreeText"/>

element **DynamicBrakeForceInhibitionArea**

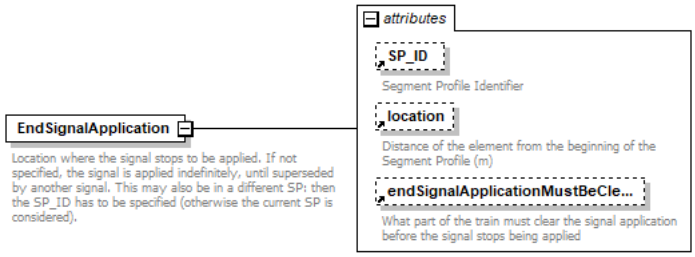
diagram	
type	**SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**
attributes	Name Type Use Default Fixed Annotation identifier **xs:string** documentation Element identifier startLocation **distanceType** documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment

	endLocation	**distanceType**		Profile (m) documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
	startEndQualifier	**derived by: xs:string**	required	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation [SUBSET-126] Dynamic Brake Force Inhibition Area			
source	<xs:element name="DynamicBrakeForceInhibitionArea" type="SP_GenericArea_ComplexType"> <xs:annotation> <xs:documentation>[SUBSET-126] Dynamic Brake Force Inhibition Area</xs:documentation> </xs:annotation> </xs:element>			

element **EarliestEndDateTime**

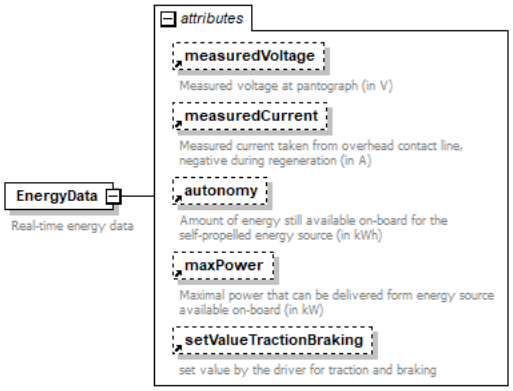
diagram	
type	**xs:dateTime**
properties	content simple
annotation	documentation Forecasted earliest time for end of interruption
source	<xs:element name="EarliestEndDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Forecasted earliest time for end of interruption</xs:documentation> </xs:annotation> </xs:element>

element **EndSignalApplication**

diagram	
properties	content complex

used by	element **SignalApplication**
attributes	Name / Type / Use / Default / Fixed / Annotation SP_ID — **derived by: xs:string** — documentation: Segment Profile Identifier location — **distanceType** — documentation: Distance of the element from the beginning of the Segment Profile (m) endSignalApplicationMustBeClearedBy — **trainPartType** — documentation: What part of the train must clear the signal application before the signal stops being applied
annotation	documentation Location where the signal stops to be applied. If not specified, the signal is applied indefinitely, until superseded by another signal. This may also be in a different SP: then the SP_ID has to be specified (otherwise the current SP is considered).
source	<xs:element name="EndSignalApplication"> <xs:annotation> <xs:documentation>Location where the signal stops to be applied. If not specified, the signal is applied indefinitely, until superseded by another signal. This may also be in a different SP: then the SP_ID has to be specified (otherwise the current SP is considered).</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="SP_ID"/> <xs:attribute ref="location"/> <xs:attribute ref="endSignalApplicationMustBeClearedBy"/> </xs:complexType> </xs:element>

element **EnergyData**

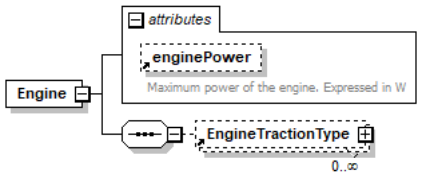
diagram	
properties	content complex
used by	element **B2G_StatusReport**
attributes	Name / Type / Use / Default / Fixed / Annotation

	measuredVoltage	**voltageType**	documentation Measured voltage at pantograph (in V)
	measuredCurrent	**currentType**	documentation Measured current taken from overhead contact line, negative during regeneration (in A)
	autonomy	**energyType**	documentation Amount of energy still available on-board for the self-propelled energy source (in kWh)
	maxPower	**powerTypekW**	documentation Maximal power that can be delivered form energy source available on-board (in kW)
	setValueTractionBraking	**xs:string**	documentation set value by the driver for traction and braking
annotation	documentation Real-time energy data		

source

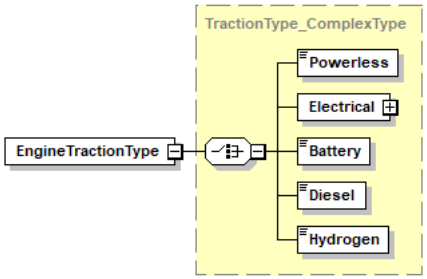
```
<xs:element name="EnergyData">
  <xs:annotation>
    <xs:documentation>Real-time energy data</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="measuredVoltage"/>
    <xs:attribute ref="measuredCurrent"/>
    <xs:attribute ref="autonomy"/>
    <xs:attribute ref="maxPower"/>
    <xs:attribute ref="setValueTractionBraking"/>
  </xs:complexType>
</xs:element>
```

element **Engine**

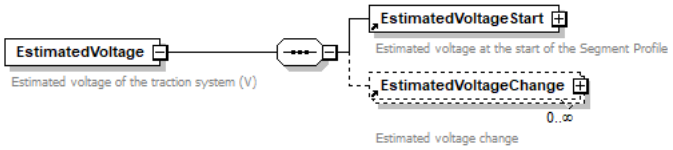
diagram	
properties	content complex
children	**EngineTractionType**

used by	complexType **TC_Features_ComplexType**
attributes	Name: enginePower; Type: **powerTypeW**; Use: ; Default: ; Fixed: ; Annotation: documentation: Maximum power of the engine. Expressed in W
source	<xs:element name="Engine"> <xs:complexType> <xs:sequence> <xs:element ref="EngineTractionType" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="enginePower"/> </xs:complexType> </xs:element>

element **EngineTractionType**

diagram	
type	**TractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Battery** **Diesel** **Hydrogen**
used by	element **Engine**
source	<xs:element name="EngineTractionType" type="TractionType_ComplexType"/>

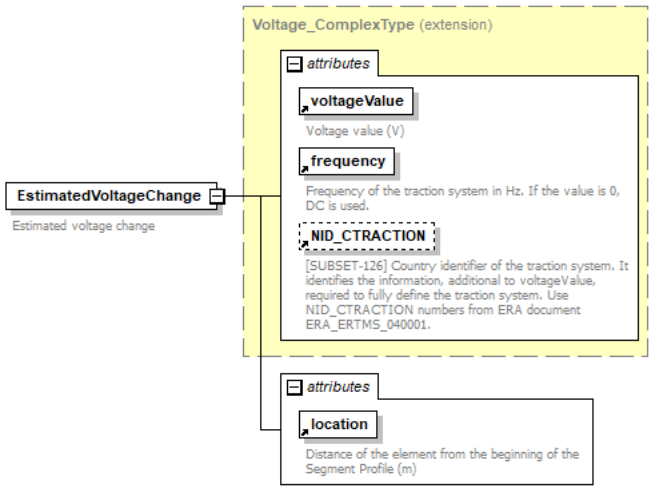
element **EstimatedVoltage**

diagram	
properties	content complex
children	**EstimatedVoltageStart** **EstimatedVoltageChange**
used by	element **SP_Characteristics**
annotation	documentation Estimated voltage of the traction system (V)
source	<xs:element name="EstimatedVoltage"> <xs:annotation> <xs:documentation>Estimated voltage of the traction system (V)</xs:documentation>

```
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="EstimatedVoltageStart"/>
      <xs:element ref="EstimatedVoltageChange" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **EstimatedVoltageChange**

diagram



type: extension of **Voltage_ComplexType**

properties: content complex

used by: element **EstimatedVoltage**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation: Voltage value (V)
frequency	**xs:decimal**	required			documentation: Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation: [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

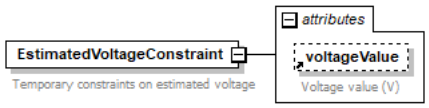
documentation
Estimated voltage change

source

```
<xs:element name="EstimatedVoltageChange">
  <xs:annotation>
    <xs:documentation>Estimated voltage change</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="Voltage_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **EstimatedVoltageConstraint**

diagram



properties — content complex

used by — element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**				documentation Voltage value (V)

annotation

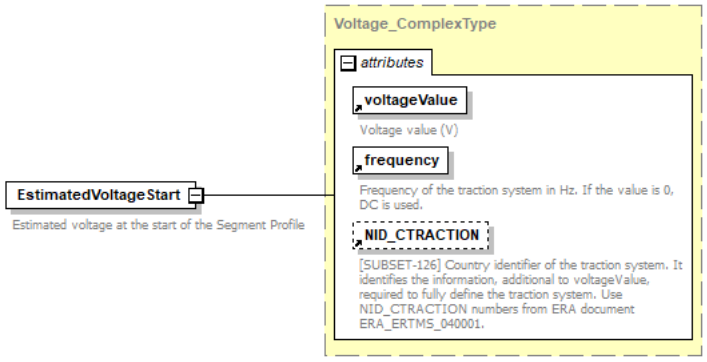
documentation
Temporary constraints on estimated voltage

source

```
<xs:element name="EstimatedVoltageConstraint">
  <xs:annotation>
    <xs:documentation>Temporary constraints on estimated voltage</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="voltageValue"/>
  </xs:complexType>
</xs:element>
```

element **EstimatedVoltageStart**

diagram



type — **Voltage_ComplexType**

properties	content complex
used by	element **EstimatedVoltage**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation Voltage value (V)
frequency	**xs:decimal**	required			documentation Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.

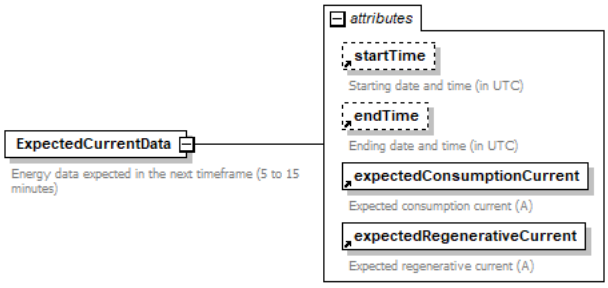
annotation	documentation Estimated voltage at the start of the Segment Profile

source

```
<xs:element name="EstimatedVoltageStart" type="Voltage_ComplexType">
  <xs:annotation>
    <xs:documentation>Estimated voltage at the start of the Segment Profile</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **ExpectedCurrentData**

diagram



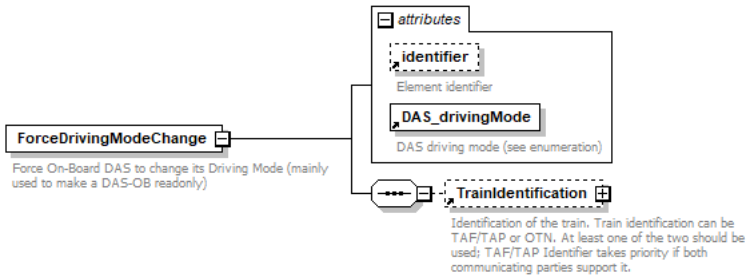
properties	content complex
used by	element **B2G_StatusReport**

attributes

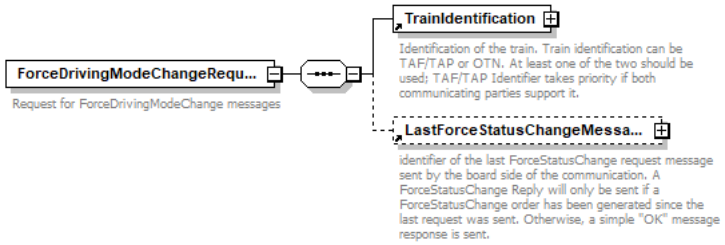
Name	Type	Use	Default	Fixed	Annotation
startTime	**UTCDateTimeType**				documentation Starting date and time (in UTC)
endTime	**UTCDateTimeType**				documentation Ending date and time (in UTC)
expectedConsumptionCurrent	**currentType**	required			documentation Expected consumption current (A)
expectedRegenerativeCurrent	**currentType**	required			documentation Expected

	regenerative current (A)
annotation	documentation Energy data expected in the next timeframe (5 to 15 minutes)
source	<xs:element name="ExpectedCurrentData"> <xs:annotation> <xs:documentation>Energy data expected in the next timeframe (5 to 15 minutes)</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="startTime"/> <xs:attribute ref="endTime"/> <xs:attribute ref="expectedConsumptionCurrent" use="required"/> <xs:attribute ref="expectedRegenerativeCurrent" use="required"/> </xs:complexType> </xs:element>

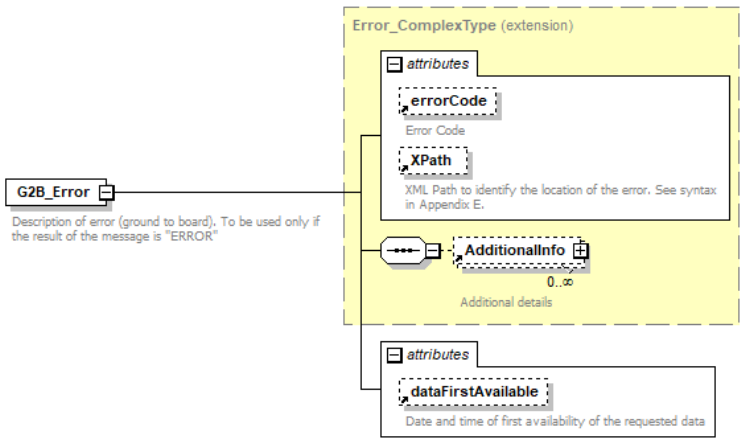
element **ForceDrivingModeChange**

diagram	
properties	content complex
children	**TrainIdentification**
used by	elements **G2B_ReplyPayload** **G2B_Request**
attributes	Name: identifier — Type: **xs:string** — Use: — Default: — Fixed: — Annotation: documentation Element identifier Name: DAS_drivingMode — Type: **derived by: xs:string** — Use: required — Default: — Fixed: — Annotation: documentation DAS driving mode (see enumeration)
annotation	documentation Force On-Board DAS to change its Driving Mode (mainly used to make a DAS-OB readonly)
source	<xs:element name="ForceDrivingModeChange"> <xs:annotation> <xs:documentation>Force On-Board DAS to change its Driving Mode (mainly used to make a DAS-OB readonly)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="DAS_drivingMode" use="required"/> </xs:complexType> </xs:element>

element **ForceDrivingModeChangeRequest**

diagram	
properties	content complex
children	**TrainIdentification** **LastForceStatusChangeMessage**
used by	element **B2G_Request**
annotation	documentation Request for ForceDrivingModeChange messages
source	<xs:element name="ForceDrivingModeChangeRequest"> <xs:annotation> <xs:documentation>Request for ForceDrivingModeChange messages</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="LastForceStatusChangeMessage" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **G2B_Error**

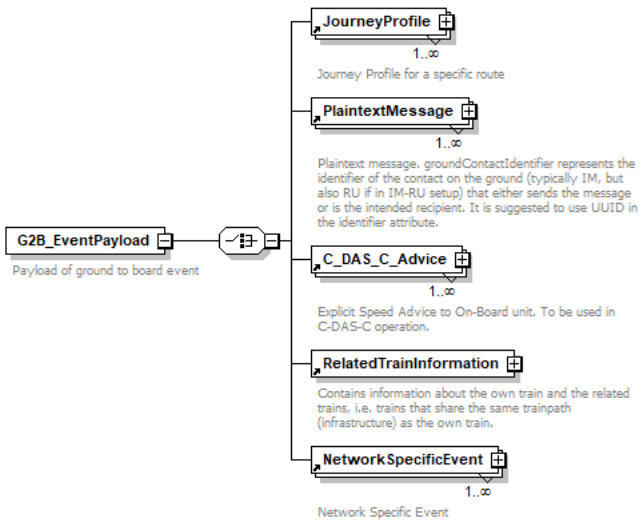
diagram	
type	extension of **Error_ComplexType**
properties	content complex
children	**AdditionalInfo**
used by	element **G2B_MessageResponse**

attributes	Name	Type	Use	Default	Fixed	Annotation
	errorCode	**xs:string**				documentation Error Code
	XPath	**xs:string**				documentation XML Path to identify the location of the error. See syntax in Appendix E.
	dataFirstAvailable	**UTCDateTimeType**				documentation Date and time of first availability of the requested data
annotation	documentation Description of error (ground to board). To be used only if the result of the message is "ERROR"					

source

```
<xs:element name="G2B_Error">
  <xs:annotation>
    <xs:documentation>Description of error (ground to board). To be used only if the result of the message is "ERROR"</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="Error_ComplexType">
        <xs:attribute ref="dataFirstAvailable"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **G2B_EventPayload**

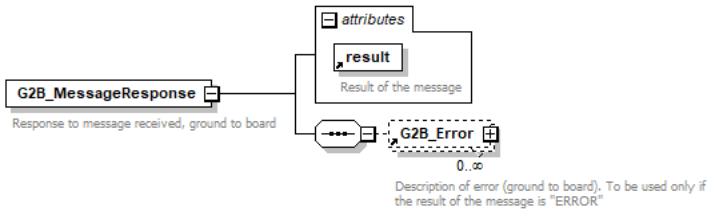
diagram	
properties	content complex	
children	**JourneyProfile PlaintextMessage C_DAS_C_Advice RelatedTrainInformation NetworkSpecificEvent**	
used by	element **SFERA_G2B_EventMessage**	
annotation	documentation Payload of ground to board event	

source

```
<xs:element name="G2B_EventPayload">
  <xs:annotation>
    <xs:documentation>Payload of ground to board event</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:choice>
      <xs:element ref="JourneyProfile" maxOccurs="unbounded"/>
      <xs:element ref="PlaintextMessage" maxOccurs="unbounded"/>
      <xs:element ref="C_DAS_C_Advice" maxOccurs="unbounded"/>
      <xs:element ref="RelatedTrainInformation"/>
      <xs:element ref="NetworkSpecificEvent" maxOccurs="unbounded"/>
    </xs:choice>
  </xs:complexType>
</xs:element>
```

element **G2B_MessageResponse**

diagram



properties

content complex

children **G2B_Error**

used by

element **G2B_ReplyPayload**

attributes

Name	Type	Use	Default	Fixed	Annotation
result	**derived by: xs:string**	required			documentation Result of the message

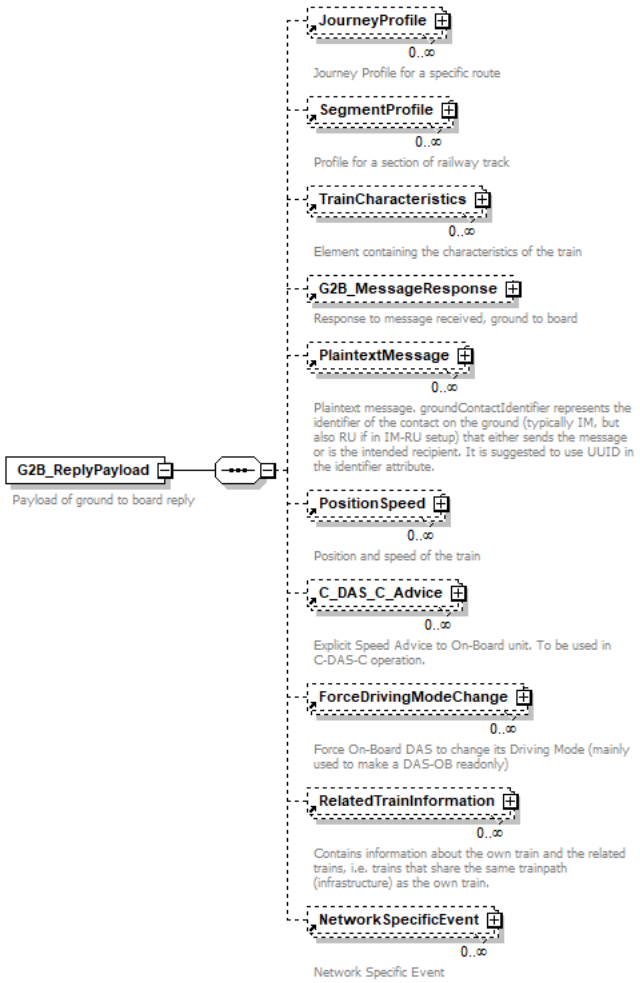
annotation

documentation
Response to message received, ground to board

source

```
<xs:element name="G2B_MessageResponse">
  <xs:annotation>
    <xs:documentation>Response to message received, ground to board</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="G2B_Error" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="result" use="required"/>
  </xs:complexType>
</xs:element>
```

element **G2B_ReplyPayload**

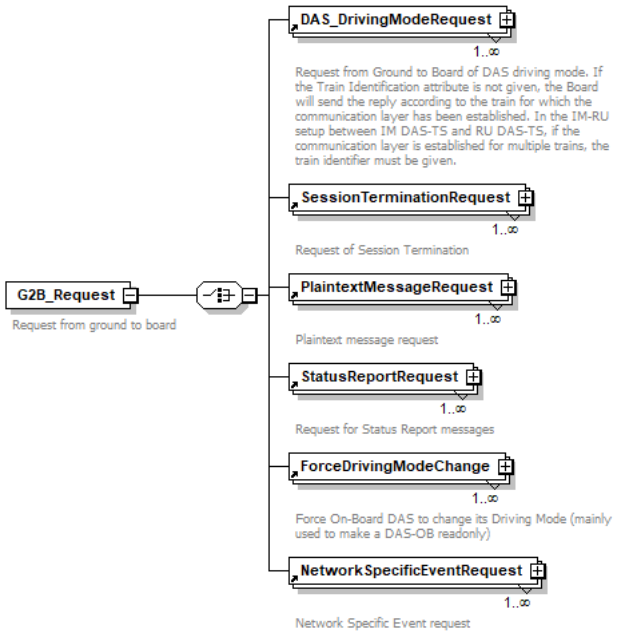
diagram	
properties	content complex
children	JourneyProfile SegmentProfile TrainCharacteristics G2B_MessageResponse PlaintextMessage PositionSpeed C_DAS_C_Advice ForceDrivingModeChange RelatedTrainInformation NetworkSpecificEvent
used by	element SFERA_G2B_ReplyMessage
annotation	documentation Payload of ground to board reply

source

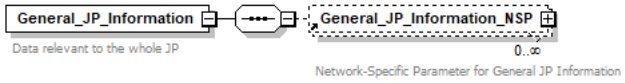
```
<xs:element name="G2B_ReplyPayload">
 <xs:annotation>
  <xs:documentation>Payload of ground to board reply</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="JourneyProfile" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="SegmentProfile" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="TrainCharacteristics" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="G2B_MessageResponse" minOccurs="0"/>
   <xs:element ref="PlaintextMessage" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="PositionSpeed" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="C_DAS_C_Advice" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="ForceDrivingModeChange" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="RelatedTrainInformation" minOccurs="0" maxOccurs="unbounded"/>
```

	<xs:element ref="NetworkSpecificEvent" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

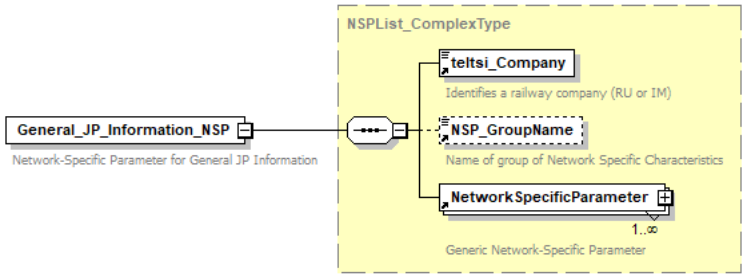
element **G2B_Request**

diagram	
properties	content complex
children	**DAS_DrivingModeRequest** **SessionTerminationRequest** **PlaintextMessageRequest** **StatusReportRequest** **ForceDrivingModeChange** **NetworkSpecificEventRequest**
used by	element **SFERA_G2B_RequestMessage**
annotation	documentation Request from ground to board
source	<xs:element name="G2B_Request"> <xs:annotation> <xs:documentation>Request from ground to board</xs:documentation> </xs:annotation> <xs:complexType> <xs:choice> <xs:element ref="DAS_DrivingModeRequest" maxOccurs="unbounded"/> <xs:element ref="SessionTerminationRequest" maxOccurs="unbounded"/> <xs:element ref="PlaintextMessageRequest" maxOccurs="unbounded"/> <xs:element ref="StatusReportRequest" maxOccurs="unbounded"/> <xs:element ref="ForceDrivingModeChange" maxOccurs="unbounded"/> <xs:element ref="NetworkSpecificEventRequest" maxOccurs="unbounded"/> </xs:choice> </xs:complexType> </xs:element>

element **General_JP_Information**

diagram	
properties	content complex
children	**General_JP_Information_NSP**
used by	element **JourneyProfile**
annotation	documentation Data relevant to the whole JP
source	<xs:element name="General_JP_Information"> <xs:annotation> <xs:documentation>Data relevant to the whole JP</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="General_JP_Information_NSP" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **General_JP_Information_NSP**

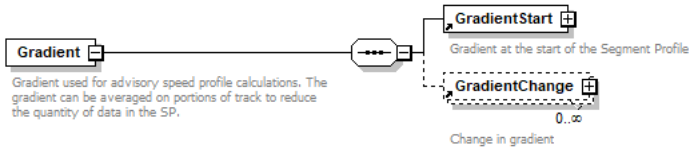
diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	element **General_JP_Information**
annotation	documentation Network-Specific Parameter for General JP Information
source	<xs:element name="General_JP_Information_NSP" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network-Specific Parameter for General JP Information</xs:documentation> </xs:annotation> </xs:element>

element **GeoLocalisationTimestamp**

diagram	
type	**UTCDateTimeType**

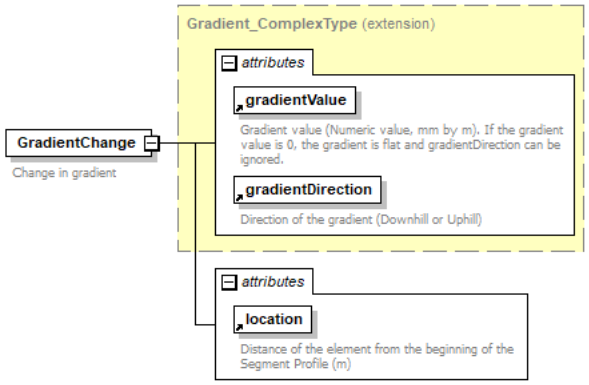
properties	content simple
used by	element **PositionSpeed**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Timestamp of the geolocalisation measurement
source	<xs:element name="GeoLocalisationTimestamp" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Timestamp of the geolocalisation measurement</xs:documentation> </xs:annotation> </xs:element>

element **Gradient**

diagram	
properties	content complex
children	**GradientStart** **GradientChange**
used by	element **SP_Characteristics**
annotation	documentation Gradient used for advisory speed profile calculations. The gradient can be averaged on portions of track to reduce the quantity of data in the SP.
source	<xs:element name="Gradient"> <xs:annotation> <xs:documentation>Gradient used for advisory speed profile calculations. The gradient can be averaged on portions of track to reduce the quantity of data in the SP.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="GradientStart"/> <xs:element ref="GradientChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **GradientChange**

diagram



type: extension of **Gradient_ComplexType**

properties: content complex

used by: element **Gradient**

attributes

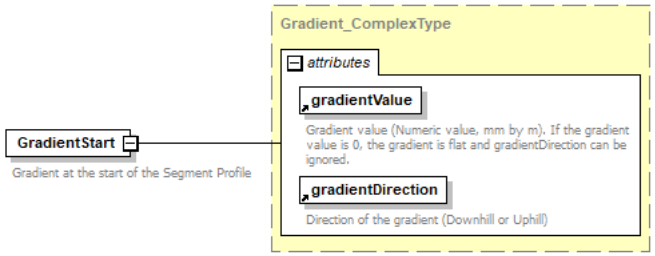
Name	Type	Use	Default	Fixed	Annotation
gradientValue	**gradientValueType**	required			documentation: Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
gradientDirection	**gradientDirectionType**	required			documentation: Direction of the gradient (Downhill or Uphill)
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation: documentation
Change in gradient

source

```
<xs:element name="GradientChange">
  <xs:annotation>
    <xs:documentation>Change in gradient</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="Gradient_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **GradientStart**

diagram	
type	**Gradient_ComplexType**
properties	content complex
used by	element **Gradient**

attributes

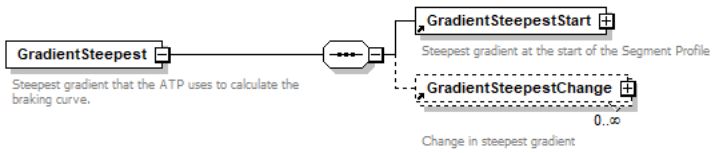
Name	Type	Use	Default	Fixed	Annotation
gradientValue	**gradientValueType**	required			documentation Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
gradientDirection	**gradientDirectionType**	required			documentation Direction of the gradient (Downhill or Uphill)

annotation
documentation
Gradient at the start of the Segment Profile

source

```
<xs:element name="GradientStart" type="Gradient_ComplexType">
  <xs:annotation>
    <xs:documentation>Gradient at the start of the Segment Profile</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **GradientSteepest**

diagram	
properties	content complex
children	**GradientSteepestStart** **GradientSteepestChange**
used by	element **SP_Characteristics**
annotation	documentation Steepest gradient that the ATP uses to calculate the braking curve.

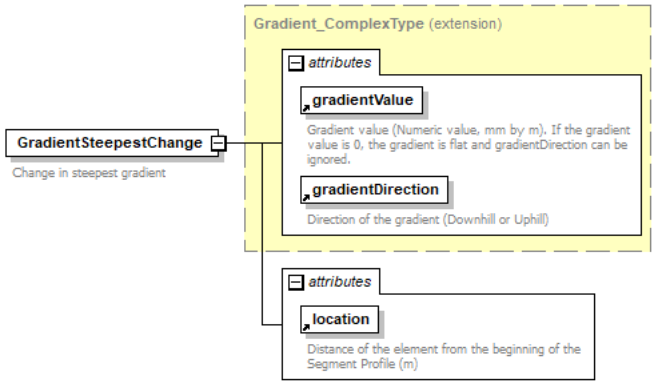
source

```
<xs:element name="GradientSteepest">
  <xs:annotation>
    <xs:documentation>Steepest gradient that the ATP uses to calculate the braking
curve.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
```

```
    <xs:sequence>
      <xs:element ref="GradientSteepestStart"/>
      <xs:element ref="GradientSteepestChange" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **GradientSteepestChange**

diagram



type: extension of **Gradient_ComplexType**

properties: content complex

used by: element **GradientSteepest**

attributes

Name	Type	Use	Default	Fixed	Annotation
gradientValue	**gradientValueType**	required			documentation: Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
gradientDirection	**gradientDirectionType**	required			documentation: Direction of the gradient (Downhill or Uphill)
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation: documentation
Change in steepest gradient

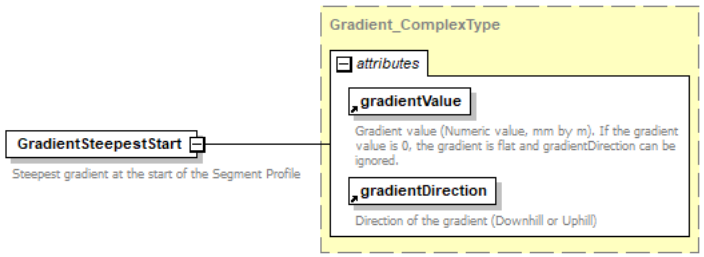
source

```
<xs:element name="GradientSteepestChange">
  <xs:annotation>
    <xs:documentation>Change in steepest gradient</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="Gradient_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
```

```
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **GradientSteepestStart**

diagram



type **Gradient_ComplexType**

properties
content complex

used by
element **GradientSteepest**

attributes

Name	Type	Use	Default	Fixed	Annotation
gradientValue	**gradientValueType**	required			documentation Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
gradientDirection	**gradientDirectionType**	required			documentation Direction of the gradient (Downhill or Uphill)

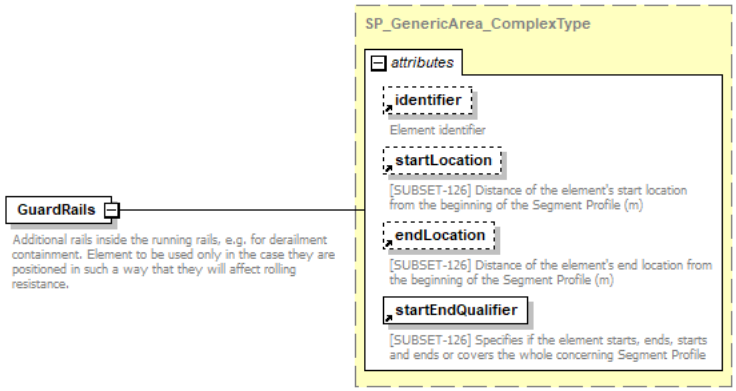
annotation
documentation
Steepest gradient at the start of the Segment Profile

source

```
<xs:element name="GradientSteepestStart" type="Gradient_ComplexType">
  <xs:annotation>
    <xs:documentation>Steepest gradient at the start of the Segment Profile</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **GuardRails**

diagram



type **SP_GenericArea_ComplexType**

properties
content complex

used by
element **SP_Areas**

attributes

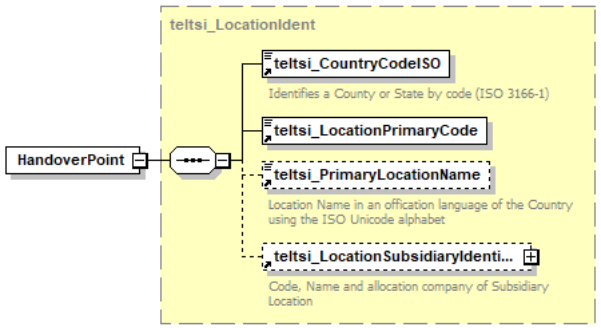
Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

annotation
documentation
Additional rails inside the running rails, e.g. for derailment containment. Element to be used only in the case they are positioned in such a way that they will affect rolling resistance.

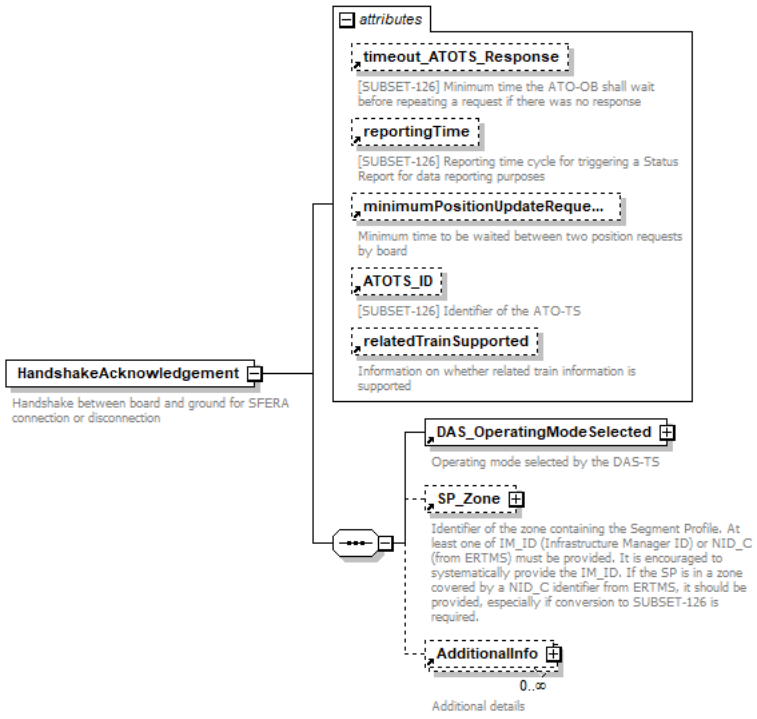
source

```
<xs:element name="GuardRails" type="SP_GenericArea_ComplexType">
  <xs:annotation>
    <xs:documentation>Additional rails inside the running rails, e.g. for derailment containment. Element to be used only in the case they are positioned in such a way that they will affect rolling resistance.</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **HandoverPoint**

diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	complexType **OTN_ID_ComplexType**
source	<xs:element name="HandoverPoint" type="teltsi_LocationIdent"/>

element **HandshakeAcknowledgement**

diagram	
properties	content complex
children	**DAS_OperatingModeSelected** **SP_Zone** **AdditionalInfo**
used by	element **SFERA_G2B_ReplyMessage**

attributes

Name	Type	Use	Default	Fixed	Annotation
timeout_ATOTS_Response	**xs:duration**				documentation [SUBSET-126] Minimum time the ATO-OB shall wait before repeating a request if there was no response
reportingTime	**xs:duration**				documentation [SUBSET-126] Reporting time cycle for triggering a Status Report for data reporting purposes
minimumPositionUpdateRequestPeriod	**xs:duration**				documentation Minimum time to be waited between two position requests by board
ATOTS_ID	**xs:string**				documentation [SUBSET-126] Identifier of the ATO-TS
relatedTrainSupported	**relatedTrainRequestType**				documentation Information on whether related train information is supported

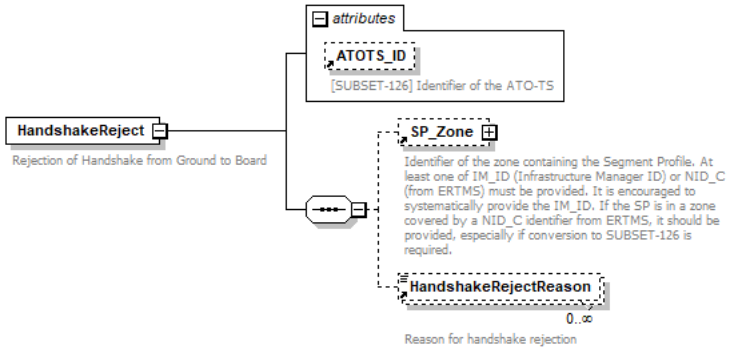
annotation

documentation
Handshake between board and ground for SFERA connection or disconnection

source

```
<xs:element name="HandshakeAcknowledgement">
  <xs:annotation>
    <xs:documentation>Handshake between board and ground for SFERA connection or disconnection</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="DAS_OperatingModeSelected"/>
      <xs:element ref="SP_Zone" minOccurs="0"/>
      <xs:element ref="AdditionalInfo" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="timeout_ATOTS_Response"/>
    <xs:attribute ref="reportingTime"/>
    <xs:attribute ref="minimumPositionUpdateRequestPeriod"/>
    <xs:attribute ref="ATOTS_ID"/>
    <xs:attribute ref="relatedTrainSupported"/>
  </xs:complexType>
</xs:element>
```

element **HandshakeReject**

diagram	
properties	content complex
children	**SP_Zone** **HandshakeRejectReason**
used by	element **SFERA_G2B_ReplyMessage**

attributes

Name	Type	Use	Default	Fixed	Annotation
ATOTS_ID	**xs:string**				documentation [SUBSET-126] Identifier of the ATO-TS

annotation

documentation
Rejection of Handshake from Ground to Board

source

```
<xs:element name="HandshakeReject">
  <xs:annotation>
    <xs:documentation>Rejection of Handshake from Ground to Board</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="SP_Zone" minOccurs="0"/>
      <xs:element ref="HandshakeRejectReason" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="ATOTS_ID"/>
  </xs:complexType>
</xs:element>
```

element **HandshakeRejectReason**

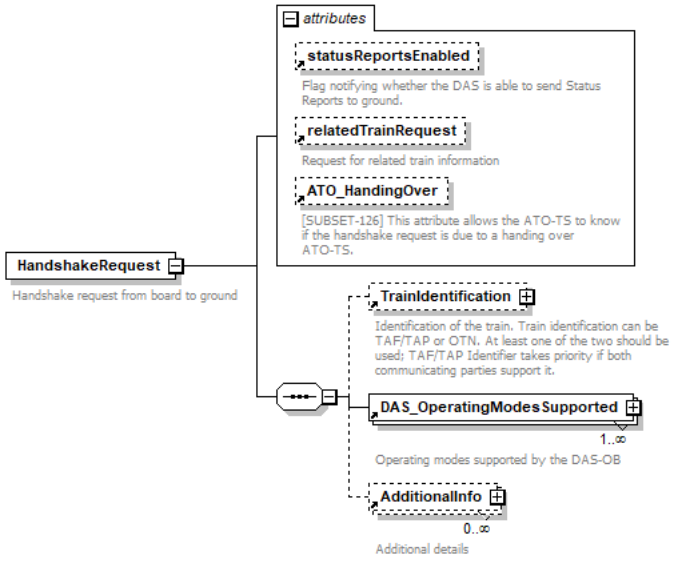
diagram	
type	restriction of **xs:string**
properties	content simple
used by	element **HandshakeReject**

facets

Kind	Value	Annotation
enumeration	SFERA version incompatible	
enumeration	ATO system version incompatible	
enumeration	Another DAS-TS/ATO-TS in charge	
enumeration	DAS-TS/ATO-TS in charge unknown	
enumeration	Architecture not supported	
enumeration	Connectivity not supported	
enumeration	Driving Modes not supported	

annotation	documentation Reason for handshake rejection
source	<xs:element name="HandshakeRejectReason"> <xs:annotation> <xs:documentation>Reason for handshake rejection</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="SFERA version incompatible"/> <xs:enumeration value="ATO system version incompatible"/> <xs:enumeration value="Another DAS-TS/ATO-TS in charge"/> <xs:enumeration value="DAS-TS/ATO-TS in charge unknown"/> <xs:enumeration value="Architecture not supported"/> <xs:enumeration value="Connectivity not supported"/> <xs:enumeration value="Driving Modes not supported"/> </xs:restriction> </xs:simpleType> </xs:element>

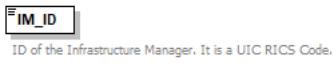
element **HandshakeRequest**

diagram	
properties	content complex
children	**TrainIdentification** **DAS_OperatingModesSupported** **AdditionalInfo**
used by	element **SFERA_B2G_RequestMessage**

attributes	Name	Type	Use	Default	Fixed	Annotation
	statusReportsEnabled	**xs:boolean**				documentation Flag notifying whether the DAS is able to send Status Reports to ground.
	relatedTrainRequest	**relatedTrainRequestType**				documentation Request for related train information
	ATO_HandingOver	**xs:boolean**				documentation

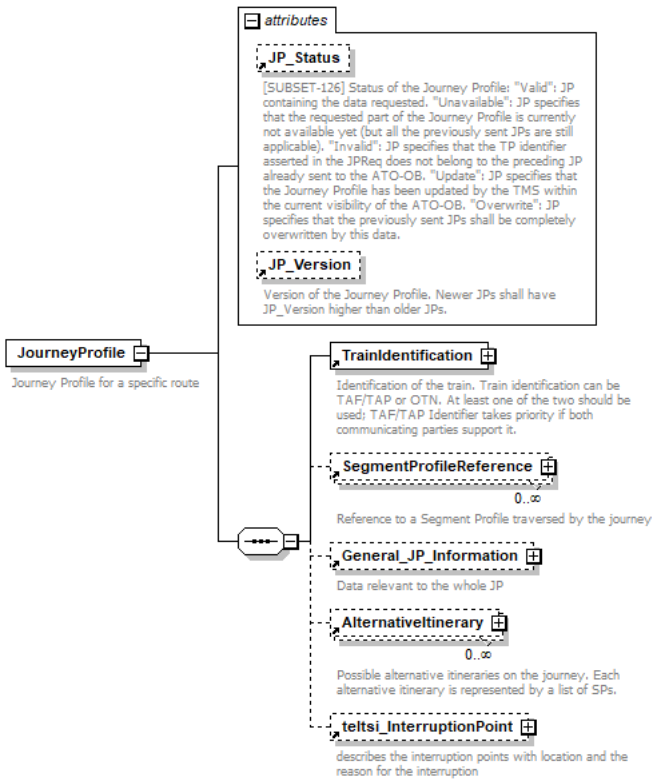
	[SUBSET-126] This attribute allows the ATO-TS to know if the handshake request is due to a handing over ATO-TS.
annotation	documentation Handshake request from board to ground
source	<xs:element name="HandshakeRequest"> <xs:annotation> <xs:documentation>Handshake request from board to ground</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> <xs:element ref="DAS_OperatingModesSupported" maxOccurs="unbounded"/> <xs:element ref="AdditionalInfo" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="statusReportsEnabled"/> <xs:attribute ref="relatedTrainRequest"/> <xs:attribute ref="ATO_HandingOver"/> </xs:complexType> </xs:element>

element **IM_ID**

diagram	
type	**teltsi_CompanyCode**
properties	content simple
used by	complexType **SP_Zone_ComplexType**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation ID of the Infrastructure Manager. It is a UIC RICS Code.
source	<xs:element name="IM_ID" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>ID of the Infrastructure Manager. It is a UIC RICS Code.</xs:documentation> </xs:annotation> </xs:element>

element **JourneyProfile**

diagram



properties

content complex

children **TrainIdentification** **SegmentProfileReference** **General_JP_Information** **AlternativeItinerary** **teltsi_InterruptionPoint**

used by

elements **G2B_EventPayload** **G2B_ReplyPayload**

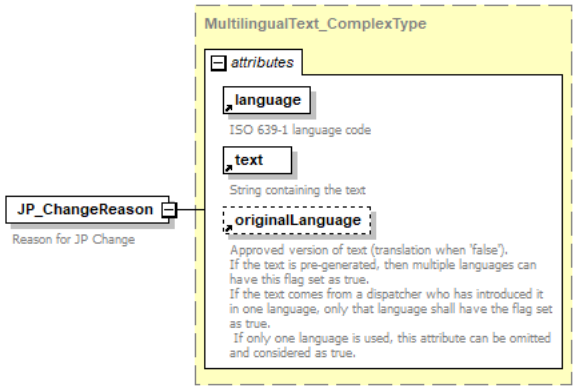
attributes

Name	Type	Use	Default	Fixed	Annotation
JP_Status	**derived by: xs:string**				documentation [SUBSET-126] Status of the Journey Profile: "Valid": JP containing the data requested. "Unavailable": JP specifies that the requested part of the Journey Profile is currently not available yet (but all the previously sent JPs are still applicable). "Invalid": JP specifies that the TP identifier asserted in the JPReq does not belong to

	JP_Version **xs:unsignedInt** documentation Version of the Journey Profile. Newer JPs shall have JP_Version higher than older JPs.	the preceding JP already sent to the ATO-OB. "Update": JP specifies that the Journey Profile has been updated by the TMS within the current visibility of the ATO-OB. "Overwrite": JP specifies that the previously sent JPs shall be completely overwritten by this data.
annotation	documentation Journey Profile for a specific route	
source	<xs:element name="JourneyProfile"> <xs:annotation> <xs:documentation>Journey Profile for a specific route</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="SegmentProfileReference" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="General_JP_Information" minOccurs="0"/> <xs:element ref="AlternativeItinerary" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="teltsi_InterruptionPoint" minOccurs="0"/> </xs:sequence> <xs:attribute ref="JP_Status"/> <xs:attribute ref="JP_Version"/> </xs:complexType> </xs:element>	

element **JP_ChangeReason**

diagram



type **MultilingualText_ComplexType**

properties
- content complex

used by
- element **ChangeReason**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

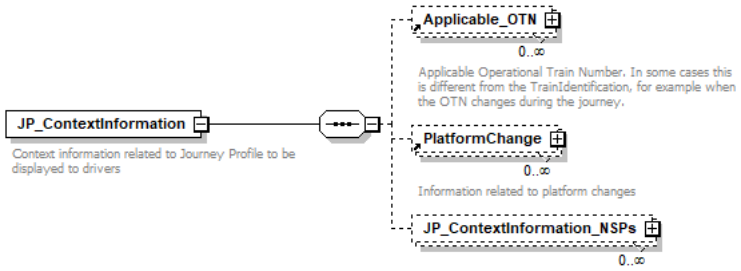
annotation
documentation
Reason for JP Change

source

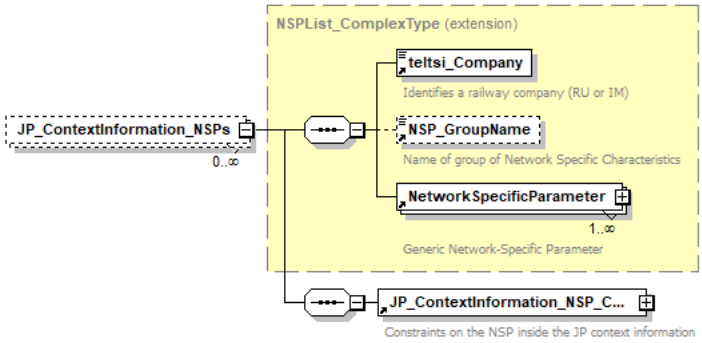
```
<xs:element name="JP_ChangeReason" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Reason for JP Change</xs:documentation>
  </xs:annotation>
```

	</xs:element>

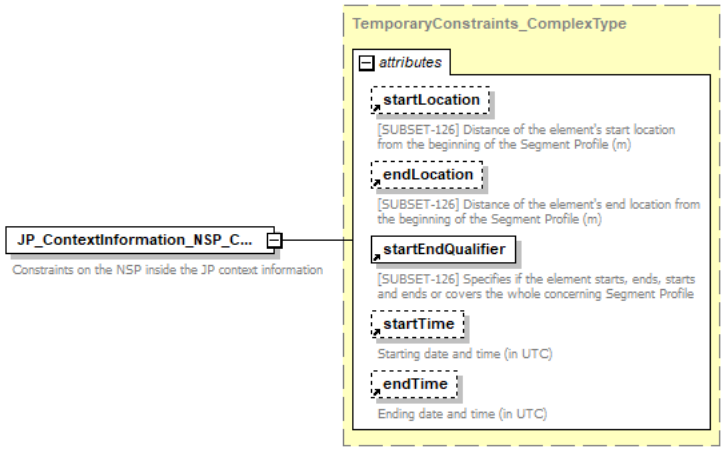
element **JP_ContextInformation**

diagram	
properties	content complex
children	**Applicable_OTN** **PlatformChange** **JP_ContextInformation_NSPs**
used by	element **SegmentProfileReference**
annotation	documentation Context information related to Journey Profile to be displayed to drivers
source	<xs:element name="JP_ContextInformation"> <xs:annotation> <xs:documentation>Context information related to Journey Profile to be displayed to drivers</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="Applicable_OTN" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="PlatformChange" minOccurs="0" maxOccurs="unbounded"/> <xs:element name="JP_ContextInformation_NSPs" minOccurs="0" maxOccurs="unbounded"> <xs:complexType> <xs:complexContent> <xs:extension base="NSPList_ComplexType"> <xs:sequence> <xs:element ref="JP_ContextInformation_NSP_Constraints"/> </xs:sequence> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element> </xs:sequence> </xs:complexType> </xs:element>

element **JP_ContextInformation/JP_ContextInformation_NSPs**

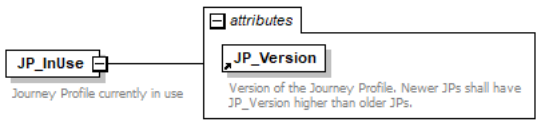
diagram	
type	extension of **NSPList_ComplexType**
properties	minOcc 0 maxOcc unbounded content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter** **JP_ContextInformation_NSP_Constraints**
source	<xs:element name="JP_ContextInformation_NSPs" minOccurs="0" maxOccurs="unbounded"> <xs:complexType> <xs:complexContent> <xs:extension base="NSPList_ComplexType"> <xs:sequence> <xs:element ref="JP_ContextInformation_NSP_Constraints"/> </xs:sequence> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **JP_ContextInformation_NSP_Constraints**

diagram	
type	**TemporaryConstraints_ComplexType**
properties	content complex
used by	element **JP_ContextInformation/JP_ContextInformation_NSPs**
attributes	Name: startLocation; Type: **distanceType**; Use; Default; Fixed; Annotation documentation

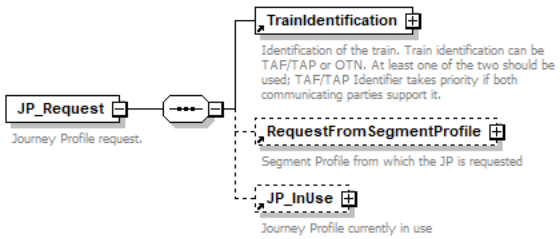
	endLocation	**distanceType**				[SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m) documentation
	startEndQualifier	**derived by: xs:string**	required			[SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
	startTime	**UTCDateTimeType**				documentation Starting date and time (in UTC)
	endTime	**UTCDateTimeType**				documentation Ending date and time (in UTC)
annotation	documentation Constraints on the NSP inside the JP context information					
source	<xs:element name="JP_ContextInformation_NSP_Constraints" type="TemporaryConstraints_ComplexType"> <xs:annotation> <xs:documentation>Constraints on the NSP inside the JP context information</xs:documentation> </xs:annotation> </xs:element>					

element **JP_InUse**

diagram						
properties	content complex					
used by	elements **B2G_StatusReport** **JP_Request**					
attributes	Name JP_Version	Type **xs:unsignedInt**	Use required	Default	Fixed	Annotation documentation Version of the Journey Profile. Newer JPs shall have JP_Version higher than

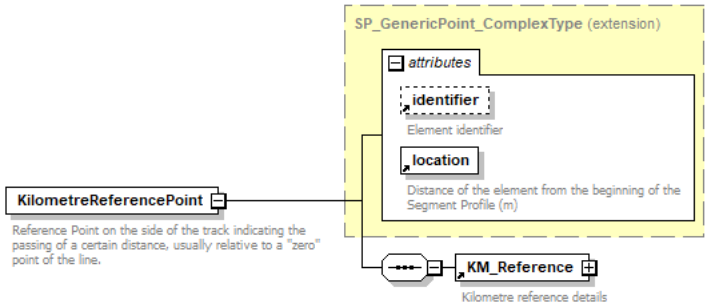
	older JPs.
annotation	documentation Journey Profile currently in use
source	<xs:element name="JP_InUse"> <xs:annotation> <xs:documentation>Journey Profile currently in use</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="JP_Version" use="required"/> </xs:complexType> </xs:element>

element **JP_Request**

diagram	
properties	content complex
children	**TrainIdentification** **RequestFromSegmentProfile** **JP_InUse**
used by	element **B2G_Request**
annotation	documentation Journey Profile request.
source	<xs:element name="JP_Request"> <xs:annotation> <xs:documentation>Journey Profile request.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="RequestFromSegmentProfile" minOccurs="0"/> <xs:element ref="JP_InUse" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **KilometreReferencePoint**

diagram



type: extension of **SP_GenericPoint_ComplexType**

properties: content complex

children: **KM_Reference**

used by: element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation: Element identifier
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

documentation

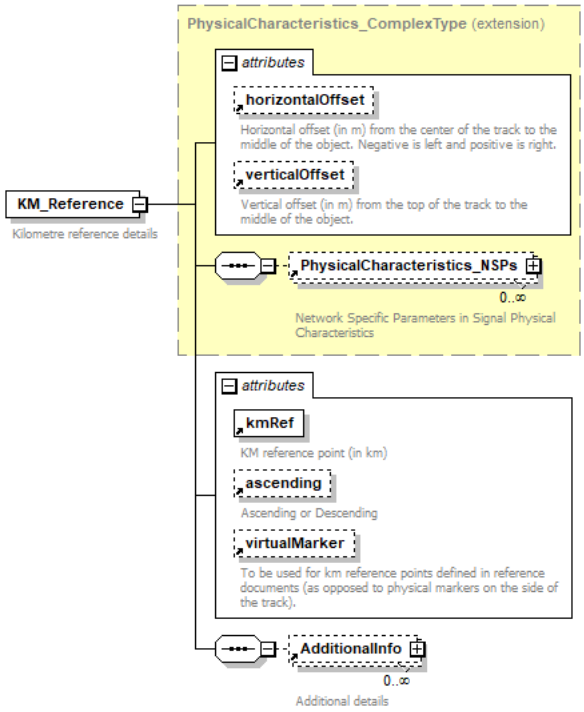
Reference Point on the side of the track indicating the passing of a certain distance, usually relative to a "zero" point of the line.

source

```
<xs:element name="KilometreReferencePoint">
 <xs:annotation>
  <xs:documentation>Reference Point on the side of the track indicating the passing of a certain distance, usually relative to a "zero" point of the line.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericPoint_ComplexType">
    <xs:sequence>
     <xs:element ref="KM_Reference"/>
    </xs:sequence>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **KM_Reference**

diagram



type: extension of **PhysicalCharacteristics_ComplexType**

properties: content complex

children: **PhysicalCharacteristics_NSPs** **AdditionalInfo**

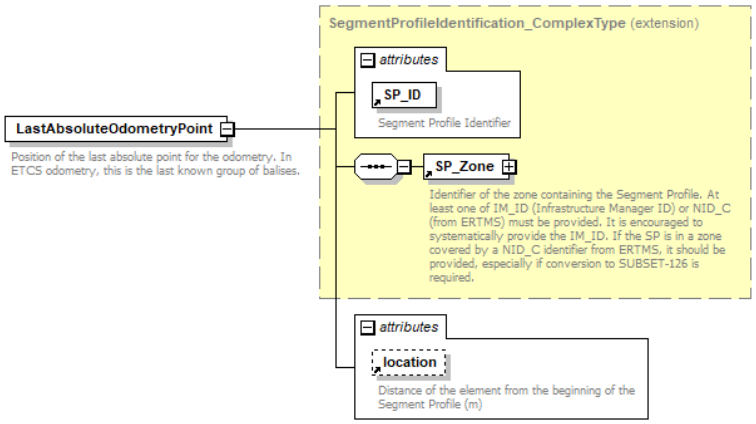
used by: element **KilometreReferencePoint**

attributes

Name	Type	Use	Default	Fixed	Annotation
horizontalOffset	**distanceType**				documentation Horizontal offset (in m) from the center of the track to the middle of the object. Negative is left and positive is right.
verticalOffset	**distanceType**				documentation Vertical offset (in m) from the top of the track to the middle of the object.
kmRef	**xs:double**	required			documentation KM reference point (in km)
ascending	**derived by: xs:string**				documentation Ascending or Descending
virtualMarker	**xs:boolean**				documentation To be used for km reference points defined in reference documents (as

	opposed to physical markers on the side of the track).
annotation	documentation Kilometre reference details
source	<xs:element name="KM_Reference"> <xs:annotation> <xs:documentation>Kilometre reference details</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="PhysicalCharacteristics_ComplexType"> <xs:sequence> <xs:element ref="AdditionalInfo" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="kmRef" use="required"/> <xs:attribute ref="ascending"/> <xs:attribute ref="virtualMarker"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

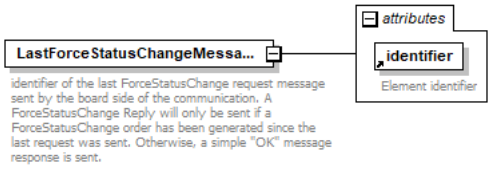
element **LastAbsoluteOdometryPoint**

diagram	
type	extension of **SegmentProfileIdentification_ComplexType**
properties	content complex
children	**SP_Zone**
used by	element **PositionSpeed**

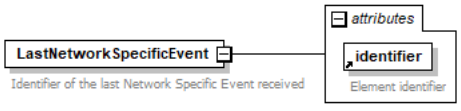
attributes	Name	Type	Use	Default	Fixed	Annotation
	SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
	location	**distanceType**				documentation Distance of the element from the beginning of

	the Segment Profile (m)
annotation	documentation Position of the last absolute point for the odometry. In ETCS odometry, this is the last known group of balises.
source	<xs:element name="LastAbsoluteOdometryPoint"> <xs:annotation> <xs:documentation>Position of the last absolute point for the odometry. In ETCS odometry, this is the last known group of balises.</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SegmentProfileIdentification_ComplexType"> <xs:attribute ref="location"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **LastForceStatusChangeMessage**

diagram	
properties	content complex
used by	element **ForceDrivingModeChangeRequest**
attributes	Name: identifier; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation Element identifier
annotation	documentation identifier of the last ForceStatusChange request message sent by the board side of the communication. A ForceStatusChange Reply will only be sent if a ForceStatusChange order has been generated since the last request was sent. Otherwise, a simple "OK" message response is sent.
source	<xs:element name="LastForceStatusChangeMessage"> <xs:annotation> <xs:documentation>identifier of the last ForceStatusChange request message sent by the board side of the communication. A ForceStatusChange Reply will only be sent if a ForceStatusChange order has been generated since the last request was sent. Otherwise, a simple "OK" message response is sent.</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="identifier" use="required"/> </xs:complexType> </xs:element>

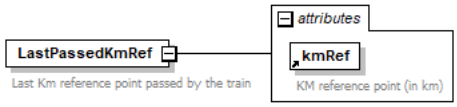
element **LastNetworkSpecificEvent**

diagram	

properties	content complex
used by	element **NetworkSpecificEventRequest**
attributes	Name: identifier; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation documentation: Element identifier
annotation	documentation: Identifier of the last Network Specific Event received

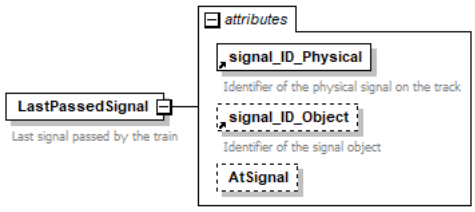
source	`<xs:element name="LastNetworkSpecificEvent">`
	`<xs:annotation>`
	`<xs:documentation>Identifier of the last Network Specific Event received</xs:documentation>`
	`</xs:annotation>`
	`<xs:complexType>`
	`<xs:attribute ref="identifier" use="required"/>`
	`</xs:complexType>`
	`</xs:element>`

element **LastPassedKmRef**

diagram	
properties	content complex
used by	element **TrainLocationInformation**
attributes	Name: kmRef; Type: **xs:double**; Use: required; Default: ; Fixed: ; Annotation documentation: KM reference point (in km)
annotation	documentation: Last Km reference point passed by the train

source	`<xs:element name="LastPassedKmRef">`
	`<xs:annotation>`
	`<xs:documentation>Last Km reference point passed by the train</xs:documentation>`
	`</xs:annotation>`
	`<xs:complexType>`
	`<xs:attribute ref="kmRef" use="required"/>`
	`</xs:complexType>`
	`</xs:element>`

element **LastPassedSignal**

diagram	
properties	content complex
used by	element **TrainLocationInformation**

attributes	Name	Type	Use	Default	Fixed	Annotation
	signal_ID_Physical	**xs:string**	required			documentation Identifier of the physical signal on the track
	signal_ID_Object	**xs:string**				documentation Identifier of the signal object
	AtSignal	**xs:boolean**				
annotation	documentation Last signal passed by the train					

source

```
<xs:element name="LastPassedSignal">
  <xs:annotation>
    <xs:documentation>Last signal passed by the train</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="signal_ID_Physical" use="required"/>
    <xs:attribute ref="signal_ID_Object"/>
    <xs:attribute name="AtSignal" type="xs:boolean">
      <xs:annotation/>
    </xs:attribute>
  </xs:complexType>
</xs:element>
```

attribute **LastPassedSignal/@AtSignal**

type	**xs:boolean**

source

```
<xs:attribute name="AtSignal" type="xs:boolean">
  <xs:annotation/>
</xs:attribute>
```

element **LastPlaintextMessage**

diagram						
properties	content complex					
used by	element **PlaintextMessageRequest**					
attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**	required			documentation Element identifier
annotation	documentation Identifier of the last PlainTextMessage message received by the board side of the communication. If this attribute is used, one or more PlainTextMessage replies will only be sent if more recent advice is available. If no more recent advice is available, a simple "OK" message response is sent. If the LastPlainTextMessage attribute is not used, all messages relevant to the train will be sent. This can be used after the DAS-OB application restarted for example.					

source

```
<xs:element name="LastPlaintextMessage">
  <xs:annotation>
```

```
    <xs:documentation>Identifier of the last PlainTextMessage message received by the board side of the communication. If this attribute is used, one or more PlainTextMessage replies will only be sent if more recent advice is available. If no more recent advice is available, a simple "OK" message response is sent. If the LastPlainTextMessage attribute is not used, all messages relevant to the train will be sent. This can be used after the DAS-OB application restarted for example.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="identifier" use="required"/>
  </xs:complexType>
</xs:element>
```


element **LastStatusReportMessage**

diagram



properties: content complex

used by: element **StatusReportRequest**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**	required			documentation Element identifier

annotation

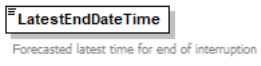
documentation
identifier of the last StatusReport message received by the ground side of the communication. One or more StatusReport replies will only be sent if more recent data is available. Otherwise, a simple "OK" message response is sent.

source


```
<xs:element name="LastStatusReportMessage">
  <xs:annotation>
    <xs:documentation>identifier of the last StatusReport message received by the ground side of the communication. One or more StatusReport replies will only be sent if more recent data is available. Otherwise, a simple "OK" message response is sent.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="identifier" use="required"/>
  </xs:complexType>
</xs:element>
```


element **LatestEndDateTime**

diagram



type: **xs:dateTime**

properties: content simple

annotation

documentation
Forecasted latest time for end of interruption

source

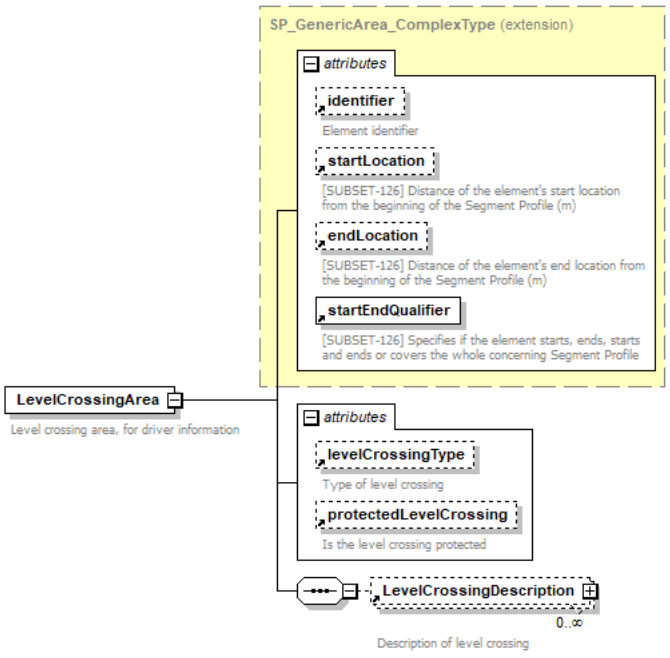

```
<xs:element name="LatestEndDateTime" type="xs:dateTime">
  <xs:annotation>
    <xs:documentation>Forecasted latest time for end of interruption</xs:documentation>
  </xs:annotation>
```

	</xs:element>

element **LeavingEnteringStatus**

diagram	
type	**LeavingEnteringStatusType**
properties	content simple
used by	element **SegmentOccupationStatus**
facets	Kind Value Annotation enumeration Entering enumeration Leaving enumeration Crossing
annotation	documentation Gives an indication if a train is entering of leaving your trainpath at a certain location
source	<xs:element name="LeavingEnteringStatus" type="LeavingEnteringStatusType"> <xs:annotation> <xs:documentation>Gives an indication if a train is entering of leaving your trainpath at a certain location</xs:documentation> </xs:annotation> </xs:element>

element **LevelCrossingArea**

diagram	
type	extension of **SP_GenericArea_ComplexType**
properties	content complex
children	**LevelCrossingDescription**

used by: element **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
levelCrossingType	**xs:string**				documentation Type of level crossing
protectedLevelCrossing	**xs:boolean**				documentation Is the level crossing protected

annotation

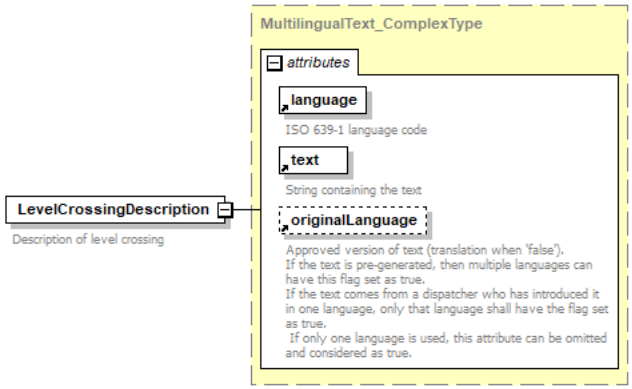
documentation
Level crossing area, for driver information

source

```
<xs:element name="LevelCrossingArea">
 <xs:annotation>
  <xs:documentation>Level crossing area, for driver information</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericArea_ComplexType">
    <xs:sequence>
     <xs:element ref="LevelCrossingDescription" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="levelCrossingType"/>
    <xs:attribute ref="protectedLevelCrossing"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **LevelCrossingDescription**

diagram



type **MultilingualText_ComplexType**

properties
content complex

used by
element **LevelCrossingArea**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

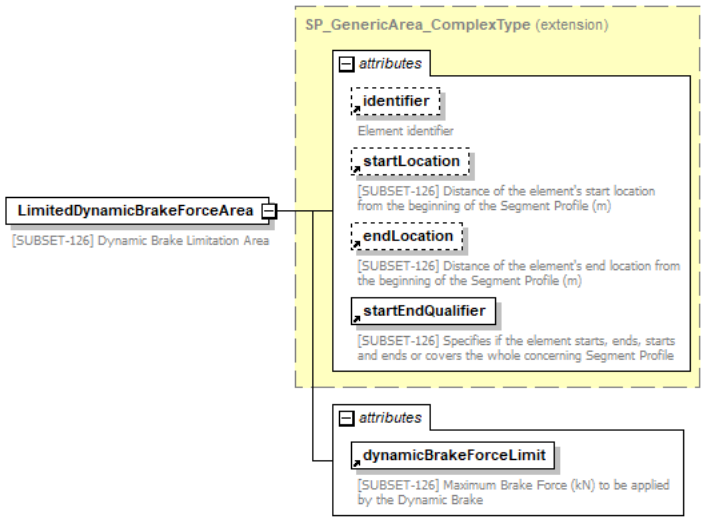
annotation
documentation
Description of level crossing

source

```
<xs:element name="LevelCrossingDescription" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Description of level crossing</xs:documentation>
  </xs:annotation>
```

	</xs:element>

element **LimitedDynamicBrakeForceArea**

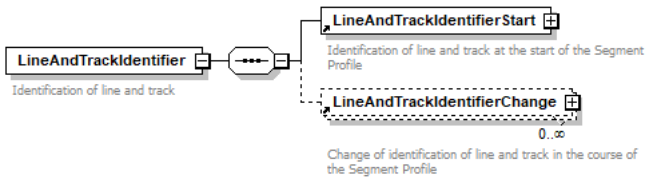
diagram	
type	extension of **SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
dynamicBrakeForceLimit	**xs:unsignedShort**	required			documentation [SUBSET-126]

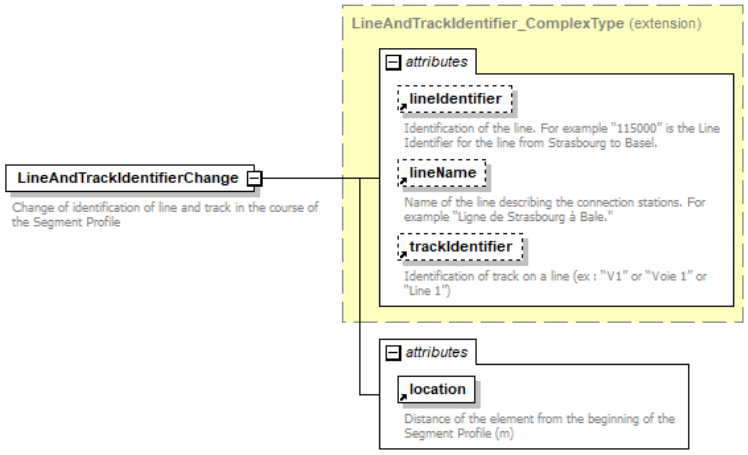
	Maximum Brake Force (kN) to be applied by the Dynamic Brake
annotation	documentation [SUBSET-126] Dynamic Brake Limitation Area
source	<xs:element name="LimitedDynamicBrakeForceArea"> <xs:annotation> <xs:documentation>[SUBSET-126] Dynamic Brake Limitation Area</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericArea_ComplexType"> <xs:attribute ref="dynamicBrakeForceLimit" use="required"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **LineAndTrackIdentifier**

diagram	
properties	content complex
children	**LineAndTrackIdentifierStart** **LineAndTrackIdentifierChange**
used by	element **SP_ContextInformation**
annotation	documentation Identification of line and track
source	<xs:element name="LineAndTrackIdentifier"> <xs:annotation> <xs:documentation>Identification of line and track</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="LineAndTrackIdentifierStart"/> <xs:element ref="LineAndTrackIdentifierChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **LineAndTrackIdentifierChange**

diagram



type: extension of **LineAndTrackIdentifier_ComplexType**

properties: content complex

used by: element **LineAndTrackIdentifier**

attributes

Name	Type	Use	Default	Fixed	Annotation
lineIdentifier	**xs:string**				documentation: Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.
lineName					documentation: Name of the line describing the connection stations. For example “Ligne de Strasbourg à Bale."
trackIdentifier					documentation: Identification of track on a line (ex : “V1” or “Voie 1” or “Line 1”)
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

documentation
Change of identification of line and track in the course of the Segment Profile

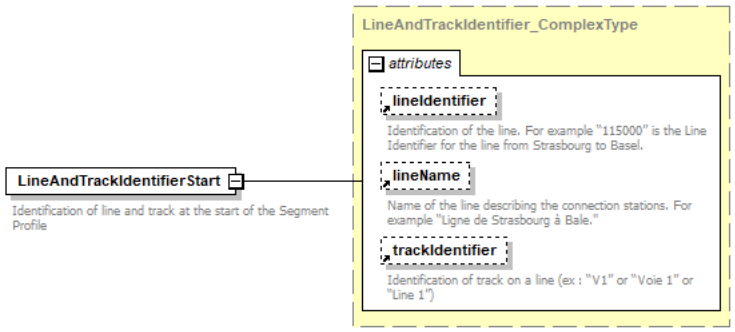
source

```
<xs:element name="LineAndTrackIdentifierChange">
  <xs:annotation>
    <xs:documentation>Change of identification of line and track in the course of the Segment Profile</xs:documentation>
  </xs:annotation>
  <xs:complexType>
```

```
    <xs:complexContent>
      <xs:extension base="LineAndTrackIdentifier_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **LineAndTrackIdentifierStart**

diagram



type **LineAndTrackIdentifier_ComplexType**

properties
- content complex

used by
- element **LineAndTrackIdentifier**

attributes

Name	Type	Use	Default	Fixed	Annotation
lineIdentifier	**xs:string**				documentation Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.
lineName					documentation Name of the line describing the connection stations. For example “Ligne de Strasbourg à Bale."
trackIdentifier					documentation Identification of track on a line (ex : “V1” or “Voie 1” or “Line 1”)

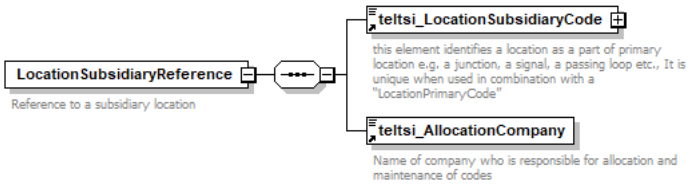
annotation
documentation
Identification of line and track at the start of the Segment Profile

source

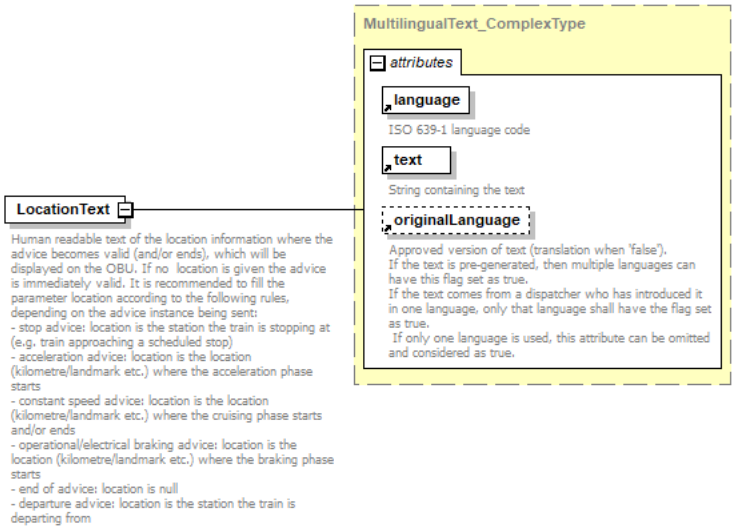
```
<xs:element name="LineAndTrackIdentifierStart" type="LineAndTrackIdentifier_ComplexType">
  <xs:annotation>
    <xs:documentation>Identification of line and track at the start of the Segment
Profile</xs:documentation>
  </xs:annotation>
```

	</xs:element>

element **LocationSubsidiaryReference**

diagram	
properties	content complex
children	**teltsi_LocationSubsidiaryCode** **teltsi_AllocationCompany**
used by	element **TAF_TAP_LocationReference**
annotation	documentation Reference to a subsidiary location
source	<xs:element name="LocationSubsidiaryReference"> <xs:annotation> <xs:documentation>Reference to a subsidiary location</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_LocationSubsidiaryCode"/> <xs:element ref="teltsi_AllocationCompany"/> </xs:sequence> </xs:complexType> </xs:element>

element **LocationText**

diagram	
type	**MultilingualText_ComplexType**
properties	content complex
used by	element **C_DAS_C_Advice**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

annotation

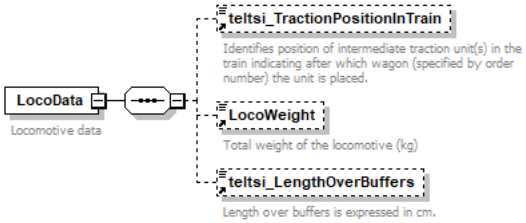
documentation

Human readable text of the location information where the advice becomes valid (and/or ends), which will be displayed on the OBU. If no location is given the advice is immediately valid. It is recommended to fill the parameter location according to the following rules, depending on the advice instance being sent:
- stop advice: location is the station the train is stopping at (e.g. train approaching a scheduled stop)
- acceleration advice: location is the location (kilometre/landmark etc.) where the acceleration phase starts
- constant speed advice: location is the location (kilometre/landmark etc.) where the cruising phase starts and/or ends
- operational/electrical braking advice: location is the location (kilometre/landmark etc.) where the braking phase starts
- end of advice: location is null
- departure advice: location is the station the train is departing from

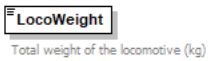
source

```
<xs:element name="LocationText" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Human readable text of the location information where the advice becomes valid (and/or ends), which will be displayed on the OBU. If no location is given the advice is immediately valid. It is recommended to fill the parameter location according to the following rules, depending on the advice instance being sent:
- stop advice: location is the station the train is stopping at (e.g. train approaching a scheduled stop)
- acceleration advice: location is the location (kilometre/landmark etc.) where the acceleration phase starts
- constant speed advice: location is the location (kilometre/landmark etc.) where the cruising phase starts and/or ends
- operational/electrical braking advice: location is the location (kilometre/landmark etc.) where the braking phase starts
- end of advice: location is null
- departure advice: location is the station the train is departing from</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **LocoData**

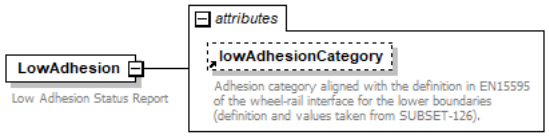
diagram	
properties	content complex
children	**teltsi_TractionPositionInTrain** **LocoWeight** **teltsi_LengthOverBuffers**
used by	element **VehicleData**
annotation	documentation Locomotive data
source	<xs:element name="LocoData"> <xs:annotation> <xs:documentation>Locomotive data</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_TractionPositionInTrain" minOccurs="0"/> <xs:element ref="LocoWeight" minOccurs="0"/> <xs:element ref="teltsi_LengthOverBuffers" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **LocoWeight**

diagram	
type	**teltsi_WeightValueKilo**
properties	content simple
used by	element **LocoData**
facets	Kind / Value / Annotation minInclusive 0 maxInclusive 999999 whiteSpace collapse
annotation	documentation Total weight of the locomotive (kg)
source	<xs:element name="LocoWeight" type="teltsi_WeightValueKilo"> <xs:annotation> <xs:documentation>Total weight of the locomotive (kg)</xs:documentation> </xs:annotation> </xs:element>

element **LowAdhesion**

diagram



properties

content complex

used by

element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
lowAdhesionCategory	**adhesionCategoryType**				documentation Adhesion category aligned with the definition in EN15595 of the wheel-rail interface for the lower boundaries (definition and values taken from SUBSET-126).

annotation

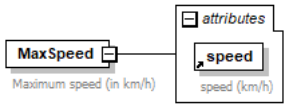
documentation
Low Adhesion Status Report

source

```
<xs:element name="LowAdhesion">
  <xs:annotation>
    <xs:documentation>Low Adhesion Status Report</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="lowAdhesionCategory"/>
  </xs:complexType>
</xs:element>
```

element **MaxSpeed**

diagram



properties

content complex

used by

element **SignalInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
speed	**speedType**	required			documentation speed (km/h)

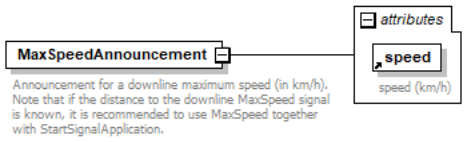
annotation

documentation
Maximum speed (in km/h)

source

```
<xs:element name="MaxSpeed">
  <xs:annotation>
    <xs:documentation>Maximum speed (in km/h)</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="speed" use="required"/>
  </xs:complexType>
</xs:element>
```

element **MaxSpeedAnnouncement**

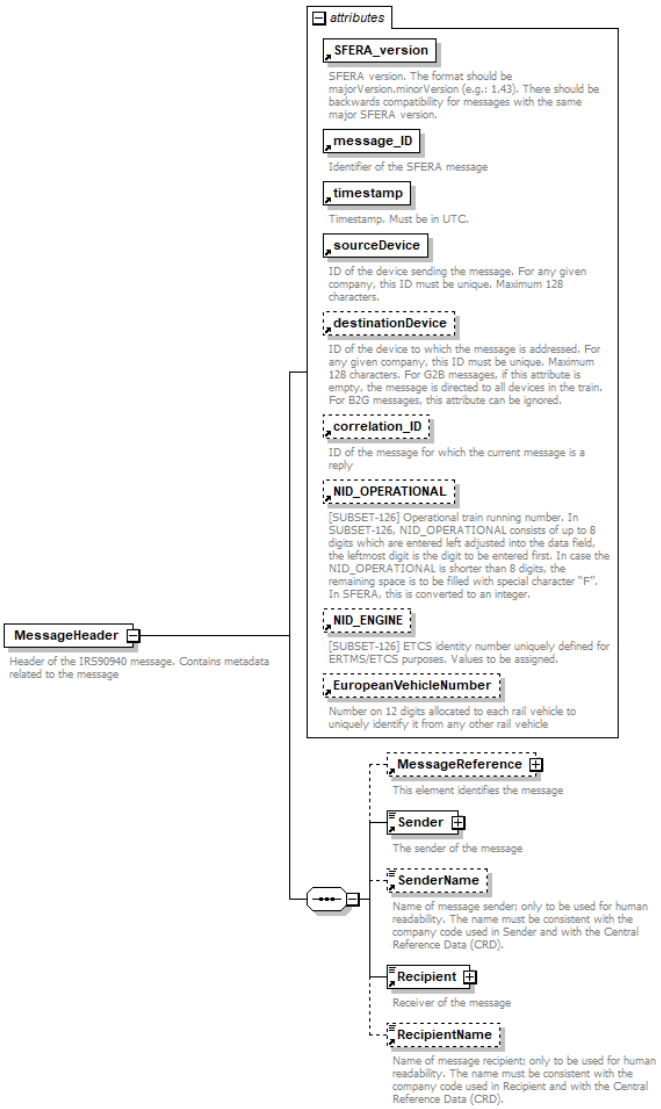
diagram	
properties	content complex
used by	element **SignalInformation**
attributes	Name: speed; Type: **speedType**; Use: required; Default: ; Fixed: ; Annotation documentation: speed (km/h)
annotation	documentation Announcement for a downline maximum speed (in km/h). Note that if the distance to the downline MaxSpeed signal is known, it is recommended to use MaxSpeed together with StartSignalApplication.
source	<xs:element name="MaxSpeedAnnouncement"> <xs:annotation> <xs:documentation>Announcement for a downline maximum speed (in km/h). Note that if the distance to the downline MaxSpeed signal is known, it is recommended to use MaxSpeed together with StartSignalApplication.</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="speed" use="required"/> </xs:complexType> </xs:element>

element **MaxSpeedTrack**

diagram	
type	**xs:string**
properties	content simple
used by	element **TrainLocationInformation**
annotation	documentation Maximum speed of the train possible at this track
source	<xs:element name="MaxSpeedTrack" type="xs:string"> <xs:annotation> <xs:documentation>Maximum speed of the train possible at this track</xs:documentation> </xs:annotation> </xs:element>

element **MessageHeader**

diagram



properties

content complex

children **MessageReference** **Sender** **SenderName** **Recipient** **RecipientName**

used by

elements **SFERA_B2G_EventMessage** **SFERA_B2G_ReplyMessage** **SFERA_B2G_RequestMessage** **SFERA_G2B_EventMessage** **SFERA_G2B_ReplyMessage** **SFERA_G2B_RequestMessage**

attributes

Name	Type	Use	Default	Fixed	Annotation
SFERA_version	**derived by: xs:string**	required			documentation: SFERA version. The format should be majorVersion.minorVersion (e.g.: 1.43). There should be backwards compatibility for messages with the same major SFERA version.
message_ID	**UUIDv4_Type**	required			documentation: Identifier of the SFERA message
timestamp	**UTCDateTimeType**	required			documentation: Timestamp. Must be in UTC.

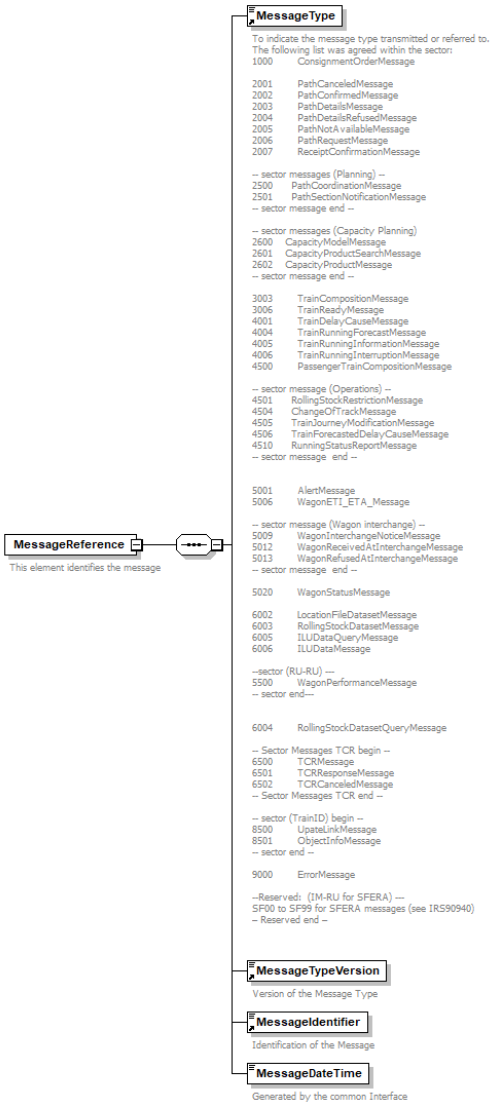
	sourceDevice	**deviceType**	required	documentation ID of the device sending the message. For any given company, this ID must be unique. Maximum 128 characters.
	destinationDevice	**deviceType**		documentation ID of the device to which the message is addressed. For any given company, this ID must be unique. Maximum 128 characters. For G2B messages, if this attribute is empty, the message is directed to all devices in the train. For B2G messages, this attribute can be ignored.
	correlation_ID	**UUIDv4_Type**		documentation ID of the message for which the current message is a reply
	NID_OPERATIONAL	**xs:unsignedInt**		documentation [SUBSET-126] Operational train running number. In SUBSET-126, NID_OPERATIONAL consists of up to 8 digits which are entered left adjusted into the data field, the leftmost digit is the digit to be entered first. In case the NID_OPERATIONAL is shorter than 8 digits, the remaining space is to be filled with special character “F”. In SFERA, this is converted to an integer.
	NID_ENGINE	**derived by:** **xs:unsignedInt**		documentation [SUBSET-126] ETCS identity number uniquely defined for ERTMS/ETCS purposes. Values to be assigned.
	EuropeanVehicleNumber	**derived by:** **xs:string**		documentation Number on 12 digits allocated to each rail vehicle to uniquely identify it from any other rail vehicle
annotation	documentation Header of the IRS90940 message. Contains metadata related to the message			
source	<xs:element name="MessageHeader"> <xs:annotation> <xs:documentation>Header of the IRS90940 message. Contains metadata related to the message</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageReference" minOccurs="0"/> <xs:element ref="Sender"/> <xs:element ref="SenderName" minOccurs="0"/> <xs:element ref="Recipient"/> <xs:element ref="RecipientName" minOccurs="0"/> </xs:sequence> <xs:attribute ref="SFERA_version" use="required"/>			

```
    <xs:attribute ref="message_ID" use="required"/>
    <xs:attribute ref="timestamp" use="required"/>
    <xs:attribute ref="sourceDevice" use="required"/>
    <xs:attribute ref="destinationDevice"/>
    <xs:attribute ref="correlation_ID"/>
    <xs:attribute ref="NID_OPERATIONAL"/>
    <xs:attribute ref="NID_ENGINE"/>
    <xs:attribute ref="EuropeanVehicleNumber"/>
  </xs:complexType>
</xs:element>
```

element **MessageIdentifier**

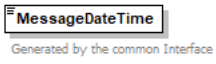
diagram	
type	**teltsi_FreeText**
properties	content simple
used by	element **MessageReference**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation Identification of the Message
source	<xs:element name="MessageIdentifier" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>Identification of the Message</xs:documentation> </xs:annotation> </xs:element>

element **MessageReference**

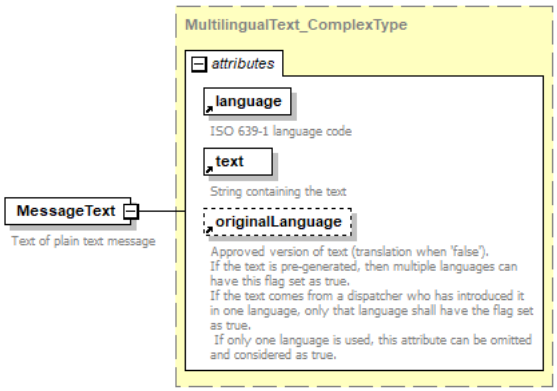
diagram	
properties	content complex
children	**MessageType** **MessageTypeVersion** **MessageIdentifier** **MessageDateTime**
used by	element **MessageHeader**
annotation	documentation This element identifies the message
source	<xs:element name="MessageReference"> <xs:annotation> <xs:documentation>This element identifies the message</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageType"/> <xs:element ref="MessageTypeVersion"/> <xs:element ref="MessageIdentifier"/> <xs:element name="MessageDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Generated by the common Interface</xs:documentation>

```
      </xs:annotation>
    </xs:element>
  </xs:sequence>
 </xs:complexType>
</xs:element>
```

element **MessageReference/MessageDateTime**

diagram	
type	**xs:dateTime**
properties	content simple
annotation	documentation Generated by the common Interface
source	<xs:element name="MessageDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Generated by the common Interface</xs:documentation> </xs:annotation> </xs:element>

element **MessageText**

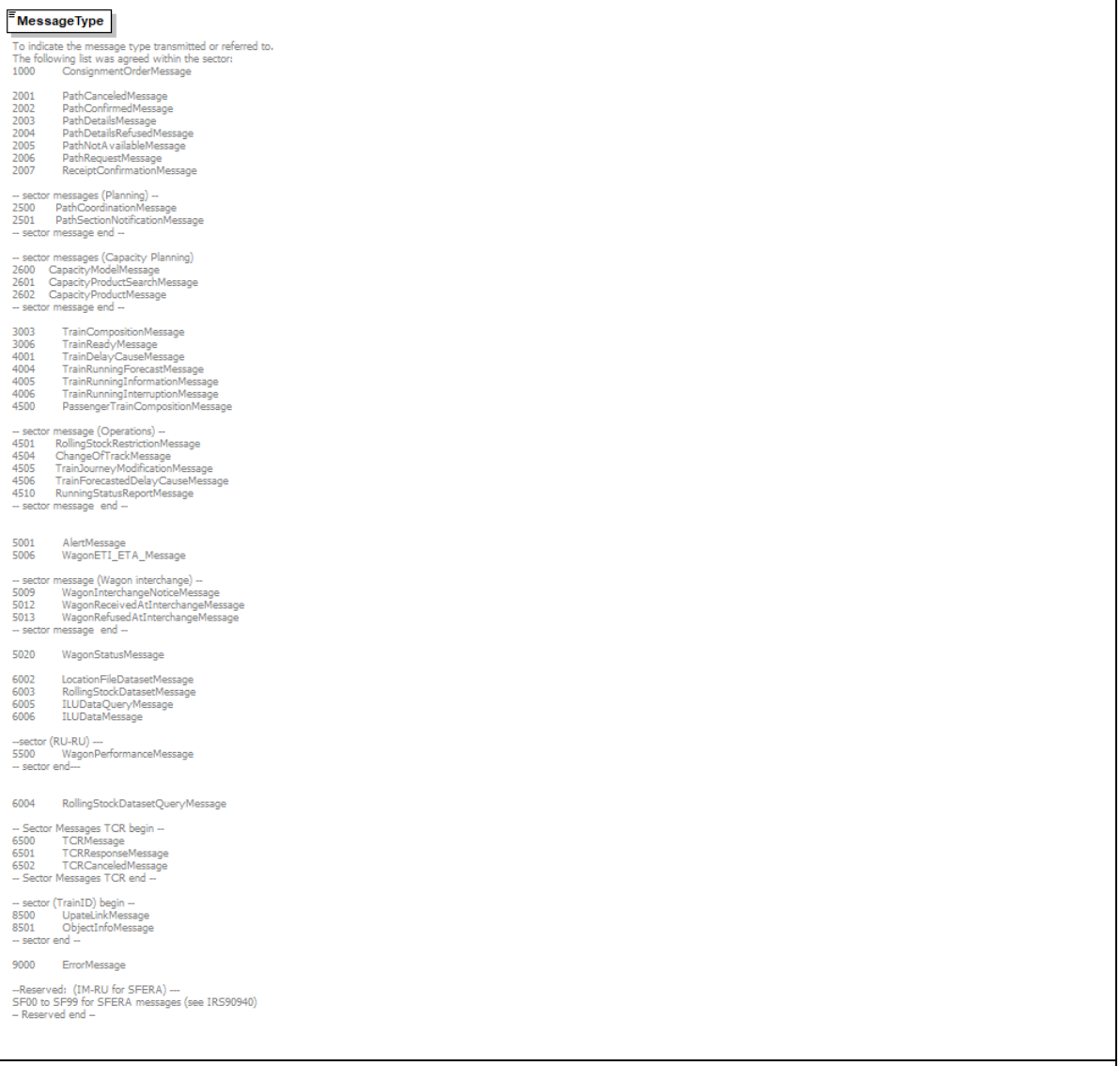
diagram	
type	**MultilingualText_ComplexType**
properties	content complex
used by	element **PlaintextMessage**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated,

		then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
annotation	documentation Text of plain text message	
source	<xs:element name="MessageText" type="MultilingualText_ComplexType"> <xs:annotation> <xs:documentation>Text of plain text message</xs:documentation> </xs:annotation> </xs:element>	

element **MessageType**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	element **MessageReference**
facets	Kind Value Annotation minLength 1 maxLength 4
annotation	documentation To indicate the message type transmitted or referred to. The following list was agreed within the sector: 1000 ConsignmentOrderMessage 2001 PathCanceledMessage 2002 PathConfirmedMessage 2003 PathDetailsMessage 2004 PathDetailsRefusedMessage 2005 PathNotAvailableMessage 2006 PathRequestMessage 2007 ReceiptConfirmationMessage -- sector messages (Planning) --

2500 PathCoordinationMessage
2501 PathSectionNotificationMessage
-- sector message end --

-- sector messages (Capacity Planning)
2600 CapacityModelMessage
2601 CapacityProductSearchMessage
2602 CapacityProductMessage
-- sector message end --

3003 TrainCompositionMessage
3006 TrainReadyMessage
4001 TrainDelayCauseMessage
4004 TrainRunningForecastMessage
4005 TrainRunningInformationMessage
4006 TrainRunningInterruptionMessage
4500 PassengerTrainCompositionMessage

-- sector message (Operations) --
4501 RollingStockRestrictionMessage
4504 ChangeOfTrackMessage
4505 TrainJourneyModificationMessage
4506 TrainForecastedDelayCauseMessage
4510 RunningStatusReportMessage
-- sector message end --

5001 AlertMessage
5006 WagonETI_ETA_Message

-- sector message (Wagon interchange) --
5009 WagonInterchangeNoticeMessage
5012 WagonReceivedAtInterchangeMessage
5013 WagonRefusedAtInterchangeMessage
-- sector message end --

5020 WagonStatusMessage

6002 LocationFileDatasetMessage
6003 RollingStockDatasetMessage
6005 ILUDataQueryMessage
6006 ILUDataMessage

--sector (RU-RU) ---
5500 WagonPerformanceMessage
-- sector end---

6004 RollingStockDatasetQueryMessage

-- Sector Messages TCR begin --
6500 TCRMessage
6501 TCRResponseMessage
6502 TCRCanceledMessage
-- Sector Messages TCR end --

-- sector (TrainID) begin --
8500 UpateLinkMessage
8501 ObjectInfoMessage
-- sector end --

9000 ErrorMessage

--Reserved: (IM-RU for SFERA) ---
SF00 to SF99 for SFERA messages (see IRS90940)
– Reserved end –

source

```
<xs:element name="MessageType">
  <xs:annotation>
    <xs:documentation>To indicate the message type transmitted or referred to. The following list
was agreed within the sector:
1000        ConsignmentOrderMessage

2001        PathCanceledMessage
2002        PathConfirmedMessage
2003        PathDetailsMessage
2004        PathDetailsRefusedMessage
2005        PathNotAvailableMessage
2006        PathRequestMessage
2007        ReceiptConfirmationMessage

-- sector messages (Planning) --
2500      PathCoordinationMessage
2501      PathSectionNotificationMessage
-- sector message end --

-- sector messages (Capacity Planning)
2600    CapacityModelMessage
2601    CapacityProductSearchMessage
2602    CapacityProductMessage
-- sector message end --

3003        TrainCompositionMessage
3006        TrainReadyMessage
4001        TrainDelayCauseMessage
4004        TrainRunningForecastMessage
4005        TrainRunningInformationMessage
4006        TrainRunningInterruptionMessage
4500        PassengerTrainCompositionMessage

-- sector message (Operations) --
4501      RollingStockRestrictionMessage
4504      ChangeOfTrackMessage
4505      TrainJourneyModificationMessage
4506      TrainForecastedDelayCauseMessage
4510      RunningStatusReportMessage
-- sector message  end --


5001        AlertMessage
5006        WagonETI_ETA_Message

-- sector message (Wagon interchange) --
5009        WagonInterchangeNoticeMessage
5012        WagonReceivedAtInterchangeMessage
5013        WagonRefusedAtInterchangeMessage
-- sector message  end --

5020        WagonStatusMessage

6002        LocationFileDatasetMessage
6003        RollingStockDatasetMessage
```

6005 ILUDataQueryMessage
6006 ILUDataMessage

--sector (RU-RU) ---
5500 WagonPerformanceMessage
-- sector end---

6004 RollingStockDatasetQueryMessage

-- Sector Messages TCR begin --
6500 TCRMessage
6501 TCRResponseMessage
6502 TCRCanceledMessage
-- Sector Messages TCR end --

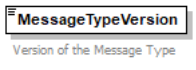
-- sector (TrainID) begin --
8500 UpateLinkMessage
8501 ObjectInfoMessage
-- sector end --

9000 ErrorMessage

--Reserved: (IM-RU for SFERA) ---
SF00 to SF99 for SFERA messages (see IRS90940)
– Reserved end –

```
                    </xs:documentation>
  </xs:annotation>
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:minLength value="1"/>
      <xs:maxLength value="4"/>
    </xs:restriction>
  </xs:simpleType>
</xs:element>
```

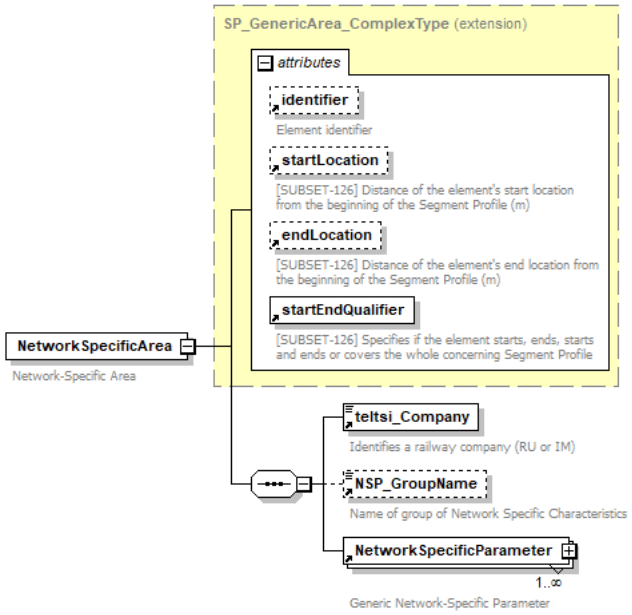
element **MessageTypeVersion**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	element **MessageReference**
facets	Kind Value Annotation maxLength 25
annotation	documentation Version of the Message Type
source	<xs:element name="MessageTypeVersion"> <xs:annotation> <xs:documentation>Version of the Message Type</xs:documentation> </xs:annotation> <xs:simpleType>

```
    <xs:restriction base="xs:string">
      <xs:maxLength value="25"/>
    </xs:restriction>
  </xs:simpleType>
</xs:element>
```

element **NetworkSpecificArea**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

children: **teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**

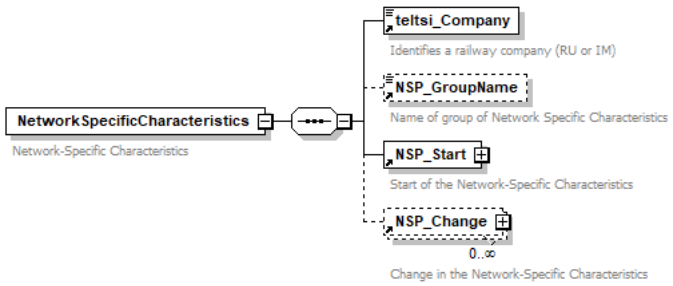
used by: elements **SP_Areas** **SP_ContextInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the

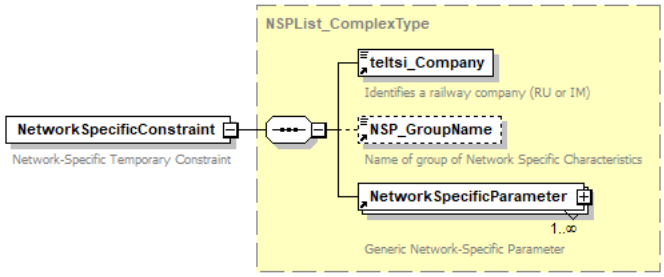
	element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation Network-Specific Area
source	<xs:element name="NetworkSpecificArea"> <xs:annotation> <xs:documentation>Network-Specific Area</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericArea_ComplexType"> <xs:sequence> <xs:element ref="teltsi_Company"/> <xs:element ref="NSP_GroupName" minOccurs="0"/> <xs:element ref="NetworkSpecificParameter" maxOccurs="unbounded"/> </xs:sequence> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **NetworkSpecificCharacteristics**

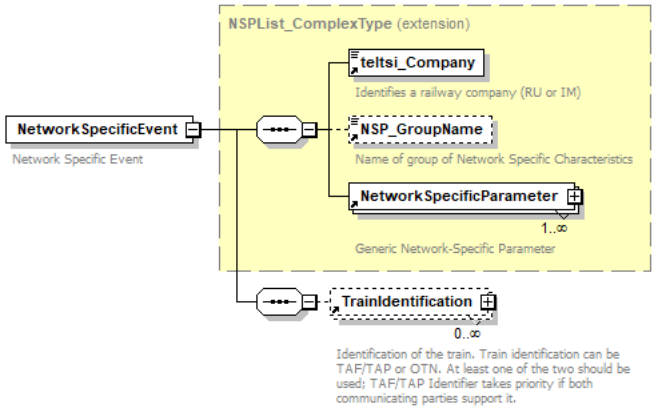
diagram	
properties	content complex
children	**teltsi_Company NSP_GroupName NSP_Start NSP_Change**
used by	elements **SP_Characteristics SP_ContextInformation**
annotation	documentation Network-Specific Characteristics
source	<xs:element name="NetworkSpecificCharacteristics"> <xs:annotation> <xs:documentation>Network-Specific Characteristics</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_Company"/> <xs:element ref="NSP_GroupName" minOccurs="0"/> <xs:element ref="NSP_Start"/> <xs:element ref="NSP_Change" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence>

	</xs:complexType> </xs:element>

element **NetworkSpecificConstraint**

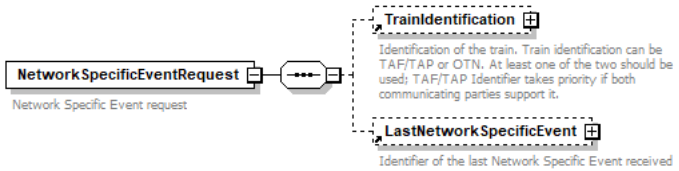
diagram	
type	NSPList_ComplexType
properties	content complex
children	teltsi_Company NSP_GroupName NetworkSpecificParameter
used by	element TemporaryConstraints
annotation	documentation Network-Specific Temporary Constraint
source	<xs:element name="NetworkSpecificConstraint" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network-Specific Temporary Constraint</xs:documentation> </xs:annotation> </xs:element>

element **NetworkSpecificEvent**

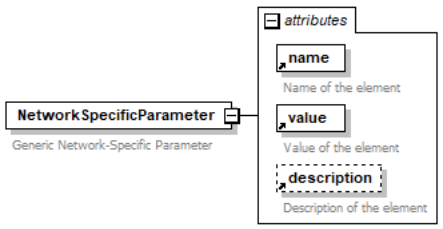
diagram	
type	extension of NSPList_ComplexType
properties	content complex
children	teltsi_Company NSP_GroupName NetworkSpecificParameter TrainIdentification
used by	elements B2G_EventPayload B2G_ReplyPayload G2B_EventPayload G2B_ReplyPayload
annotation	documentation Network Specific Event
source	<xs:element name="NetworkSpecificEvent">

```
  <xs:annotation>
    <xs:documentation>Network Specific Event</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="NSPList_ComplexType">
        <xs:sequence>
          <xs:element ref="TrainIdentification" minOccurs="0" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **NetworkSpecificEventRequest**

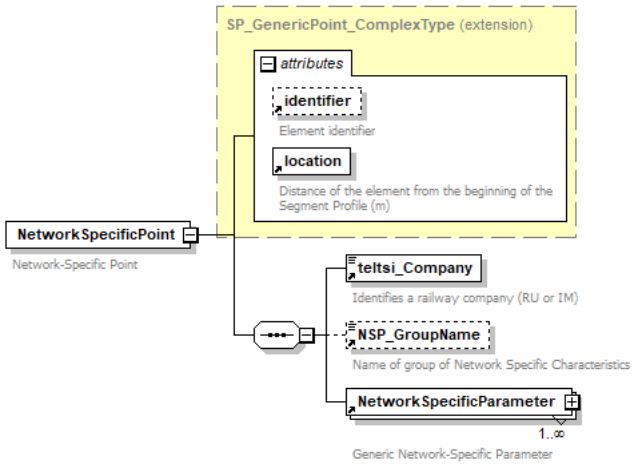
diagram	
properties	content complex
children	**TrainIdentification** **LastNetworkSpecificEvent**
used by	elements **B2G_Request** **G2B_Request**
annotation	documentation Network Specific Event request
source	<xs:element name="NetworkSpecificEventRequest"> <xs:annotation> <xs:documentation>Network Specific Event request</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> <xs:element ref="LastNetworkSpecificEvent" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **NetworkSpecificParameter**

diagram	
properties	content complex
used by	elements **NetworkSpecificArea** **NetworkSpecificPoint** **NSP_Change** **NSP_Start** complexType **NSPList_ComplexType**

attributes	Name	Type	Use	Default	Fixed	Annotation
	name	**xs:string**	required			documentation Name of the element
	value	**xs:string**	required			documentation Value of the element
	description	**xs:string**				documentation Description of the element
annotation	documentation Generic Network-Specific Parameter					
source	<xs:element name="NetworkSpecificParameter"> <xs:annotation> <xs:documentation>Generic Network-Specific Parameter</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="name" use="required"/> <xs:attribute ref="value" use="required"/> <xs:attribute ref="description"/> </xs:complexType> </xs:element>					

element **NetworkSpecificPoint**

diagram	
type	extension of **SP_GenericPoint_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	elements **SP_ContextInformation** **SP_Points**

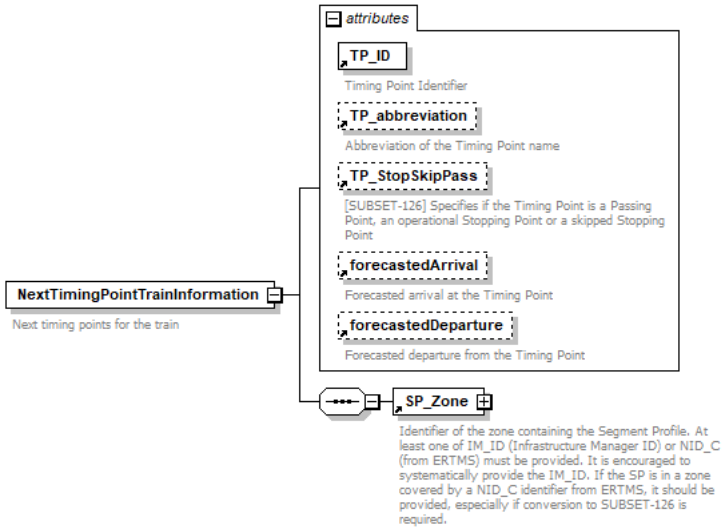
attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**				documentation Element identifier
	location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
annotation	documentation Network-Specific Point					

source

```
<xs:element name="NetworkSpecificPoint">
 <xs:annotation>
  <xs:documentation>Network-Specific Point</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericPoint_ComplexType">
    <xs:sequence>
     <xs:element ref="teltsi_Company"/>
     <xs:element ref="NSP_GroupName" minOccurs="0"/>
     <xs:element ref="NetworkSpecificParameter" maxOccurs="unbounded"/>
    </xs:sequence>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **NextTimingPointTrainInformation**

diagram



properties

content complex

children

SP_Zone

used by

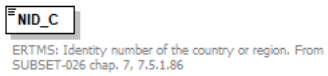
element **TrainLocationInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
TP_ID	**xs:string**	required			documentation Timing Point Identifier
TP_abbreviation	**xs:string**				documentation Abbreviation of the Timing Point name
TP_StopSkipPass	**derived by: xs:string**				documentation [SUBSET-126] Specifies if the Timing Point is a Passing Point, an operational Stopping Point

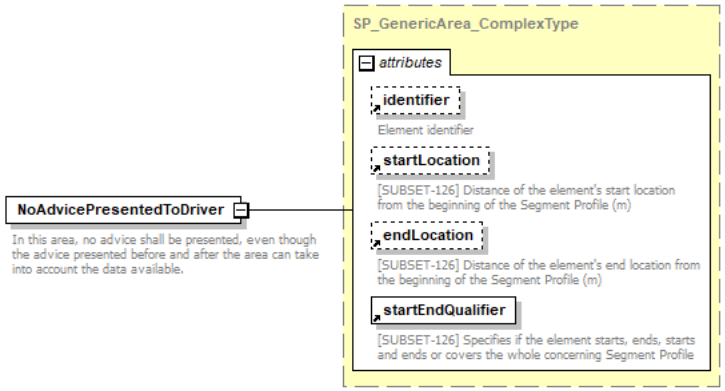
	forecastedArrival **UTCDateTimeType** or a skipped Stopping Point documentation Forecasted arrival at the Timing Point forecastedDeparture **UTCDateTimeType** documentation Forecasted departure from the Timing Point
annotation	documentation Next timing points for the train
source	<xs:element name="NextTimingPointTrainInformation"> <xs:annotation> <xs:documentation>Next timing points for the train</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="SP_Zone"/> </xs:sequence> <xs:attribute ref="TP_ID" use="required"/> <xs:attribute ref="TP_abbreviation"/> <xs:attribute ref="TP_StopSkipPass"/> <xs:attribute ref="forecastedArrival"/> <xs:attribute ref="forecastedDeparture"/> </xs:complexType> </xs:element>

element **NID_C**

diagram	
type	**NID_C_Type**
properties	content simple
used by	elements **ATOTS_ContactInfo** **BaliseGroup** complexType **SP_Zone_ComplexType**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1023
annotation	documentation ERTMS: Identity number of the country or region. From SUBSET-026 chap. 7, 7.5.1.86
source	<xs:element name="NID_C" type="NID_C_Type"> <xs:annotation> <xs:documentation>ERTMS: Identity number of the country or region. From SUBSET-026 chap. 7, 7.5.1.86</xs:documentation> </xs:annotation> </xs:element>

element **NoAdvicePresentedToDriver**

diagram



type **SP_GenericArea_ComplexType**

properties
- content complex

used by
- element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

annotation

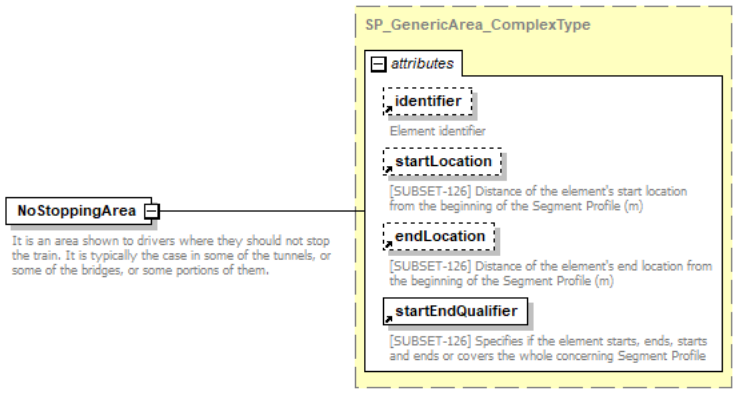
documentation
In this area, no advice shall be presented, even though the advice presented before and after the area can take into account the data available.

source

```
<xs:element name="NoAdvicePresentedToDriver" type="SP_GenericArea_ComplexType">
  <xs:annotation>
    <xs:documentation>In this area, no advice shall be presented, even though the advice presented before and after the area can take into account the data available.</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **NoStoppingArea**

diagram



type **SP_GenericArea_ComplexType**

properties
content complex

used by
element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

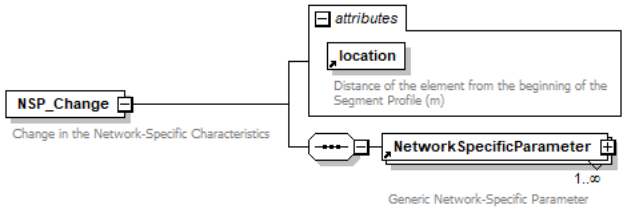
annotation
documentation
It is an area shown to drivers where they should not stop the train. It is typically the case in some of the tunnels, or some of the bridges, or some portions of them.

source

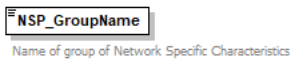
```
<xs:element name="NoStoppingArea" type="SP_GenericArea_ComplexType">
  <xs:annotation>
    <xs:documentation>It is an area shown to drivers where they should not stop the train. It is typically the case in some of the tunnels, or some of the bridges, or some portions of them.</xs:documentation>
```

	</xs:annotation> </xs:element>

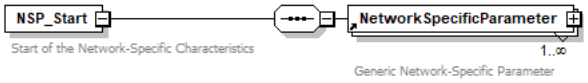
element **NSP_Change**

diagram	
properties	content complex
children	**NetworkSpecificParameter**
used by	element **NetworkSpecificCharacteristics**
attributes	Name: location; Type: **distanceType**; Use: required; Default: ; Fixed: ; Annotation: documentation: Distance of the element from the beginning of the Segment Profile (m)
annotation	documentation Change in the Network-Specific Characteristics
source	<xs:element name="NSP_Change"> <xs:annotation> <xs:documentation>Change in the Network-Specific Characteristics</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="NetworkSpecificParameter" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="location" use="required"/> </xs:complexType> </xs:element>

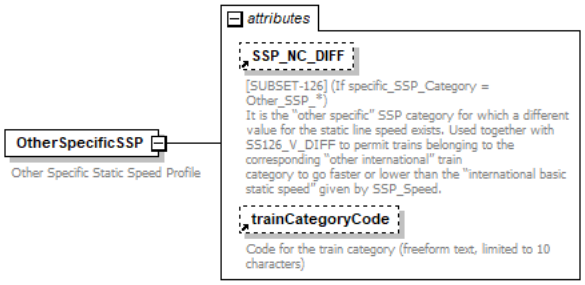
element **NSP_GroupName**

diagram	
used by	elements **NetworkSpecificArea** **NetworkSpecificCharacteristics** **NetworkSpecificPoint** complexType **NSPList_ComplexType**
annotation	documentation Name of group of Network Specific Characteristics
source	<xs:element name="NSP_GroupName"> <xs:annotation> <xs:documentation>Name of group of Network Specific Characteristics</xs:documentation> </xs:annotation> </xs:element>

element **NSP_Start**

diagram	
properties	content complex
children	**NetworkSpecificParameter**
used by	element **NetworkSpecificCharacteristics**
annotation	documentation Start of the Network-Specific Characteristics
source	<xs:element name="NSP_Start"> <xs:annotation> <xs:documentation>Start of the Network-Specific Characteristics</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="NetworkSpecificParameter" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **OtherSpecificSSP**

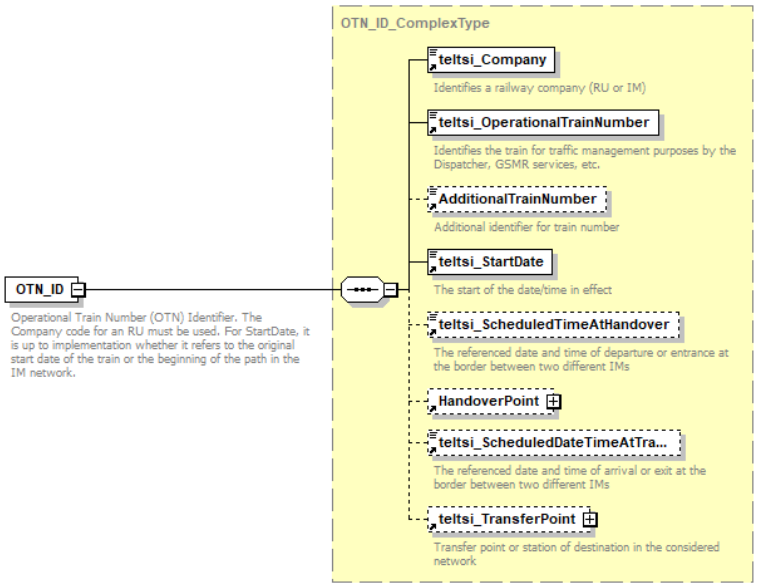
diagram	
properties	content complex
used by	complexType **SpecificSSP_ComplexType**

attributes

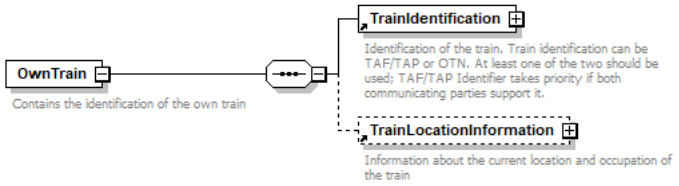
Name	Type	Use	Default	Fixed	Annotation
SSP_NC_DIFF	**NC_TrainType**				documentation [SUBSET-126] (If specific_SSP_Category = Other_SSP_*) It is the "other specific" SSP category for which a different value for the static line speed exists. Used together with SS126_V_DIFF to permit trains belonging to the corresponding "other international" train category to go faster or lower than the "international basic static speed" given by SSP_Speed.
trainCategoryCode	**derived by:** **xs:string**				documentation Code for the train

	category (freeform text, limited to 10 characters)
annotation	documentation Other Specific Static Speed Profile
source	<xs:element name="OtherSpecificSSP"> <xs:annotation> <xs:documentation>Other Specific Static Speed Profile</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="SSP_NC_DIFF"/> <xs:attribute ref="trainCategoryCode"/> </xs:complexType> </xs:element>

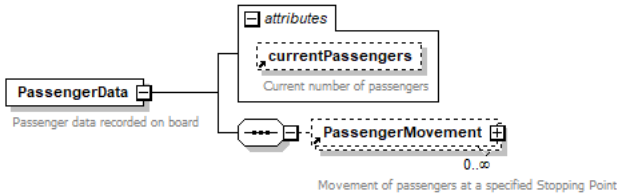
element **OTN_ID**

diagram	
type	**OTN_ID_ComplexType**
properties	content complex
children	**teltsi_Company teltsi_OperationalTrainNumber AdditionalTrainNumber teltsi_StartDate teltsi_ScheduledTimeAtHandover HandoverPoint teltsi_ScheduledDateTimeAtTransfer teltsi_TransferPoint**
used by	complexType **TrainIdentification_ComplexType**
annotation	documentation Operational Train Number (OTN) Identifier. The Company code for an RU must be used. For StartDate, it is up to implementation whether it refers to the original start date of the train or the beginning of the path in the IM network.
source	<xs:element name="OTN_ID" type="OTN_ID_ComplexType"> <xs:annotation> <xs:documentation>Operational Train Number (OTN) Identifier. The Company code for an RU must be used. For StartDate, it is up to implementation whether it refers to the original start date of the train or the beginning of the path in the IM network.</xs:documentation> </xs:annotation> </xs:element>

element **OwnTrain**

diagram	
properties	content complex
children	**TrainIdentification** **TrainLocationInformation**
used by	element **RelatedTrainInformation**
annotation	documentation Contains the identification of the own train
source	<xs:element name="OwnTrain"> <xs:annotation> <xs:documentation>Contains the identification of the own train</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="TrainLocationInformation" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

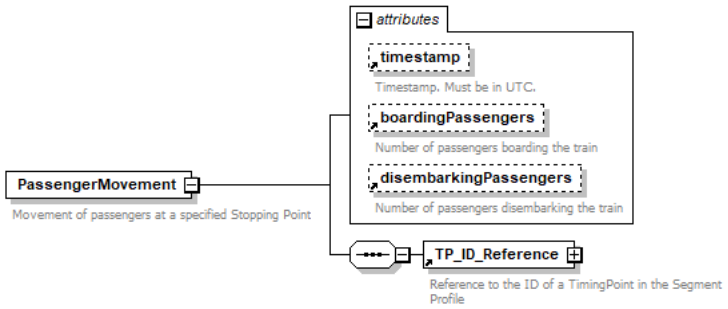
element **PassengerData**

diagram	
properties	content complex
children	**PassengerMovement**
used by	element **B2G_StatusReport**
attributes	Name: currentPassengers; Type: **xs:unsignedInt**; Use: ; Default: ; Fixed: ; Annotation: documentation: Current number of passengers
annotation	documentation Passenger data recorded on board
source	<xs:element name="PassengerData"> <xs:annotation> <xs:documentation>Passenger data recorded on board</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="PassengerMovement" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence>

```
    <xs:attribute ref="currentPassengers"/>
  </xs:complexType>
</xs:element>
```

element **PassengerMovement**

diagram



properties: content complex

children: **TP_ID_Reference**

used by: element **PassengerData**

attributes

Name	Type	Use	Default	Fixed	Annotation
timestamp	**UTCDateTimeType**				documentation: Timestamp. Must be in UTC.
boardingPassengers	**xs:unsignedInt**				documentation: Number of passengers boarding the train
disembarkingPassengers	**xs:unsignedInt**				documentation: Number of passengers disembarking the train

annotation

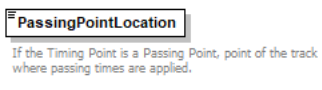
documentation
Movement of passengers at a specified Stopping Point

source

```
<xs:element name="PassengerMovement">
  <xs:annotation>
    <xs:documentation>Movement of passengers at a specified Stopping Point</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="TP_ID_Reference"/>
    </xs:sequence>
    <xs:attribute ref="timestamp"/>
    <xs:attribute ref="boardingPassengers"/>
    <xs:attribute ref="disembarkingPassengers"/>
  </xs:complexType>
</xs:element>
```

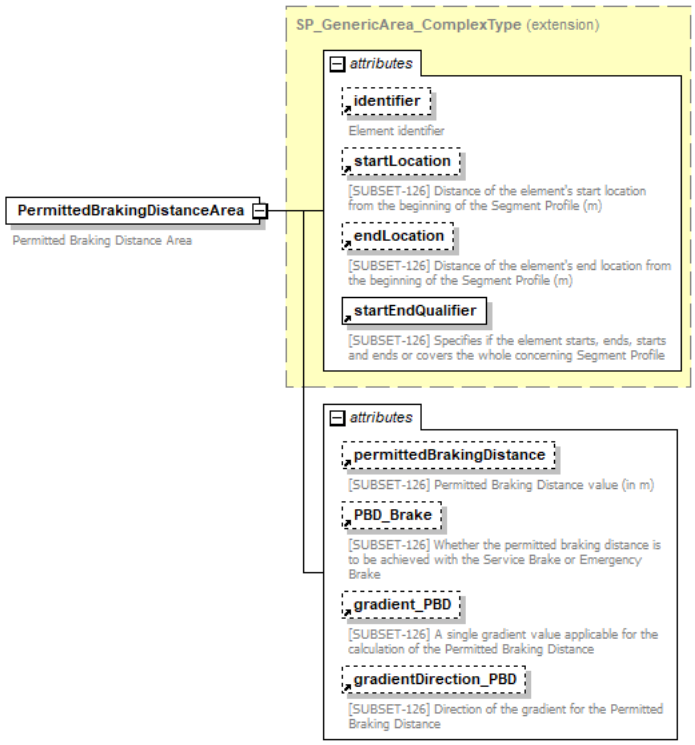
element **PassingPointLocation**

diagram



type	**xs:boolean**
properties	content simple
used by	element **TimingPoint**
annotation	documentation If the Timing Point is a Passing Point, point of the track where passing times are applied.
source	<xs:element name="PassingPointLocation" type="xs:boolean"> <xs:annotation> <xs:documentation>If the Timing Point is a Passing Point, point of the track where passing times are applied.</xs:documentation> </xs:annotation> </xs:element>

element **PermittedBrakingDistanceArea**

diagram	
type	extension of **SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**
attributes	Name / Type / Use / Default / Fixed / Annotation identifier — **xs:string** — documentation: Element identifier startLocation — **distanceType** — documentation: [SUBSET-126] Distance of the element's start location from the beginning of the Segment

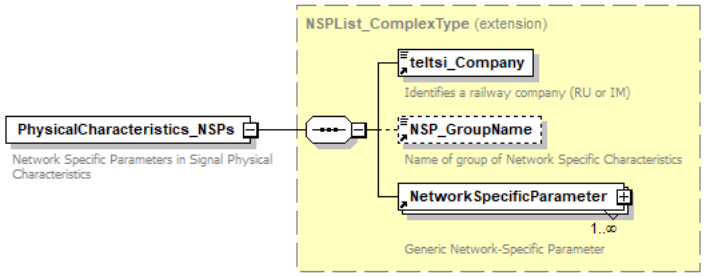
	endLocation	**distanceType**		Profile (m) documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
	startEndQualifier	**derived by: xs:string**	required	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
	permittedBrakingDistance	**distanceType**		documentation [SUBSET-126] Permitted Braking Distance value (in m)
	PBD_Brake	**derived by: xs:string**		documentation [SUBSET-126] Whether the permitted braking distance is to be achieved with the Service Brake or Emergency Brake
	gradient_PBD	**gradientValueType**		documentation [SUBSET-126] A single gradient value applicable for the calculation of the Permitted Braking Distance
	gradientDirection_PBD	**gradientDirectionType**		documentation [SUBSET-126] Direction of the gradient for the Permitted Braking Distance
annotation	documentation Permitted Braking Distance Area			

source

```
<xs:element name="PermittedBrakingDistanceArea">
 <xs:annotation>
  <xs:documentation>Permitted Braking Distance Area</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericArea_ComplexType">
    <xs:attribute ref="permittedBrakingDistance"/>
    <xs:attribute ref="PBD_Brake"/>
```

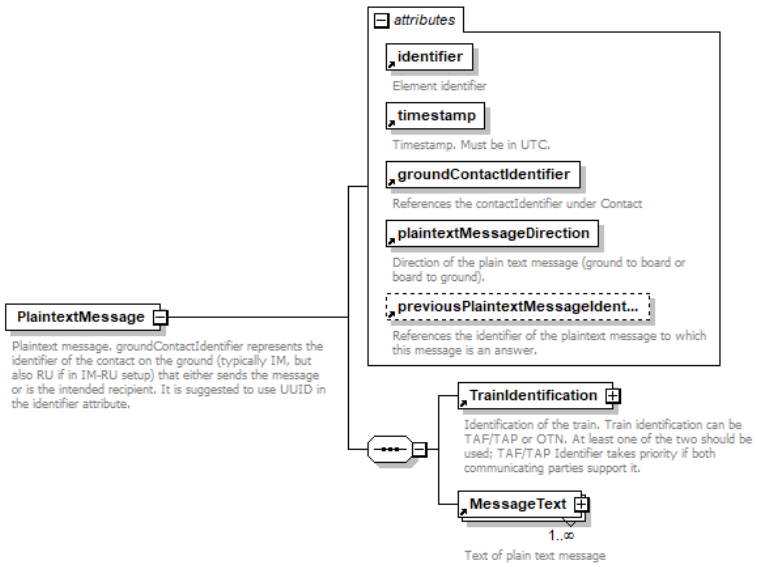
```
      <xs:attribute ref="gradient_PBD"/>
      <xs:attribute ref="gradientDirection_PBD"/>
    </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **PhysicalCharacteristics_NSPs**

diagram	
type	extension of **NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	complexType **PhysicalCharacteristics_ComplexType**
annotation	documentation Network Specific Parameters in Signal Physical Characteristics
source	<xs:element name="PhysicalCharacteristics_NSPs"> <xs:annotation> <xs:documentation>Network Specific Parameters in Signal Physical Characteristics</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="NSPList_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

element **PlaintextMessage**

diagram



properties

content complex

children **TrainIdentification** **MessageText**

used by

elements **B2G_EventPayload** **B2G_ReplyPayload** **G2B_EventPayload** **G2B_ReplyPayload**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**	required			documentation Element identifier
timestamp	**UTCDateTimeType**	required			documentation Timestamp. Must be in UTC.
groundContactIdentifier	**xs:string**	required			documentation References the contactIdentifier under Contact
plaintextMessageDirection	**derived by: xs:string**	required			documentation Direction of the plain text message (ground to board or board to ground).
previousPlaintextMessageIdentifier					documentation References the identifier of the plaintext message to which this message is an answer.

annotation

documentation

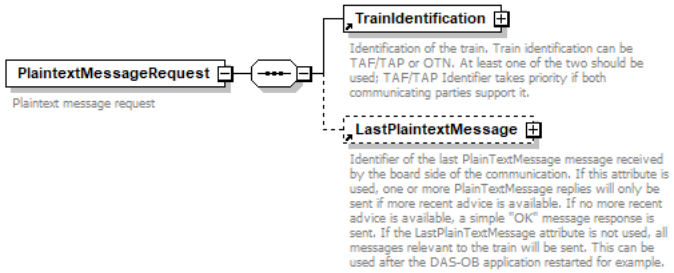
Plaintext message. groundContactIdentifier represents the identifier of the contact on the ground (typically IM, but also RU if in IM-RU setup) that either sends the message or is the intended recipient. It is suggested to use UUID in the identifier attribute.

source

```
<xs:element name="PlaintextMessage">
 <xs:annotation>
  <xs:documentation>Plaintext message. groundContactIdentifier represents the identifier of the contact on the ground (typically IM, but also RU if in IM-RU setup) that either sends the message or is the intended recipient. It is suggested to use UUID in the identifier attribute.</xs:documentation>
 </xs:annotation>
```

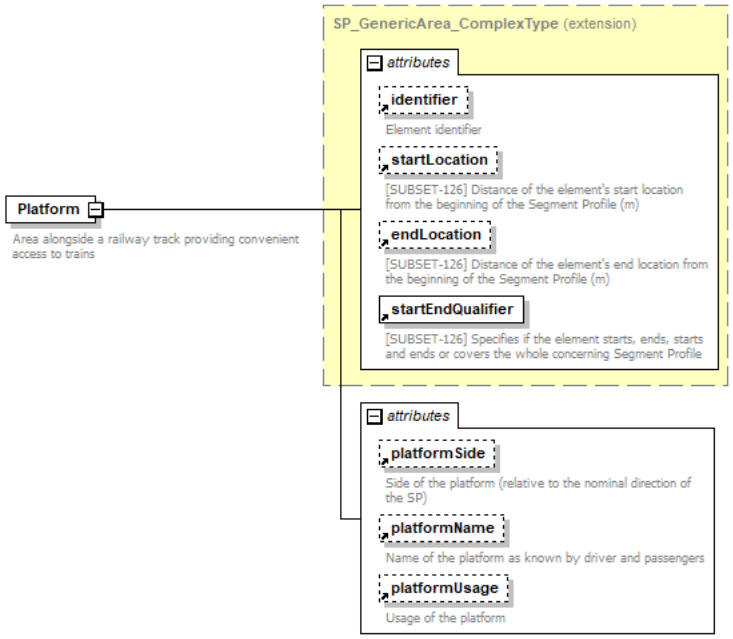
```
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="TrainIdentification"/>
      <xs:element ref="MessageText" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="identifier" use="required"/>
    <xs:attribute ref="timestamp" use="required"/>
    <xs:attribute ref="groundContactIdentifier" use="required"/>
    <xs:attribute ref="plaintextMessageDirection" use="required"/>
    <xs:attribute ref="previousPlaintextMessageIdentifier"/>
  </xs:complexType>
</xs:element>
```

element **PlaintextMessageRequest**

diagram	
properties	content complex
children	**TrainIdentification** **LastPlaintextMessage**
used by	elements **B2G_Request** **G2B_Request**
annotation	documentation Plaintext message request
source	<xs:element name="PlaintextMessageRequest"> <xs:annotation> <xs:documentation>Plaintext message request</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="LastPlaintextMessage" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **Platform**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

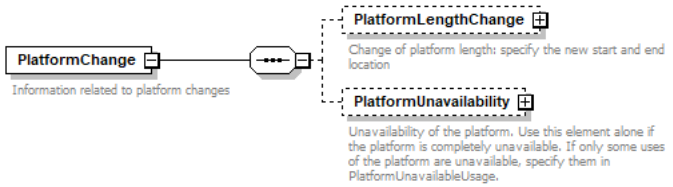
used by: element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
platformSide	**derived by: xs:string**				documentation Side of the

	platformName **xs:string** platformUsage **derived by: xs:string**	platform (relative to the nominal direction of the SP) documentation Name of the platform as known by driver and passengers documentation Usage of the platform
annotation	documentation Area alongside a railway track providing convenient access to trains	
source	<xs:element name="Platform"> <xs:annotation> <xs:documentation>Area alongside a railway track providing convenient access to trains</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericArea_ComplexType"> <xs:attribute ref="platformSide"/> <xs:attribute ref="platformName"/> <xs:attribute ref="platformUsage"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>	

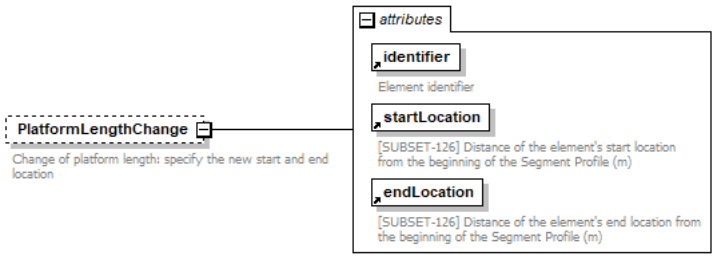
element **PlatformChange**

diagram	
properties	content complex
children	**PlatformLengthChange** **PlatformUnavailability**
used by	element **JP_ContextInformation**
annotation	documentation Information related to platform changes
source	<xs:element name="PlatformChange"> <xs:annotation> <xs:documentation>Information related to platform changes</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element name="PlatformLengthChange" minOccurs="0"> <xs:annotation> <xs:documentation>Change of platform length: specify the new start and end location</xs:documentation>

```
      </xs:annotation>
      <xs:complexType>
        <xs:attribute ref="identifier" use="required"/>
        <xs:attribute ref="startLocation" use="required"/>
        <xs:attribute ref="endLocation" use="required"/>
      </xs:complexType>
    </xs:element>
    <xs:element name="PlatformUnavailability" minOccurs="0">
      <xs:annotation>
        <xs:documentation>Unavailability of the platform. Use this element alone if the platform is
completely unavailable. If only some uses of the platform are unavailable, specify them in
PlatformUnavailableUsage.</xs:documentation>
      </xs:annotation>
      <xs:complexType>
        <xs:sequence>
          <xs:element name="PlatformUnavailableUsage" minOccurs="0" maxOccurs="unbounded">
            <xs:annotation>
              <xs:documentation>The platform is unavailable for this use</xs:documentation>
            </xs:annotation>
            <xs:complexType>
              <xs:attribute ref="platformUsage" use="required"/>
            </xs:complexType>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
 </xs:complexType>
</xs:element>
```

element **PlatformChange/PlatformLengthChange**

diagram



properties

minOcc	0
maxOcc	1
content	complex

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**	required			documentation Element identifier
startLocation	**distanceType**	required			documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**	required			documentation

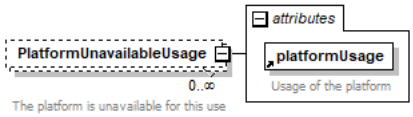
	[SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
annotation	documentation Change of platform length: specify the new start and end location
source	<xs:element name="PlatformLengthChange" minOccurs="0"> <xs:annotation> <xs:documentation>Change of platform length: specify the new start and end location</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="identifier" use="required"/> <xs:attribute ref="startLocation" use="required"/> <xs:attribute ref="endLocation" use="required"/> </xs:complexType> </xs:element>

element **PlatformChange/PlatformUnavailability**

diagram	
properties	minOcc 0 maxOcc 1 content complex
children	**PlatformUnavailableUsage**
annotation	documentation Unavailability of the platform. Use this element alone if the platform is completely unavailable. If only some uses of the platform are unavailable, specify them in PlatformUnavailableUsage.
source	<xs:element name="PlatformUnavailability" minOccurs="0"> <xs:annotation> <xs:documentation>Unavailability of the platform. Use this element alone if the platform is completely unavailable. If only some uses of the platform are unavailable, specify them in PlatformUnavailableUsage.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element name="PlatformUnavailableUsage" minOccurs="0" maxOccurs="unbounded"> <xs:annotation> <xs:documentation>The platform is unavailable for this use</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="platformUsage" use="required"/> </xs:complexType> </xs:element> </xs:sequence> </xs:complexType> </xs:element>

element **PlatformChange/PlatformUnavailability/PlatformUnavailableUsage**

diagram



properties

minOcc	0
maxOcc	unbounded
content	complex

attributes

Name	Type	Use	Default	Fixed	Annotation
platformUsage	**derived by: xs:string**	required			documentation: Usage of the platform

annotation

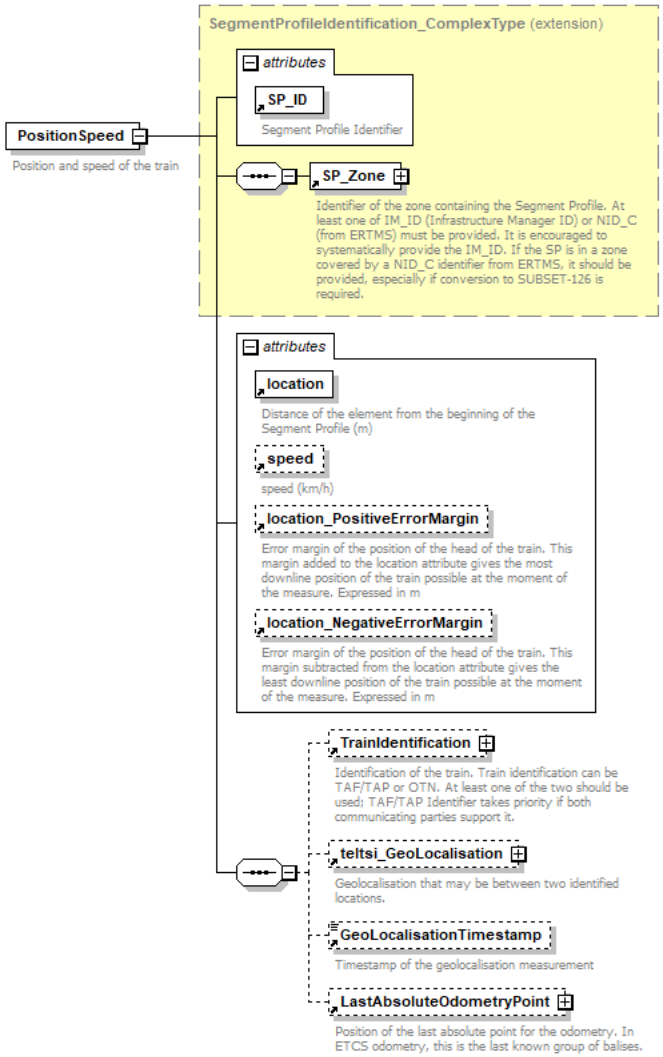
documentation
The platform is unavailable for this use

source

```
<xs:element name="PlatformUnavailableUsage" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>The platform is unavailable for this use</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="platformUsage" use="required"/>
  </xs:complexType>
</xs:element>
```

element **PositionSpeed**

diagram



type: extension of **SegmentProfileIdentification_ComplexType**

properties: content complex

children: **SP_Zone** **TrainIdentification** **teltsi_GeoLocalisation** **GeoLocalisationTimestamp** **LastAbsoluteOdometryPoint**

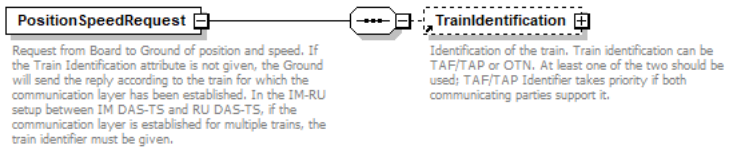
used by: elements **B2G_StatusReport** **G2B_ReplyPayload** **TrainLocationInformation**

attributes

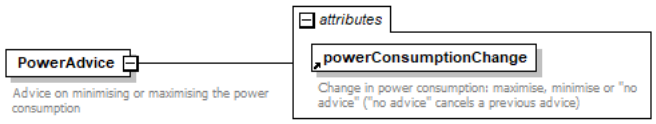
Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
speed	**speedType**				documentation speed (km/h)
location_PositiveErrorMargin	**distanceType**				documentation Error margin of the position

	location_NegativeErrorMargin **distanceType**	of the head of the train. This margin added to the location attribute gives the most downline position of the train possible at the moment of the measure. Expressed in m documentation Error margin of the position of the head of the train. This margin subtracted from the location attribute gives the least downline position of the train possible at the moment of the measure. Expressed in m
annotation	documentation Position and speed of the train	
source	<xs:element name="PositionSpeed"> <xs:annotation> <xs:documentation>Position and speed of the train</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SegmentProfileIdentification_ComplexType"> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> <xs:element ref="teltsi_GeoLocalisation" minOccurs="0"/> <xs:element ref="GeoLocalisationTimestamp" minOccurs="0"/> <xs:element ref="LastAbsoluteOdometryPoint" minOccurs="0"/> </xs:sequence> <xs:attribute ref="location" use="required"/> <xs:attribute ref="speed"/> <xs:attribute ref="location_PositiveErrorMargin"/> <xs:attribute ref="location_NegativeErrorMargin"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>	

element **PositionSpeedRequest**

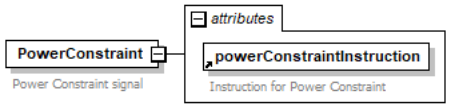
diagram	
properties	content complex
children	**TrainIdentification**
used by	element **B2G_Request**
annotation	documentation Request from Board to Ground of position and speed. If the Train Identification attribute is not given, the Ground will send the reply according to the train for which the communication layer has been established. In the IM-RU setup between IM DAS-TS and RU DAS-TS, if the communication layer is established for multiple trains, the train identifier must be given.
source	<xs:element name="PositionSpeedRequest"> <xs:annotation> <xs:documentation>Request from Board to Ground of position and speed. If the Train Identification attribute is not given, the Ground will send the reply according to the train for which the communication layer has been established. In the IM-RU setup between IM DAS-TS and RU DAS-TS, if the communication layer is established for multiple trains, the train identifier must be given.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **PowerAdvice**

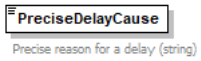
diagram						
properties	content complex					
used by	element **TemporaryConstraints**					
attributes	Name powerConsumptionChange	Type **derived by: xs:string**	Use required	Default	Fixed	Annotation documentation Change in power consumption: maximise, minimise or "no advice" ("no advice" cancels a previous advice)
annotation	documentation Advice on minimising or maximising the power consumption					
source	<xs:element name="PowerAdvice"> <xs:annotation> <xs:documentation>Advice on minimising or maximising the power consumption</xs:documentation>					

```
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="powerConsumptionChange" use="required"/>
  </xs:complexType>
</xs:element>
```

element **PowerConstraint**

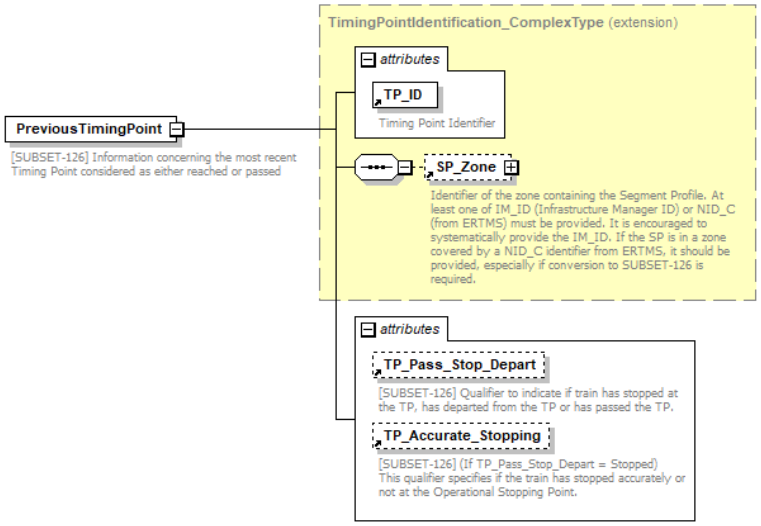
diagram	
properties	content complex
used by	element **SignalInformation**
attributes	Name: powerConstraintInstruction; Type: **derived by: xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation Instruction for Power Constraint
annotation	documentation Power Constraint signal
source	<xs:element name="PowerConstraint"> <xs:annotation> <xs:documentation>Power Constraint signal</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="powerConstraintInstruction" use="required"/> </xs:complexType> </xs:element>

element **PreciseDelayCause**

diagram	
type	**xs:string**
properties	content simple
used by	elements **DelayNotice** **TrainReadyStatus**
annotation	documentation Precise reason for a delay (string)
source	<xs:element name="PreciseDelayCause" type="xs:string"> <xs:annotation> <xs:documentation>Precise reason for a delay (string)</xs:documentation> </xs:annotation> </xs:element>

element **PreviousTimingPoint**

diagram



type: extension of **TimingPointIdentification_ComplexType**

properties: content complex

children: **SP_Zone**

used by: element **B2G_StatusReport**

attributes

Name	Type	Use	Default	Fixed	Annotation
TP_ID	**xs:string**	required			documentation: Timing Point Identifier
TP_Pass_Stop_Depart	**derived by: xs:string**				documentation: [SUBSET-126] Qualifier to indicate if train has stopped at the TP, has departed from the TP or has passed the TP.
TP_Accurate_Stopping	**derived by: xs:string**				documentation: [SUBSET-126] (If TP_Pass_Stop_Depart = Stopped) This qualifier specifies if the train has stopped accurately or not at the Operational Stopping Point.

annotation

documentation
[SUBSET-126] Information concerning the most recent Timing Point considered as either reached or passed

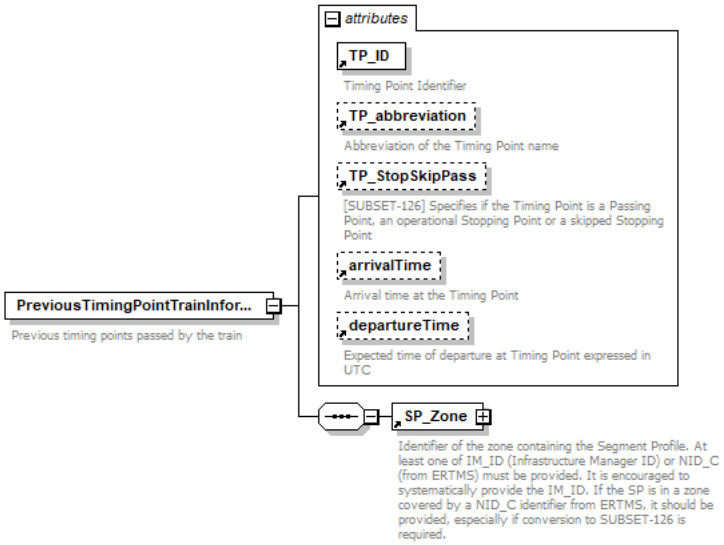
source

```
<xs:element name="PreviousTimingPoint">
 <xs:annotation>
  <xs:documentation>[SUBSET-126] Information concerning the most recent Timing Point considered as either reached or passed</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="TimingPointIdentification_ComplexType">
    <xs:attribute ref="TP_Pass_Stop_Depart"/>
    <xs:attribute ref="TP_Accurate_Stopping"/>
   </xs:extension>
  </xs:complexContent>
```

```
  </xs:complexType>
</xs:element>
```

element **PreviousTimingPointTrainInformation**

diagram



properties

content complex

children **SP_Zone**

used by

element **TrainLocationInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
TP_ID	**xs:string**	required			documentation Timing Point Identifier
TP_abbreviation	**xs:string**				documentation Abbreviation of the Timing Point name
TP_StopSkipPass	**derived by: xs:string**				documentation [SUBSET-126] Specifies if the Timing Point is a Passing Point, an operational Stopping Point or a skipped Stopping Point
arrivalTime	**UTCDateTimeType**				documentation Arrival time at the Timing Point
departureTime	**UTCDateTimeType**				documentation Expected time of departure at Timing Point expressed in UTC

annotation

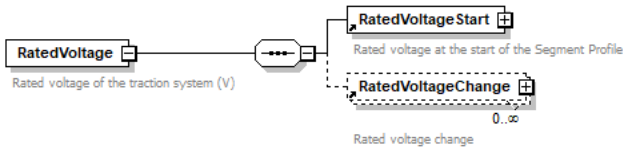
documentation
Previous timing points passed by the train

source

```
<xs:element name="PreviousTimingPointTrainInformation">
  <xs:annotation>
```

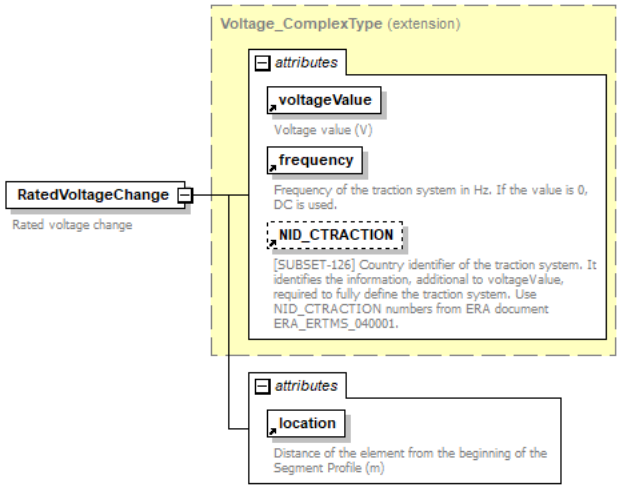
	<xs:documentation>Previous timing points passed by the train</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="SP_Zone"/> </xs:sequence> <xs:attribute ref="TP_ID" use="required"/> <xs:attribute ref="TP_abbreviation"/> <xs:attribute ref="TP_StopSkipPass"/> <xs:attribute ref="arrivalTime"/> <xs:attribute ref="departureTime"/> </xs:complexType> </xs:element>

element **RatedVoltage**

diagram	
properties	content complex
children	**RatedVoltageStart** **RatedVoltageChange**
used by	element **SP_Characteristics**
annotation	documentation Rated voltage of the traction system (V)
source	<xs:element name="RatedVoltage"> <xs:annotation> <xs:documentation>Rated voltage of the traction system (V)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="RatedVoltageStart"/> <xs:element ref="RatedVoltageChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **RatedVoltageChange**

diagram



type: extension of **Voltage_ComplexType**

properties: content complex

used by: element **RatedVoltage**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation: Voltage value (V)
frequency	**xs:decimal**	required			documentation: Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation: [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

documentation
Rated voltage change

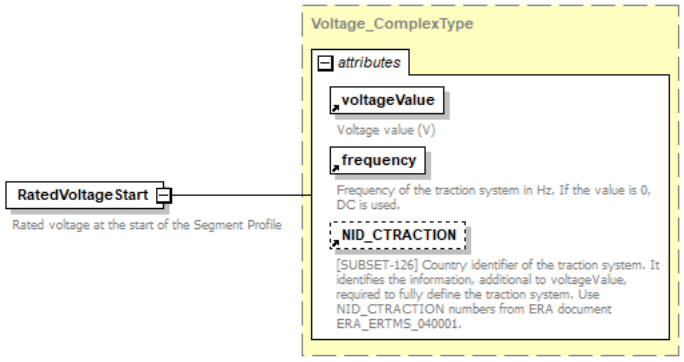
source

```
<xs:element name="RatedVoltageChange">
 <xs:annotation>
  <xs:documentation>Rated voltage change</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="Voltage_ComplexType">
    <xs:attribute ref="location" use="required"/>
   </xs:extension>
  </xs:complexContent>
```

```
  </xs:complexType>
</xs:element>
```

element **RatedVoltageStart**

diagram



type **Voltage_ComplexType**

properties
content complex

used by
element **RatedVoltage**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation Voltage value (V)
frequency	**xs:decimal**	required			documentation Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.

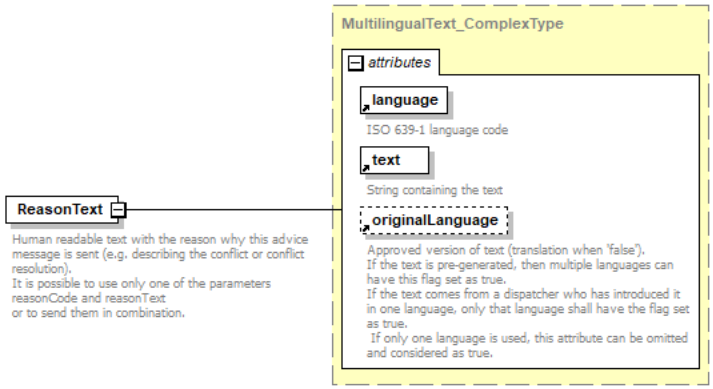
annotation
documentation
Rated voltage at the start of the Segment Profile

source

```
<xs:element name="RatedVoltageStart" type="Voltage_ComplexType">
  <xs:annotation>
    <xs:documentation>Rated voltage at the start of the Segment Profile</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **ReasonText**

diagram



type **MultilingualText_ComplexType**

properties
content complex

used by
complexType **AdvisedSpeed_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

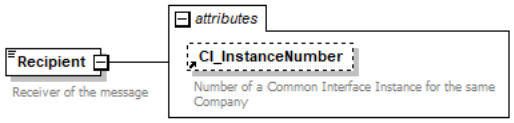
annotation
documentation
Human readable text with the reason why this advice message is sent (e.g. describing the conflict or conflict resolution).
It is possible to use only one of the parameters reasonCode and reasonText
or to send them in combination.

source

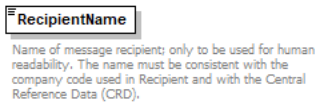
```
<xs:element name="ReasonText" type="MultilingualText_ComplexType">
 <xs:annotation>
  <xs:documentation>Human readable text with the reason why this advice message is sent (e.g.
```

	describing the conflict or conflict resolution). It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination.</xs:documentation> </xs:annotation> </xs:element>

element **Recipient**

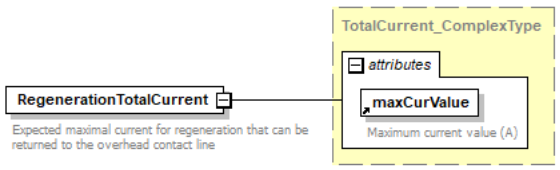
diagram	
type	extension of **teltsi_CompanyCode**
properties	content complex
used by	element **MessageHeader**
facets	Kind / Value / Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
attributes	Name: CI_InstanceNumber; Type: teltsi_Numeric2-2; Use; Default; Fixed; Annotation: documentation: Number of a Common Interface Instance for the same Company
annotation	documentation Receiver of the message
source	<xs:element name="Recipient"> <xs:annotation> <xs:documentation>Receiver of the message</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="teltsi_CompanyCode"> <xs:attribute ref="CI_InstanceNumber"/> </xs:extension> </xs:simpleContent> </xs:complexType> </xs:element>

element **RecipientName**

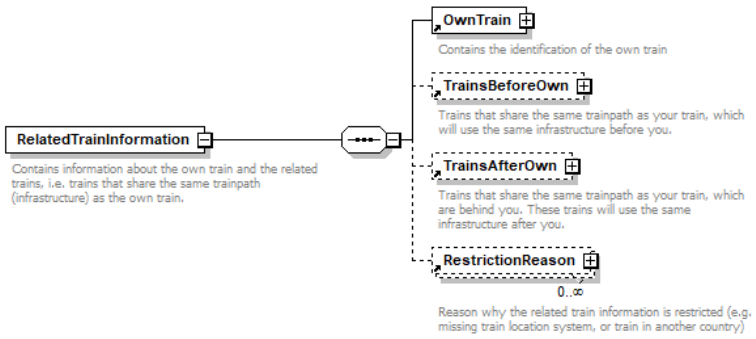
diagram	
type	**xs:string**
properties	content simple
used by	element **MessageHeader**

annotation	documentation Name of message recipient; only to be used for human readability. The name must be consistent with the company code used in Recipient and with the Central Reference Data (CRD).
source	<xs:element name="RecipientName" type="xs:string"> <xs:annotation> <xs:documentation>Name of message recipient; only to be used for human readability. The name must be consistent with the company code used in Recipient and with the Central Reference Data (CRD).</xs:documentation> </xs:annotation> </xs:element>

element **RegenerationTotalCurrent**

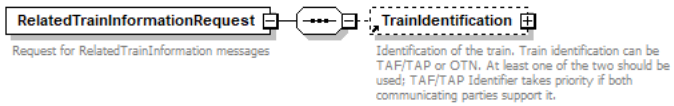
diagram	
type	**TotalCurrent_ComplexType**
properties	content complex
used by	element **TemporaryConstraints**
attributes	Name: maxCurValue; Type: **currentType**; Use: required; Default: ; Fixed: ; Annotation: documentation Maximum current value (A)
annotation	documentation Expected maximal current for regeneration that can be returned to the overhead contact line
source	<xs:element name="RegenerationTotalCurrent" type="TotalCurrent_ComplexType"> <xs:annotation> <xs:documentation>Expected maximal current for regeneration that can be returned to the overhead contact line</xs:documentation> </xs:annotation> </xs:element>

element **RelatedTrainInformation**

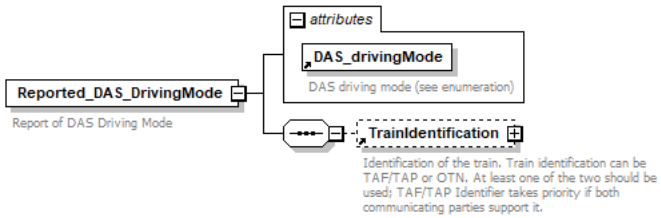
diagram	
properties	content complex
children	**OwnTrain** **TrainsBeforeOwn** **TrainsAfterOwn** **RestrictionReason**

used by	elements **G2B_EventPayload** **G2B_ReplyPayload**
annotation	documentation Contains information about the own train and the related trains, i.e. trains that share the same trainpath (infrastructure) as the own train.
source	<xs:element name="RelatedTrainInformation"> <xs:annotation> <xs:documentation>Contains information about the own train and the related trains, i.e. trains that share the same trainpath (infrastructure) as the own train.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="OwnTrain"/> <xs:element ref="TrainsBeforeOwn" minOccurs="0"/> <xs:element ref="TrainsAfterOwn" minOccurs="0"/> <xs:element ref="RestrictionReason" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **RelatedTrainInformationRequest**

diagram	
properties	content complex
children	**TrainIdentification**
used by	element **B2G_Request**
annotation	documentation Request for RelatedTrainInformation messages
source	<xs:element name="RelatedTrainInformationRequest"> <xs:annotation> <xs:documentation>Request for RelatedTrainInformation messages</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **Reported_DAS_DrivingMode**

diagram	
properties	content complex

children	**TrainIdentification**
used by	elements **B2G_EventPayload** **B2G_ReplyPayload**

attributes	Name	Type	Use	Default	Fixed	Annotation
	DAS_drivingMode	**derived by: xs:string**	required			documentation DAS driving mode (see enumeration)

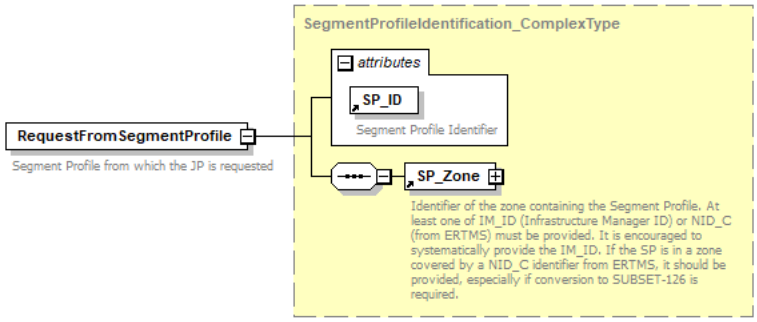
annotation	documentation Report of DAS Driving Mode

source

```
<xs:element name="Reported_DAS_DrivingMode">
  <xs:annotation>
    <xs:documentation>Report of DAS Driving Mode</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="TrainIdentification" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute ref="DAS_drivingMode" use="required"/>
  </xs:complexType>
</xs:element>
```

element **RequestFromSegmentProfile**

diagram



type	**SegmentProfileIdentification_ComplexType**
properties	content complex
children	**SP_Zone**
used by	element **JP_Request**

attributes	Name	Type	Use	Default	Fixed	Annotation
	SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier

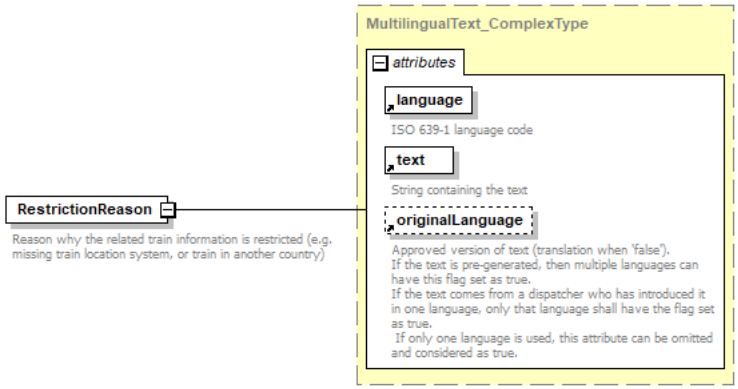
annotation	documentation Segment Profile from which the JP is requested

source

```
<xs:element name="RequestFromSegmentProfile"
type="SegmentProfileIdentification_ComplexType">
  <xs:annotation>
    <xs:documentation>Segment Profile from which the JP is requested</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **RestrictionReason**

diagram



type **MultilingualText_ComplexType**

properties
- content complex

used by
- element **RelatedTrainInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

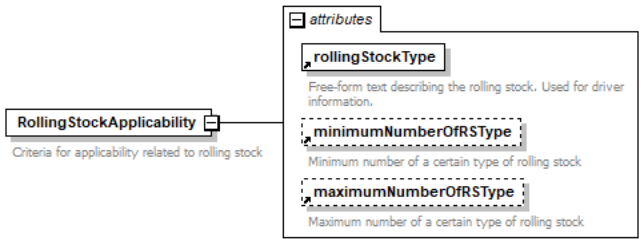
annotation
documentation
Reason why the related train information is restricted (e.g. missing train location system, or train in another country)

source

```
<xs:element name="RestrictionReason" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Reason why the related train information is restricted (e.g. missing train location system, or train in another country)</xs:documentation>
  </xs:annotation>
```

	</xs:element>

element **RollingStockApplicability**

diagram	
properties	content complex
used by	element **StoppingPointApplicability**

attributes

Name	Type	Use	Default	Fixed	Annotation
rollingStockType	**xs:string**	required			documentation Free-form text describing the rolling stock. Used for driver information.
minimumNumberOfRSType	**xs:unsignedByte**				documentation Minimum number of a certain type of rolling stock
maximumNumberOfRSType	**xs:unsignedByte**				documentation Maximum number of a certain type of rolling stock

annotation

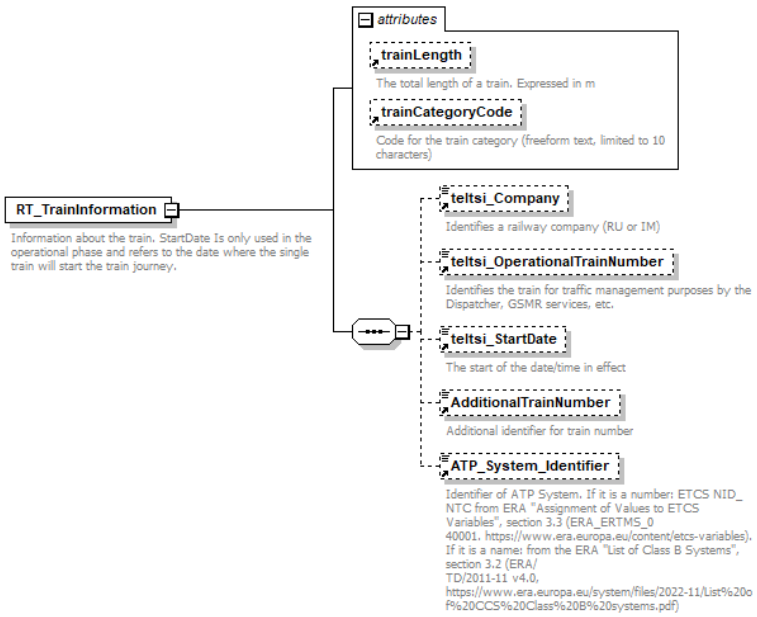
documentation
Criteria for applicability related to rolling stock

source

```
<xs:element name="RollingStockApplicability">
  <xs:annotation>
    <xs:documentation>Criteria for applicability related to rolling stock</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="rollingStockType" use="required"/>
    <xs:attribute ref="minimumNumberOfRSType"/>
    <xs:attribute ref="maximumNumberOfRSType"/>
  </xs:complexType>
</xs:element>
```

element **RT_TrainInformation**

diagram



properties

content complex

children **teltsi_Company** **teltsi_OperationalTrainNumber** **teltsi_StartDate** **AdditionalTrainNumber** **ATP_System_Identifier**

used by

complexType **TrainType**

attributes

Name	Type	Use	Default	Fixed	Annotation
trainLength	**xs:integer**				documentation The total length of a train. Expressed in m
trainCategoryCode	**derived by: xs:string**				documentation Code for the train category (freeform text, limited to 10 characters)

annotation

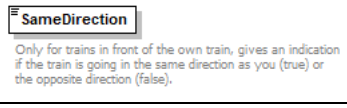
documentation
Information about the train. StartDate Is only used in the operational phase and refers to the date where the single train will start the train journey.

source

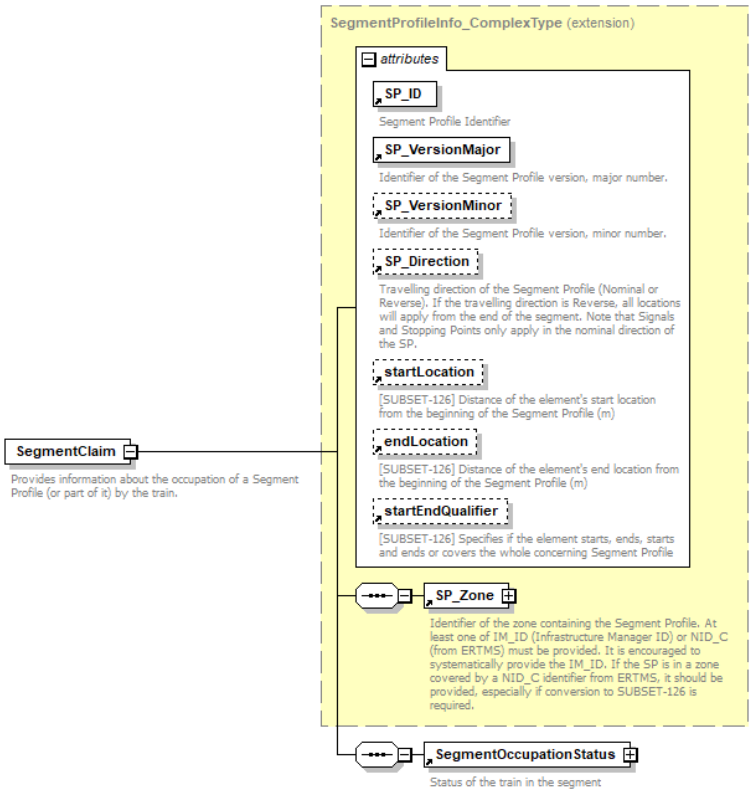
```
<xs:element name="RT_TrainInformation">
 <xs:annotation>
  <xs:documentation>Information about the train. StartDate Is only used in the operational phase and refers to the date where the single train will start the train journey.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="teltsi_Company" minOccurs="0"/>
   <xs:element ref="teltsi_OperationalTrainNumber" minOccurs="0"/>
   <xs:element ref="teltsi_StartDate" minOccurs="0"/>
   <xs:element ref="AdditionalTrainNumber" minOccurs="0"/>
   <xs:element ref="ATP_System_Identifier" minOccurs="0"/>
  </xs:sequence>
  <xs:attribute ref="trainLength"/>
  <xs:attribute ref="trainCategoryCode"/>
 </xs:complexType>
```

	</xs:element>

element **SameDirection**

diagram	
type	**xs:boolean**
properties	content simple
used by	element **SegmentOccupationStatus**
annotation	documentation Only for trains in front of the own train, gives an indication if the train is going in the same direction as you (true) or the opposite direction (false).
source	<xs:element name="SameDirection" type="xs:boolean"> <xs:annotation> <xs:documentation>Only for trains in front of the own train, gives an indication if the train is going in the same direction as you (true) or the opposite direction (false).</xs:documentation> </xs:annotation> </xs:element>

element **SegmentClaim**

diagram	
type	extension of **SegmentProfileInfo_ComplexType**
properties	content complex

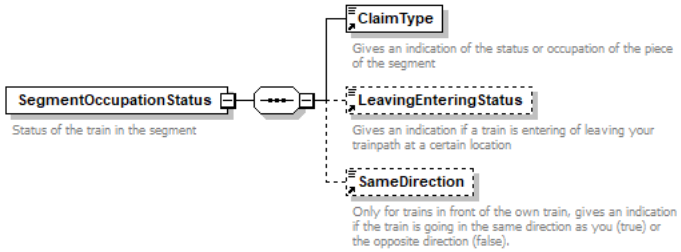
children	**SP_Zone** **SegmentOccupationStatus**
used by	element **TrainLocationInformation**

attributes

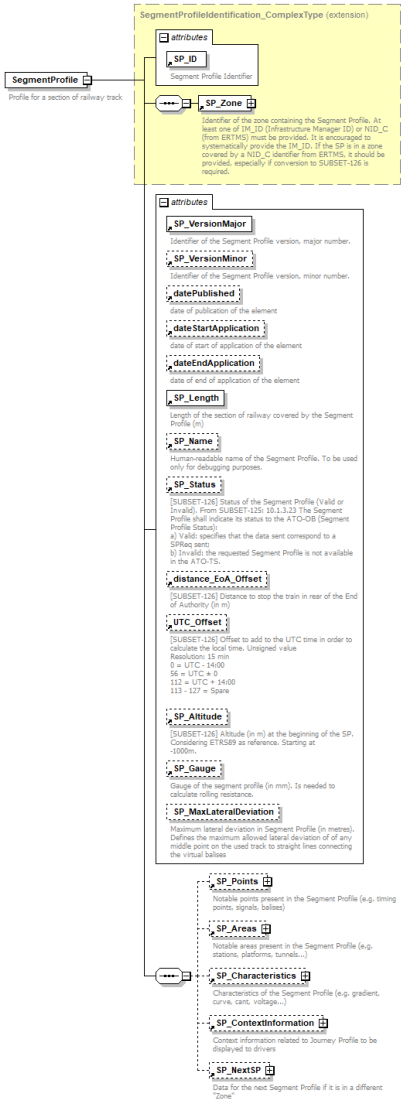
Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**	required			documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment Profile version, minor number.
SP_Direction	**derived by: xs:string**				documentation Travelling direction of the Segment Profile (Nominal or Reverse). If the travelling direction is Reverse, all locations will apply from the end of the segment. Note that Signals and Stopping Points only apply in the nominal direction of the SP.
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**				documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment

	Profile
annotation	documentation Provides information about the occupation of a Segment Profile (or part of it) by the train.
source	<xs:element name="SegmentClaim"> <xs:annotation> <xs:documentation>Provides information about the occupation of a Segment Profile (or part of it) by the train.</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SegmentProfileInfo_ComplexType"> <xs:sequence> <xs:element ref="SegmentOccupationStatus"/> </xs:sequence> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **SegmentOccupationStatus**

diagram	
properties	content complex
children	**ClaimType** **LeavingEnteringStatus** **SameDirection**
used by	element **SegmentClaim**
annotation	documentation Status of the train in the segment
source	<xs:element name="SegmentOccupationStatus"> <xs:annotation> <xs:documentation>Status of the train in the segment</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="ClaimType"/> <xs:element ref="LeavingEnteringStatus" minOccurs="0"/> <xs:element ref="SameDirection" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **SegmentProfile**

diagram	
type	extension of **SegmentProfileIdentification_ComplexType**
properties	content complex
children	**SP_Zone** **SP_Points** **SP_Areas** **SP_Characteristics** **SP_ContextInformation** **SP_NextSP**
used by	element **G2B_ReplyPayload**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**	required			documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment Profile

				version, minor number.
	datePublished	**UTCDateTimeType**		documentation date of publication of the element
	dateStartApplication	**UTCDateTimeType**		documentation date of start of application of the element
	dateEndApplication	**UTCDateTimeType**		documentation date of end of application of the element
	SP_Length	**distanceType**	required	documentation Length of the section of railway covered by the Segment Profile (m)
	SP_Name	**xs:string**		documentation Human-readable name of the Segment Profile. To be used only for debugging purposes.
	SP_Status	**derived by: xs:string**		documentation [SUBSET-126] Status of the Segment Profile (Valid or Invalid). From SUBSET-125: 10.1.3.23 The Segment Profile shall indicate its status to the ATO-OB (Segment Profile Status): a) Valid: specifies that the data sent correspond to a SPReq sent; b) Invalid: the requested Segment Profile is not available in the ATO-TS.
	distance_EoA_Offset	**distanceType**		documentation [SUBSET-126] Distance to stop the train in rear of the End of Authority (in m)
	UTC_Offset	**xs:unsignedByte**		documentation [SUBSET-126] Offset to add to the UTC time in order

			to calculate the local time. Unsigned value Resolution: 15 min 0 = UTC - 14:00 56 = UTC ± 0 112 = UTC + 14:00 113 - 127 = Spare
	SP_Altitude	**xs:decimal**	documentation [SUBSET-126] Altitude (in m) at the beginning of the SP. Considering ETRS89 as reference. Starting at -1000m.
	SP_Gauge	**xs:int**	documentation Gauge of the segment profile (in mm). Is needed to calculate rolling resistance.
	SP_MaxLateralDeviation	**xs:double**	documentation Maximum lateral deviation in Segment Profile (in metres). Defines the maximum allowed lateral deviation of of any middle point on the used track to straight lines connecting the virtual balises
annotation	documentation Profile for a section of railway track		
source	<xs:element name="SegmentProfile"> <xs:annotation> <xs:documentation>Profile for a section of railway track</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SegmentProfileIdentification_ComplexType"> <xs:sequence> <xs:element ref="SP_Points" minOccurs="0"/> <xs:element ref="SP_Areas" minOccurs="0"/> <xs:element ref="SP_Characteristics" minOccurs="0"/> <xs:element ref="SP_ContextInformation" minOccurs="0"/> <xs:element ref="SP_NextSP" minOccurs="0"/>		

```
        </xs:sequence>
        <xs:attribute ref="SP_VersionMajor" use="required"/>
        <xs:attribute ref="SP_VersionMinor"/>
        <xs:attribute ref="datePublished"/>
        <xs:attribute ref="dateStartApplication"/>
        <xs:attribute ref="dateEndApplication"/>
        <xs:attribute ref="SP_Length" use="required"/>
        <xs:attribute ref="SP_Name"/>
        <xs:attribute ref="SP_Status"/>
        <xs:attribute ref="distance_EoA_Offset"/>
        <xs:attribute ref="UTC_Offset"/>
        <xs:attribute ref="SP_Altitude"/>
        <xs:attribute ref="SP_Gauge"/>
        <xs:attribute name="SP_MaxLateralDeviation" type="xs:double">
          <xs:annotation>
            <xs:documentation>Maximum lateral deviation in Segment Profile (in metres). Defines the
maximum allowed lateral deviation of of any middle point on the used track to straight lines
connecting the virtual balises</xs:documentation>
          </xs:annotation>
        </xs:attribute>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

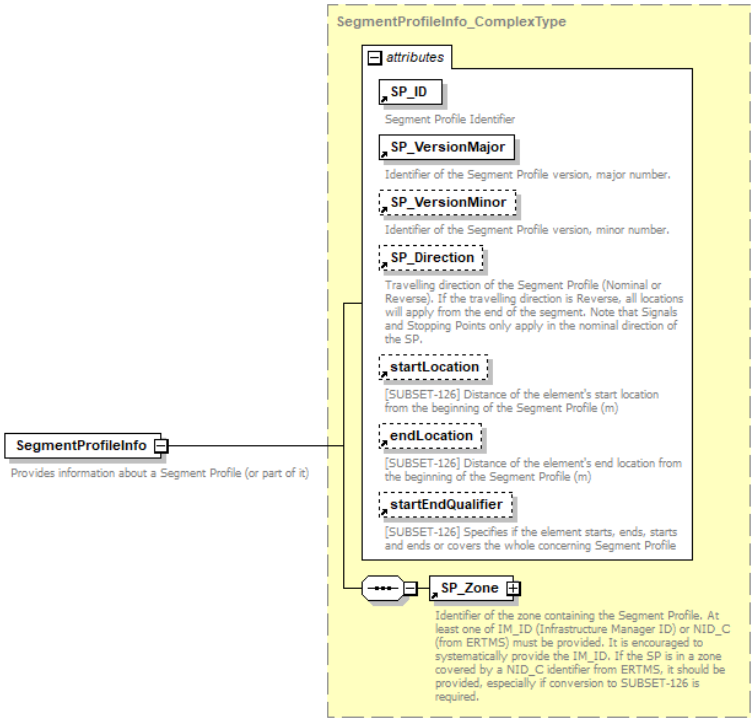
attribute **SegmentProfile/@SP_MaxLateralDeviation**

type	**xs:double**
annotation	documentation Maximum lateral deviation in Segment Profile (in metres). Defines the maximum allowed lateral deviation of of any middle point on the used track to straight lines connecting the virtual balises

source	`<xs:attribute name="SP_MaxLateralDeviation" type="xs:double">`
	`  <xs:annotation>`
	`  <xs:documentation>Maximum lateral deviation in Segment Profile (in metres). Defines the maximum allowed lateral deviation of of any middle point on the used track to straight lines connecting the virtual balises</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

element **SegmentProfileInfo**

diagram



type **SegmentProfileInfo_ComplexType**

properties content complex

children **SP_Zone**

used by element **C_DAS_C_Advice**

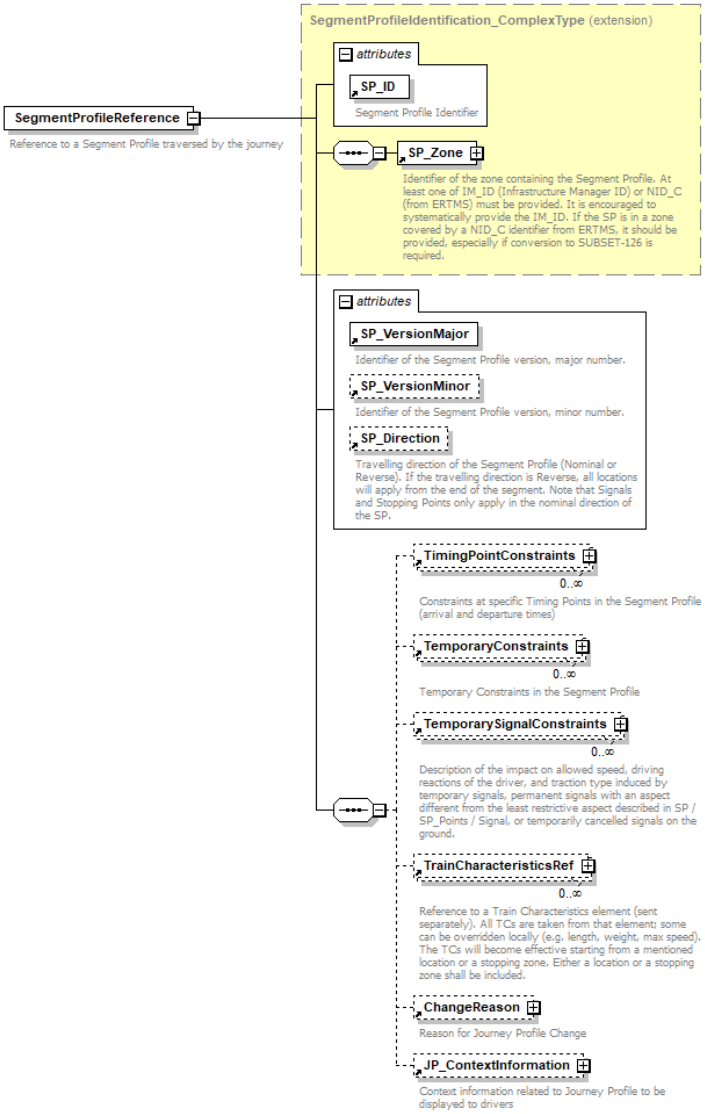
attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**	required			documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment Profile version, minor number.
SP_Direction	**derived by: xs:string**				documentation Travelling direction of the Segment Profile (Nominal or Reverse). If the travelling direction is Reverse, all locations will apply from the end of the segment. Note

			that Signals and Stopping Points only apply in the nominal direction of the SP.
	startLocation	**distanceType**	documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**	documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
	startEndQualifier	**derived by:** **xs:string**	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation Provides information about a Segment Profile (or part of it)		
source	<xs:element name="SegmentProfileInfo" type="SegmentProfileInfo_ComplexType"> <xs:annotation> <xs:documentation>Provides information about a Segment Profile (or part of it)</xs:documentation> </xs:annotation> </xs:element>		

element **SegmentProfileReference**

diagram



type: extension of **SegmentProfileIdentification_ComplexType**

properties: content complex

children: **SP_Zone** **TimingPointConstraints** **TemporaryConstraints** **TemporarySignalConstraints** **TrainCharacteristicsRef** **ChangeReason** **JP_ContextInformation**

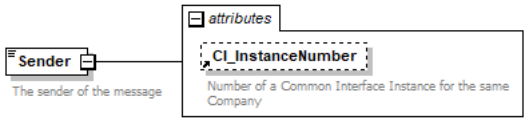
used by: elements **AlternativeItinerary** **JourneyProfile**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**	required			documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment

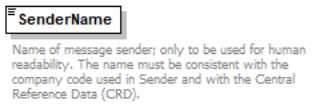
	SP_Direction **derived by: xs:string**	Profile version, minor number. documentation Travelling direction of the Segment Profile (Nominal or Reverse). If the travelling direction is Reverse, all locations will apply from the end of the segment. Note that Signals and Stopping Points only apply in the nominal direction of the SP.
annotation	documentation Reference to a Segment Profile traversed by the journey	
source	<xs:element name="SegmentProfileReference"> <xs:annotation> <xs:documentation>Reference to a Segment Profile traversed by the journey</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SegmentProfileIdentification_ComplexType"> <xs:sequence> <xs:element ref="TimingPointConstraints" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TemporaryConstraints" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TemporarySignalConstraints" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TrainCharacteristicsRef" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="ChangeReason" minOccurs="0"/> <xs:element ref="JP_ContextInformation" minOccurs="0"/> </xs:sequence> <xs:attribute ref="SP_VersionMajor" use="required"/> <xs:attribute ref="SP_VersionMinor"/> <xs:attribute ref="SP_Direction"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>	

element **Sender**

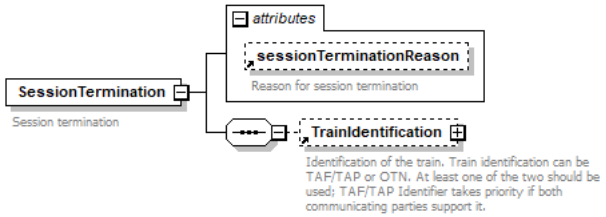
diagram	
type	extension of **teltsi_CompanyCode**
properties	content complex

used by	element **MessageHeader**
facets	Kind / Value / Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
attributes	Name: CI_InstanceNumber Type: **teltsi_Numeric2-2** Use: Default: Fixed: Annotation: documentation: Number of a Common Interface Instance for the same Company
annotation	documentation The sender of the message
source	<xs:element name="Sender"> <xs:annotation> <xs:documentation>The sender of the message</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="teltsi_CompanyCode"> <xs:attribute ref="CI_InstanceNumber"/> </xs:extension> </xs:simpleContent> </xs:complexType> </xs:element>

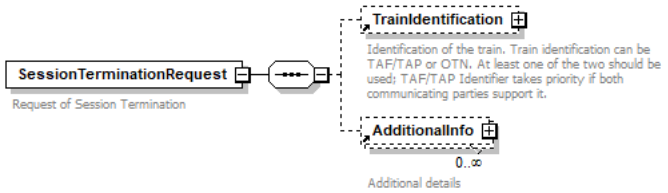
element **SenderName**

diagram	
type	**xs:string**
properties	content simple
used by	element **MessageHeader**
annotation	documentation Name of message sender; only to be used for human readability. The name must be consistent with the company code used in Sender and with the Central Reference Data (CRD).
source	<xs:element name="SenderName" type="xs:string"> <xs:annotation> <xs:documentation>Name of message sender; only to be used for human readability. The name must be consistent with the company code used in Sender and with the Central Reference Data (CRD).</xs:documentation> </xs:annotation> </xs:element>

element **SessionTermination**

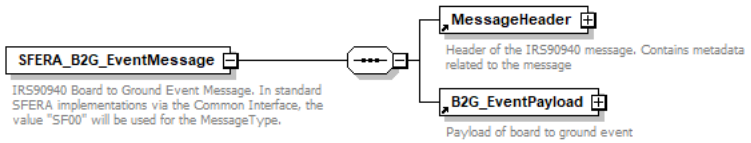
diagram	
properties	content complex
children	**TrainIdentification**
used by	elements **B2G_EventPayload** **B2G_ReplyPayload**
attributes	Name: sessionTerminationReason; Type: **derived by: xs:string**; Use: ; Default: ; Fixed: ; Annotation: documentation: Reason for session termination
annotation	documentation Session termination
source	<xs:element name="SessionTermination"> <xs:annotation> <xs:documentation>Session termination</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> </xs:sequence> <xs:attribute ref="sessionTerminationReason"/> </xs:complexType> </xs:element>

element **SessionTerminationRequest**

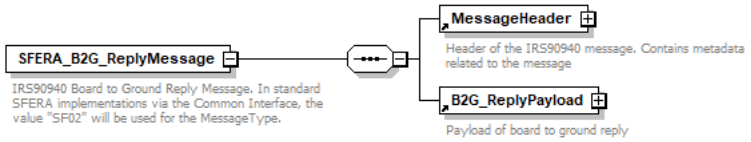
diagram	
properties	content complex
children	**TrainIdentification** **AdditionalInfo**
used by	element **G2B_Request**
annotation	documentation Request of Session Termination
source	<xs:element name="SessionTerminationRequest"> <xs:annotation> <xs:documentation>Request of Session Termination</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification" minOccurs="0"/> <xs:element ref="AdditionalInfo" minOccurs="0" maxOccurs="unbounded"/>

```
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **SFERA_B2G_EventMessage**

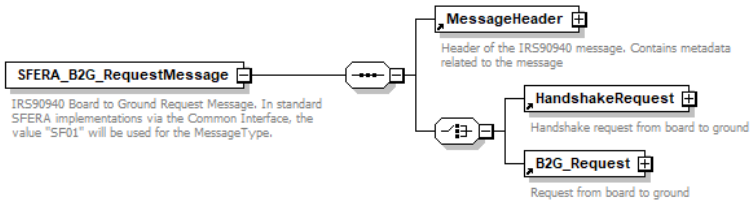
diagram	
properties	content complex
children	**MessageHeader** **B2G_EventPayload**
annotation	documentation IRS90940 Board to Ground Event Message. In standard SFERA implementations via the Common Interface, the value "SF00" will be used for the MessageType.
source	<xs:element name="SFERA_B2G_EventMessage"> <xs:annotation> <xs:documentation>IRS90940 Board to Ground Event Message. In standard SFERA implementations via the Common Interface, the value "SF00" will be used for the MessageType.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageHeader"/> <xs:element ref="B2G_EventPayload"/> </xs:sequence> </xs:complexType> </xs:element>

element **SFERA_B2G_ReplyMessage**

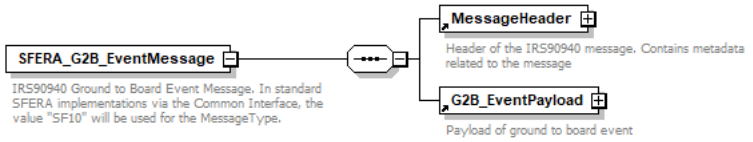
diagram	
properties	content complex
children	**MessageHeader** **B2G_ReplyPayload**
annotation	documentation IRS90940 Board to Ground Reply Message. In standard SFERA implementations via the Common Interface, the value "SF02" will be used for the MessageType.
source	<xs:element name="SFERA_B2G_ReplyMessage"> <xs:annotation> <xs:documentation>IRS90940 Board to Ground Reply Message. In standard SFERA implementations via the Common Interface, the value "SF02" will be used for the MessageType.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageHeader"/> <xs:element ref="B2G_ReplyPayload"/>

```
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **SFERA_B2G_RequestMessage**

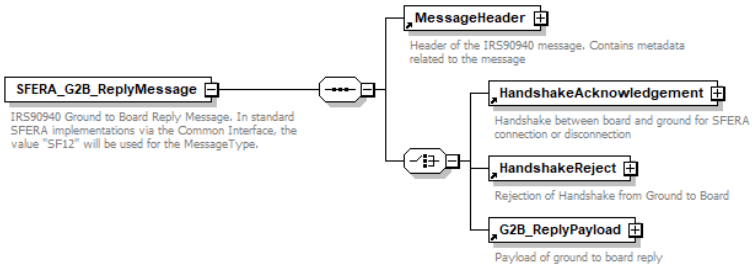
diagram	
properties	content complex
children	**MessageHeader** **HandshakeRequest** **B2G_Request**
annotation	documentation IRS90940 Board to Ground Request Message. In standard SFERA implementations via the Common Interface, the value "SF01" will be used for the MessageType.
source	<xs:element name="SFERA_B2G_RequestMessage"> <xs:annotation> <xs:documentation>IRS90940 Board to Ground Request Message. In standard SFERA implementations via the Common Interface, the value "SF01" will be used for the MessageType.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageHeader"/> <xs:choice> <xs:element ref="HandshakeRequest"/> <xs:element ref="B2G_Request"/> </xs:choice> </xs:sequence> </xs:complexType> </xs:element>

element **SFERA_G2B_EventMessage**

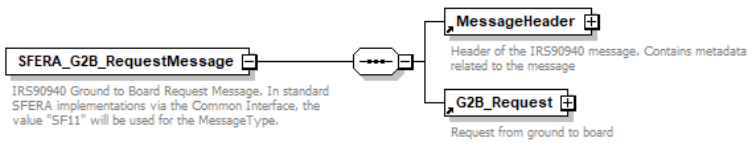
diagram	
properties	content complex
children	**MessageHeader** **G2B_EventPayload**
annotation	documentation IRS90940 Ground to Board Event Message. In standard SFERA implementations via the Common Interface, the value "SF10" will be used for the MessageType.
source	<xs:element name="SFERA_G2B_EventMessage"> <xs:annotation> <xs:documentation>IRS90940 Ground to Board Event Message. In standard SFERA implementations via the Common Interface, the value "SF10" will be used for the MessageType.</xs:documentation>

```
  </xs:annotation>
  <xs:complexType>
   <xs:sequence>
    <xs:element ref="MessageHeader"/>
    <xs:element ref="G2B_EventPayload"/>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **SFERA_G2B_ReplyMessage**

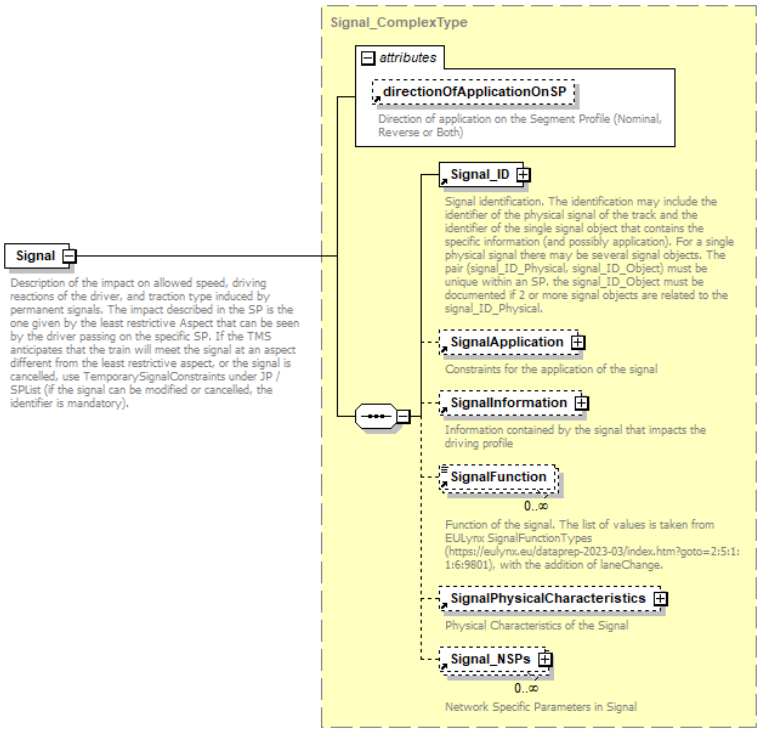
diagram	
properties	content complex
children	**MessageHeader** **HandshakeAcknowledgement** **HandshakeReject** **G2B_ReplyPayload**
annotation	documentation IRS90940 Ground to Board Reply Message. In standard SFERA implementations via the Common Interface, the value "SF12" will be used for the MessageType.
source	<xs:element name="SFERA_G2B_ReplyMessage"> <xs:annotation> <xs:documentation>IRS90940 Ground to Board Reply Message. In standard SFERA implementations via the Common Interface, the value "SF12" will be used for the MessageType.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageHeader"/> <xs:choice> <xs:element ref="HandshakeAcknowledgement"/> <xs:element ref="HandshakeReject"/> <xs:element ref="G2B_ReplyPayload"/> </xs:choice> </xs:sequence> </xs:complexType> </xs:element>

element **SFERA_G2B_RequestMessage**

diagram	
properties	content complex
children	**MessageHeader** **G2B_Request**

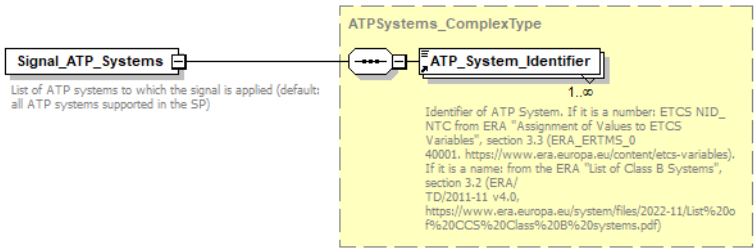
annotation	documentation IRS90940 Ground to Board Request Message. In standard SFERA implementations via the Common Interface, the value "SF11" will be used for the MessageType.
source	<xs:element name="SFERA_G2B_RequestMessage"> <xs:annotation> <xs:documentation>IRS90940 Ground to Board Request Message. In standard SFERA implementations via the Common Interface, the value "SF11" will be used for the MessageType.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="MessageHeader"/> <xs:element ref="G2B_Request"/> </xs:sequence> </xs:complexType> </xs:element>

element **Signal**

diagram	
type	**Signal_ComplexType**
properties	content complex
children	**Signal_ID SignalApplication SignalInformation SignalFunction SignalPhysicalCharacteristics Signal_NSPs**
used by	element **SP_Points**
attributes	Name: directionOfApplicationOnSP Type: **derived by: xs:string** Use: Default: Fixed: Annotation: documentation: Direction of application on the Segment Profile (Nominal,

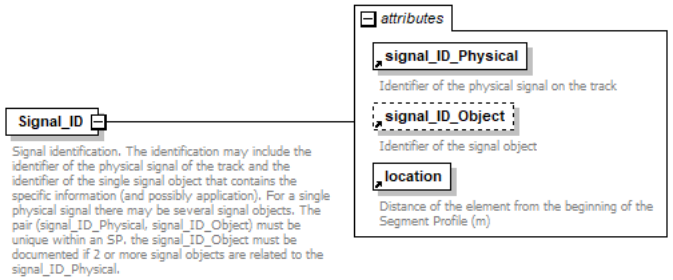
	Reverse or Both)
annotation	documentation Description of the impact on allowed speed, driving reactions of the driver, and traction type induced by permanent signals. The impact described in the SP is the one given by the least restrictive Aspect that can be seen by the driver passing on the specific SP. If the TMS anticipates that the train will meet the signal at an aspect different from the least restrictive aspect, or the signal is cancelled, use TemporarySignalConstraints under JP / SPList (if the signal can be modified or cancelled, the identifier is mandatory).
source	<xs:element name="Signal" type="Signal_ComplexType"> <xs:annotation> <xs:documentation>Description of the impact on allowed speed, driving reactions of the driver, and traction type induced by permanent signals. The impact described in the SP is the one given by the least restrictive Aspect that can be seen by the driver passing on the specific SP. If the TMS anticipates that the train will meet the signal at an aspect different from the least restrictive aspect, or the signal is cancelled, use TemporarySignalConstraints under JP / SPList (if the signal can be modified or cancelled, the identifier is mandatory).</xs:documentation> </xs:annotation> </xs:element>

element **Signal_ATP_Systems**

diagram	
type	**ATPSystems_ComplexType**
properties	content complex
children	**ATP_System_Identifier**
used by	element **SignalApplication**
annotation	documentation List of ATP systems to which the signal is applied (default: all ATP systems supported in the SP)
source	<xs:element name="Signal_ATP_Systems" type="ATPSystems_ComplexType"> <xs:annotation> <xs:documentation>List of ATP systems to which the signal is applied (default: all ATP systems supported in the SP)</xs:documentation> </xs:annotation> </xs:element>

element **Signal_ID**

diagram



properties

content complex

used by

complexType **Signal_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
signal_ID_Physical	**xs:string**	required			documentation Identifier of the physical signal on the track
signal_ID_Object	**xs:string**				documentation Identifier of the signal object
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)

annotation

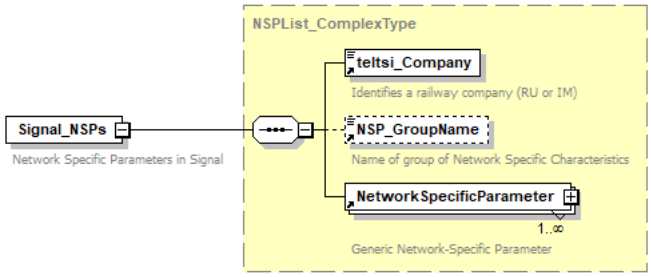
documentation

Signal identification. The identification may include the identifier of the physical signal of the track and the identifier of the single signal object that contains the specific information (and possibly application). For a single physical signal there may be several signal objects. The pair (signal_ID_Physical, signal_ID_Object) must be unique within an SP. the signal_ID_Object must be documented if 2 or more signal objects are related to the signal_ID_Physical.

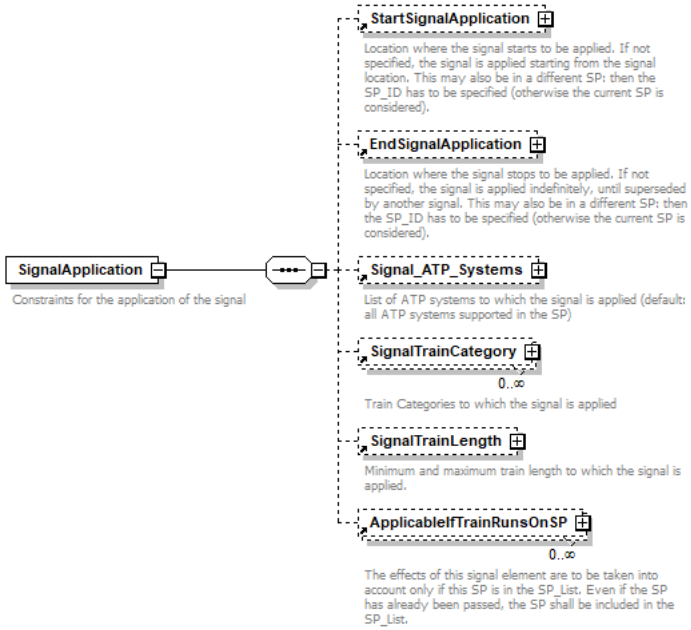
source

```
<xs:element name="Signal_ID">
 <xs:annotation>
  <xs:documentation>Signal identification. The identification may include the identifier of the physical signal of the track and the identifier of the single signal object that contains the specific information (and possibly application). For a single physical signal there may be several signal objects. The pair (signal_ID_Physical, signal_ID_Object) must be unique within an SP. the signal_ID_Object must be documented if 2 or more signal objects are related to the signal_ID_Physical.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:attribute ref="signal_ID_Physical" use="required"/>
  <xs:attribute ref="signal_ID_Object"/>
  <xs:attribute ref="location" use="required"/>
 </xs:complexType>
</xs:element>
```

element **Signal_NSPs**

diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	complexType **Signal_ComplexType**
annotation	documentation Network Specific Parameters in Signal
source	<xs:element name="Signal_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Signal</xs:documentation> </xs:annotation> </xs:element>

element **SignalApplication**

diagram	
properties	content complex
children	**StartSignalApplication** **EndSignalApplication** **Signal_ATP_Systems** **SignalTrainCategory** **SignalTrainLength** **ApplicableIfTrainRunsOnSP**
used by	complexType **Signal_ComplexType**
annotation	documentation Constraints for the application of the signal

source

```
<xs:element name="SignalApplication">
  <xs:annotation>
    <xs:documentation>Constraints for the application of the signal</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="StartSignalApplication" minOccurs="0"/>
      <xs:element ref="EndSignalApplication" minOccurs="0"/>
      <xs:element ref="Signal_ATP_Systems" minOccurs="0"/>
      <xs:element ref="SignalTrainCategory" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="SignalTrainLength" minOccurs="0"/>
      <xs:element ref="ApplicableIfTrainRunsOnSP" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **SignalFunction**

diagram



type: restriction of **xs:string**

properties: content simple

used by: complexType **Signal_ComplexType**

facets

Kind	Value	Annotation
enumeration	entry	
enumeration	exit	
enumeration	intermediate	
enumeration	advance	
enumeration	block	
enumeration	protection	
enumeration	trainProtection	
enumeration	groupMain	
enumeration	groupIntermediate	
enumeration	groupMainAndIntermediate	
enumeration	standAloneIndicator	
enumeration	trainDestinationOnlySignal	
enumeration	intermediateExitSignal	
enumeration	entryAndExit	
enumeration	entryAndBlock	
enumeration	further	
enumeration	lxProtectionSignal	
enumeration	distantOrRepeater	
enumeration	laneChange	

annotation

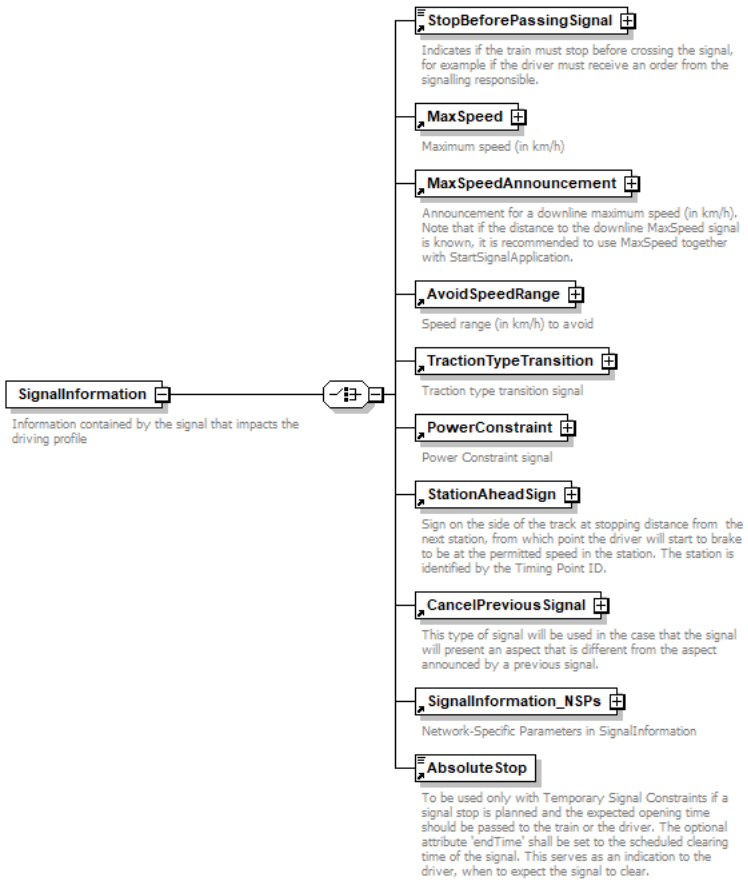
documentation
Function of the signal. The list of values is taken from EULynx SignalFunctionTypes (https://eulynx.eu/dataprep-2023-03/index.htm?goto=2:5:1:1:6:9801), with the addition of laneChange.

source

```
<xs:element name="SignalFunction">
  <xs:annotation>
```

```
    <xs:documentation>Function of the signal. The list of values is taken from EULynx SignalFunctionTypes (https://eulynx.eu/dataprep-2023-03/index.htm?goto=2:5:1:1:6:9801), with the addition of laneChange.</xs:documentation>
  </xs:annotation>
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:enumeration value="entry"/>
      <xs:enumeration value="exit"/>
      <xs:enumeration value="intermediate"/>
      <xs:enumeration value="advance"/>
      <xs:enumeration value="block"/>
      <xs:enumeration value="protection"/>
      <xs:enumeration value="trainProtection"/>
      <xs:enumeration value="groupMain"/>
      <xs:enumeration value="groupIntermediate"/>
      <xs:enumeration value="groupMainAndIntermediate"/>
      <xs:enumeration value="standAloneIndicator"/>
      <xs:enumeration value="trainDestinationOnlySignal"/>
      <xs:enumeration value="intermediateExitSignal"/>
      <xs:enumeration value="entryAndExit"/>
      <xs:enumeration value="entryAndBlock"/>
      <xs:enumeration value="further"/>
      <xs:enumeration value="lxProtectionSignal"/>
      <xs:enumeration value="distantOrRepeater"/>
      <xs:enumeration value="laneChange"/>
    </xs:restriction>
  </xs:simpleType>
</xs:element>
```

element **SignalInformation**

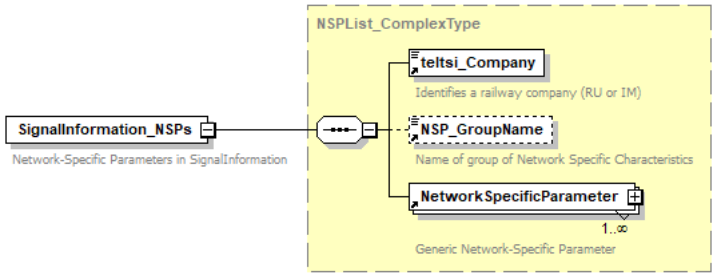
diagram	
properties	content complex
children	**StopBeforePassingSignal** **MaxSpeed** **MaxSpeedAnnouncement** **AvoidSpeedRange** **TractionTypeTransition** **PowerConstraint** **StationAheadSign** **CancelPreviousSignal** **SignalInformation_NSPs** **AbsoluteStop**
used by	complexType **Signal_ComplexType**
annotation	documentation Information contained by the signal that impacts the driving profile

source

```
<xs:element name="SignalInformation">
  <xs:annotation>
    <xs:documentation>Information contained by the signal that impacts the driving
profile</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:choice>
      <xs:element ref="StopBeforePassingSignal"/>
      <xs:element ref="MaxSpeed"/>
      <xs:element ref="MaxSpeedAnnouncement"/>
      <xs:element ref="AvoidSpeedRange"/>
      <xs:element ref="TractionTypeTransition"/>
      <xs:element ref="PowerConstraint"/>
      <xs:element ref="StationAheadSign"/>
      <xs:element ref="CancelPreviousSignal"/>
      <xs:element ref="SignalInformation_NSPs"/>
      <xs:element ref="AbsoluteStop"/>
    </xs:choice>
```

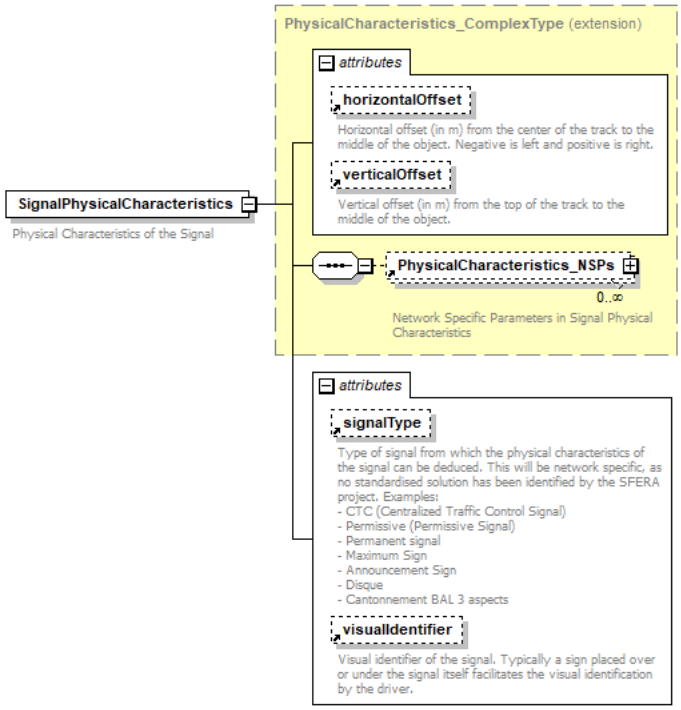
	</xs:complexType> </xs:element>

element **SignalInformation_NSPs**

diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	element **SignalInformation**
annotation	documentation Network-Specific Parameters in SignalInformation
source	<xs:element name="SignalInformation_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network-Specific Parameters in SignalInformation</xs:documentation> </xs:annotation> </xs:element>

element **SignalPhysicalCharacteristics**

diagram



type: extension of **PhysicalCharacteristics_ComplexType**

properties: content complex

children: **PhysicalCharacteristics_NSPs**

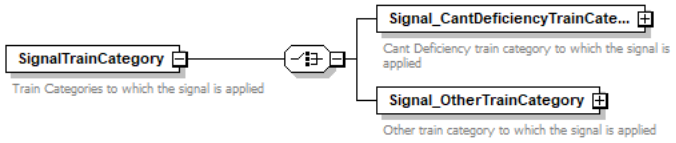
used by: complexType **Signal_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
horizontalOffset	**distanceType**				documentation Horizontal offset (in m) from the center of the track to the middle of the object. Negative is left and positive is right.
verticalOffset	**distanceType**				documentation Vertical offset (in m) from the top of the track to the middle of the object.
signalType	**xs:string**				documentation Type of signal from which the physical characteristics of the signal can be deduced. This will be network specific, as no standardised solution has been identified by the SFERA

	visualIdentifier **xs:string** documentation Visual identifier of the signal. Typically a sign placed over or under the signal itself facilitates the visual identification by the driver.	project. Examples: - CTC (Centralized Traffic Control Signal) - Permissive (Permissive Signal) - Permanent signal - Maximum Sign - Announcement Sign - Disque - Cantonnement BAL 3 aspects
annotation	documentation Physical Characteristics of the Signal	
source	<xs:element name="SignalPhysicalCharacteristics"> <xs:annotation> <xs:documentation>Physical Characteristics of the Signal</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="PhysicalCharacteristics_ComplexType"> <xs:attribute ref="signalType"/> <xs:attribute ref="visualIdentifier"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>	

element **SignalTrainCategory**

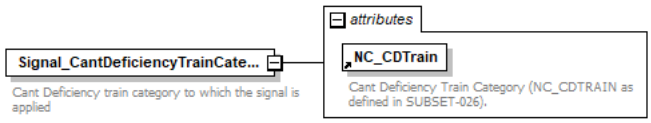
diagram	
properties	content complex
children	**Signal_CantDeficiencyTrainCategory** **Signal_OtherTrainCategory**
used by	element **SignalApplication**
annotation	documentation Train Categories to which the signal is applied

source

```
<xs:element name="SignalTrainCategory">
  <xs:annotation>
    <xs:documentation>Train Categories to which the signal is applied</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:choice>
      <xs:element name="Signal_CantDeficiencyTrainCategory">
        <xs:annotation>
          <xs:documentation>Cant Deficiency train category to which the signal is
applied</xs:documentation>
        </xs:annotation>
        <xs:complexType>
          <xs:attribute ref="NC_CDTrain" use="required"/>
        </xs:complexType>
      </xs:element>
      <xs:element name="Signal_OtherTrainCategory">
        <xs:annotation>
          <xs:documentation>Other train category to which the signal is applied</xs:documentation>
        </xs:annotation>
        <xs:complexType>
          <xs:attribute ref="NC_Train"/>
          <xs:attribute ref="trainCategoryCode"/>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:complexType>
</xs:element>
```

element **SignalTrainCategory/Signal_CantDeficiencyTrainCategory**

diagram



properties

content complex

attributes

Name	Type	Use	Default	Fixed	Annotation
NC_CDTrain	**NC_CDTrainType**	required			documentation Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).

annotation

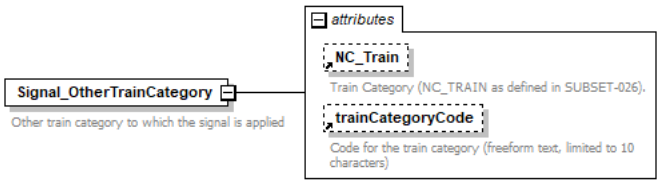
documentation
Cant Deficiency train category to which the signal is applied

source

```
<xs:element name="Signal_CantDeficiencyTrainCategory">
  <xs:annotation>
    <xs:documentation>Cant Deficiency train category to which the signal is
applied</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="NC_CDTrain" use="required"/>
  </xs:complexType>
</xs:element>
```

element **SignalTrainCategory/Signal_OtherTrainCategory**

diagram



properties

content complex

attributes

Name	Type	Use	Default	Fixed	Annotation
NC_Train	**NC_TrainType**				documentation Train Category (NC_TRAIN as defined in SUBSET-026).
trainCategoryCode	**derived by: xs:string**				documentation Code for the train category (freeform text, limited to 10 characters)

annotation

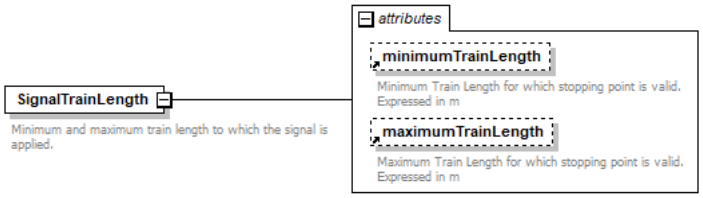
documentation
Other train category to which the signal is applied

source

```
<xs:element name="Signal_OtherTrainCategory">
  <xs:annotation>
    <xs:documentation>Other train category to which the signal is applied</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="NC_Train"/>
    <xs:attribute ref="trainCategoryCode"/>
  </xs:complexType>
</xs:element>
```

element **SignalTrainLength**

diagram



properties

content complex

used by

element **SignalApplication**

attributes

Name	Type	Use	Default	Fixed	Annotation
minimumTrainLength	**xs:double**				documentation Minimum Train Length for which stopping point is valid. Expressed in m

	maximumTrainLength **xs:double**	documentation Maximum Train Length for which stopping point is valid. Expressed in m
annotation	documentation Minimum and maximum train length to which the signal is applied.	
source	<xs:element name="SignalTrainLength"> <xs:annotation> <xs:documentation>Minimum and maximum train length to which the signal is applied.</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="minimumTrainLength"/> <xs:attribute ref="maximumTrainLength"/> </xs:complexType> </xs:element>	

element **SP_Areas**

diagram	
properties	content complex
children	**Platform Tunnel NoStoppingArea AxleLoadSpeedProfile PermittedBrakingDistanceArea SwitchOffRegenerativeBrakeArea SwitchOffEddyCurrentBrakeArea SwitchOffEddyCurrentEmergencyBrakeArea SwitchOffMagneticShoeBrakeArea DAS_ModesUnavailabilityArea DynamicBrakeForceInhibitionArea LimitedDynamicBrakeForceArea TAF_TAP_Location GuardRails BatteryChargingArea NoAdvicePresentedToDriver NetworkSpecificArea**
used by	element **SegmentProfile**
annotation	documentation Notable areas present in the Segment Profile (e.g. stations, platforms, tunnels...)
source	<xs:element name="SP_Areas"> <xs:annotation> <xs:documentation>Notable areas present in the Segment Profile (e.g. stations, platforms, tunnels...)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="Platform" minOccurs="0" maxOccurs="unbounded"/>

```
      <xs:element ref="Tunnel" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="NoStoppingArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="AxleLoadSpeedProfile" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="PermittedBrakingDistanceArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="SwitchOffRegenerativeBrakeArea" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="SwitchOffEddyCurrentBrakeArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="SwitchOffEddyCurrentEmergencyBrakeArea" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="SwitchOffMagneticShoeBrakeArea" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="DAS_ModesUnavailabilityArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="DynamicBrakeForceInhibitionArea" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="LimitedDynamicBrakeForceArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="TAF_TAP_Location" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="GuardRails" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="BatteryChargingArea" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="NoAdvicePresentedToDriver" minOccurs="0" maxOccurs="unbounded"/>
      <xs:element ref="NetworkSpecificArea" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

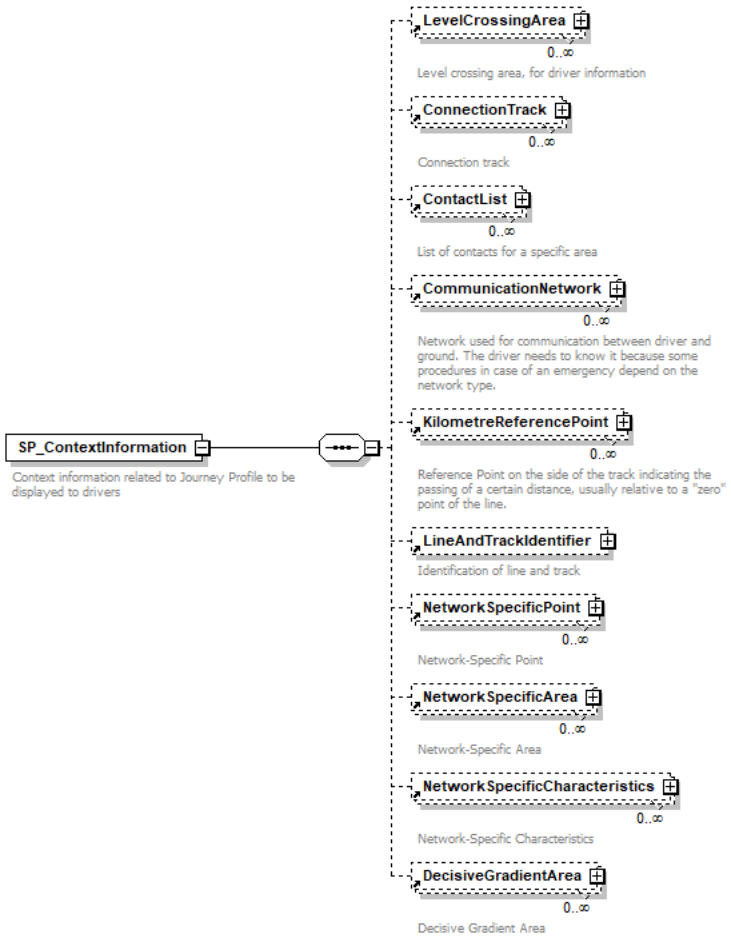
element **SP_Characteristics**

diagram



properties	content complex
children	**StaticSpeedProfile** **Gradient** **GradientSteepest** **Curves** **SuperElevation** **RatedVoltage** **EstimatedVoltage** **CurrentLimitation** **ATP_Systems** **NetworkSpecificCharacteristics**
used by	element **SegmentProfile**
annotation	documentation Characteristics of the Segment Profile (e.g. gradient, curve, cant, voltage...)
source	<xs:element name="SP_Characteristics"> <xs:annotation> <xs:documentation>Characteristics of the Segment Profile (e.g. gradient, curve, cant, voltage...)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="StaticSpeedProfile" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="Gradient" minOccurs="0"/> <xs:element ref="GradientSteepest" minOccurs="0"/> <xs:element ref="Curves" minOccurs="0"/> <xs:element ref="SuperElevation" minOccurs="0"/> <xs:element ref="RatedVoltage" minOccurs="0"/> <xs:element ref="EstimatedVoltage" minOccurs="0"/> <xs:element ref="CurrentLimitation" minOccurs="0"/> <xs:element ref="ATP_Systems" minOccurs="0"/> <xs:element ref="NetworkSpecificCharacteristics" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

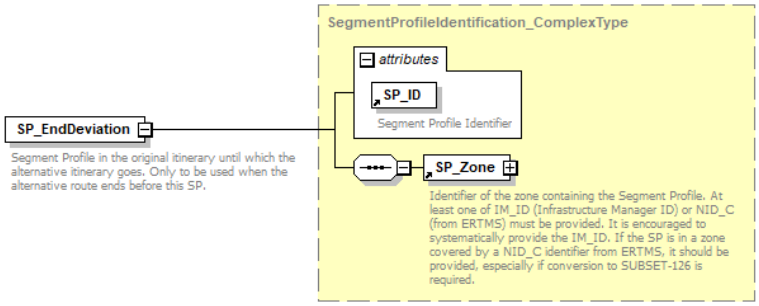
element **SP_ContextInformation**

diagram	
properties	content complex
children	**LevelCrossingArea** **ConnectionTrack** **ContactList** **CommunicationNetwork** **KilometreReferencePoint** **LineAndTrackIdentifier** **NetworkSpecificPoint** **NetworkSpecificArea** **NetworkSpecificCharacteristics** **DecisiveGradientArea**
used by	element **SegmentProfile**
annotation	documentation Context information related to Journey Profile to be displayed to drivers
source	<xs:element name="SP_ContextInformation"> <xs:annotation> <xs:documentation>Context information related to Journey Profile to be displayed to drivers</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="LevelCrossingArea" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="ConnectionTrack" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="ContactList" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="CommunicationNetwork" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="KilometreReferencePoint" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="LineAndTrackIdentifier" minOccurs="0"/> <xs:element ref="NetworkSpecificPoint" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="NetworkSpecificArea" minOccurs="0" maxOccurs="unbounded"/>

```
    <xs:element ref="NetworkSpecificCharacteristics" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element ref="DecisiveGradientArea" minOccurs="0" maxOccurs="unbounded"/>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **SP_EndDeviation**

diagram



type: **SegmentProfileIdentification_ComplexType**

properties: content complex

children: **SP_Zone**

used by: element **AlternativeItinerary**

attributes	Name	Type	Use	Default	Fixed	Annotation
	SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier

annotation

documentation

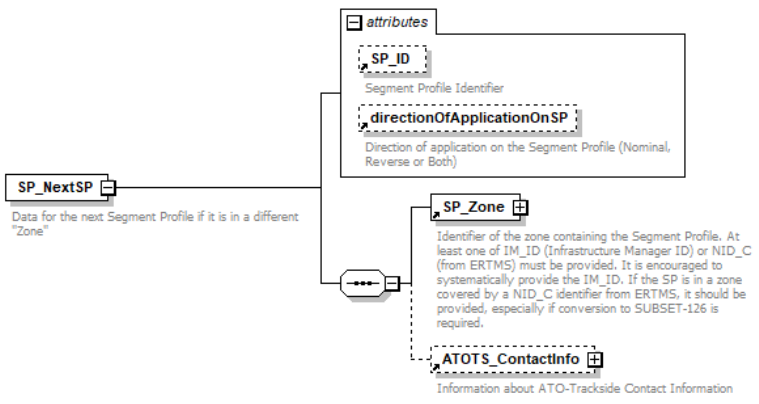
Segment Profile in the original itinerary until which the alternative itinerary goes. Only to be used when the alternative route ends before this SP.

source

```
<xs:element name="SP_EndDeviation" type="SegmentProfileIdentification_ComplexType">
 <xs:annotation>
  <xs:documentation>Segment Profile in the original itinerary until which the alternative itinerary goes. Only to be used when the alternative route ends before this SP.</xs:documentation>
 </xs:annotation>
</xs:element>
```

element **SP_NextSP**

diagram



properties	content complex
children	**SP_Zone** **ATOTS_ContactInfo**
used by	element **SegmentProfile**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**				documentation Segment Profile Identifier
directionOfApplicationOnSP	**derived by: xs:string**				documentation Direction of application on the Segment Profile (Nominal, Reverse or Both)

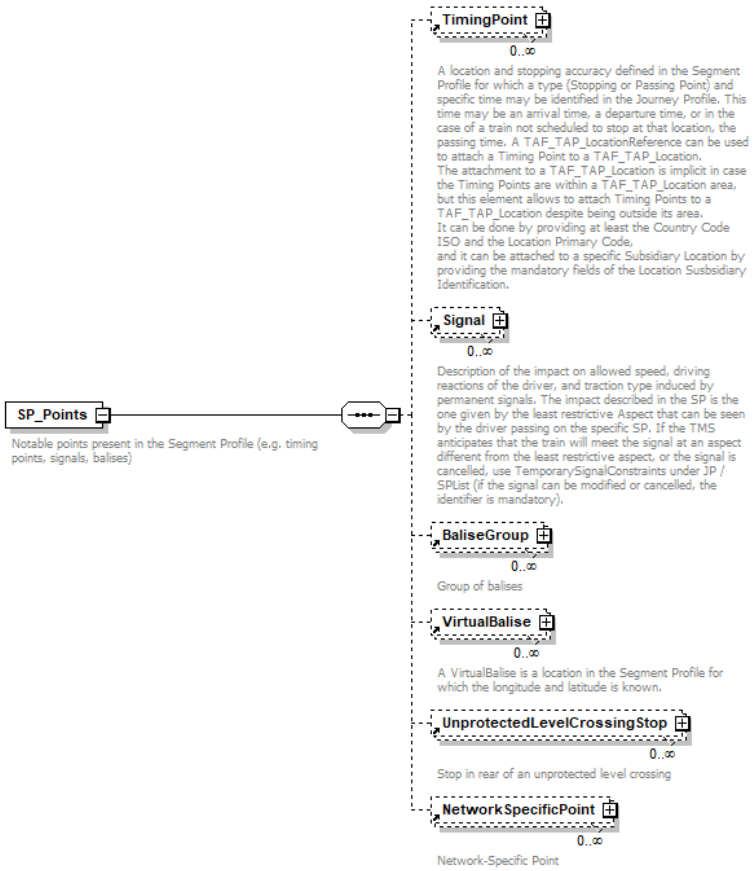
annotation

documentation
Data for the next Segment Profile if it is in a different "Zone"

source

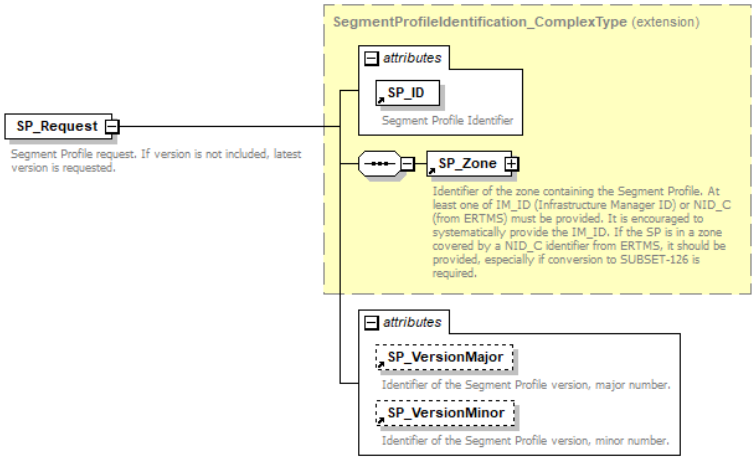
```
<xs:element name="SP_NextSP">
  <xs:annotation>
    <xs:documentation>Data for the next Segment Profile if it is in a different
"Zone"</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="SP_Zone"/>
      <xs:element ref="ATOTS_ContactInfo" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute ref="SP_ID"/>
    <xs:attribute ref="directionOfApplicationOnSP"/>
  </xs:complexType>
</xs:element>
```

element **SP_Points**

diagram	
properties	content complex
children	**TimingPoint** **Signal** **BaliseGroup** **VirtualBalise** **UnprotectedLevelCrossingStop** **NetworkSpecificPoint**
used by	element **SegmentProfile**
annotation	documentation Notable points present in the Segment Profile (e.g. timing points, signals, balises)
source	<xs:element name="SP_Points"> <xs:annotation> <xs:documentation>Notable points present in the Segment Profile (e.g. timing points, signals, balises)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TimingPoint" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="Signal" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="BaliseGroup" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="VirtualBalise" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="UnprotectedLevelCrossingStop" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="NetworkSpecificPoint" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **SP_Request**

diagram



type: extension of **SegmentProfileIdentification_ComplexType**

properties: content complex

children: **SP_Zone**

used by: element **B2G_Request**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**				documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment Profile version, minor number.

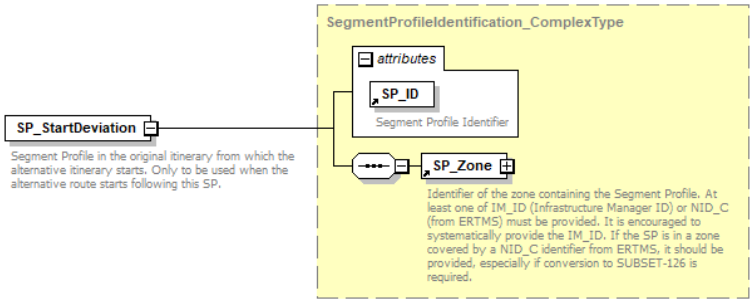
annotation

documentation
Segment Profile request. If version is not included, latest version is requested.

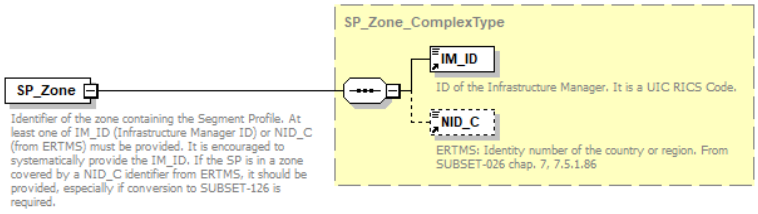
source

```
<xs:element name="SP_Request">
  <xs:annotation>
    <xs:documentation>Segment Profile request. If version is not included, latest version is
requested.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SegmentProfileIdentification_ComplexType">
        <xs:attribute ref="SP_VersionMajor"/>
        <xs:attribute ref="SP_VersionMinor"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **SP_StartDeviation**

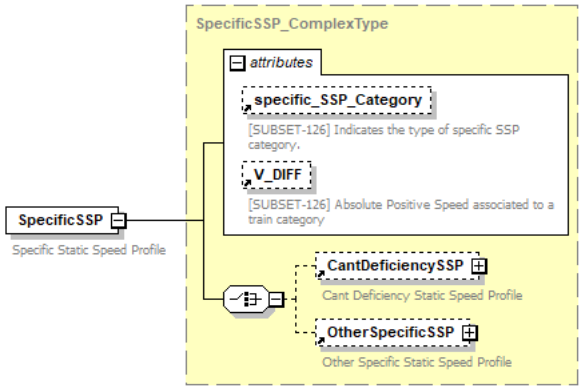
diagram	
type	SegmentProfileIdentification_ComplexType
properties	content complex
children	SP_Zone
used by	element AlternativeItinerary
attributes	Name: SP_ID; Type: derived by: xs:string; Use: required; Default: ; Fixed: ; Annotation: documentation Segment Profile Identifier
annotation	documentation Segment Profile in the original itinerary from which the alternative itinerary starts. Only to be used when the alternative route starts following this SP.
source	<xs:element name="SP_StartDeviation" type="SegmentProfileIdentification_ComplexType"> <xs:annotation> <xs:documentation>Segment Profile in the original itinerary from which the alternative itinerary starts. Only to be used when the alternative route starts following this SP.</xs:documentation> </xs:annotation> </xs:element>

element **SP_Zone**

diagram	
type	SP_Zone_ComplexType
properties	content complex
children	IM_ID NID_C
used by	elements ApplicableIfTrainRunsOnSP HandshakeAcknowledgement HandshakeReject NextTimingPointTrainInformation PreviousTimingPointTrainInformation SP_NextSP TimingPointEstimation complexTypes SegmentProfileIdentification_ComplexType TimingPointIdentification_ComplexType TimingPointZone_ComplexType
annotation	documentation Identifier of the zone containing the Segment Profile. At least one of IM_ID (Infrastructure Manager ID) or NID_C (from ERTMS) must be provided. It is encouraged to systematically provide the IM_ID. If the SP is in a zone covered by a NID_C identifier from ERTMS, it should be provided, especially if conversion to SUBSET-126 is required.
source	<xs:element name="SP_Zone" type="SP_Zone_ComplexType"> <xs:annotation>

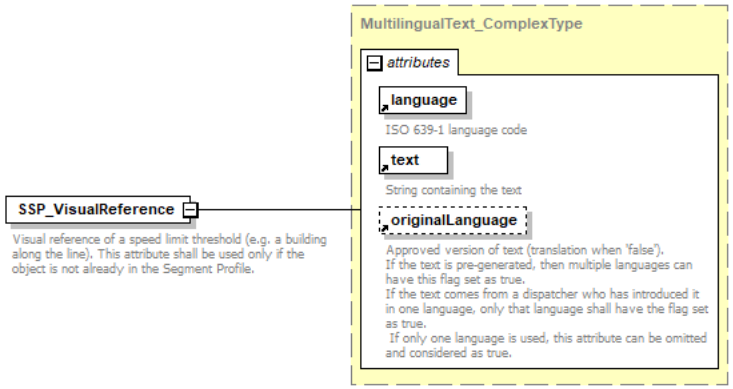
	<xs:documentation>Identifier of the zone containing the Segment Profile. At least one of IM_ID (Infrastructure Manager ID) or NID_C (from ERTMS) must be provided. It is encouraged to systematically provide the IM_ID. If the SP is in a zone covered by a NID_C identifier from ERTMS, it should be provided, especially if conversion to SUBSET-126 is required.</xs:documentation> </xs:annotation> </xs:element>

element **SpecificSSP**

diagram	
type	**SpecificSSP_ComplexType**
properties	content complex
children	**CantDeficiencySSP** **OtherSpecificSSP**
used by	complexType **SSP_ComplexType**
attributes	Name / Type / Use / Default / Fixed / Annotation specific_SSP_Category — **derived by: xs:string** — documentation [SUBSET-126] Indicates the type of specific SSP category. V_DIFF — **speedType** — documentation [SUBSET-126] Absolute Positive Speed associated to a train category
annotation	documentation Specific Static Speed Profile
source	<xs:element name="SpecificSSP" type="SpecificSSP_ComplexType"> <xs:annotation> <xs:documentation>Specific Static Speed Profile</xs:documentation> </xs:annotation> </xs:element>

element **SSP_VisualReference**

diagram



type **MultilingualText_ComplexType**

properties
content complex

used by
complexType **SSP_ComplexType**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

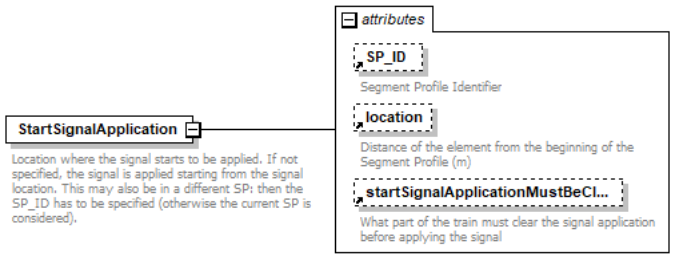
annotation
documentation
Visual reference of a speed limit threshold (e.g. a building along the line). This attribute shall be used only if the object is not already in the Segment Profile.

source

```
<xs:element name="SSP_VisualReference" type="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Visual reference of a speed limit threshold (e.g. a building along the line).
This attribute shall be used only if the object is not already in the Segment
```

	Profile.</xs:documentation> </xs:annotation> </xs:element>

element **StartSignalApplication**

diagram	
properties	content complex
used by	element **SignalApplication**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**				documentation Segment Profile Identifier
location	**distanceType**				documentation Distance of the element from the beginning of the Segment Profile (m)
startSignalApplicationMustBeClearedBy	**trainPartType**				documentation What part of the train must clear the signal application before applying the signal

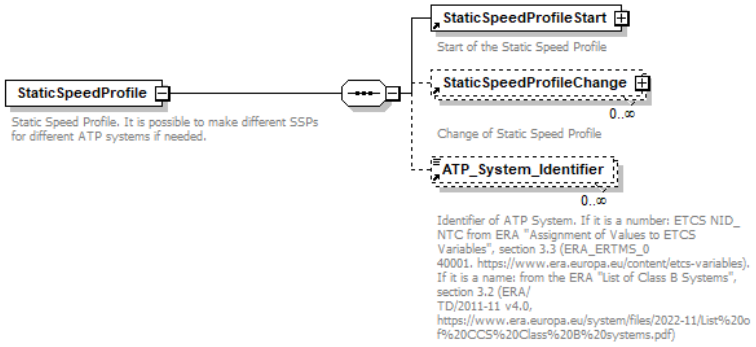
annotation

documentation
Location where the signal starts to be applied. If not specified, the signal is applied starting from the signal location. This may also be in a different SP: then the SP_ID has to be specified (otherwise the current SP is considered).

source

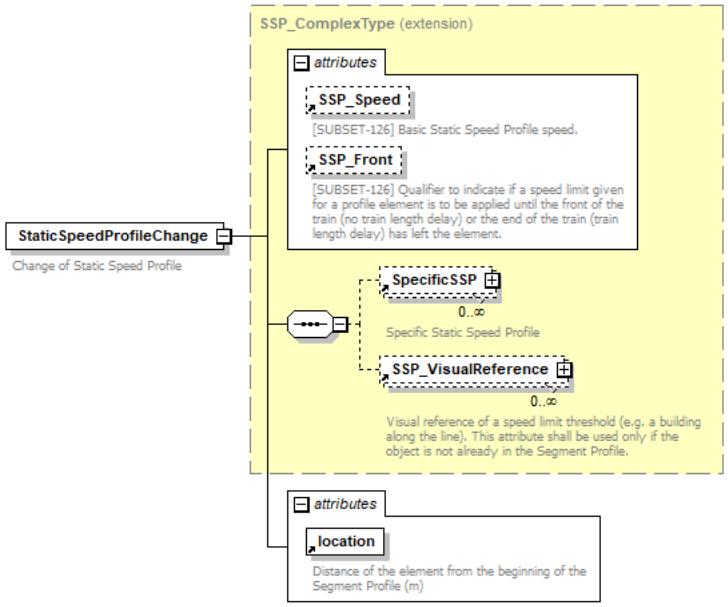
```
<xs:element name="StartSignalApplication">
 <xs:annotation>
  <xs:documentation>Location where the signal starts to be applied. If not specified, the signal is applied starting from the signal location. This may also be in a different SP: then the SP_ID has to be specified (otherwise the current SP is considered).</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:attribute ref="SP_ID"/>
  <xs:attribute ref="location"/>
  <xs:attribute ref="startSignalApplicationMustBeClearedBy"/>
 </xs:complexType>
</xs:element>
```

element **StaticSpeedProfile**

diagram	
properties	content complex
children	**StaticSpeedProfileStart** **StaticSpeedProfileChange** **ATP_System_Identifier**
used by	element **SP_Characteristics**
annotation	documentation Static Speed Profile. It is possible to make different SSPs for different ATP systems if needed.
source	<xs:element name="StaticSpeedProfile"> <xs:annotation> <xs:documentation>Static Speed Profile. It is possible to make different SSPs for different ATP systems if needed.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="StaticSpeedProfileStart"/> <xs:element ref="StaticSpeedProfileChange" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="ATP_System_Identifier" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **StaticSpeedProfileChange**

diagram



type: extension of **SSP_ComplexType**

properties: content complex

children: **SpecificSSP** **SSP_VisualReference**

used by: element **StaticSpeedProfile**

attributes

Name	Type	Use	Default	Fixed	Annotation
SSP_Speed	**speedType**				documentation [SUBSET-126] Basic Static Speed Profile speed.
SSP_Front	**xs:boolean**				documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (no train length delay) or the end of the train (train length delay) has left the element.
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)

annotation: documentation
Change of Static Speed Profile

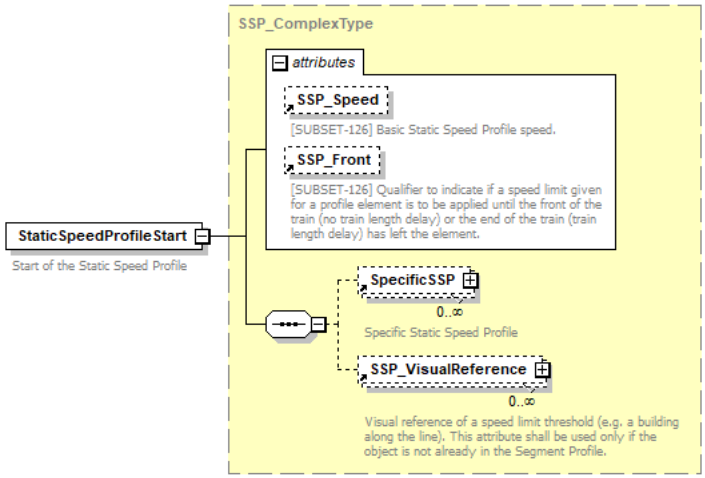
source

```
<xs:element name="StaticSpeedProfileChange">
 <xs:annotation>
```

```
    <xs:documentation>Change of Static Speed Profile</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SSP_ComplexType">
        <xs:attribute ref="location" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **StaticSpeedProfileStart**

diagram



type	**SSP_ComplexType**
properties	content complex
children	**SpecificSSP** **SSP_VisualReference**
used by	element **StaticSpeedProfile**

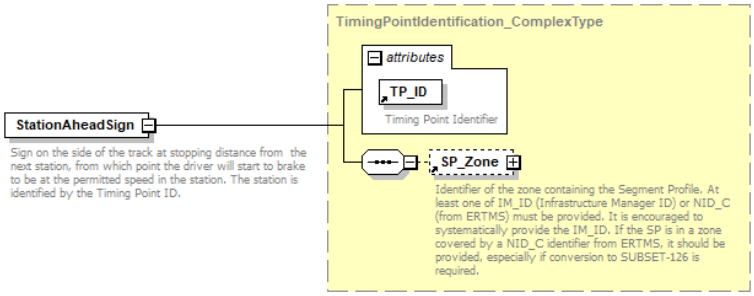
attributes

Name	Type	Use	Default	Fixed	Annotation
SSP_Speed	**speedType**				documentation [SUBSET-126] Basic Static Speed Profile speed.
SSP_Front	**xs:boolean**				documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (no train length delay) or the end of the train (train length delay) has left the element.

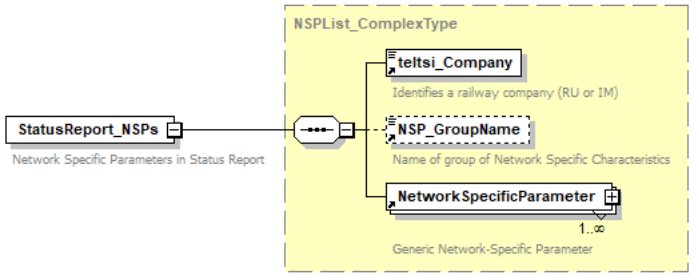
annotation documentation

	Start of the Static Speed Profile
source	<xs:element name="StaticSpeedProfileStart" type="SSP_ComplexType"> <xs:annotation> <xs:documentation>Start of the Static Speed Profile</xs:documentation> </xs:annotation> </xs:element>

element **StationAheadSign**

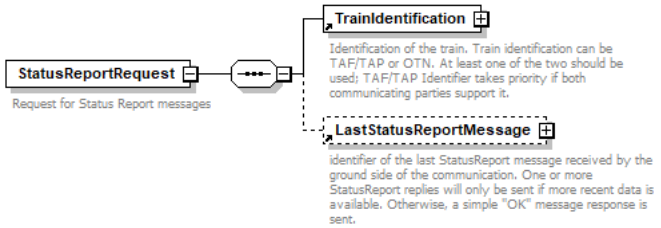
diagram	
type	**TimingPointIdentification_ComplexType**
properties	content complex
children	**SP_Zone**
used by	element **SignalInformation**
attributes	Name: TP_ID; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation: Timing Point Identifier
annotation	documentation Sign on the side of the track at stopping distance from the next station, from which point the driver will start to brake to be at the permitted speed in the station. The station is identified by the Timing Point ID.
source	<xs:element name="StationAheadSign" type="TimingPointIdentification_ComplexType"> <xs:annotation> <xs:documentation>Sign on the side of the track at stopping distance from the next station, from which point the driver will start to brake to be at the permitted speed in the station. The station is identified by the Timing Point ID.</xs:documentation> </xs:annotation> </xs:element>

element **StatusReport_NSPs**

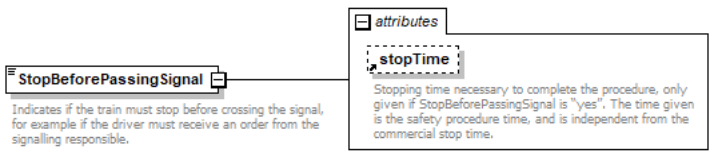
diagram	
type	**NSPList_ComplexType**
properties	content complex

children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	element **B2G_StatusReport**
annotation	documentation Network Specific Parameters in Status Report
source	<xs:element name="StatusReport_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Status Report</xs:documentation> </xs:annotation> </xs:element>

element **StatusReportRequest**

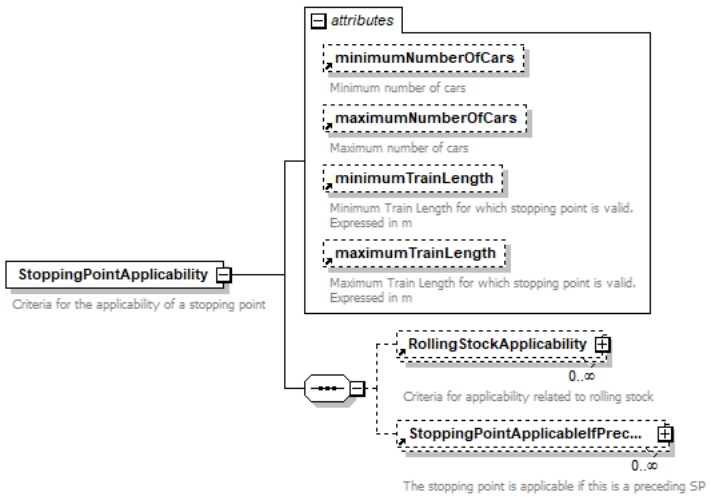
diagram	
properties	content complex
children	**TrainIdentification** **LastStatusReportMessage**
used by	element **G2B_Request**
annotation	documentation Request for Status Report messages
source	<xs:element name="StatusReportRequest"> <xs:annotation> <xs:documentation>Request for Status Report messages</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TrainIdentification"/> <xs:element ref="LastStatusReportMessage" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **StopBeforePassingSignal**

diagram	
type	extension of **xs:boolean**
properties	content complex
used by	element **SignalInformation**
attributes	Name: stopTime Type: **xs:duration** Use: Default: Fixed: Annotation: documentation Stopping time necessary

	to complete the procedure, only given if StopBeforePassingSignal is “yes”. The time given is the safety procedure time, and is independent from the commercial stop time.
annotation	documentation Indicates if the train must stop before crossing the signal, for example if the driver must receive an order from the signalling responsible.
source	<xs:element name="StopBeforePassingSignal"> <xs:annotation> <xs:documentation>Indicates if the train must stop before crossing the signal, for example if the driver must receive an order from the signalling responsible.</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="xs:boolean"> <xs:attribute ref="stopTime"/> </xs:extension> </xs:simpleContent> </xs:complexType> </xs:element>

element **StoppingPointApplicability**

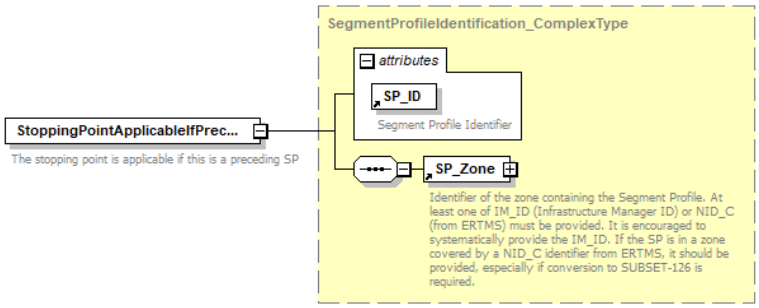
diagram	
properties	content complex
children	**RollingStockApplicability** **StoppingPointApplicableIfPrecedingSP**
used by	element **StoppingPointLocation**

attributes

Name	Type	Use	Default	Fixed	Annotation
minimumNumberOfCars	**xs:short**				documentation Minimum number of cars
maximumNumberOfCars	**xs:short**				documentation Maximum number of cars
minimumTrainLength	**xs:double**				documentation

	maximumTrainLength **xs:double**	Minimum Train Length for which stopping point is valid. Expressed in m documentation Maximum Train Length for which stopping point is valid. Expressed in m
annotation	documentation Criteria for the applicability of a stopping point	
source	<xs:element name="StoppingPointApplicability"> <xs:annotation> <xs:documentation>Criteria for the applicability of a stopping point</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="RollingStockApplicability" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="StoppingPointApplicableIfPrecedingSP" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="minimumNumberOfCars"/> <xs:attribute ref="maximumNumberOfCars"/> <xs:attribute ref="minimumTrainLength"/> <xs:attribute ref="maximumTrainLength"/> </xs:complexType> </xs:element>	

element **StoppingPointApplicableIfPrecedingSP**

diagram	
type	**SegmentProfileIdentification_ComplexType**
properties	content complex
children	**SP_Zone**
used by	element **StoppingPointApplicability**
attributes	Name: SP_ID; Type: **derived by: xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation Segment Profile Identifier
annotation	documentation The stopping point is applicable if this is a preceding SP

source

```
<xs:element name="StoppingPointApplicableIfPrecedingSP"
type="SegmentProfileIdentification_ComplexType">
  <xs:annotation>
    <xs:documentation>The stopping point is applicable if this is a preceding SP</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **StoppingPointDepartureDetails**

diagram



properties

content complex

used by

element **TimingPointConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
departureTime	**UTCDateTimeType**	required			documentation Expected time of departure at Timing Point expressed in UTC
earliestDepartureTime	**UTCDateTimeType**				documentation Earliest departure time desired, expressed in UTC.
plannedDepartureTime	**UTCDateTimeType**				documentation The original planned departure time from the Timing Point expressed in UTC. Useful for driver information.
leaveASAP	**xs:boolean**				documentation Leave as soon as possible, taking into

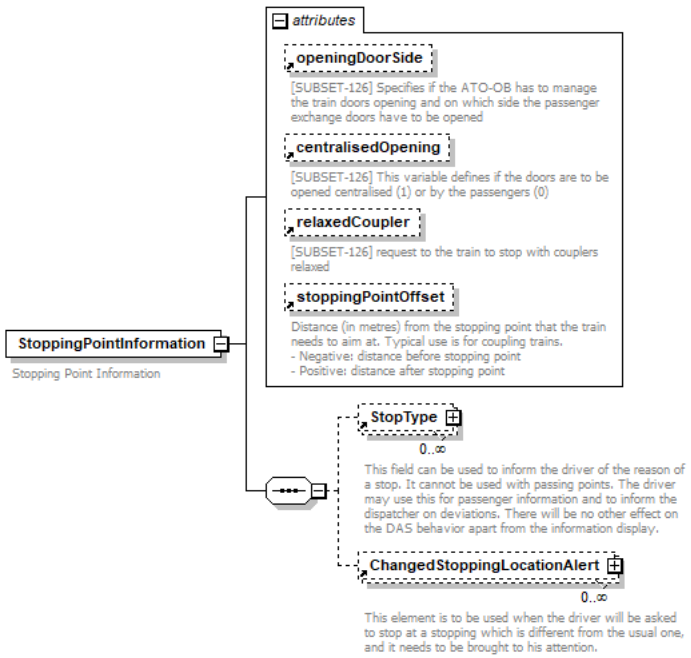
			account also the other variables (boolean).
	departureAuthorisationType	**xs:string**	documentation Type of authorisation for departure (e.g. SMS)
	trainHold	**xs:boolean**	documentation [SUBSET-126] The variable defines if the train is requested to be held at the Stopping Point (1) or not (0)
	minimumDwellTime	**xs:duration**	documentation [SUBSET-126] Minimum dwell time at given Stopping Point (in seconds)
	automaticClosing	**xs:boolean**	documentation [SUBSET-126] This variable defines if the ATO-OB has to manage the train doors closing (1) or not (0)
annotation	documentation Departure details for the Stopping Point		

source

```
<xs:element name="StoppingPointDepartureDetails">
  <xs:annotation>
    <xs:documentation>Departure details for the Stopping Point</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="departureTime" use="required"/>
    <xs:attribute ref="earliestDepartureTime"/>
    <xs:attribute ref="plannedDepartureTime"/>
    <xs:attribute ref="leaveASAP"/>
    <xs:attribute ref="departureAuthorisationType"/>
    <xs:attribute ref="trainHold"/>
    <xs:attribute ref="minimumDwellTime"/>
    <xs:attribute ref="automaticClosing"/>
  </xs:complexType>
</xs:element>
```

element **StoppingPointInformation**

diagram



properties

content complex

children **StopType** **ChangedStoppingLocationAlert**

used by

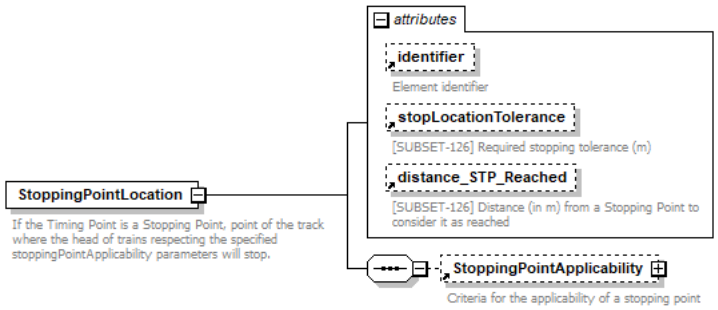
element **TimingPointConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
openingDoorSide	**derived by: xs:string**				documentation [SUBSET-126] Specifies if the ATO-OB has to manage the train doors opening and on which side the passenger exchange doors have to be opened
centralisedOpening	**xs:boolean**				documentation [SUBSET-126] This variable defines if the doors are to be opened centralised (1) or by the passengers (0)
relaxedCoupler	**xs:boolean**				documentation [SUBSET-126] request to the train to stop with couplers relaxed
stoppingPointOffset	**distanceType**				documentation Distance (in metres) from the stopping point that the train needs to

	aim at. Typical use is for coupling trains. - Negative: distance before stopping point - Positive: distance after stopping point
annotation	documentation Stopping Point Information
source	<xs:element name="StoppingPointInformation"> <xs:annotation> <xs:documentation>Stopping Point Information</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="StopType" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="ChangedStoppingLocationAlert" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="openingDoorSide"/> <xs:attribute ref="centralisedOpening"/> <xs:attribute ref="relaxedCoupler"/> <xs:attribute ref="stoppingPointOffset"/> </xs:complexType> </xs:element>

element **StoppingPointLocation**

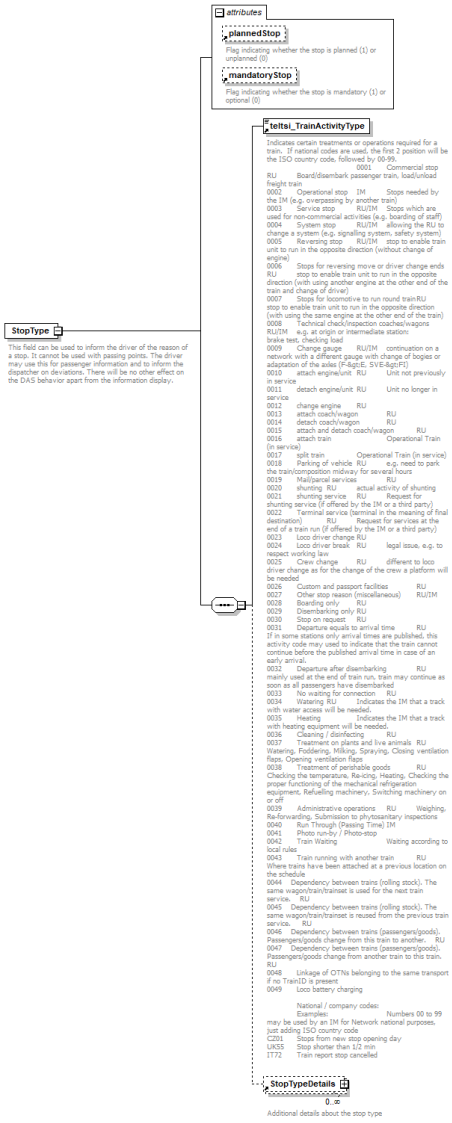
diagram	
properties	content complex
children	**StoppingPointApplicability**
used by	element **TimingPoint**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
stopLocationTolerance	**distanceType**				documentation [SUBSET-126] Required stopping tolerance (m)
distance_STP_Reached	**distanceType**				documentation [SUBSET-126] Distance (in m) from a Stopping Point

	to consider it as reached
annotation	documentation If the Timing Point is a Stopping Point, point of the track where the head of trains respecting the specified stoppingPointApplicability parameters will stop.
source	<xs:element name="StoppingPointLocation"> <xs:annotation> <xs:documentation>If the Timing Point is a Stopping Point, point of the track where the head of trains respecting the specified stoppingPointApplicability parameters will stop.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="StoppingPointApplicability" minOccurs="0"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="stopLocationTolerance"/> <xs:attribute ref="distance_STP_Reached"/> </xs:complexType> </xs:element>

element **StopType**

diagram	
properties	content complex
children	**teltsi_TrainActivityType** **StopTypeDetails**
used by	element **StoppingPointInformation**

attributes

Name	Type	Use	Default	Fixed	Annotation
plannedStop	**xs:boolean**				documentation Flag indicating whether the stop is planned (1) or unplanned (0)
mandatoryStop	**xs:boolean**				documentation Flag indicating whether the stop is mandatory (1) or optional (0)

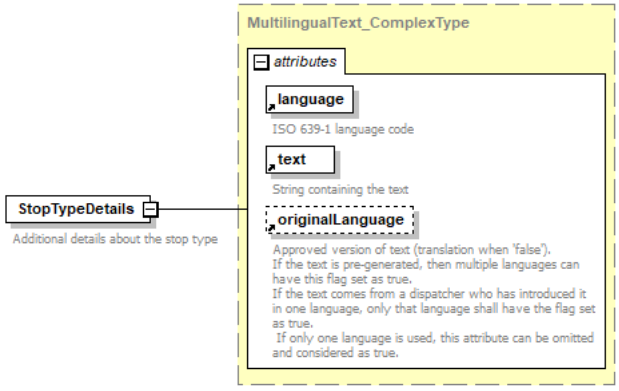
annotation

documentation

This field can be used to inform the driver of the reason of a stop. It cannot be used with passing points. The driver may use this for passenger information and to inform the dispatcher on deviations. There will be no other effect on the DAS

	behavior apart from the information display.
source	<xs:element name="StopType"> <xs:annotation> <xs:documentation>This field can be used to inform the driver of the reason of a stop. It cannot be used with passing points. The driver may use this for passenger information and to inform the dispatcher on deviations. There will be no other effect on the DAS behavior apart from the information display.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_TrainActivityType"/> <xs:element ref="StopTypeDetails" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="plannedStop"/> <xs:attribute ref="mandatoryStop"/> </xs:complexType> </xs:element>

element **StopTypeDetails**

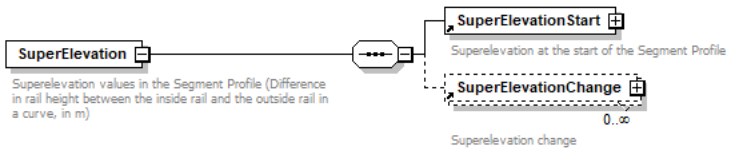
diagram	
type	**MultilingualText_ComplexType**
properties	content complex
used by	element **StopType**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text

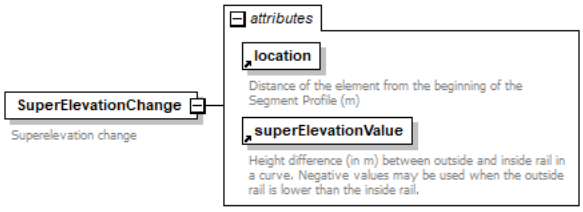
	comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
annotation	documentation Additional details about the stop type
source	<xs:element name="StopTypeDetails" type="MultilingualText_ComplexType"> <xs:annotation> <xs:documentation>Additional details about the stop type</xs:documentation> </xs:annotation> </xs:element>

element **SuperElevation**

diagram	
properties	content complex
children	**SuperElevationStart** **SuperElevationChange**
used by	element **SP_Characteristics**
annotation	documentation Superelevation values in the Segment Profile (Difference in rail height between the inside rail and the outside rail in a curve, in m)
source	<xs:element name="SuperElevation"> <xs:annotation> <xs:documentation>Superelevation values in the Segment Profile (Difference in rail height between the inside rail and the outside rail in a curve, in m)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="SuperElevationStart"/> <xs:element ref="SuperElevationChange" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **SuperElevationChange**

diagram



properties

content complex

used by

element **SuperElevation**

attributes

Name	Type	Use	Default	Fixed	Annotation
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
superElevationValue	**xs:decimal**	required			documentation Height difference (in m) between outside and inside rail in a curve. Negative values may be used when the outside rail is lower than the inside rail.

annotation

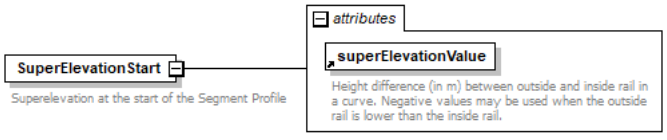
documentation
Superelevation change

source

```
<xs:element name="SuperElevationChange">
  <xs:annotation>
    <xs:documentation>Superelevation change</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="location" use="required"/>
    <xs:attribute ref="superElevationValue" use="required"/>
  </xs:complexType>
</xs:element>
```

element **SuperElevationStart**

diagram



properties

content complex

used by

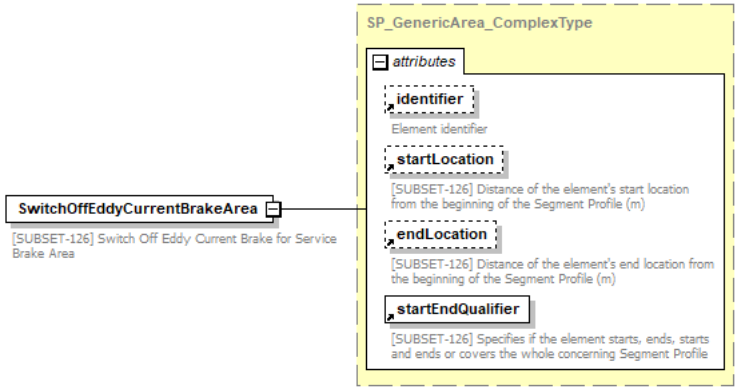
element **SuperElevation**

attributes

Name	Type	Use	Default	Fixed	Annotation
superElevationValue	**xs:decimal**	required			documentation Height difference (in m) between

	outside and inside rail in a curve. Negative values may be used when the outside rail is lower than the inside rail.
annotation	documentation Superelevation at the start of the Segment Profile
source	<xs:element name="SuperElevationStart"> <xs:annotation> <xs:documentation>Superelevation at the start of the Segment Profile</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="superElevationValue" use="required"/> </xs:complexType> </xs:element>

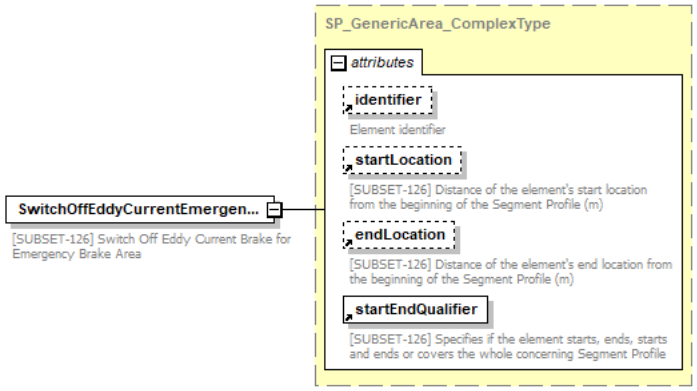
element **SwitchOffEddyCurrentBrakeArea**

diagram	
type	**SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**

attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**				documentation Element identifier
	startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of

	startEndQualifier	**derived by: xs:string**	required	the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation [SUBSET-126] Switch Off Eddy Current Brake for Service Brake Area			
source	<xs:element name="SwitchOffEddyCurrentBrakeArea" type="SP_GenericArea_ComplexType"> <xs:annotation> <xs:documentation>[SUBSET-126] Switch Off Eddy Current Brake for Service Brake Area</xs:documentation> </xs:annotation> </xs:element>			

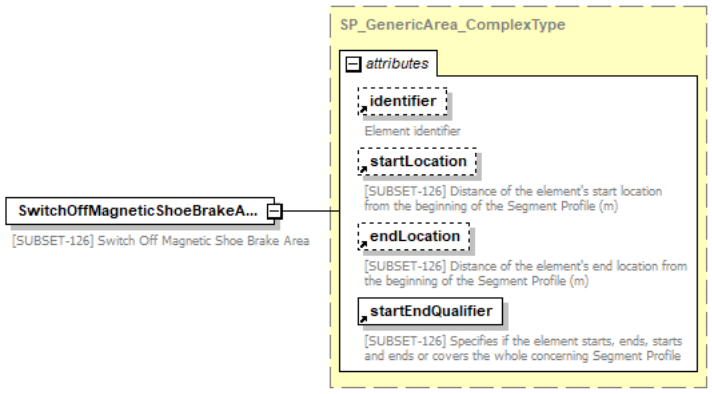
element **SwitchOffEddyCurrentEmergencyBrakeArea**

diagram	
type	**SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**

attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**				documentation Element identifier
	startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the

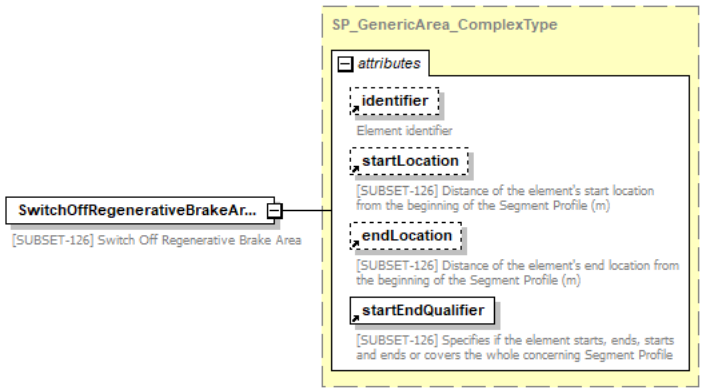
	startEndQualifier	**derived by: xs:string**	required			beginning of the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation [SUBSET-126] Switch Off Eddy Current Brake for Emergency Brake Area					
source	<xs:element name="SwitchOffEddyCurrentEmergencyBrakeArea" type="SP_GenericArea_ComplexType"> <xs:annotation> <xs:documentation>[SUBSET-126] Switch Off Eddy Current Brake for Emergency Brake Area</xs:documentation> </xs:annotation> </xs:element>					

element **SwitchOffMagneticShoeBrakeArea**

diagram						
type	**SP_GenericArea_ComplexType**					
properties	content complex					
used by	element **SP_Areas**					
attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**				documentation Element identifier
	startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's

	startEndQualifier	**derived by: xs:string**	required			end location from the beginning of the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation [SUBSET-126] Switch Off Magnetic Shoe Brake Area					
source	<xs:element name="SwitchOffMagneticShoeBrakeArea" type="SP_GenericArea_ComplexType"> <xs:annotation> <xs:documentation>[SUBSET-126] Switch Off Magnetic Shoe Brake Area</xs:documentation> </xs:annotation> </xs:element>					

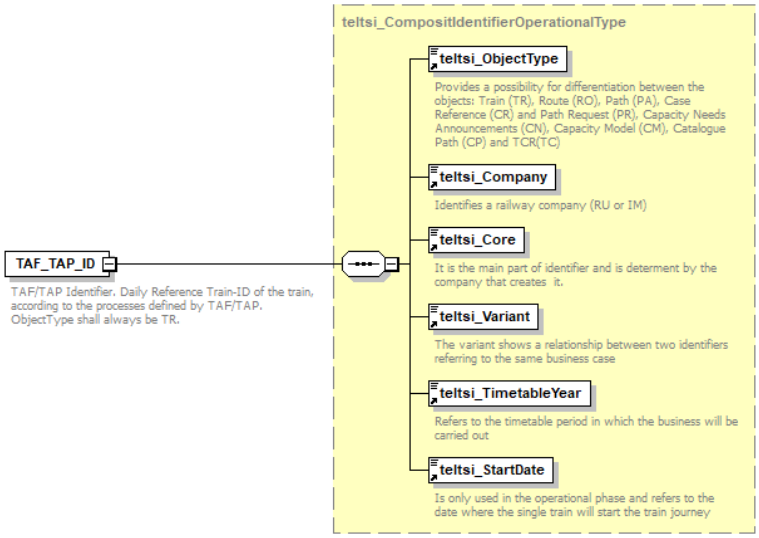
element **SwitchOffRegenerativeBrakeArea**

diagram	
type	**SP_GenericArea_ComplexType**
properties	content complex
used by	element **SP_Areas**

attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**				documentation Element identifier
	startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's

	startEndQualifier	**derived by: xs:string**	required	end location from the beginning of the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation [SUBSET-126] Switch Off Regenerative Brake Area			
source	<xs:element name="SwitchOffRegenerativeBrakeArea" type="SP_GenericArea_ComplexType"> <xs:annotation> <xs:documentation>[SUBSET-126] Switch Off Regenerative Brake Area</xs:documentation> </xs:annotation> </xs:element>			

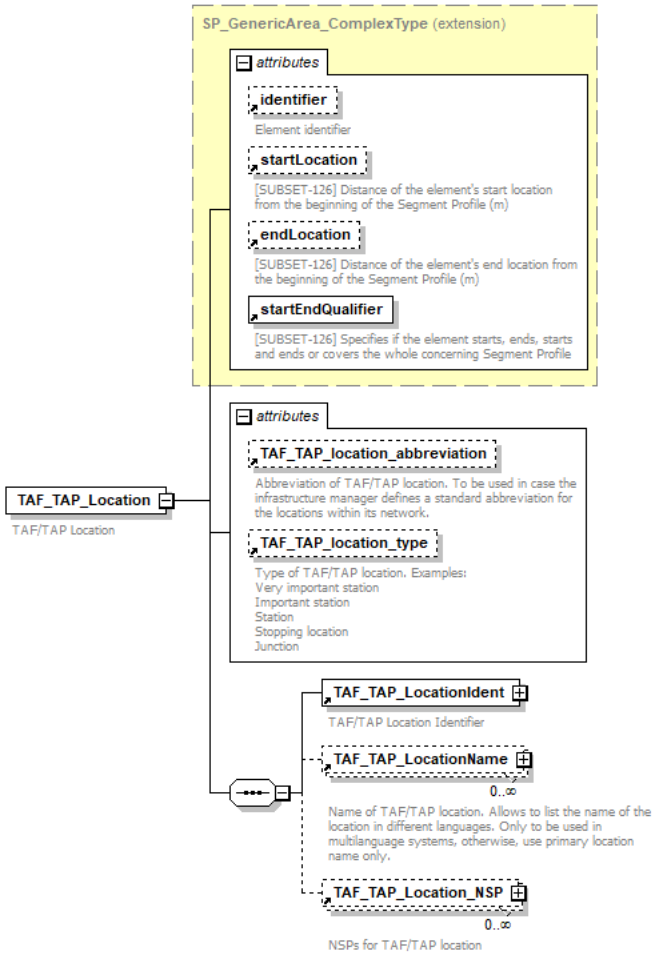
element **TAF_TAP_ID**

diagram	
type	**teltsi_CompositIdentifierOperationalType**
properties	content complex
children	**teltsi_ObjectType** **teltsi_Company** **teltsi_Core** **teltsi_Variant** **teltsi_TimetableYear** **teltsi_StartDate**
used by	complexType **TrainIdentification_ComplexType**
annotation	documentation TAF/TAP Identifier. Daily Reference Train-ID of the train, according to the processes defined by TAF/TAP. ObjectType shall always be TR.
source	<xs:element name="TAF_TAP_ID" type="teltsi_CompositIdentifierOperationalType"> <xs:annotation> <xs:documentation>TAF/TAP Identifier. Daily Reference Train-ID of the train, according to the processes defined by TAF/TAP. ObjectType shall always be TR.</xs:documentation>

```
  </xs:annotation>
</xs:element>
```

element **TAF_TAP_Location**

diagram



type	extension of **SP_GenericArea_ComplexType**
properties	content complex
children	**TAF_TAP_LocationIdent** **TAF_TAP_LocationName** **TAF_TAP_Location_NSP**
used by	element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126]

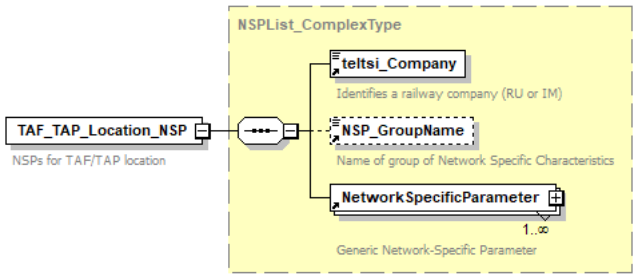
	startEndQualifier	**derived by: xs:string**	required	Distance of the element's end location from the beginning of the Segment Profile (m) documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
	TAF_TAP_location_abbreviation			documentation Abbreviation of TAF/TAP location. To be used in case the infrastructure manager defines a standard abbreviation for the locations within its network.
	TAF_TAP_location_type			documentation Type of TAF/TAP location. Examples: Very important station Important station Station Stopping location Junction
annotation	documentation TAF/TAP Location			

source

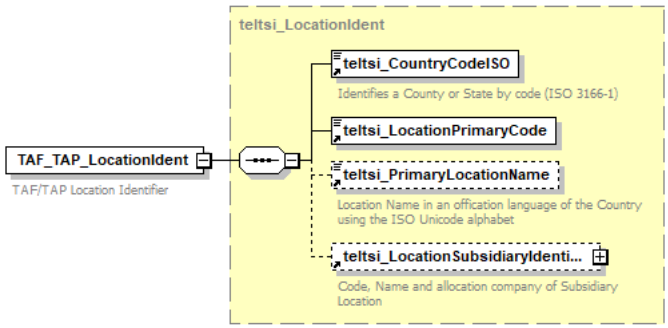
```
<xs:element name="TAF_TAP_Location">
 <xs:annotation>
  <xs:documentation>TAF/TAP Location</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericArea_ComplexType">
    <xs:sequence>
     <xs:element ref="TAF_TAP_LocationIdent"/>
     <xs:element ref="TAF_TAP_LocationName" minOccurs="0" maxOccurs="unbounded"/>
     <xs:element ref="TAF_TAP_Location_NSP" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="TAF_TAP_location_abbreviation"/>
    <xs:attribute ref="TAF_TAP_location_type"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
```

	</xs:element>

element **TAF_TAP_Location_NSP**

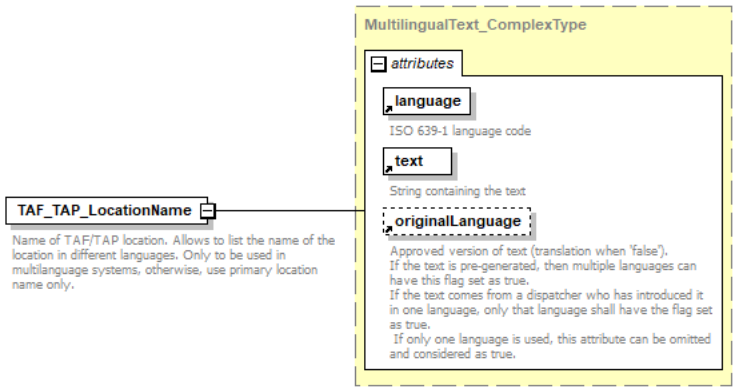
diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	element **TAF_TAP_Location**
annotation	documentation NSPs for TAF/TAP location
source	<xs:element name="TAF_TAP_Location_NSP" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>NSPs for TAF/TAP location</xs:documentation> </xs:annotation> </xs:element>

element **TAF_TAP_LocationIdent**

diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	element **TAF_TAP_Location**
annotation	documentation TAF/TAP Location Identifier
source	<xs:element name="TAF_TAP_LocationIdent" type="teltsi_LocationIdent"> <xs:annotation> <xs:documentation>TAF/TAP Location Identifier</xs:documentation> </xs:annotation>

	</xs:element>

element **TAF_TAP_LocationName**

diagram	
type	**MultilingualText_ComplexType**
properties	content complex
used by	element **TAF_TAP_Location**

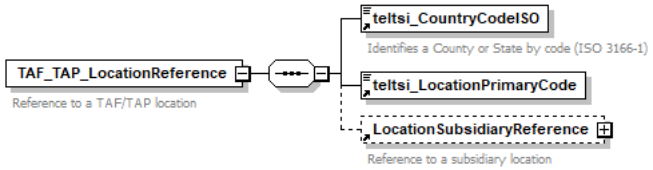
attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

annotation	documentation Name of TAF/TAP location. Allows to list the name of the location in different languages. Only to be used in multilanguage systems, otherwise, use primary location name only.

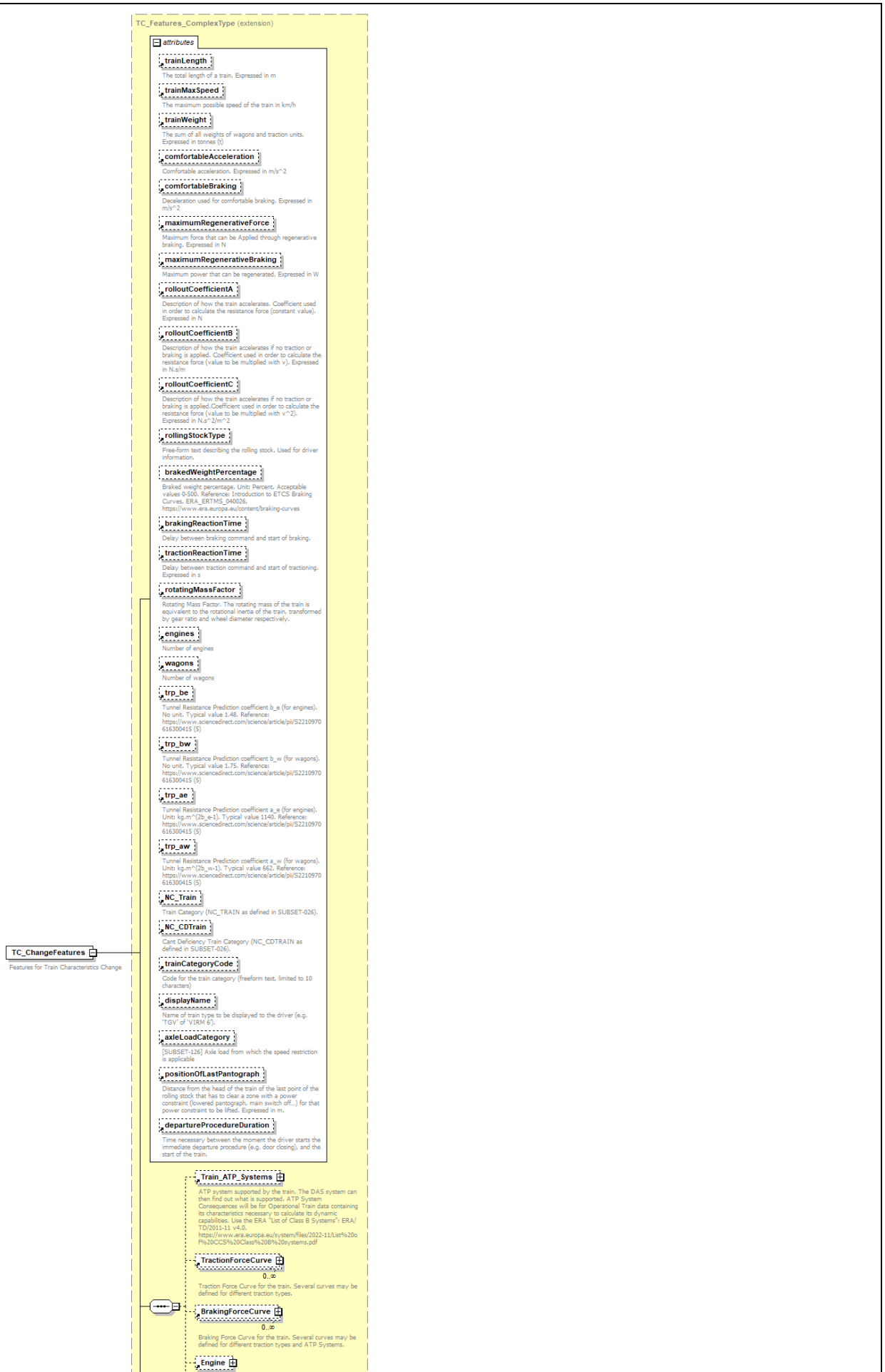
source	<xs:element name="TAF_TAP_LocationName" type="MultilingualText_ComplexType"> <xs:annotation> <xs:documentation>Name of TAF/TAP location. Allows to list the name of the location in different languages. Only to be used in multilanguage systems, otherwise, use primary location name only.</xs:documentation> </xs:annotation> </xs:element>

element **TAF_TAP_LocationReference**

diagram	
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **LocationSubsidiaryReference**
used by	element **TimingPoint**
annotation	documentation Reference to a TAF/TAP location
source	<xs:element name="TAF_TAP_LocationReference"> <xs:annotation> <xs:documentation>Reference to a TAF/TAP location</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_CountryCodeISO"/> <xs:element ref="teltsi_LocationPrimaryCode"/> <xs:element ref="LocationSubsidiaryReference" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **TC_ChangeFeatures**

diagram


TC_Features_ComplexType (extension)
attributes
trainLength
The total length of a train. Expressed in m
trainMaxSpeed
The maximum possible speed of the train in km/h
trainWeight
The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration
Comfortable acceleration. Expressed in m/s^2
comfortableBraking
Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce
Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking
Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA
Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB
Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC
Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType
Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage
Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026. https://www.era.europa.eu/content/braking-curves
brakingReactionTime
Delay between braking command and start of braking.
tractionReactionTime
Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor
Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines
Number of engines
wagons
Number of wagons
trp_be
Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_bw
Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_ae
Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_aw
Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
NC_Train
Train Category (NC_TRAIN as defined in SUBSET-026).
NC_CDTrain
Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
trainCategoryCode
Code for the train category (freeform text, limited to 10 characters)
displayName
Name of train type to be displayed to the driver (e.g. 'TGV' or 'VIRM 6').
axleLoadCategory
[SUBSET-126] Axle load from which the speed restriction is applicable
positionOfLastPantograph
Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off...) for that power constraint to be lifted. Expressed in m.
departureProcedureDuration
Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
TC_ChangeFeatures
Features for Train Characteristics Change
Train_ATP_Systems
ATP system supported by the train. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities. Use the ERA "List of Class B Systems": ERA/TD/2011-11 v4.0. https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf
TractionForceCurve
0..∞
Traction Force Curve for the train. Several curves may be defined for different traction types.
BrakingForceCurve
0..∞
Braking Force Curve for the train. Several curves may be defined for different traction types and ATP Systems.
Engine

type	extension of **TC_Features_ComplexType**
properties	content complex
children	**Train_ATP_Systems** **TractionForceCurve** **BrakingForceCurve** **Engine** **VehicleData** **TC_Features_NSPs** **UnavailableTractionType**
used by	element **TrainCharacteristicsChange**

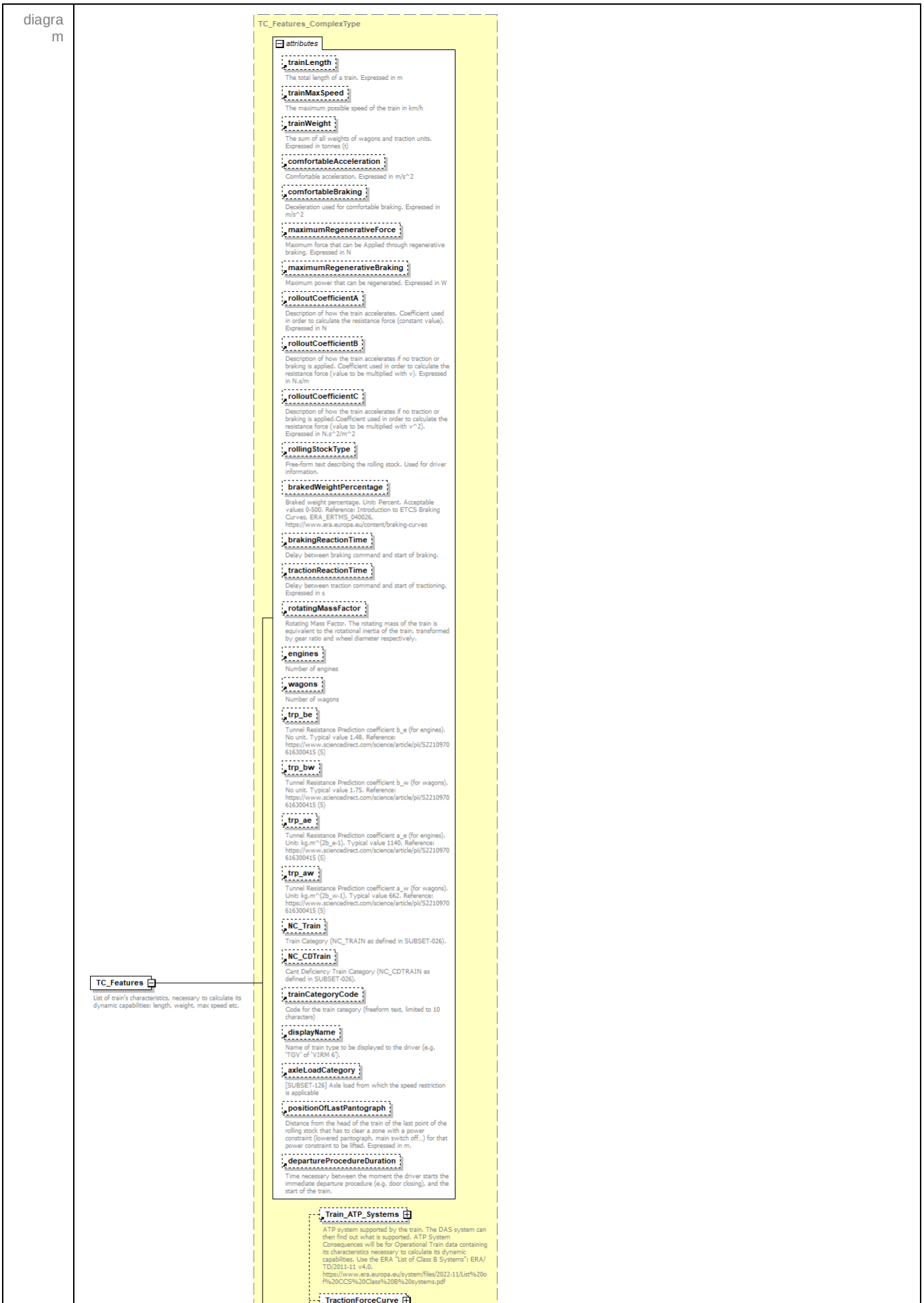
attributes

Name	Type	Use	Default	Fixed	Annotation
trainLength	**xs:integer**	optional			documentation The total length of a train. Expressed in m
trainMaxSpeed	**xs:integer**	optional			documentation The maximum possible speed of the train in km/h
trainWeight	**xs:int**	optional			documentation The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration	**xs:double**	optional			documentation Comfortable acceleration. Expressed in m/s^2
comfortableBraking	**xs:double**	optional			documentation Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce	**xs:double**	optional			documentation Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking	**xs:double**	optional			documentation Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA	**xs:double**	optional			documentation Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType	**xs:string**	optional			documentation Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage	**xs:decimal**				documentation Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves
brakingReactionTime	**xs:duration**	optional			documentation Delay between braking command and start of braking.
tractionReactionTime	**xs:int**	optional			documentation Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor	**xs:decimal**				documentation Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines	**xs:int**				documentation Number of engines
wagons	**xs:int**				documentation Number of wagons
trp_be	**xs:decimal**				documentation Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S221

	trp_bw	**xs:decimal**	0970616300415 (5) documentation Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	trp_ae	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	trp_aw	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	NC_Train	**NC_TrainType**	documentation Train Category (NC_TRAIN as defined in SUBSET-026).
	NC_CDTrain	**NC_CDTrainType**	documentation Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
	trainCategoryCode	**derived by: xs:string**	documentation Code for the train category (freeform text, limited to 10 characters)
	displayName	**xs:string**	documentation Name of train type to be displayed to the driver (e.g. 'TGV' of 'VIRM 6').
	axleLoadCategory	**derived by: xs:string**	documentation [SUBSET-126] Axle load from which the speed restriction is applicable
	positionOfLastPantograph	**distanceType**	documentation Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off…) for that power constraint to be lifted. Expressed in m.
	departureProcedureDuration	**xs:duration**	documentation Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
	remainingEnginePower	**percentageType**	documentation Percentage of power available due to engine problems
annotation	documentation Features for Train Characteristics Change		
source			

```
<xs:element name="TC_ChangeFeatures">
 <xs:annotation>
  <xs:documentation>Features for Train Characteristics Change</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="TC_Features_ComplexType">
    <xs:sequence>
     <xs:element ref="UnavailableTractionType" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="remainingEnginePower"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **TC_Features**


diagram
TC_Features_ComplexType
attributes
trainLength
The total length of a train. Expressed in m
trainMaxSpeed
The maximum possible speed of the train in km/h
trainWeight
The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration
Comfortable acceleration. Expressed in m/s^2
comfortableBraking
Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce
Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking
Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA
Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB
Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC
Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType
Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage
Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves
brakingReactionTime
Delay between braking command and start of braking.
tractionReactionTime
Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor
Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines
Number of engines
wagons
Number of wagons
trp_be
Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_bw
Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_ae
Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_aw
Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
NC_Train
Train Category (NC_TRAIN as defined in SUBSET-026).
NC_CDTrain
Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
trainCategoryCode
Code for the train category (freeform text, limited to 10 characters)
displayName
Name of train type to be displayed to the driver (e.g. 'TGV' of 'VIRM 6').
axleLoadCategory
[SUBSET-126] Axle load from which the speed restriction is applicable
positionOfLastPantograph
Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off...) for that power constraint to be lifted. Expressed in m.
departureProcedureDuration
Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
Train_ATP_Systems
ATP system supported by the train. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities. Use the ERA "List of Class B Systems": ERA/TD/2011-11 v4.0. https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf
TractionForceCurve
TC_Features
List of train's characteristics, necessary to calculate its dynamic capabilities: length, weight, max speed etc.

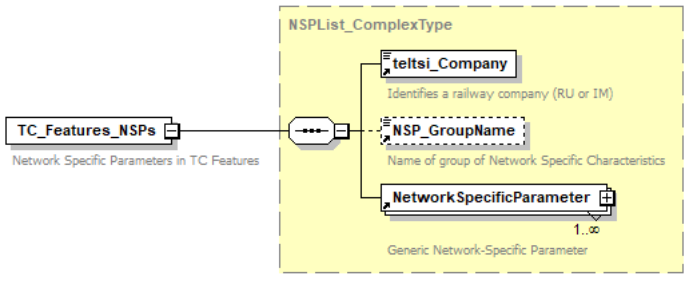
type	**TC_Features_ComplexType**
properties	content complex
children	**Train_ATP_Systems** **TractionForceCurve** **BrakingForceCurve** **Engine** **VehicleData** **TC_Features_NSPs**
used by	element **TrainCharacteristics**

attributes

Name	Type	Use	Default	Fixed	Annotation
trainLength	**xs:integer**	optional			documentation The total length of a train. Expressed in m
trainMaxSpeed	**xs:integer**	optional			documentation The maximum possible speed of the train in km/h
trainWeight	**xs:int**	optional			documentation The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration	**xs:double**	optional			documentation Comfortable acceleration. Expressed in m/s^2
comfortableBraking	**xs:double**	optional			documentation Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce	**xs:double**	optional			documentation Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking	**xs:double**	optional			documentation Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA	**xs:double**	optional			documentation Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType	**xs:string**	optional			documentation Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage	**xs:decimal**				documentation Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves
brakingReactionTime	**xs:duration**	optional			documentation Delay between braking command and start of braking.
tractionReactionTime	**xs:int**	optional			documentation Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor	**xs:decimal**				documentation Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines	**xs:int**				documentation Number of engines
wagons	**xs:int**				documentation Number of wagons
trp_be	**xs:decimal**				documentation Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)

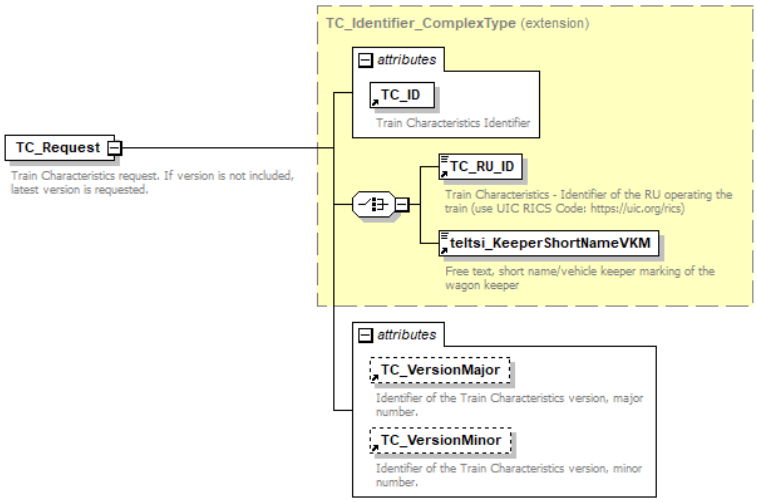
	trp_bw	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	trp_ae	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	trp_aw	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	NC_Train	**NC_TrainType**	documentation Train Category (NC_TRAIN as defined in SUBSET-026).
	NC_CDTrain	**NC_CDTrainType**	documentation Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
	trainCategoryCode	**derived by: xs:string**	documentation Code for the train category (freeform text, limited to 10 characters)
	displayName	**xs:string**	documentation Name of train type to be displayed to the driver (e.g. 'TGV' of 'VIRM 6').
	axleLoadCategory	**derived by: xs:string**	documentation [SUBSET-126] Axle load from which the speed restriction is applicable
	positionOfLastPantograph	**distanceType**	documentation Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off…) for that power constraint to be lifted. Expressed in m.
	departureProcedureDuration	**xs:duration**	documentation Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
annotation	documentation List of train's characteristics, necessary to calculate its dynamic capabilities: length, weight, max speed etc.		
source	<xs:element name="TC_Features" type="TC_Features_ComplexType"> <xs:annotation> <xs:documentation>List of train's characteristics, necessary to calculate its dynamic capabilities: length, weight, max speed etc.</xs:documentation> </xs:annotation> </xs:element>		

element **TC_Features_NSPs**

diagram	

type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Network Specific Parameters in TC Features
source	<xs:element name="TC_Features_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in TC Features</xs:documentation> </xs:annotation> </xs:element>

element **TC_Request**

diagram	
type	extension of **TC_Identifier_ComplexType**
properties	content complex
children	**TC_RU_ID** **teltsi_KeeperShortNameVKM**
used by	element **B2G_Request**

attributes

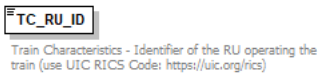
Name	Type	Use	Default	Fixed	Annotation
TC_ID	**xs:string**	required			documentation Train Characteristics Identifier
TC_VersionMajor	**xs:string**				documentation Identifier of the Train Characteristics version, major number.
TC_VersionMinor	**xs:string**				documentation Identifier of the Train Characteristics version, minor number.

annotation	documentation Train Characteristics request. If version is not included, latest version is requested.

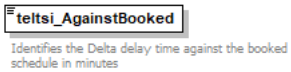
source

```
<xs:element name="TC_Request">
  <xs:annotation>
    <xs:documentation>Train Characteristics request. If version is not included, latest version is requested.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="TC_Identifier_ComplexType">
        <xs:attribute ref="TC_VersionMajor"/>
        <xs:attribute ref="TC_VersionMinor"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **TC_RU_ID**

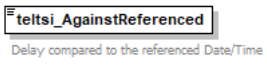
diagram	
type	**teltsi_CompanyCode**
properties	content simple
used by	complexType **TC_Identifier_ComplexType**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation Train Characteristics - Identifier of the RU operating the train (use UIC RICS Code: https://uic.org/rics)
source	<xs:element name="TC_RU_ID" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>Train Characteristics - Identifier of the RU operating the train (use UIC RICS Code: https://uic.org/rics)</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_AgainstBooked**

diagram	
type	**teltsi_DeltaTime**
properties	content simple
used by	element **teltsi_TrainDelay**
facets	Kind Value Annotation length 5
annotation	documentation Identifies the Delta delay time against the booked schedule in minutes
source	<xs:element name="teltsi_AgainstBooked" type="teltsi_DeltaTime"> <xs:annotation>

	<xs:documentation>Identifies the Delta delay time against the booked schedule in minutes</xs:documentation> </xs:annotation> </xs:element>

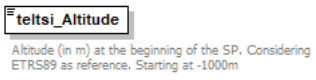
element **teltsi_AgainstReferenced**

diagram	
type	**teltsi_DeltaTime**
properties	content simple
used by	element **teltsi_TrainDelay**
facets	Kind Value Annotation length 5
annotation	documentation Delay compared to the referenced Date/Time
source	<xs:element name="teltsi_AgainstReferenced" type="teltsi_DeltaTime"> <xs:annotation> <xs:documentation>Delay compared to the referenced Date/Time</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_AllocationCompany**

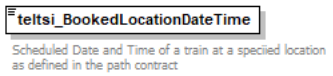
diagram	
type	**teltsi_CompanyCode**
properties	content simple
used by	elements **LocationSubsidiaryReference** **teltsi_LocationSubsidiaryIdentification**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation Name of company who is responsible for allocation and maintenance of codes
source	<xs:element name="teltsi_AllocationCompany" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>Name of company who is responsible for allocation and maintenance of codes</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_Altitude**

diagram	

type	**xs:integer**
properties	content simple
used by	element **teltsi_GeographicCoordinates**
annotation	documentation Altitude (in m) at the beginning of the SP. Considering ETRS89 as reference. Starting at -1000m
source	<xs:element name="teltsi_Altitude" type="xs:integer"> <xs:annotation> <xs:documentation>Altitude (in m) at the beginning of the SP. Considering ETRS89 as reference. Starting at -1000m</xs:documentation> </xs:annotation> </xs:element>

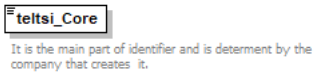
element **teltsi_BookedLocationDateTime**

diagram	
type	**xs:dateTime**
properties	content simple
used by	element **teltsi_InterruptionPoint**
annotation	documentation Scheduled Date and Time of a train at a speciied location as defined in the path contract
source	<xs:element name="teltsi_BookedLocationDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Scheduled Date and Time of a train at a speciied location as defined in the path contract</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_Company**

diagram	
type	**teltsi_CompanyCode**
properties	content simple
used by	elements **NetworkSpecificArea** **NetworkSpecificCharacteristics** **NetworkSpecificPoint** **RT_TrainInformation** complexTypes **NSPList_ComplexType** **OTN_ID_ComplexType** **teltsi_CompositIdentifierOperationalType**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation Identifies a railway company (RU or IM)
source	<xs:element name="teltsi_Company" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>Identifies a railway company (RU or IM)</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_Core**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	complexType **teltsi_CompositIdentifierOperationalType**
facets	Kind Value Annotation minLength 12 maxLength 12 whiteSpace replace pattern [\-*0-9A-Z]{12}
annotation	documentation It is the main part of identifier and is determent by the company that creates it.
source	<xs:element name="teltsi_Core"> <xs:annotation> <xs:documentation>It is the main part of identifier and is determent by the company that creates it.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:minLength value="12"/> <xs:whiteSpace value="replace"/> <xs:maxLength value="12"/> <xs:pattern value="[\-*0-9A-Z]{12}"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_CountryCodeISO**

diagram	
type	extension of **teltsi_CountryIdentISO**
properties	content complex
used by	element **TAF_TAP_LocationReference** complexType **teltsi_LocationIdent**
facets	Kind Value Annotation minLength 2 maxLength 2
annotation	documentation Identifies a County or State by code (ISO 3166-1)
source	<xs:element name="teltsi_CountryCodeISO"> <xs:annotation> <xs:documentation>Identifies a County or State by code (ISO 3166-1)</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="teltsi_CountryIdentISO"/> </xs:simpleContent>

```
  </xs:complexType>
</xs:element>
```

element **teltsi_DelayCause**

diagram



type **teltsi_DelayCode**

properties content simple

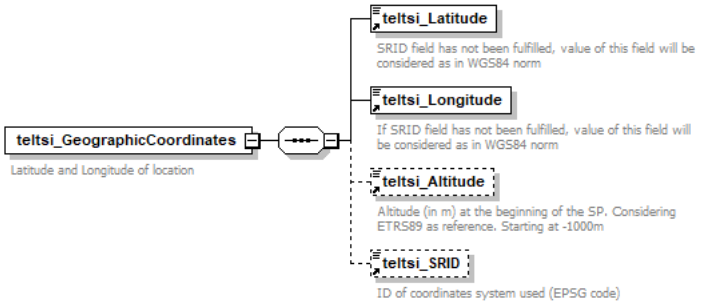
used by elements **DelayNotice** **TrainReadyStatus**

facets

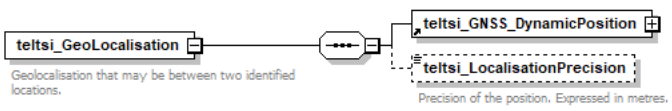
Kind	Value	Annotation
enumeration	11	documentation Formation of trains if managed by infrastructure manager
enumeration	10	documentation Timetable compilation
enumeration	12	documentation Mistakes in operational procedures
enumeration	13	documentation Wrong application of priority rules
enumeration	14	
enumeration	18	documentation Staff
enumeration	19	documentation Other causes related to the operational planning and management
enumeration	20	documentation Signalling installations
enumeration	21	documentation Signalling installations at level crossings
enumeration	22	documentation Telecommunication installations
enumeration	23	documentation Power supply equipment
enumeration	24	documentation Track
enumeration	25	documentation Structures
enumeration	28	documentation Staff
enumeration	29	documentation Other causes related to infrastructure installations
enumeration	31	documentation Irregularities in execution of construction work
enumeration	30	documentation Planned construction work
enumeration	32	documentation Speed restrictions due to defective track
enumeration	39	documentation Other causes related to Civil engineering
enumeration	40	documentation Delay caused by next IM
enumeration	41	documentation Delay caused by previous IM
enumeration	50	documentation exceeding the stop time
enumeration	51	documentation Request of the RU
enumeration	52	documentation Loading operations
enumeration	53	documentation Loading irregularities
enumeration	54	documentation Commercial preparation of the train

	enumeration	58	documentation Staff
	enumeration	59	documentation Other causes related to commercial causes
	enumeration	60	documentation Roster planning/re-rostering
	enumeration	61	documentation Formation of trains by the RU
	enumeration	62	documentation Problems affecting coaches
	enumeration	63	documentation Problems affecting wagons
	enumeration	64	documentation Problems affecting traction units
	enumeration	68	documentation Staff
	enumeration	69	documentation Other causes related to Rolling Stock
	enumeration	70	documentation Delay caused by next RU
	enumeration	71	documentation Delay caused by previous RU
	enumeration	80	documentation Strike
	enumeration	81	documentation Administrative formalites
	enumeration	82	documentation Outside influence
	enumeration	83	documentation Effects of weather and natural causes
	enumeration	84	documentation Delay caused by external reasons on the next network
	enumeration	89	documentation Other external causes
	enumeration	90	documentation Dangerous incidents, accidents and hazards
	enumeration	91	documentation Track occupation caused by the lateness of the same train
	enumeration	92	documentation Track occupation caused by the lateness of another train
	enumeration	93	documentation Turn round
	enumeration	94	documentation Connection
	enumeration	95	documentation Further investigation needed
annotation	documentation This element identifies the reason for a delay (modified DelayReason)		
source	<xs:element name="teltsi_DelayCause" type="teltsi_DelayCode"> <xs:annotation> <xs:documentation>This element identifies the reason for a delay (modified DelayReason)</xs:documentation> </xs:annotation> </xs:element>		

element **teltsi_GeographicCoordinates**

diagram	
properties	content complex
children	**teltsi_Latitude** **teltsi_Longitude** **teltsi_Altitude** **teltsi_SRID**
used by	element **teltsi_GNSS_DynamicPosition**
annotation	documentation Latitude and Longitude of location
source	<xs:element name="teltsi_GeographicCoordinates"> <xs:annotation> <xs:documentation>Latitude and Longitude of location</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_Latitude"> <xs:annotation> <xs:documentation>SRID field has not been fulfilled, value of this field will be considered as in WGS84 norm</xs:documentation> </xs:annotation> </xs:element> <xs:element ref="teltsi_Longitude"> <xs:annotation> <xs:documentation>If SRID field has not been fulfilled, value of this field will be considered as in WGS84 norm</xs:documentation> </xs:annotation> </xs:element> <xs:element ref="teltsi_Altitude" minOccurs="0"/> <xs:element ref="teltsi_SRID" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **teltsi_GeoLocalisation**

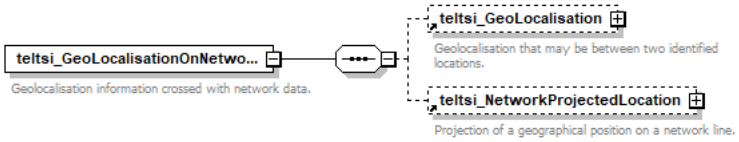
diagram	
properties	content complex
children	**teltsi_GNSS_DynamicPosition** **teltsi_LocalisationPrecision**
used by	elements **PositionSpeed** **teltsi_GeoLocalisationOnNetwork**
annotation	documentation Geolocalisation that may be between two identified locations.

source	<xs:element name="teltsi_GeoLocalisation"> <xs:annotation> <xs:documentation>Geolocalisation that may be between two identified locations. </xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_GNSS_DynamicPosition"/> <xs:element name="teltsi_LocalisationPrecision" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Precision of the position. Expressed in metres.</xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType> </xs:element>

element **teltsi_GeoLocalisation/teltsi_LocalisationPrecision**

diagram	
type	**xs:float**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Precision of the position. Expressed in metres.
source	<xs:element name="teltsi_LocalisationPrecision" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Precision of the position. Expressed in metres.</xs:documentation> </xs:annotation> </xs:element>

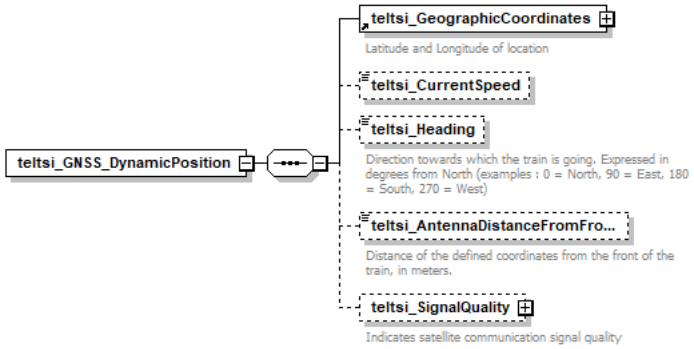
element **teltsi_GeoLocalisationOnNetwork**

diagram	
properties	content complex
children	**teltsi_GeoLocalisation** **teltsi_NetworkProjectedLocation**
used by	element **teltsi_InterruptionPoint**
annotation	documentation Geolocalisation information crossed with network data.
source	<xs:element name="teltsi_GeoLocalisationOnNetwork"> <xs:annotation> <xs:documentation>Geolocalisation information crossed with network data. </xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence>

```
    <xs:element ref="teltsi_GeoLocalisation" minOccurs="0"/>
    <xs:element ref="teltsi_NetworkProjectedLocation" minOccurs="0"/>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **teltsi_GNSS_DynamicPosition**

diagram



properties: content complex

children: **teltsi_GeographicCoordinates** **teltsi_CurrentSpeed** **teltsi_Heading** **teltsi_AntennaDistanceFromFrontOfTrain** **teltsi_SignalQuality**

used by: element **teltsi_GeoLocalisation**

source

```
<xs:element name="teltsi_GNSS_DynamicPosition">
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="teltsi_GeographicCoordinates"/>
   <xs:element name="teltsi_CurrentSpeed" type="teltsi_Speed" minOccurs="0"/>
   <xs:element name="teltsi_Heading" type="xs:integer" minOccurs="0">
    <xs:annotation>
     <xs:documentation>Direction towards which the train is going. Expressed in degrees from North (examples : 0 = North, 90 = East, 180 = South, 270 = West)</xs:documentation>
    </xs:annotation>
   </xs:element>
   <xs:element name="teltsi_AntennaDistanceFromFrontOfTrain" type="xs:float" minOccurs="0">
    <xs:annotation>
     <xs:documentation>Distance of the defined coordinates from the front of the train, in meters.</xs:documentation>
    </xs:annotation>
   </xs:element>
   <xs:element name="teltsi_SignalQuality" minOccurs="0">
    <xs:annotation>
     <xs:documentation>Indicates satellite communication signal quality</xs:documentation>
    </xs:annotation>
    <xs:complexType>
     <xs:sequence>
      <xs:element name="teltsi_SatelliteHDOP" type="xs:decimal">
       <xs:annotation>
        <xs:documentation>HDOP = Horizontal Dilution Of Precision. Parameter used to describe the strength of the current satellite configuration, or geometry, and it's impact on the accuracy of the data collected by a GNSS receiver at the time of use. HDOP provides a measure of accuracy in 2-D position (e.g. Latitude and Longitude). Smaller value indicates better precision. Values less than 1
```

	are the best. Values from 2 - 5 are good. Values greater 10 indicate bad quality </xs:documentation> </xs:annotation> </xs:element> <xs:element name="teltsi_NumberOfVisibleSatellites" type="xs:integer"> <xs:annotation> <xs:documentation>Indicates the number of satellites being used in the position solution. Greater value indicates better quality.</xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType> </xs:element> </xs:sequence> </xs:complexType> </xs:element>

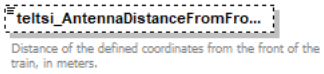
element **teltsi_GNSS_DynamicPosition/teltsi_CurrentSpeed**

diagram	teltsi_CurrentSpeed
type	**teltsi_Speed**
properties	minOcc 0 maxOcc 1 content simple
facets	Kind Value Annotation minInclusive 001 maxInclusive 999
source	<xs:element name="teltsi_CurrentSpeed" type="teltsi_Speed" minOccurs="0"/>

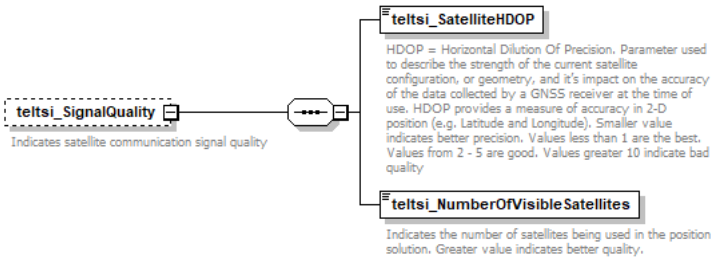
element **teltsi_GNSS_DynamicPosition/teltsi_Heading**

diagram	teltsi_Heading Direction towards which the train is going. Expressed in degrees from North (examples : 0 = North, 90 = East, 180 = South, 270 = West)
type	**xs:integer**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Direction towards which the train is going. Expressed in degrees from North (examples : 0 = North, 90 = East, 180 = South, 270 = West)
source	<xs:element name="teltsi_Heading" type="xs:integer" minOccurs="0"> <xs:annotation> <xs:documentation>Direction towards which the train is going. Expressed in degrees from North (examples : 0 = North, 90 = East, 180 = South, 270 = West)</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_GNSS_DynamicPosition/teltsi_AntennaDistanceFromFrontOfTrain**

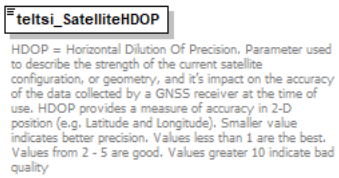
diagram	
type	**xs:float**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Distance of the defined coordinates from the front of the train, in meters.
source	<xs:element name="teltsi_AntennaDistanceFromFrontOfTrain" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Distance of the defined coordinates from the front of the train, in meters.</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_GNSS_DynamicPosition/teltsi_SignalQuality**

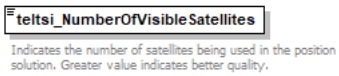
diagram	
properties	minOcc 0 maxOcc 1 content complex
children	**teltsi_SatelliteHDOP** **teltsi_NumberOfVisibleSatellites**
annotation	documentation Indicates satellite communication signal quality
source	<xs:element name="teltsi_SignalQuality" minOccurs="0"> <xs:annotation> <xs:documentation>Indicates satellite communication signal quality</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element name="teltsi_SatelliteHDOP" type="xs:decimal"> <xs:annotation> <xs:documentation>HDOP = Horizontal Dilution Of Precision. Parameter used to describe the strength of the current satellite configuration, or geometry, and it's impact on the accuracy of the data collected by a GNSS receiver at the time of use. HDOP provides a measure of accuracy in 2-D position (e.g. Latitude and Longitude). Smaller value indicates better precision. Values less than 1 are the best. Values from 2 - 5 are good. Values greater 10 indicate bad quality </xs:documentation> </xs:annotation> </xs:element> <xs:element name="teltsi_NumberOfVisibleSatellites" type="xs:integer"> <xs:annotation> <xs:documentation>Indicates the number of satellites being used in the position solution. Greater value indicates better quality.</xs:documentation> </xs:annotation>

```
    </xs:element>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

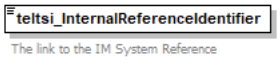
element **teltsi_GNSS_DynamicPosition/teltsi_SignalQuality/teltsi_SatelliteHDOP**

diagram	
type	**xs:decimal**
properties	content simple
annotation	documentation HDOP = Horizontal Dilution Of Precision. Parameter used to describe the strength of the current satellite configuration, or geometry, and it's impact on the accuracy of the data collected by a GNSS receiver at the time of use. HDOP provides a measure of accuracy in 2-D position (e.g. Latitude and Longitude). Smaller value indicates better precision. Values less than 1 are the best. Values from 2 - 5 are good. Values greater 10 indicate bad quality
source	<xs:element name="teltsi_SatelliteHDOP" type="xs:decimal"> <xs:annotation> <xs:documentation>HDOP = Horizontal Dilution Of Precision. Parameter used to describe the strength of the current satellite configuration, or geometry, and it's impact on the accuracy of the data collected by a GNSS receiver at the time of use. HDOP provides a measure of accuracy in 2-D position (e.g. Latitude and Longitude). Smaller value indicates better precision. Values less than 1 are the best. Values from 2 - 5 are good. Values greater 10 indicate bad quality</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_GNSS_DynamicPosition/teltsi_SignalQuality/teltsi_NumberOfVisibleSatellites**

diagram	
type	**xs:integer**
properties	content simple
annotation	documentation Indicates the number of satellites being used in the position solution. Greater value indicates better quality.
source	<xs:element name="teltsi_NumberOfVisibleSatellites" type="xs:integer"> <xs:annotation> <xs:documentation>Indicates the number of satellites being used in the position solution. Greater value indicates better quality.</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_InternalReferenceIdentifier**

diagram	
type	**teltsi_FreeText**
properties	content simple
used by	element **teltsi_InterruptionPoint/teltsi_Interruption**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation The link to the IM System Reference
source	<xs:element name="teltsi_InternalReferenceIdentifier" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>The link to the IM System Reference</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_InterruptionDateTime**

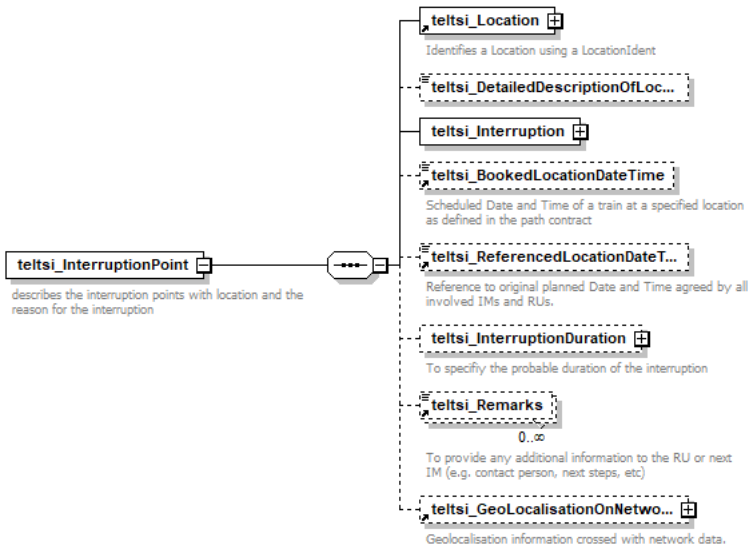
diagram	
type	**xs:dateTime**
properties	content simple
used by	element **teltsi_InterruptionPoint/teltsi_Interruption**
annotation	documentation Date and Time when the Train was interrupted
source	<xs:element name="teltsi_InterruptionDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Date and Time when the Train was interrupted</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_InterruptionDescription**

diagram	
type	**teltsi_FreeText**
properties	content simple
used by	element **teltsi_InterruptionPoint/teltsi_Interruption**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation The free text description of an interruption
source	<xs:element name="teltsi_InterruptionDescription" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>The free text description of an interruption</xs:documentation>

	</xs:annotation> </xs:element>

element **teltsi_InterruptionPoint**

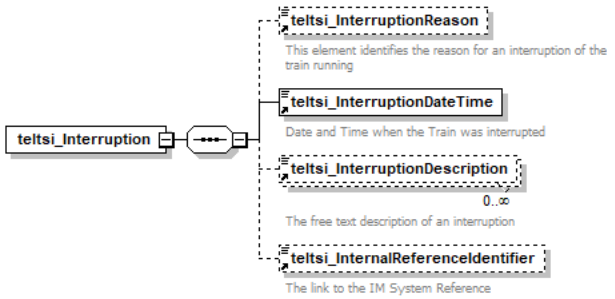
diagram	
properties	content complex
children	**teltsi_Location** **teltsi_DetailedDescriptionOfLocation** **teltsi_Interruption** **teltsi_BookedLocationDateTime** **teltsi_ReferencedLocationDateTime** **teltsi_InterruptionDuration** **teltsi_Remarks** **teltsi_GeoLocalisationOnNetwork**
used by	elements **B2G_StatusReport** **JourneyProfile**
annotation	documentation describes the interruption points with location and the reason for the interruption
source	<xs:element name="teltsi_InterruptionPoint"> <xs:annotation> <xs:documentation>describes the interruption points with location and the reason for the interruption</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_Location"/> <xs:element name="teltsi_DetailedDescriptionOfLocation" type="teltsi_FreeText" minOccurs="0"/> <xs:element name="teltsi_Interruption"> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_InterruptionReason" minOccurs="0"/> <xs:element ref="teltsi_InterruptionDateTime"/> <xs:element ref="teltsi_InterruptionDescription" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="teltsi_InternalReferenceIdentifier" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element> <xs:element ref="teltsi_BookedLocationDateTime" minOccurs="0"> <xs:annotation> <xs:documentation>Scheduled Date and Time of a train at a specified location as defined in the path contract</xs:documentation>

```
      </xs:annotation>
    </xs:element>
    <xs:element ref="teltsi_ReferencedLocationDateTime" minOccurs="0"/>
    <xs:element name="teltsi_InterruptionDuration" minOccurs="0">
     <xs:annotation>
      <xs:documentation>To specifiy the probable duration of the interruption</xs:documentation>
     </xs:annotation>
     <xs:complexType>
      <xs:sequence>
       <xs:element name="teltsi_EarliestEndDateTime" type="xs:dateTime" minOccurs="0">
        <xs:annotation>
         <xs:documentation>Forecasted earliest time for end of interruption</xs:documentation>
        </xs:annotation>
       </xs:element>
       <xs:element name="teltsi_LatestEndDateTime" type="xs:dateTime" minOccurs="0">
        <xs:annotation>
         <xs:documentation>Forecasted latest time for end of interruption</xs:documentation>
        </xs:annotation>
       </xs:element>
      </xs:sequence>
     </xs:complexType>
    </xs:element>
    <xs:element ref="teltsi_Remarks" minOccurs="0" maxOccurs="unbounded">
     <xs:annotation>
      <xs:documentation>To provide any additional information to the RU or next IM (e.g. contact
person, next steps, etc)</xs:documentation>
     </xs:annotation>
    </xs:element>
    <xs:element ref="teltsi_GeoLocalisationOnNetwork" minOccurs="0"/>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

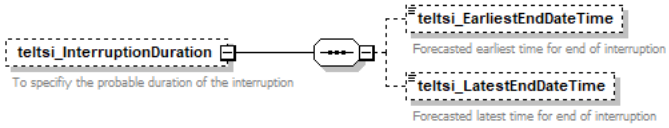
element **teltsi_InterruptionPoint/teltsi_DetailedDescriptionOfLocation**

diagram	teltsi_DetailedDescriptionOfLoc...
type	**teltsi_FreeText**
properties	minOcc 0 maxOcc 1 content simple
facets	Kind Value Annotation minLength 1 maxLength 255
source	`<xs:element name="teltsi_DetailedDescriptionOfLocation" type="teltsi_FreeText" minOccurs="0"/>`

element **teltsi_InterruptionPoint/teltsi_Interruption**

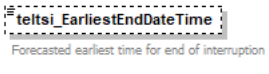
diagram	
properties	content complex
children	**teltsi_InterruptionReason** **teltsi_InterruptionDateTime** **teltsi_InterruptionDescription** **teltsi_InternalReferenceIdentifier**
source	<xs:element name="teltsi_Interruption"> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_InterruptionReason" minOccurs="0"/> <xs:element ref="teltsi_InterruptionDateTime"/> <xs:element ref="teltsi_InterruptionDescription" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="teltsi_InternalReferenceIdentifier" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **teltsi_InterruptionPoint/teltsi_InterruptionDuration**

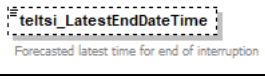
diagram	
properties	minOcc 0 maxOcc 1 content complex
children	**teltsi_EarliestEndDateTime** **teltsi_LatestEndDateTime**
annotation	documentation To specifiy the probable duration of the interruption
source	<xs:element name="teltsi_InterruptionDuration" minOccurs="0"> <xs:annotation> <xs:documentation>To specifiy the probable duration of the interruption</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element name="teltsi_EarliestEndDateTime" type="xs:dateTime" minOccurs="0"> <xs:annotation> <xs:documentation>Forecasted earliest time for end of interruption</xs:documentation> </xs:annotation> </xs:element> <xs:element name="teltsi_LatestEndDateTime" type="xs:dateTime" minOccurs="0"> <xs:annotation> <xs:documentation>Forecasted latest time for end of interruption</xs:documentation> </xs:annotation>

```
    </xs:element>
   </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **teltsi_InterruptionPoint/teltsi_InterruptionDuration/teltsi_EarliestEndDateTime**

diagram	
type	**xs:dateTime**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Forecasted earliest time for end of interruption
source	<xs:element name="teltsi_EarliestEndDateTime" type="xs:dateTime" minOccurs="0"> <xs:annotation> <xs:documentation>Forecasted earliest time for end of interruption</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_InterruptionPoint/teltsi_InterruptionDuration/teltsi_LatestEndDateTime**

diagram	
type	**xs:dateTime**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Forecasted latest time for end of interruption
source	<xs:element name="teltsi_LatestEndDateTime" type="xs:dateTime" minOccurs="0"> <xs:annotation> <xs:documentation>Forecasted latest time for end of interruption</xs:documentation> </xs:annotation> </xs:element>

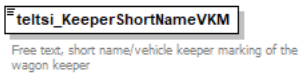
element **teltsi_InterruptionReason**

diagram	
type	**teltsi_DelayCode**
properties	content simple
used by	element **teltsi_InterruptionPoint/teltsi_Interruption**
facets	Kind Value Annotation enumeration 11 documentation Formation of trains if managed by infrastructure manager enumeration 10 documentation Timetable compilation enumeration 12 documentation Mistakes in operational procedures

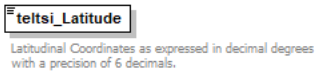
	enumeration	13	documentation Wrong application of priority rules
	enumeration	14	
	enumeration	18	documentation Staff
	enumeration	19	documentation Other causes related to the operational planning and management
	enumeration	20	documentation Signalling installations
	enumeration	21	documentation Signalling installations at level crossings
	enumeration	22	documentation Telecommunication installations
	enumeration	23	documentation Power supply equipment
	enumeration	24	documentation Track
	enumeration	25	documentation Structures
	enumeration	28	documentation Staff
	enumeration	29	documentation Other causes related to infrastructure installations
	enumeration	31	documentation Irregularities in execution of construction work
	enumeration	30	documentation Planned construction work
	enumeration	32	documentation Speed restrictions due to defective track
	enumeration	39	documentation Other causes related to Civil engineering
	enumeration	40	documentation Delay caused by next IM
	enumeration	41	documentation Delay caused by previous IM
	enumeration	50	documentation exceeding the stop time
	enumeration	51	documentation Request of the RU
	enumeration	52	documentation Loading operations
	enumeration	53	documentation Loading irregularities
	enumeration	54	documentation Commercial preparation of the train
	enumeration	58	documentation Staff
	enumeration	59	documentation Other causes related to commercial causes
	enumeration	60	documentation Roster planning/re-rostering
	enumeration	61	documentation Formation of trains by the RU
	enumeration	62	documentation Problems affecting coaches
	enumeration	63	documentation Problems affecting wagons
	enumeration	64	documentation Problems affecting traction units
	enumeration	68	documentation Staff
	enumeration	69	documentation Other causes related to Rolling Stock
	enumeration	70	documentation Delay caused by next RU
	enumeration	71	documentation Delay caused by previous RU
	enumeration	80	documentation Strike

	enumeration 81 documentation Administrative formalites enumeration 82 documentation Outside influence enumeration 83 documentation Effects of weather and natural causes enumeration 84 documentation Delay caused by external reasons on the next network enumeration 89 documentation Other external causes enumeration 90 documentation Dangerous incidents, accidents and hazards enumeration 91 documentation Track occupation caused by the lateness of the same train enumeration 92 documentation Track occupation caused by the lateness of another train enumeration 93 documentation Turn round enumeration 94 documentation Connection enumeration 95 documentation Further investigation needed
annotation	documentation This element identifies the reason for an interruption of the train running
source	<xs:element name="teltsi_InterruptionReason" type="teltsi_DelayCode"> <xs:annotation> <xs:documentation>This element identifies the reason for an interruption of the train running</xs:documentation> </xs:annotation> </xs:element>

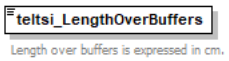
element **teltsi_KeeperShortNameVKM**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	complexType **TC_Identifier_ComplexType**
facets	Kind Value Annotation maxLength 10
annotation	documentation Free text, short name/vehicle keeper marking of the wagon keeper
source	<xs:element name="teltsi_KeeperShortNameVKM"> <xs:annotation> <xs:documentation>Free text, short name/vehicle keeper marking of the wagon keeper</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:maxLength value="10"/> </xs:restriction> </xs:simpleType> </xs:element>

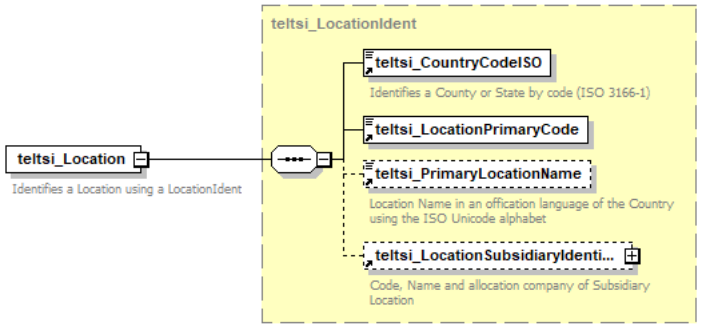
element **teltsi_Latitude**

diagram	
type	**xs:float**
properties	content simple
used by	element **teltsi_GeographicCoordinates**
annotation	documentation Latitudinal Coordinates as expressed in decimal degrees with a precision of 6 decimals.
source	<xs:element name="teltsi_Latitude" type="xs:float"> <xs:annotation> <xs:documentation>Latitudinal Coordinates as expressed in decimal degrees with a precision of 6 decimals.</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_LengthOverBuffers**

diagram	
type	restriction of **xs:integer**
properties	content simple
used by	elements **LocoData** **WagonData**
facets	Kind Value Annotation minInclusive 1 maxInclusive 999999
annotation	documentation Length over buffers is expressed in cm.
source	<xs:element name="teltsi_LengthOverBuffers"> <xs:annotation> <xs:documentation>Length over buffers is expressed in cm.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:integer"> <xs:minInclusive value="1"/> <xs:maxInclusive value="999999"/> </xs:restriction> </xs:simpleType> </xs:element>

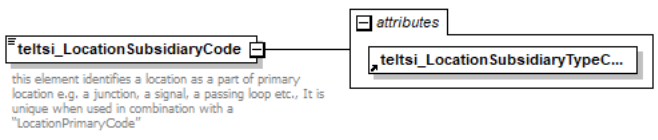
element **teltsi_Location**

diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	element **teltsi_InterruptionPoint**
annotation	documentation Identifies a Location using a LocationIdent
source	<xs:element name="teltsi_Location" type="teltsi_LocationIdent"> <xs:annotation> <xs:documentation>Identifies a Location using a LocationIdent</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_LocationPrimaryCode**

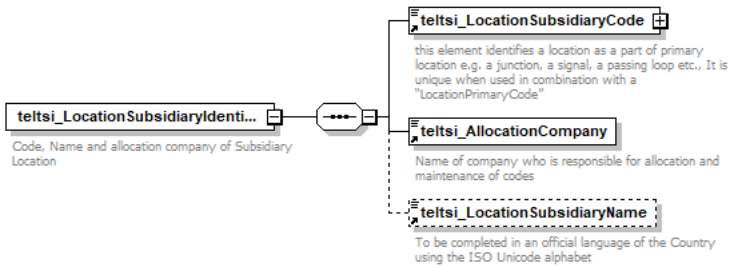
diagram	
type	**teltsi_Numeric1-5**
properties	content simple
used by	element **TAF_TAP_LocationReference** complexType **teltsi_LocationIdent**
facets	Kind Value Annotation minInclusive 1 maxInclusive 99999
source	<xs:element name="teltsi_LocationPrimaryCode" type="teltsi_Numeric1-5"/>

element **teltsi_LocationSubsidiaryCode**

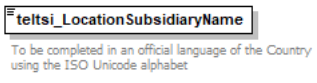
diagram	
type	extension of **teltsi_String1-10**
properties	content complex
used by	elements **LocationSubsidiaryReference** **teltsi_LocationSubsidiaryIdentification**

facets	Kind Value Annotation minLength 1 maxLength 10
attributes	Name Type Use Default Fixed Annotation teltsi_LocationSubsidiaryTypeCode **derived by: xs:token** required
annotation	documentation this element identifies a location as a part of primary location e.g. a junction, a signal, a passing loop etc., It is unique when used in combination with a “LocationPrimaryCode”
source	<xs:element name="teltsi_LocationSubsidiaryCode"> <xs:annotation> <xs:documentation>this element identifies a location as a part of primary location e.g. a junction, a signal, a passing loop etc., It is unique when used in combination with a “LocationPrimaryCode”</xs:documentation> </xs:annotation> <xs:complexType> <xs:simpleContent> <xs:extension base="teltsi_String1-10"> <xs:attribute ref="teltsi_LocationSubsidiaryTypeCode" use="required"/> </xs:extension> </xs:simpleContent> </xs:complexType> </xs:element>

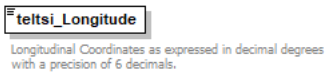
element **teltsi_LocationSubsidiaryIdentification**

diagram	
properties	content complex
children	**teltsi_LocationSubsidiaryCode** **teltsi_AllocationCompany** **teltsi_LocationSubsidiaryName**
used by	complexType **teltsi_LocationIdent**
annotation	documentation Code, Name and allocation company of Subsidiary Location
source	<xs:element name="teltsi_LocationSubsidiaryIdentification"> <xs:annotation> <xs:documentation>Code, Name and allocation company of Subsidiary Location</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_LocationSubsidiaryCode"/> <xs:element ref="teltsi_AllocationCompany"/> <xs:element ref="teltsi_LocationSubsidiaryName" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

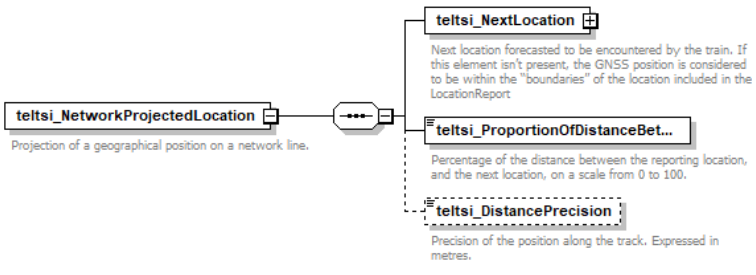
element **teltsi_LocationSubsidiaryName**

diagram	
type	**teltsi_FreeText**
properties	content simple
used by	element **teltsi_LocationSubsidiaryIdentification**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation To be completed in an official language of the Country using the ISO Unicode alphabet
source	<xs:element name="teltsi_LocationSubsidiaryName" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>To be completed in an official language of the Country using the ISO Unicode alphabet</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_Longitude**

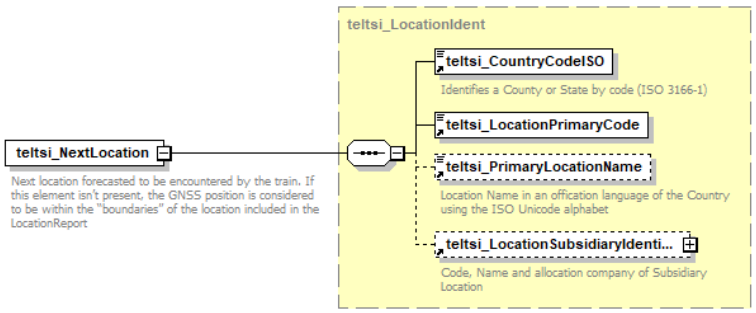
diagram	
type	**xs:float**
properties	content simple
used by	element **teltsi_GeographicCoordinates**
annotation	documentation Longitudinal Coordinates as expressed in decimal degrees with a precision of 6 decimals.
source	<xs:element name="teltsi_Longitude" type="xs:float"> <xs:annotation> <xs:documentation>Longitudinal Coordinates as expressed in decimal degrees with a precision of 6 decimals.</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_NetworkProjectedLocation**

diagram	

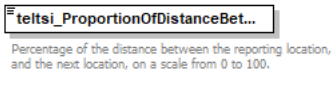
properties	content complex
children	**teltsi_NextLocation** **teltsi_ProportionOfDistanceBetweenLocations** **teltsi_DistancePrecision**
used by	element **teltsi_GeoLocalisationOnNetwork**
annotation	documentation Projection of a geographical position on a network line.
source	<xs:element name="teltsi_NetworkProjectedLocation"> <xs:annotation> <xs:documentation>Projection of a geographical position on a network line. </xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element name="teltsi_NextLocation" type="teltsi_LocationIdent"> <xs:annotation> <xs:documentation>Next location forecasted to be encountered by the train. If this element isn't present, the GNSS position is considered to be within the "boundaries" of the location included in the LocationReport</xs:documentation> </xs:annotation> </xs:element> <xs:element name="teltsi_ProportionOfDistanceBetweenLocations" type="teltsi_Percentage"> <xs:annotation> <xs:documentation>Percentage of the distance between the reporting location, and the next location, on a scale from 0 to 100. </xs:documentation> </xs:annotation> </xs:element> <xs:element name="teltsi_DistancePrecision" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Precision of the position along the track. Expressed in metres.</xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType> </xs:element>

element **teltsi_NetworkProjectedLocation/teltsi_NextLocation**

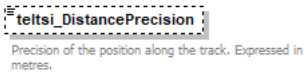
diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
annotation	documentation Next location forecasted to be encountered by the train. If this element isn't present, the GNSS position is considered to

	be within the “boundaries” of the location included in the LocationReport
source	<xs:element name="teltsi_NextLocation" type="teltsi_LocationIdent"> <xs:annotation> <xs:documentation>Next location forecasted to be encountered by the train. If this element isn’t present, the GNSS position is considered to be within the “boundaries” of the location included in the LocationReport</xs:documentation> </xs:annotation> </xs:element>

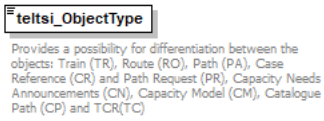
element **teltsi_NetworkProjectedLocation/teltsi_ProportionOfDistanceBetweenLocations**

diagram	
type	**teltsi_Percentage**
properties	content simple
facets	Kind Value Annotation minInclusive 0 maxInclusive 100
annotation	documentation Percentage of the distance between the reporting location, and the next location, on a scale from 0 to 100.
source	<xs:element name="teltsi_ProportionOfDistanceBetweenLocations" type="teltsi_Percentage"> <xs:annotation> <xs:documentation>Percentage of the distance between the reporting location, and the next location, on a scale from 0 to 100. </xs:documentation> </xs:annotation> </xs:element>

element **teltsi_NetworkProjectedLocation/teltsi_DistancePrecision**

diagram	
type	**xs:float**
properties	minOcc 0 maxOcc 1 content simple
annotation	documentation Precision of the position along the track. Expressed in metres.
source	<xs:element name="teltsi_DistancePrecision" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Precision of the position along the track. Expressed in metres.</xs:documentation> </xs:annotation> </xs:element>

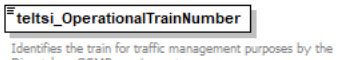
element **teltsi_ObjectType**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	complexType **teltsi_CompositIdentifierOperationalType**
facets	Kind / Value / Annotation minLength 2 maxLength 2 pattern [0-9A-Z]{2} enumeration TR enumeration RO enumeration PA enumeration CR enumeration PR enumeration CN enumeration CM enumeration CP enumeration TC
annotation	documentation Provides a possibility for differentiation between the objects: Train (TR), Route (RO), Path (PA), Case Reference (CR) and Path Request (PR), Capacity Needs Announcements (CN), Capacity Model (CM), Catalogue Path (CP) and TCR(TC)

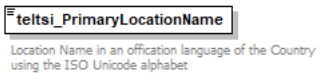
source

```
<xs:element name="teltsi_ObjectType">
 <xs:annotation>
  <xs:documentation>Provides a possibility for differentiation between the objects: Train (TR), Route (RO), Path (PA), Case Reference (CR) and Path Request (PR), Capacity Needs Announcements (CN), Capacity Model (CM), Catalogue Path (CP) and TCR(TC)</xs:documentation>
 </xs:annotation>
 <xs:simpleType>
  <xs:restriction base="xs:string">
   <xs:minLength value="2"/>
   <xs:maxLength value="2"/>
   <xs:pattern value="[0-9A-Z]{2}"/>
   <xs:enumeration value="TR"/>
   <xs:enumeration value="RO"/>
   <xs:enumeration value="PA"/>
   <xs:enumeration value="CR"/>
   <xs:enumeration value="PR"/>
   <xs:enumeration value="CN"/>
   <xs:enumeration value="CM"/>
   <xs:enumeration value="CP"/>
   <xs:enumeration value="TC"/>
  </xs:restriction>
 </xs:simpleType>
</xs:element>
```

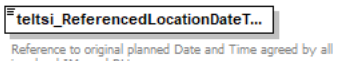
element **teltsi_OperationalTrainNumber**

diagram	
type	**teltsi_String1-8**
properties	content simple
used by	element **RT_TrainInformation** complexType **OTN_ID_ComplexType**
facets	Kind Value Annotation minLength 1 maxLength 8
annotation	documentation Identifies the train for traffic management purposes by the Dispatcher, GSMR services, etc.
source	<xs:element name="teltsi_OperationalTrainNumber" type="teltsi_String1-8"> <xs:annotation> <xs:documentation>Identifies the train for traffic management purposes by the Dispatcher, GSMR services, etc.</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_PrimaryLocationName**

diagram	
type	**teltsi_FreeText**
properties	content simple
used by	complexType **teltsi_LocationIdent**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation Location Name in an offication language of the Country using the ISO Unicode alphabet
source	<xs:element name="teltsi_PrimaryLocationName" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>Location Name in an offication language of the Country using the ISO Unicode alphabet</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_ReferencedLocationDateTime**

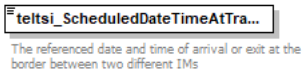
diagram	
type	**xs:dateTime**
properties	content simple
used by	element **teltsi_InterruptionPoint**
annotation	documentation Reference to original planned Date and Time agreed by all involved IMs and RUs.

source	<xs:element name="teltsi_ReferencedLocationDateTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>Reference to original planned Date and Time agreed by all involved IMs and RUs. </xs:documentation> </xs:annotation> </xs:element>

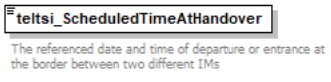
element **teltsi_Remarks**

diagram	
type	**teltsi_FreeText**
properties	content simple
used by	element **teltsi_InterruptionPoint**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation Free Form Text
source	<xs:element name="teltsi_Remarks" type="teltsi_FreeText"> <xs:annotation> <xs:documentation>Free Form Text</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_ScheduledDateTimeAtTransfer**

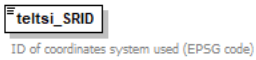
diagram	
type	**xs:dateTime**
properties	content simple
used by	complexType **OTN_ID_ComplexType**
annotation	documentation The referenced date and time of arrival or exit at the border between two different IMs
source	<xs:element name="teltsi_ScheduledDateTimeAtTransfer" type="xs:dateTime"> <xs:annotation> <xs:documentation>The referenced date and time of arrival or exit at the border between two different IMs</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_ScheduledTimeAtHandover**

diagram	
type	**xs:dateTime**

properties	content simple
used by	complexType **OTN_ID_ComplexType**
annotation	documentation The referenced date and time of departure or entrance at the border between two different IMs
source	<xs:element name="teltsi_ScheduledTimeAtHandover" type="xs:dateTime"> <xs:annotation> <xs:documentation>The referenced date and time of departure or entrance at the border between two different IMs</xs:documentation> </xs:annotation> </xs:element>

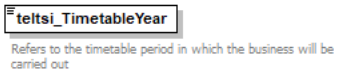
element **teltsi_SRID**

diagram	
type	restriction of **xs:string**
properties	content simple
used by	element **teltsi_GeographicCoordinates**
facets	Kind Value Annotation pattern [0-9]{4}
annotation	documentation ID of coordinates system used (EPSG code)
source	<xs:element name="teltsi_SRID"> <xs:annotation> <xs:documentation>ID of coordinates system used (EPSG code)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:pattern value="[0-9]{4}"/> </xs:restriction> </xs:simpleType> </xs:element>

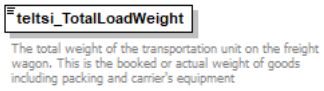
element **teltsi_StartDate**

diagram	
type	**xs:date**
properties	content simple
used by	element **RT_TrainInformation** complexTypes **OTN_ID_ComplexType** **teltsi_CompositIdentifierOperationalType**
annotation	documentation The start of the date/time in effect
source	<xs:element name="teltsi_StartDate" type="xs:date"> <xs:annotation> <xs:documentation>The start of the date/time in effect</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_TimetableYear**

diagram	
type	restriction of **xs:integer**
properties	content simple
used by	complexType **teltsi_CompositIdentifierOperationalType**
facets	Kind Value Annotation minInclusive 2012 maxInclusive 2097
annotation	documentation Refers to the timetable period in which the business will be carried out
source	<xs:element name="teltsi_TimetableYear"> <xs:annotation> <xs:documentation>Refers to the timetable period in which the business will be carried out</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:integer"> <xs:minInclusive value="2012"/> <xs:maxInclusive value="2097"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_TotalLoadWeight**

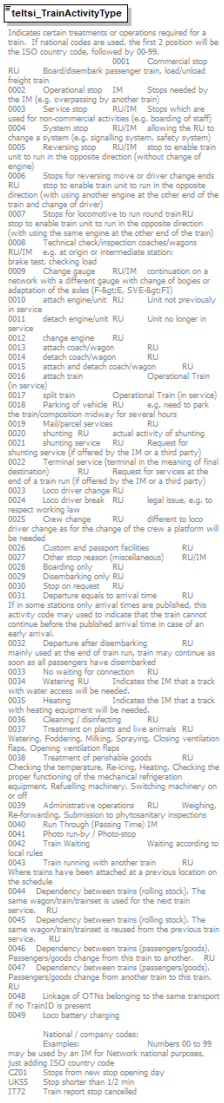
diagram	
type	**teltsi_WeightValueKilo**
properties	content simple
used by	element **WagonData**
facets	Kind Value Annotation minInclusive 0 maxInclusive 999999 whiteSpace collapse
annotation	documentation The total weight of the transportation unit on the freight wagon. This is the booked or actual weight of goods including packing and carrier's equipment
source	<xs:element name="teltsi_TotalLoadWeight" type="teltsi_WeightValueKilo"> <xs:annotation> <xs:documentation>The total weight of the transportation unit on the freight wagon. This is the booked or actual weight of goods including packing and carrier's equipment</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_TractionPositionInTrain**

diagram	
type	restriction of **xs:integer**
properties	content simple
used by	element **LocoData**
facets	Kind / Value / Annotation minInclusive 01 maxInclusive 99
annotation	documentation Identifies position of intermediate traction unit(s) in the train indicating after which wagon (specified by order number) the unit is placed.
source	<xs:element name="teltsi_TractionPositionInTrain"> <xs:annotation> <xs:documentation>Identifies position of intermediate traction unit(s) in the train indicating after which wagon (specified by order number) the unit is placed.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:integer"> <xs:minInclusive value="01"/> <xs:maxInclusive value="99"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_TrainActivityType**

diagram



type

restriction of **xs:string**

properties

content simple

used by

element **StopType**

facets

Kind	Value	Annotation
minLength	4	
maxLength	4	

annotation

documentation

Indicates certain treatments or operations required for a train. If national codes are used, the first 2 position will be the ISO country code, followed by 00-99.

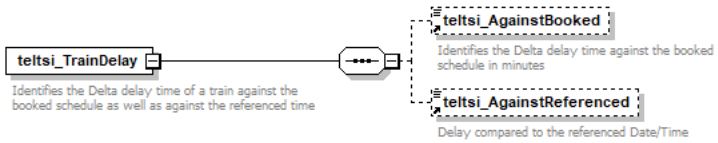
		0001	Commercial stop	RU Board/disembark passenger train, load/unload freight train
0002	Operational stop	IM	Stops needed by the IM (e.g. overpassing by another train)	
0003	Service stop	RU/IM	Stops which are used for non-commercial activities (e.g. boarding of staff)	
0004	System stop	RU/IM	allowing the RU to change a system (e.g. signalling system, safety system)	
0005	Reversing stop	RU/IM	stop to enable train unit to run in the opposite direction (without change of engine)	
0006	Stops for reversing move or driver change ends	RU	stop to enable train unit to run in the opposite direction (with using another engine at the other end of the train and change of driver)	
0007	Stops for locomotive to run round train	RU	stop to enable train unit to run in the opposite direction (with	

	using the same engine at the other end of the train) 0008 Technical check/inspection coaches/wagons RU/IM e.g. at origin or intermediate station: brake test, checking load 0009 Change gauge RU/IM continuation on a network with a different gauge with change of bogies or adaptation of the axles (F->E, SVE->FI) 0010 attach engine/unit RU Unit not previously in service 0011 detach engine/unit RU Unit no longer in service 0012 change engine RU 0013 attach coach/wagon RU 0014 detach coach/wagon RU 0015 attach and detach coach/wagon RU 0016 attach train Operational Train (in service) 0017 split train Operational Train (in service) 0018 Parking of vehicle RU e.g. need to park the train/composition midway for several hours 0019 Mail/parcel services RU 0020 shunting RU actual activity of shunting 0021 shunting service RU Request for shunting service (if offered by the IM or a third party) 0022 Terminal service (terminal in the meaning of final destination) RU Request for services at the end of a train run (if offered by the IM or a third party) 0023 Loco driver change RU 0024 Loco driver break RU legal issue, e.g. to respect working law 0025 Crew change RU different to loco driver change as for the change of the crew a platform will be needed 0026 Custom and passport facilities RU 0027 Other stop reason (miscellaneous) RU/IM 0028 Boarding only RU 0029 Disembarking only RU 0030 Stop on request RU 0031 Departure equals to arrival time RU If in some stations only arrival times are published, this activity code may used to indicate that the train cannot continue before the published arrival time in case of an early arrival. 0032 Departure after disembarking RU mainly used at the end of train run, train may continue as soon as all passengers have disembarked 0033 No waiting for connection RU 0034 Watering RU Indicates the IM that a track with water access will be needed. 0035 Heating Indicates the IM that a track with heating equipment will be needed. 0036 Cleaning / disinfecting RU 0037 Treatment on plants and live animals RU Watering, Foddering, Milking, Spraying, Closing ventilation flaps, Opening ventilation flaps 0038 Treatment of perishable goods RU Checking the temperature, Re-icing, Heating, Checking the proper functioning of the mechanical refrigeration equipment, Refuelling machinery, Switching machinery on or off 0039 Administrative operations RU Weighing, Re-forwarding, Submission to phytosanitary inspections 0040 Run Through (Passing Time) IM 0041 Photo run-by / Photo-stop 0042 Train Waiting Waiting according to local rules 0043 Train running with another train RU Where trains have been attached at a previous location on the schedule 0044 Dependency between trains (rolling stock). The same wagon/train/trainset is used for the next train service. RU 0045 Dependency between trains (rolling stock). The same wagon/train/trainset is reused from the previous train service. RU 0046 Dependency between trains (passengers/goods). Passengers/goods change from this train to another. RU 0047 Dependency between trains (passengers/goods). Passengers/goods change from another train to this train. RU 0048 Linkage of OTNs belonging to the same transport if no TrainID is present 0049 Loco battery charging National / company codes: Examples: Numbers 00 to 99 may be used by an IM for Network national purposes, just adding ISO country code CZ01 Stops from new stop opening day UK55 Stop shorter than 1/2 min IT72 Train report stop cancelled
source	<xs:element name="teltsi_TrainActivityType"> <xs:annotation> <xs:documentation>Indicates certain treatments or operations required for a train. If national codes are used, the first 2 position will be the ISO country code, followed by 00-99. 0001 Commercial stop RU Board/disembark passenger train,

	load/unload freight train 0002 Operational stop IM Stops needed by the IM (e.g. overpassing by another train) 0003 Service stop RU/IM Stops which are used for non-commercial activities (e.g. boarding of staff) 0004 System stop RU/IM allowing the RU to change a system (e.g. signalling system, safety system) 0005 Reversing stop RU/IM stop to enable train unit to run in the opposite direction (without change of engine) 0006 Stops for reversing move or driver change ends RU stop to enable train unit to run in the opposite direction (with using another engine at the other end of the train and change of driver) 0007 Stops for locomotive to run round train RU stop to enable train unit to run in the opposite direction (with using the same engine at the other end of the train) 0008 Technical check/inspection coaches/wagons RU/IM e.g. at origin or intermediate station: brake test, checking load 0009 Change gauge RU/IM continuation on a network with a different gauge with change of bogies or adaptation of the axles (F->E, SVE->FI) 0010 attach engine/unit RU Unit not previously in service 0011 detach engine/unit RU Unit no longer in service 0012 change engine RU 0013 attach coach/wagon RU 0014 detach coach/wagon RU 0015 attach and detach coach/wagon RU 0016 attach train Operational Train (in service) 0017 split train Operational Train (in service) 0018 Parking of vehicle RU e.g. need to park the train/composition midway for several hours 0019 Mail/parcel services RU 0020 shunting RU actual activity of shunting 0021 shunting service RU Request for shunting service (if offered by the IM or a third party) 0022 Terminal service (terminal in the meaning of final destination) RU Request for services at the end of a train run (if offered by the IM or a third party) 0023 Loco driver change RU 0024 Loco driver break RU legal issue, e.g. to respect working law 0025 Crew change RU different to loco driver change as for the change of the crew a platform will be needed 0026 Custom and passport facilities RU 0027 Other stop reason (miscellaneous) RU/IM 0028 Boarding only RU 0029 Disembarking only RU 0030 Stop on request RU 0031 Departure equals to arrival time RU If in some stations only arrival times are published, this activity code may used to indicate that the train cannot continue before the published arrival time in case of an early arrival. 0032 Departure after disembarking RU mainly used at the end of train run, train may continue as soon as all passengers have disembarked 0033 No waiting for connection RU 0034 Watering RU Indicates the IM that a track with water access will be needed. 0035 Heating Indicates the IM that a track with heating equipment will be needed. 0036 Cleaning / disinfecting RU 0037 Treatment on plants and live animals RU Watering, Foddering, Milking, Spraying, Closing ventilation flaps, Opening ventilation flaps 0038 Treatment of perishable goods RU Checking the temperature, Re-icing, Heating, Checking the proper functioning of the mechanical refrigeration equipment, Refuelling machinery, Switching machinery on or off

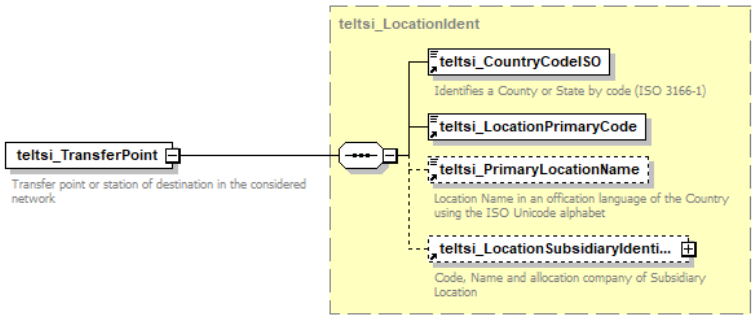
	0039 Administrative operations RU Weighing, Re-forwarding, Submission to phytosanitary inspections 0040 Run Through (Passing Time) IM 0041 Photo run-by / Photo-stop 0042 Train Waiting Waiting according to local rules 0043 Train running with another train RU Where trains have been attached at a previous location on the schedule 0044 Dependency between trains (rolling stock). The same wagon/train/trainset is used for the next train service. RU 0045 Dependency between trains (rolling stock). The same wagon/train/trainset is reused from the previous train service. RU 0046 Dependency between trains (passengers/goods). Passengers/goods change from this train to another. RU 0047 Dependency between trains (passengers/goods). Passengers/goods change from another train to this train. RU 0048 Linkage of OTNs belonging to the same transport if no TrainID is present 0049 Loco battery charging National / company codes: Examples: Numbers 00 to 99 may be used by an IM for Network national purposes, just adding ISO country code CZ01 Stops from new stop opening day UK55 Stop shorter than 1/2 min IT72 Train report stop cancelled </xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:minLength value="4"/> <xs:maxLength value="4"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_TrainDelay**

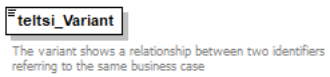
diagram	
properties	content complex
children	**teltsi_AgainstBooked** **teltsi_AgainstReferenced**
used by	element **TrainReadyStatus**
annotation	documentation Identifies the Delta delay time of a train against the booked schedule as well as against the referenced time
source	<xs:element name="teltsi_TrainDelay"> <xs:annotation> <xs:documentation>Identifies the Delta delay time of a train against the booked schedule as well as against the referenced time </xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence>

```
      <xs:element ref="teltsi_AgainstBooked" minOccurs="0"/>
      <xs:element ref="teltsi_AgainstReferenced" minOccurs="0">
        <xs:annotation>
          <xs:documentation>Delay compared to the referenced Date/Time</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **teltsi_TransferPoint**

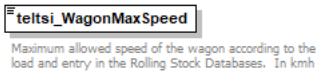
diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	complexType **OTN_ID_ComplexType**
annotation	documentation Transfer point or station of destination in the considered network
source	<xs:element name="teltsi_TransferPoint" type="teltsi_LocationIdent"> <xs:annotation> <xs:documentation>Transfer point or station of destination in the considered network</xs:documentation> </xs:annotation> </xs:element>

element **teltsi_Variant**

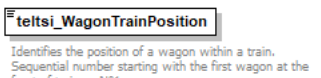
diagram	
type	restriction of **xs:string**
properties	content simple
used by	complexType **teltsi_CompositIdentifierOperationalType**
facets	Kind Value Annotation minLength 2 maxLength 2 pattern [0-9A-Z]{2}
annotation	documentation

	The variant shows a relationship between two identifiers referring to the same business case
source	<xs:element name="teltsi_Variant"> <xs:annotation> <xs:documentation>The variant shows a relationship between two identifiers referring to the same business case</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:minLength value="2"/> <xs:maxLength value="2"/> <xs:pattern value="[0-9A-Z]{2}"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_WagonMaxSpeed**

diagram	
type	restriction of **xs:int**
properties	content simple
used by	element **WagonData**
facets	Kind Value Annotation minInclusive 001 maxInclusive 999
annotation	documentation Maximum allowed speed of the wagon according to the load and entry in the Rolling Stock Databases. In kmh
source	<xs:element name="teltsi_WagonMaxSpeed"> <xs:annotation> <xs:documentation>Maximum allowed speed of the wagon according to the load and entry in the Rolling Stock Databases. In kmh</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:int"> <xs:minInclusive value="001"/> <xs:maxInclusive value="999"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_WagonTrainPosition**

diagram	
type	restriction of **xs:int**
properties	content simple
used by	element **WagonData**

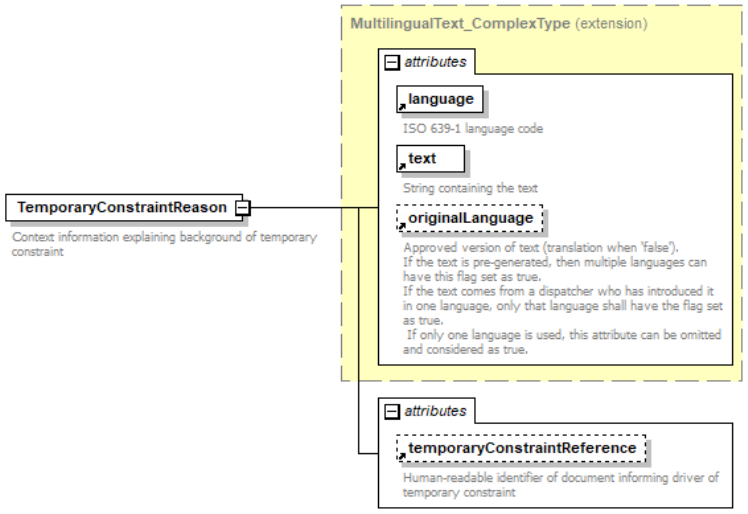
facets	Kind Value Annotation minInclusive 1 maxInclusive 999
annotation	documentation Identifies the position of a wagon within a train. Sequential number starting with the first wagon at the front of train as N°1.
source	<xs:element name="teltsi_WagonTrainPosition"> <xs:annotation> <xs:documentation>Identifies the position of a wagon within a train. Sequential number starting with the first wagon at the front of train as N°1.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:int"> <xs:minInclusive value="1"/> <xs:maxInclusive value="999"/> </xs:restriction> </xs:simpleType> </xs:element>

element **teltsi_WagonWeightEmpty**

diagram	
type	**teltsi_WeightValueKilo**
properties	content simple
used by	element **WagonData**
facets	Kind Value Annotation minInclusive 0 maxInclusive 999999 whiteSpace collapse
annotation	documentation The weight of an empty wagon according to the entry in the rolling stock database
source	<xs:element name="teltsi_WagonWeightEmpty" type="teltsi_WeightValueKilo"> <xs:annotation> <xs:documentation>The weight of an empty wagon according to the entry in the rolling stock database</xs:documentation> </xs:annotation> </xs:element>

element **TemporaryConstraintReason**

diagram



type: extension of **MultilingualText_ComplexType**

properties: content complex

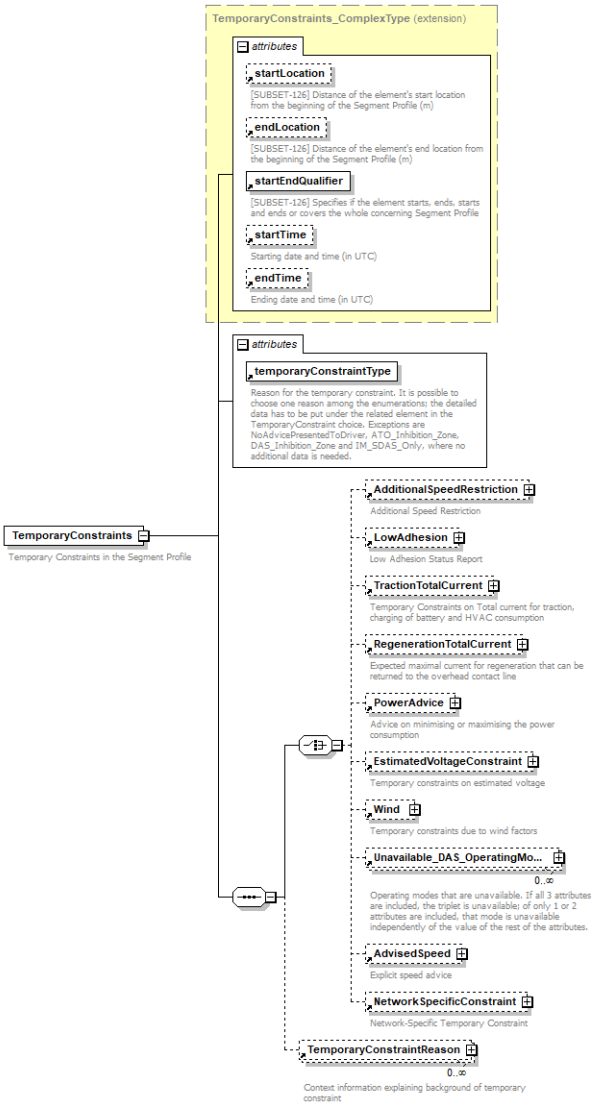
used by: element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
temporaryConstraintReference	**xs:string**				documentation Human-readable

	identifier of document informing driver of temporary constraint
annotation	documentation Context information explaining background of temporary constraint
source	<xs:element name="TemporaryConstraintReason"> <xs:annotation> <xs:documentation>Context information explaining background of temporary constraint</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="MultilingualText_ComplexType"> <xs:attribute ref="temporaryConstraintReference"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **TemporaryConstraints**

diagram	
type	extension of **TemporaryConstraints_ComplexType**
properties	content complex
children	**AdditionalSpeedRestriction** **LowAdhesion** **TractionTotalCurrent** **RegenerationTotalCurrent** **PowerAdvice** **EstimatedVoltageConstraint** **Wind** **Unavailable_DAS_OperatingModes** **AdvisedSpeed** **NetworkSpecificConstraint** **TemporaryConstraintReason**
used by	element **SegmentProfileReference**

attributes

Name	Type	Use	Default	Fixed	Annotation
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if

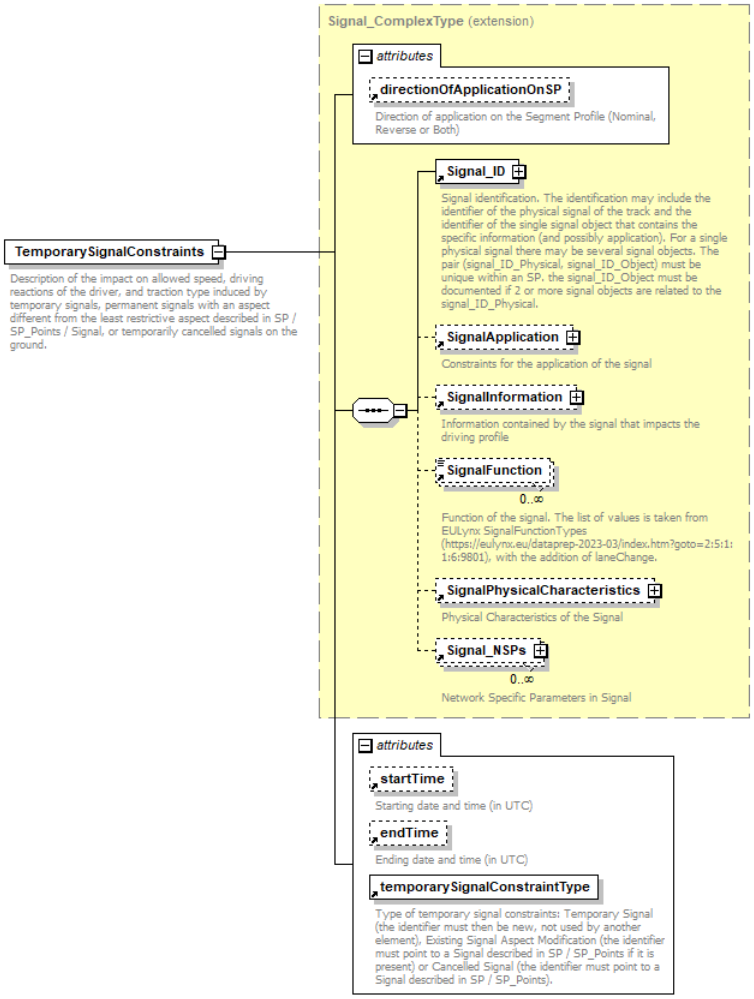
				the element starts, ends, starts and ends or covers the whole concerning Segment Profile
	startTime	**UTCDateTimeType**		documentation Starting date and time (in UTC)
	endTime	**UTCDateTimeType**		documentation Ending date and time (in UTC)
	temporaryConstraintType	**derived by: xs:string**	required	documentation Reason for the temporary constraint. It is possible to choose one reason among the enumerations; the detailed data has to be put under the related element in the TemporaryConstraint choice. Exceptions are NoAdvicePresentedToDriver, ATO_Inhibition_Zone, DAS_Inhibition_Zone and IM_SDAS_Only, where no additional data is needed.
annotation	documentation Temporary Constraints in the Segment Profile			

source

```
<xs:element name="TemporaryConstraints">
 <xs:annotation>
  <xs:documentation>Temporary Constraints in the Segment Profile</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="TemporaryConstraints_ComplexType">
    <xs:sequence>
     <xs:choice>
      <xs:element ref="AdditionalSpeedRestriction" minOccurs="0"/>
      <xs:element ref="LowAdhesion" minOccurs="0"/>
      <xs:element ref="TractionTotalCurrent" minOccurs="0"/>
      <xs:element ref="RegenerationTotalCurrent" minOccurs="0"/>
      <xs:element ref="PowerAdvice" minOccurs="0"/>
      <xs:element ref="EstimatedVoltageConstraint" minOccurs="0"/>
      <xs:element ref="Wind" minOccurs="0"/>
      <xs:element ref="Unavailable_DAS_OperatingModes" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="AdvisedSpeed" minOccurs="0"/>
      <xs:element ref="NetworkSpecificConstraint" minOccurs="0"/>
     </xs:choice>
     <xs:element ref="TemporaryConstraintReason" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attribute ref="temporaryConstraintType" use="required"/>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **TemporarySignalConstraints**

diagram



type: extension of **Signal_ComplexType**

properties: content complex

children: **Signal_ID** **SignalApplication** **SignalInformation** **SignalFunction** **SignalPhysicalCharacteristics** **Signal_NSPs**

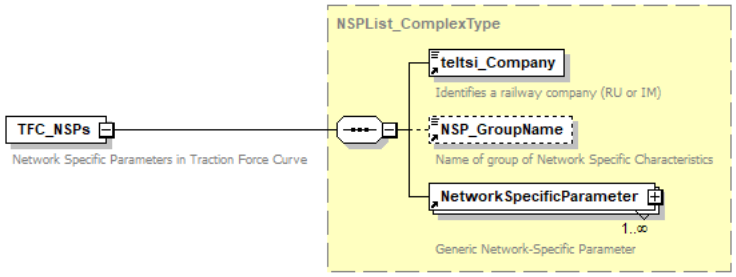
used by: element **SegmentProfileReference**

attributes

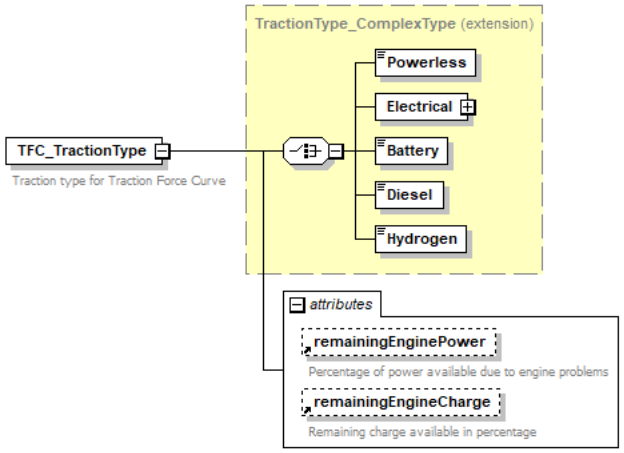
Name	Type	Use	Default	Fixed	Annotation
directionOfApplicationOnSP	**derived by: xs:string**				documentation Direction of application on the Segment Profile (Nominal, Reverse or Both)
startTime	**UTCDateTimeType**				documentation Starting date and time (in UTC)
endTime	**UTCDateTimeType**				documentation Ending date and time (in UTC)
temporarySignalConstraintType	**derived by: xs:string**	required			documentation Type of

	temporary signal constraints: Temporary Signal (the identifier must then be new, not used by another element), Existing Signal Aspect Modification (the identifier must point to a Signal described in SP / SP_Points if it is present) or Cancelled Signal (the identifier must point to a Signal described in SP / SP_Points).
annotation	documentation Description of the impact on allowed speed, driving reactions of the driver, and traction type induced by temporary signals, permanent signals with an aspect different from the least restrictive aspect described in SP / SP_Points / Signal, or temporarily cancelled signals on the ground.
source	<xs:element name="TemporarySignalConstraints"> <xs:annotation> <xs:documentation>Description of the impact on allowed speed, driving reactions of the driver, and traction type induced by temporary signals, permanent signals with an aspect different from the least restrictive aspect described in SP / SP_Points / Signal, or temporarily cancelled signals on the ground.</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="Signal_ComplexType"> <xs:attribute ref="startTime"/> <xs:attribute ref="endTime"/> <xs:attribute ref="temporarySignalConstraintType" use="required"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

element **TFC_NSPs**

diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	complexType **TractionForceCurve_ComplexType**
annotation	documentation Network Specific Parameters in Traction Force Curve
source	<xs:element name="TFC_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Traction Force Curve</xs:documentation> </xs:annotation> </xs:element>

element **TFC_TractionType**

diagram	
type	extension of **TractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Battery** **Diesel** **Hydrogen**
used by	complexType **TractionForceCurve_ComplexType**

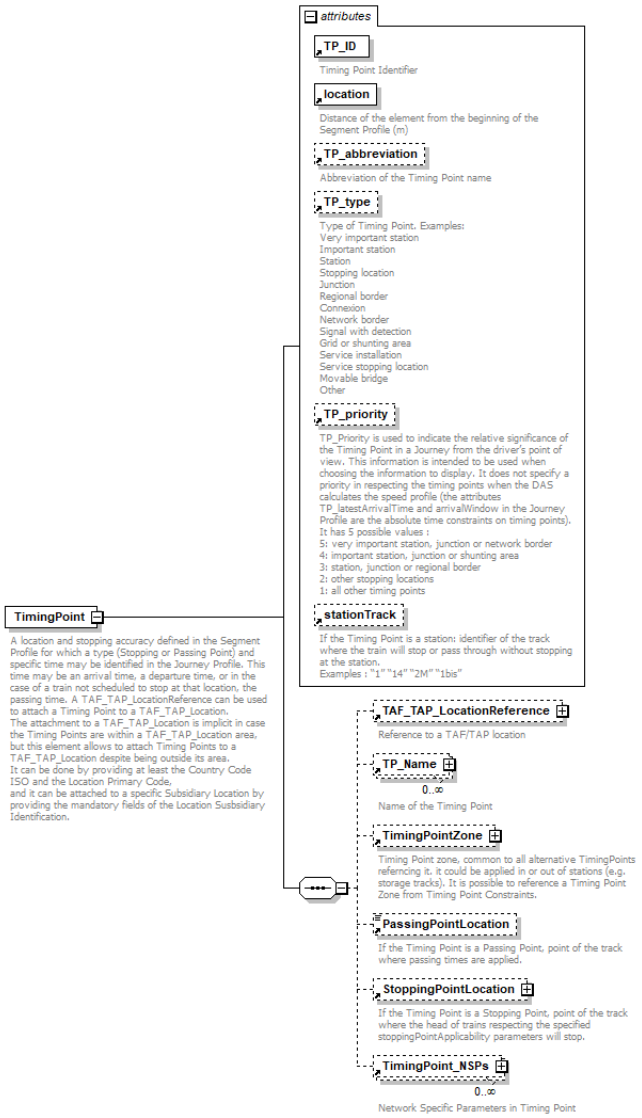
attributes

Name	Type	Use	Default	Fixed	Annotation
remainingEnginePower	**percentageType**				documentation Percentage of power available due to engine problems

	remainingEngineCharge **percentageType**	documentation Remaining charge available in percentage
annotation	documentation Traction type for Traction Force Curve	
source	<xs:element name="TFC_TractionType"> <xs:annotation> <xs:documentation>Traction type for Traction Force Curve</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="TractionType_ComplexType"> <xs:attribute ref="remainingEnginePower"/> <xs:attribute ref="remainingEngineCharge"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>	

element **TimingPoint**

diagram



properties

content complex

children **TAF_TAP_LocationReference TP_Name TimingPointZone PassingPointLocation StoppingPointLocation TimingPoint_NSPs**

used by

element **SP_Points**

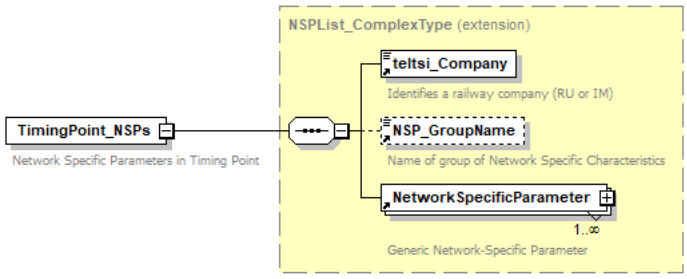
attributes

Name	Type	Use	Default	Fixed	Annotation
TP_ID	**xs:string**	required			documentation Timing Point Identifier
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)
TP_abbreviation	**xs:string**				documentation Abbreviation of the Timing Point name
TP_type	**xs:string**				documentation Type of Timing Point. Examples:

			Very important station Important station Station Stopping location Junction Regional border Connexion Network border Signal with detection Grid or shunting area Service installation Service stopping location Movable bridge Other
	TP_priority	**derived by: xs:unsignedByte**	documentation TP_Priority is used to indicate the relative significance of the Timing Point in a Journey from the driver's point of view. This information is intended to be used when choosing the information to display. It does not specify a priority in respecting the timing points when the DAS calculates the speed profile (the attributes TP_latestArrivalTime and arrivalWindow in the Journey Profile are the absolute time constraints on timing points). It has 5 possible values : 5: very important station, junction or network border 4: important station, junction or shunting area 3: station, junction or regional border 2: other stopping locations 1: all other timing points
	stationTrack	**xs:string**	documentation If the Timing Point is a station: identifier of the track where the train will stop or pass through without stopping at the station. Examples : "1" "14" "2M" "1bis"
annotation	documentation A location and stopping accuracy defined in the Segment Profile for which a type (Stopping or Passing Point) and specific time may be identified in the Journey Profile. This time may be an arrival time, a departure time, or in the case of a train not scheduled to stop at that location, the passing time. A TAF_TAP_LocationReference can be used to attach a Timing		

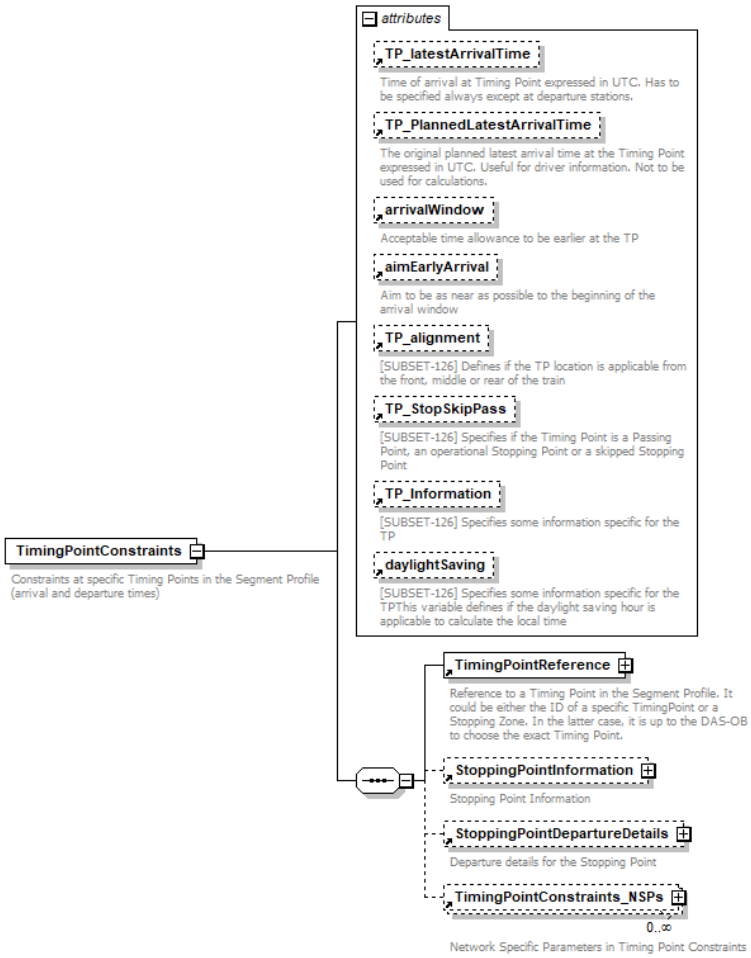
	Point to a TAF_TAP_Location. The attachment to a TAF_TAP_Location is implicit in case the Timing Points are within a TAF_TAP_Location area, but this element allows to attach Timing Points to a TAF_TAP_Location despite being outside its area. It can be done by providing at least the Country Code ISO and the Location Primary Code, and it can be attached to a specific Subsidiary Location by providing the mandatory fields of the Location Susbsidiary Identification.
source	<xs:element name="TimingPoint"> <xs:annotation> <xs:documentation>A location and stopping accuracy defined in the Segment Profile for which a type (Stopping or Passing Point) and specific time may be identified in the Journey Profile. This time may be an arrival time, a departure time, or in the case of a train not scheduled to stop at that location, the passing time. A TAF_TAP_LocationReference can be used to attach a Timing Point to a TAF_TAP_Location. The attachment to a TAF_TAP_Location is implicit in case the Timing Points are within a TAF_TAP_Location area, but this element allows to attach Timing Points to a TAF_TAP_Location despite being outside its area. It can be done by providing at least the Country Code ISO and the Location Primary Code, and it can be attached to a specific Subsidiary Location by providing the mandatory fields of the Location Susbsidiary Identification.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TAF_TAP_LocationReference" minOccurs="0"/> <xs:element ref="TP_Name" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TimingPointZone" minOccurs="0"/> <xs:element ref="PassingPointLocation" minOccurs="0"/> <xs:element ref="StoppingPointLocation" minOccurs="0"/> <xs:element ref="TimingPoint_NSPs" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="TP_ID" use="required"/> <xs:attribute ref="location" use="required"/> <xs:attribute ref="TP_abbreviation"/> <xs:attribute ref="TP_type"/> <xs:attribute ref="TP_priority"/> <xs:attribute ref="stationTrack"/> </xs:complexType> </xs:element>

element **TimingPoint_NSPs**

diagram	
type	extension of **NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**

used by	element **TimingPoint**
annotation	documentation Network Specific Parameters in Timing Point
source	<xs:element name="TimingPoint_NSPs"> <xs:annotation> <xs:documentation>Network Specific Parameters in Timing Point</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="NSPList_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

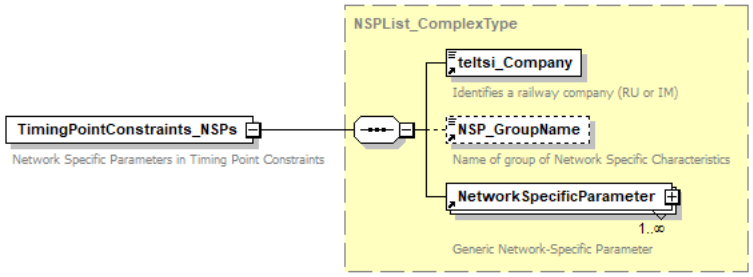
element **TimingPointConstraints**

diagram	
properties	content complex
children	**TimingPointReference** **StoppingPointInformation** **StoppingPointDepartureDetails** **TimingPointConstraints_NSPs**
used by	element **SegmentProfileReference**
attributes	Name: TP_latestArrivalTime Type: **UTCDateTimeType** Use: Default: Fixed: Annotation documentation: Time of arrival

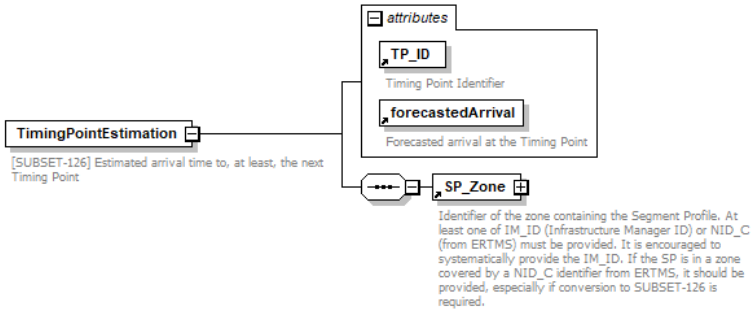
				at Timing Point expressed in UTC. Has to be specified always except at departure stations.
	TP_PlannedLatestArrivalTime	**UTCDateTimeType**		documentation The original planned latest arrival time at the Timing Point expressed in UTC. Useful for driver information. Not to be used for calculations.
	arrivalWindow	**xs:duration**	optional	documentation Acceptable time allowance to be earlier at the TP
	aimEarlyArrival	**xs:boolean**		documentation Aim to be as near as possible to the beginning of the arrival window
	TP_alignment	**derived by: xs:string**		documentation [SUBSET-126] Defines if the TP location is applicable from the front, middle or rear of the train
	TP_StopSkipPass	**derived by: xs:string**		documentation [SUBSET-126] Specifies if the Timing Point is a Passing Point, an operational Stopping Point or a skipped Stopping Point
	TP_Information	**derived by: xs:string**		documentation [SUBSET-126] Specifies some information specific for the TP
	daylightSaving	**xs:boolean**		documentation [SUBSET-126] Specifies some information specific for the TPThis variable defines if the daylight saving hour is

	applicable to calculate the local time
annotation	documentation Constraints at specific Timing Points in the Segment Profile (arrival and departure times)
source	<xs:element name="TimingPointConstraints"> <xs:annotation> <xs:documentation>Constraints at specific Timing Points in the Segment Profile (arrival and departure times)</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TimingPointReference"/> <xs:element ref="StoppingPointInformation" minOccurs="0"/> <xs:element ref="StoppingPointDepartureDetails" minOccurs="0"/> <xs:element ref="TimingPointConstraints_NSPs" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="TP_latestArrivalTime"/> <xs:attribute ref="TP_PlannedLatestArrivalTime"/> <xs:attribute ref="arrivalWindow" use="optional"/> <xs:attribute ref="aimEarlyArrival"/> <xs:attribute ref="TP_alignment"/> <xs:attribute ref="TP_StopSkipPass"/> <xs:attribute ref="TP_Information"/> <xs:attribute ref="daylightSaving"/> </xs:complexType> </xs:element>

element **TimingPointConstraints_NSPs**

diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company NSP_GroupName NetworkSpecificParameter**
used by	element **TimingPointConstraints**
annotation	documentation Network Specific Parameters in Timing Point Constraints
source	<xs:element name="TimingPointConstraints_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Timing Point Constraints</xs:documentation> </xs:annotation> </xs:element>

element **TimingPointEstimation**

diagram	
properties	content complex
children	**SP_Zone**
used by	element **B2G_StatusReport**

attributes

Name	Type	Use	Default	Fixed	Annotation
TP_ID	**xs:string**	required			documentation Timing Point Identifier
forecastedArrival	**UTCDateTimeType**	required			documentation Forecasted arrival at the Timing Point

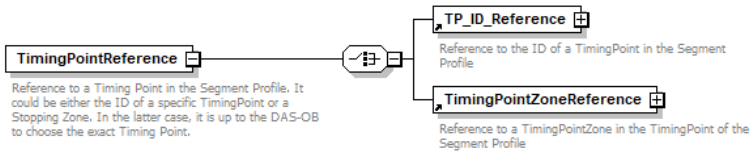
annotation

documentation
[SUBSET-126] Estimated arrival time to, at least, the next Timing Point

source

```
<xs:element name="TimingPointEstimation">
  <xs:annotation>
    <xs:documentation>[SUBSET-126] Estimated arrival time to, at least, the next Timing Point</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="SP_Zone"/>
    </xs:sequence>
    <xs:attribute ref="TP_ID" use="required"/>
    <xs:attribute ref="forecastedArrival" use="required"/>
  </xs:complexType>
</xs:element>
```

element **TimingPointReference**

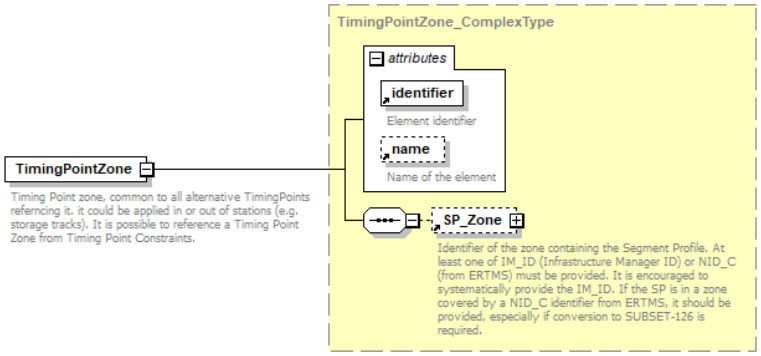
diagram	
properties	content complex
children	**TP_ID_Reference** **TimingPointZoneReference**
used by	element **TimingPointConstraints**
annotation	documentation Reference to a Timing Point in the Segment Profile. It could be either the ID of a specific TimingPoint or a Stopping Zone. In the latter case, it is up to the DAS-OB to choose the exact Timing Point.

source

```
<xs:element name="TimingPointReference">
 <xs:annotation>
  <xs:documentation>Reference to a Timing Point in the Segment Profile. It could be either the ID of a specific TimingPoint or a Stopping Zone. In the latter case, it is up to the DAS-OB to choose the exact Timing Point.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:choice>
   <xs:element ref="TP_ID_Reference"/>
   <xs:element ref="TimingPointZoneReference"/>
  </xs:choice>
 </xs:complexType>
</xs:element>
```

element **TimingPointZone**

diagram



type **TimingPointZone_ComplexType**

properties
content complex

children **SP_Zone**

used by
element **TimingPoint**

attributes

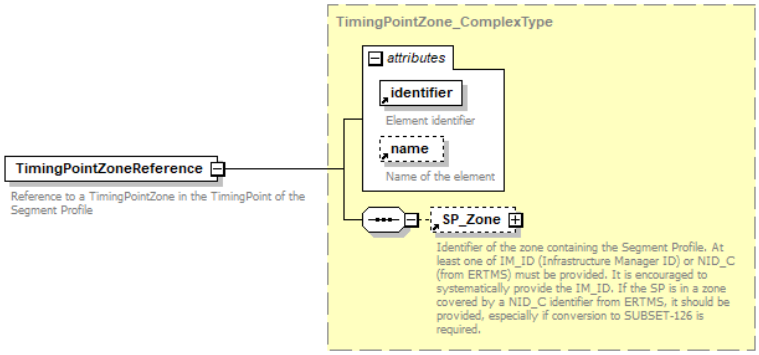
Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**	required			documentation Element identifier
name	**xs:string**				documentation Name of the element

annotation
documentation
Timing Point zone, common to all alternative TimingPoints referncing it. it could be applied in or out of stations (e.g. storage tracks). It is possible to reference a Timing Point Zone from Timing Point Constraints.

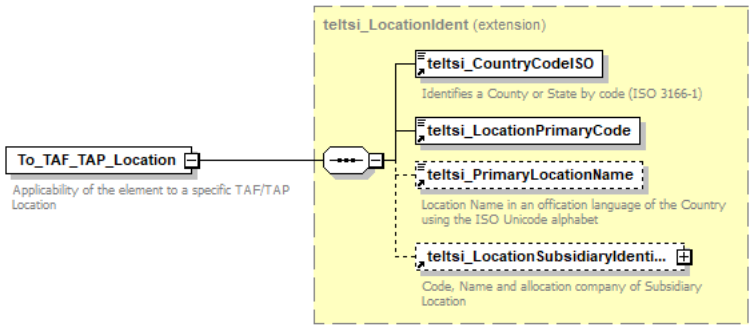
source

```
<xs:element name="TimingPointZone" type="TimingPointZone_ComplexType">
 <xs:annotation>
  <xs:documentation>Timing Point zone, common to all alternative TimingPoints referncing it. it could be applied in or out of stations (e.g. storage tracks). It is possible to reference a Timing Point Zone from Timing Point Constraints.</xs:documentation>
 </xs:annotation>
</xs:element>
```

element **TimingPointZoneReference**

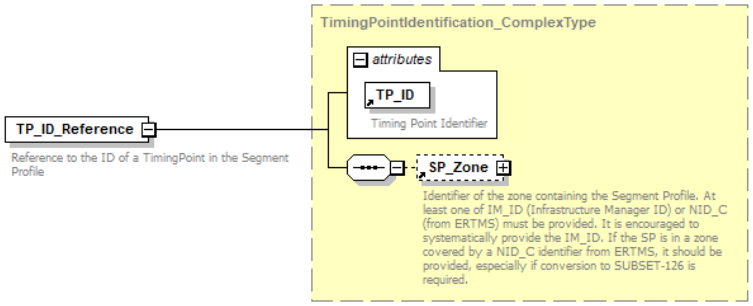
diagram	
type	**TimingPointZone_ComplexType**
properties	content complex
children	**SP_Zone**
used by	elements **TimingPointReference** **TrainCharacteristicsRef**
attributes	Name: identifier — Type: **xs:string** — Use: required — Default — Fixed — Annotation: documentation: Element identifier Name: name — Type: **xs:string** — Annotation: documentation: Name of the element
annotation	documentation Reference to a TimingPointZone in the TimingPoint of the Segment Profile
source	<xs:element name="TimingPointZoneReference" type="TimingPointZone_ComplexType"> <xs:annotation> <xs:documentation>Reference to a TimingPointZone in the TimingPoint of the Segment Profile</xs:documentation> </xs:annotation> </xs:element>

element **To_TAF_TAP_Location**

diagram	
type	extension of **teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	elements **CommunicationNetwork** **ContactList** **DecisiveGradientArea**

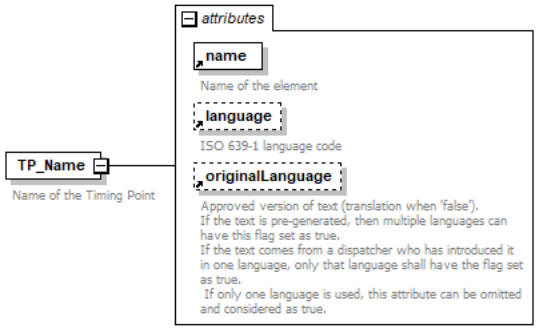
annotation	documentation Applicability of the element to a specific TAF/TAP Location
source	<xs:element name="To_TAF_TAP_Location"> <xs:annotation> <xs:documentation>Applicability of the element to a specific TAF/TAP Location</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="teltsi_LocationIdent"/> </xs:complexContent> </xs:complexType> </xs:element>

element **TP_ID_Reference**

diagram	
type	**TimingPointIdentification_ComplexType**
properties	content complex
children	**SP_Zone**
used by	elements **PassengerMovement** **TimingPointReference**
attributes	Name: TP_ID; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation documentation: Timing Point Identifier
annotation	documentation Reference to the ID of a TimingPoint in the Segment Profile
source	<xs:element name="TP_ID_Reference" type="TimingPointIdentification_ComplexType"> <xs:annotation> <xs:documentation>Reference to the ID of a TimingPoint in the Segment Profile</xs:documentation> </xs:annotation> </xs:element>

element **TP_Name**

diagram



properties

content complex

used by

element **TimingPoint**

attributes

Name	Type	Use	Default	Fixed	Annotation
name	**xs:string**	required			documentation Name of the element
language	**derived by: xs:string**				documentation ISO 639-1 language code
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

annotation

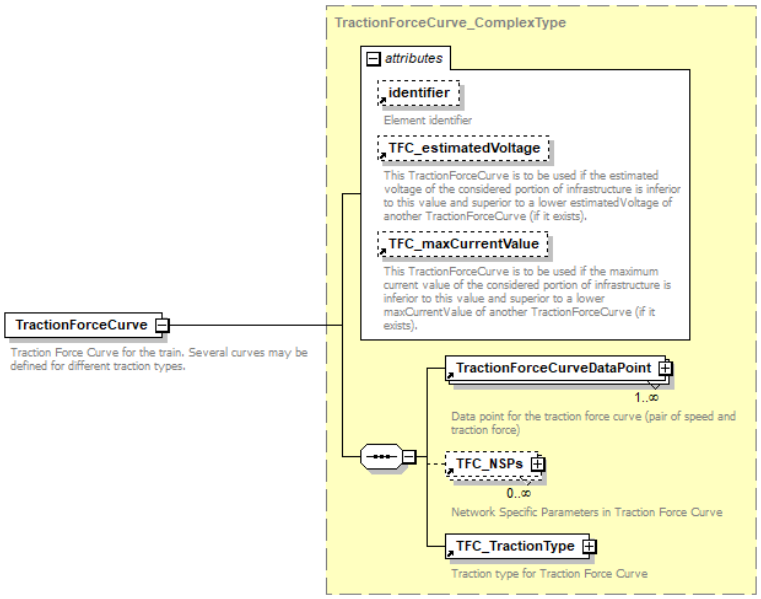
documentation
Name of the Timing Point

source

```
<xs:element name="TP_Name">
  <xs:annotation>
    <xs:documentation>Name of the Timing Point</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute ref="name" use="required"/>
    <xs:attribute ref="language"/>
    <xs:attribute ref="originalLanguage"/>
  </xs:complexType>
```

	</xs:element>

element **TractionForceCurve**

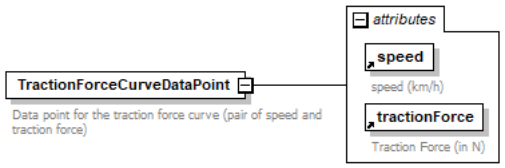
diagram	
type	**TractionForceCurve_ComplexType**
properties	content complex
children	**TractionForceCurveDataPoint** **TFC_NSPs** **TFC_TractionType**
used by	complexType **TC_Features_ComplexType**

attributes

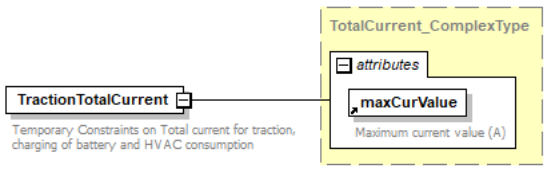
Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
TFC_estimatedVoltage	**voltageType**				documentation This TractionForceCurve is to be used if the estimated voltage of the considered portion of infrastructure is inferior to this value and superior to a lower estimatedVoltage of another TractionForceCurve (if it exists).
TFC_maxCurrentValue	**currentType**				documentation This TractionForceCurve is to be used if the maximum current value of the considered portion of infrastructure is inferior to this value and superior to a lower maxCurrentValue of another

	TractionForceCurve (if it exists).
annotation	documentation Traction Force Curve for the train. Several curves may be defined for different traction types.
source	<xs:element name="TractionForceCurve" type="TractionForceCurve_ComplexType"> <xs:annotation> <xs:documentation>Traction Force Curve for the train. Several curves may be defined for different traction types.</xs:documentation> </xs:annotation> </xs:element>

element **TractionForceCurveDataPoint**

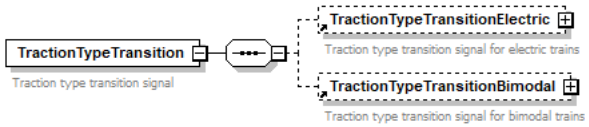
diagram	
properties	content complex
used by	complexType **TractionForceCurve_ComplexType**
attributes	Name: speed — Type: **speedType** — Use: required — Annotation: documentation: speed (km/h) Name: tractionForce — Type: **forceType** — Use: required — Annotation: documentation: Traction Force (in N)
annotation	documentation Data point for the traction force curve (pair of speed and traction force)
source	<xs:element name="TractionForceCurveDataPoint"> <xs:annotation> <xs:documentation>Data point for the traction force curve (pair of speed and traction force)</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="speed" use="required"/> <xs:attribute ref="tractionForce" use="required"/> </xs:complexType> </xs:element>

element **TractionTotalCurrent**

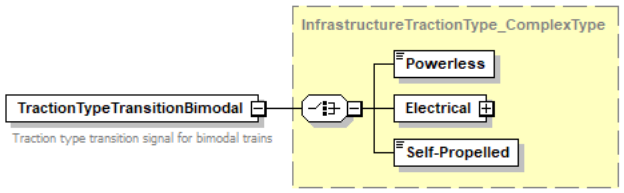
diagram	
type	**TotalCurrent_ComplexType**
properties	content complex
used by	element **TemporaryConstraints**
attributes	Name — Type — Use — Default — Fixed — Annotation

	maxCurValue **currentType** required documentation Maximum current value (A)
annotation	documentation Temporary Constraints on Total current for traction, charging of battery and HVAC consumption
source	<xs:element name="TractionTotalCurrent" type="TotalCurrent_ComplexType"> <xs:annotation> <xs:documentation>Temporary Constraints on Total current for traction, charging of battery and HVAC consumption</xs:documentation> </xs:annotation> </xs:element>

element **TractionTypeTransition**

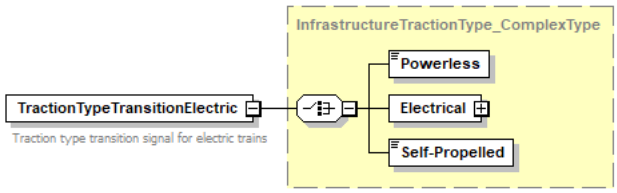
diagram	
properties	content complex
children	**TractionTypeTransitionElectric** **TractionTypeTransitionBimodal**
used by	element **SignalInformation**
annotation	documentation Traction type transition signal
source	<xs:element name="TractionTypeTransition"> <xs:annotation> <xs:documentation>Traction type transition signal</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="TractionTypeTransitionElectric" minOccurs="0"/> <xs:element ref="TractionTypeTransitionBimodal" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **TractionTypeTransitionBimodal**

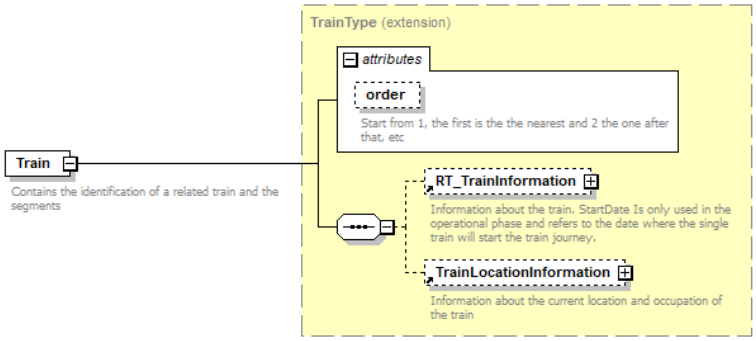
diagram	
type	**InfrastructureTractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Self-Propelled**
used by	element **TractionTypeTransition**
annotation	documentation Traction type transition signal for bimodal trains

source	<xs:element name="TractionTypeTransitionBimodal" type="InfrastructureTractionType_ComplexType"> <xs:annotation> <xs:documentation>Traction type transition signal for bimodal trains</xs:documentation> </xs:annotation> </xs:element>

element **TractionTypeTransitionElectric**

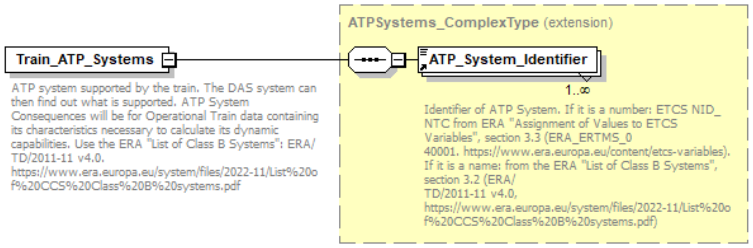
diagram	
type	**InfrastructureTractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Self-Propelled**
used by	element **TractionTypeTransition**
annotation	documentation Traction type transition signal for electric trains
source	<xs:element name="TractionTypeTransitionElectric" type="InfrastructureTractionType_ComplexType"> <xs:annotation> <xs:documentation>Traction type transition signal for electric trains</xs:documentation> </xs:annotation> </xs:element>

element **Train**

diagram	
type	extension of **TrainType**
properties	content complex
children	**RT_TrainInformation** **TrainLocationInformation**
used by	elements **TrainsAfterOwn** **TrainsBeforeOwn**
attributes	Name Type Use Default Fixed Annotation order documentation Start from 1, the first is the

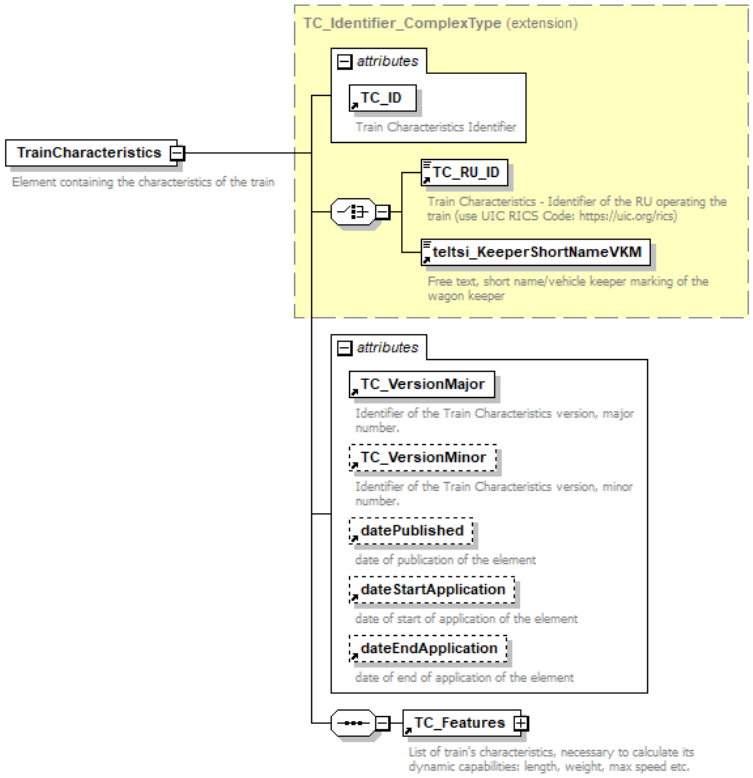
	the nearest and 2 the one after that, etc
annotation	documentation Contains the identification of a related train and the segments
source	<xs:element name="Train"> <xs:annotation> <xs:documentation>Contains the identification of a related train and the segments</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="TrainType"/> </xs:complexContent> </xs:complexType> </xs:element>

element **Train_ATP_Systems**

diagram	
type	extension of **ATPSystems_ComplexType**
properties	content complex
children	**ATP_System_Identifier**
used by	complexType **TC_Features_ComplexType**
annotation	documentation ATP system supported by the train. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities. Use the ERA "List of Class B Systems": ERA/ TD/2011-11 v4.0. https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf
source	<xs:element name="Train_ATP_Systems"> <xs:annotation> <xs:documentation>ATP system supported by the train. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities. Use the ERA "List of Class B Systems": ERA/ TD/2011-11 v4.0. https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="ATPSystems_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

element **TrainCharacteristics**

diagram



type: extension of **TC_Identifier_ComplexType**

properties: content complex

children: **TC_RU_ID** **teltsi_KeeperShortNameVKM** **TC_Features**

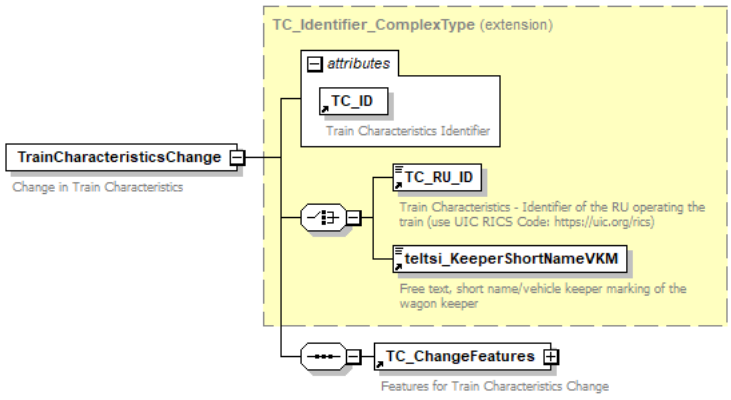
used by: element **G2B_ReplyPayload**

attributes

Name	Type	Use	Default	Fixed	Annotation
TC_ID	**xs:string**	required			documentation Train Characteristics Identifier
TC_VersionMajor	**xs:string**	required			documentation Identifier of the Train Characteristics version, major number.
TC_VersionMinor	**xs:string**				documentation Identifier of the Train Characteristics version, minor number.
datePublished	**UTCDateTimeType**				documentation date of publication of the element
dateStartApplication	**UTCDateTimeType**				documentation date of start of application of the element
dateEndApplication	**UTCDateTimeType**				documentation date of end of application of

	the element
annotation	documentation Element containing the characteristics of the train
source	<xs:element name="TrainCharacteristics"> <xs:annotation> <xs:documentation>Element containing the characteristics of the train</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="TC_Identifier_ComplexType"> <xs:sequence> <xs:element ref="TC_Features"/> </xs:sequence> <xs:attribute ref="TC_VersionMajor" use="required"/> <xs:attribute ref="TC_VersionMinor"/> <xs:attribute ref="datePublished"/> <xs:attribute ref="dateStartApplication"/> <xs:attribute ref="dateEndApplication"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>

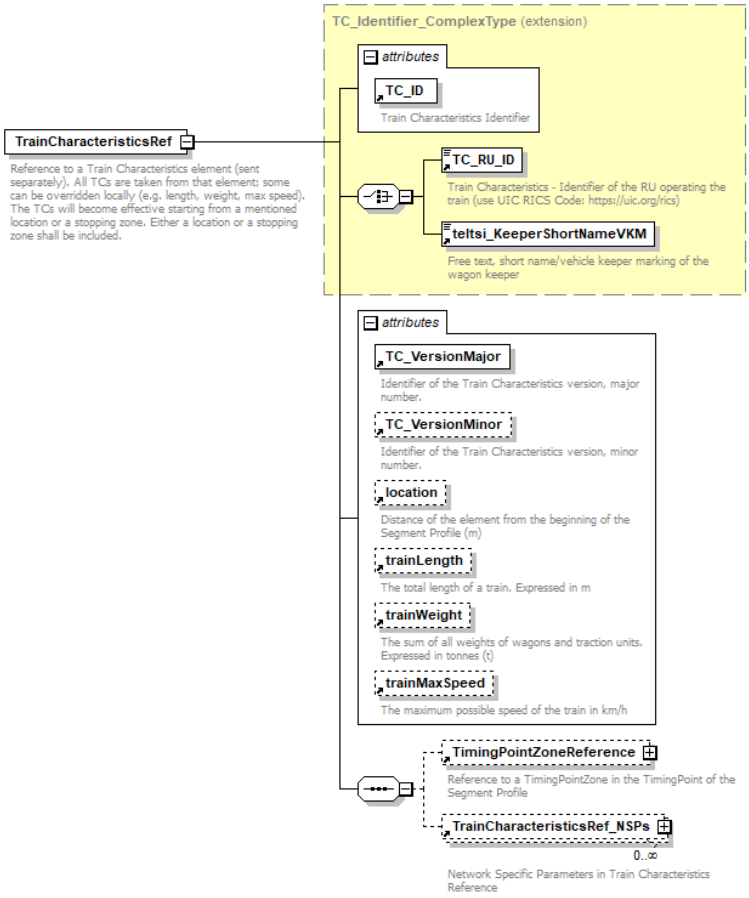
element **TrainCharacteristicsChange**

diagram	
type	extension of **TC_Identifier_ComplexType**
properties	content complex
children	**TC_RU_ID** **teltsi_KeeperShortNameVKM** **TC_ChangeFeatures**
used by	element **B2G_StatusReport**
attributes	Name: TC_ID; Type: **xs:string**; Use: required; Default: ; Fixed: ; Annotation: documentation Train Characteristics Identifier
annotation	documentation Change in Train Characteristics
source	<xs:element name="TrainCharacteristicsChange"> <xs:annotation> <xs:documentation>Change in Train Characteristics</xs:documentation> </xs:annotation>

```
  <xs:complexType>
   <xs:complexContent>
    <xs:extension base="TC_Identifier_ComplexType">
     <xs:sequence>
      <xs:element ref="TC_ChangeFeatures"/>
     </xs:sequence>
    </xs:extension>
   </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **TrainCharacteristicsRef**

diagram



type	extension of **TC_Identifier_ComplexType**
properties	content complex
children	**TC_RU_ID** **teltsi_KeeperShortNameVKM** **TimingPointZoneReference** **TrainCharacteristicsRef_NSPs**
used by	element **SegmentProfileReference**

attributes

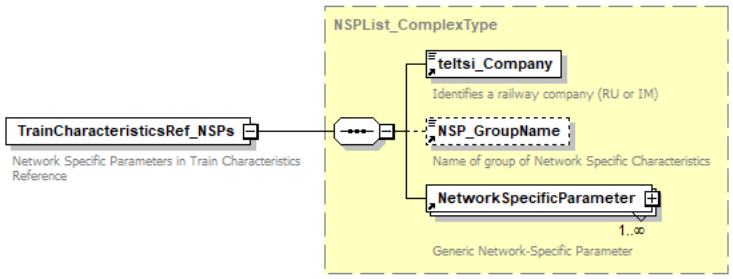
Name	Type	Use	Default	Fixed	Annotation
TC_ID	**xs:string**	required			documentation Train Characteristics Identifier
TC_VersionMajor	**xs:string**	required			documentation Identifier of the

	TC_VersionMinor	**xs:string**	Train Characteristics version, major number. documentation Identifier of the Train Characteristics version, minor number.
	location	**distanceType**	documentation Distance of the element from the beginning of the Segment Profile (m)
	trainLength	**xs:integer**	documentation The total length of a train. Expressed in m
	trainWeight	**xs:int**	documentation The sum of all weights of wagons and traction units. Expressed in tonnes (t)
	trainMaxSpeed	**xs:integer**	documentation The maximum possible speed of the train in km/h
annotation	documentation Reference to a Train Characteristics element (sent separately). All TCs are taken from that element; some can be overridden locally (e.g. length, weight, max speed). The TCs will become effective starting from a mentioned location or a stopping zone. Either a location or a stopping zone shall be included.		

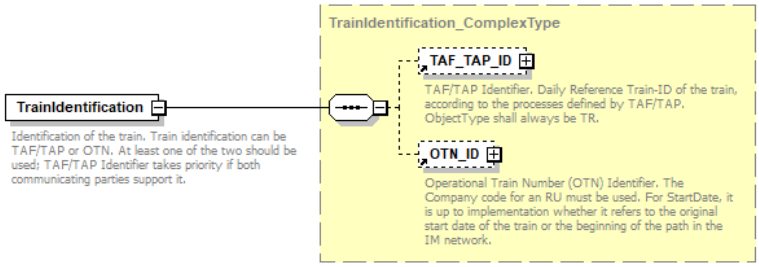
source

```
<xs:element name="TrainCharacteristicsRef">
  <xs:annotation>
    <xs:documentation>Reference to a Train Characteristics element (sent separately). All TCs are taken from that element; some can be overridden locally (e.g. length, weight, max speed). The TCs will become effective starting from a mentioned location or a stopping zone. Either a location or a stopping zone shall be included.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="TC_Identifier_ComplexType">
        <xs:sequence>
          <xs:element ref="TimingPointZoneReference" minOccurs="0"/>
          <xs:element ref="TrainCharacteristicsRef_NSPs" minOccurs="0" maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute ref="TC_VersionMajor" use="required"/>
        <xs:attribute ref="TC_VersionMinor"/>
        <xs:attribute ref="location"/>
        <xs:attribute ref="trainLength"/>
        <xs:attribute ref="trainWeight"/>
        <xs:attribute ref="trainMaxSpeed"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

element **TrainCharacteristicsRef_NSPs**

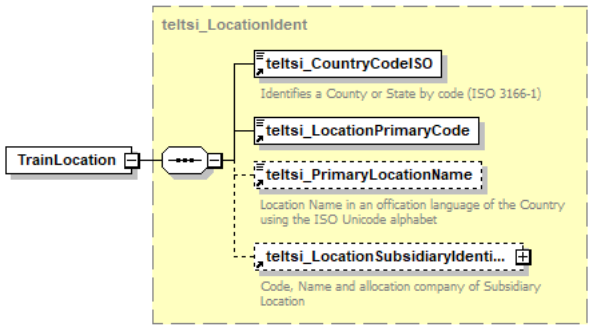
diagram	
type	**NSPList_ComplexType**
properties	content complex
children	**teltsi_Company** **NSP_GroupName** **NetworkSpecificParameter**
used by	element **TrainCharacteristicsRef**
annotation	documentation Network Specific Parameters in Train Characteristics Reference
source	<xs:element name="TrainCharacteristicsRef_NSPs" type="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Network Specific Parameters in Train Characteristics Reference</xs:documentation> </xs:annotation> </xs:element>

element **TrainIdentification**

diagram	
type	**TrainIdentification_ComplexType**
properties	content complex
children	**TAF_TAP_ID** **OTN_ID**
used by	elements **B2G_StatusReport** **C_DAS_C_Advice** **C_DAS_C_AdviceRequest** **DAS_DrivingModeRequest** **ForceDrivingModeChange** **ForceDrivingModeChangeRequest** **HandshakeRequest** **JourneyProfile** **JP_Request** **NetworkSpecificEvent** **NetworkSpecificEventRequest** **OwnTrain** **PlaintextMessage** **PlaintextMessageRequest** **PositionSpeed** **PositionSpeedRequest** **RelatedTrainInformationRequest** **Reported_DAS_DrivingMode** **SessionTermination** **SessionTerminationRequest** **StatusReportRequest**
annotation	documentation Identification of the train. Train identification can be TAF/TAP or OTN. At least one of the two should be used; TAF/TAP Identifier takes priority if both communicating parties support it.
source	<xs:element name="TrainIdentification" type="TrainIdentification_ComplexType"> <xs:annotation> <xs:documentation>Identification of the train. Train identification can be TAF/TAP or OTN. At least one of the two should be used; TAF/TAP Identifier takes priority if both communicating parties

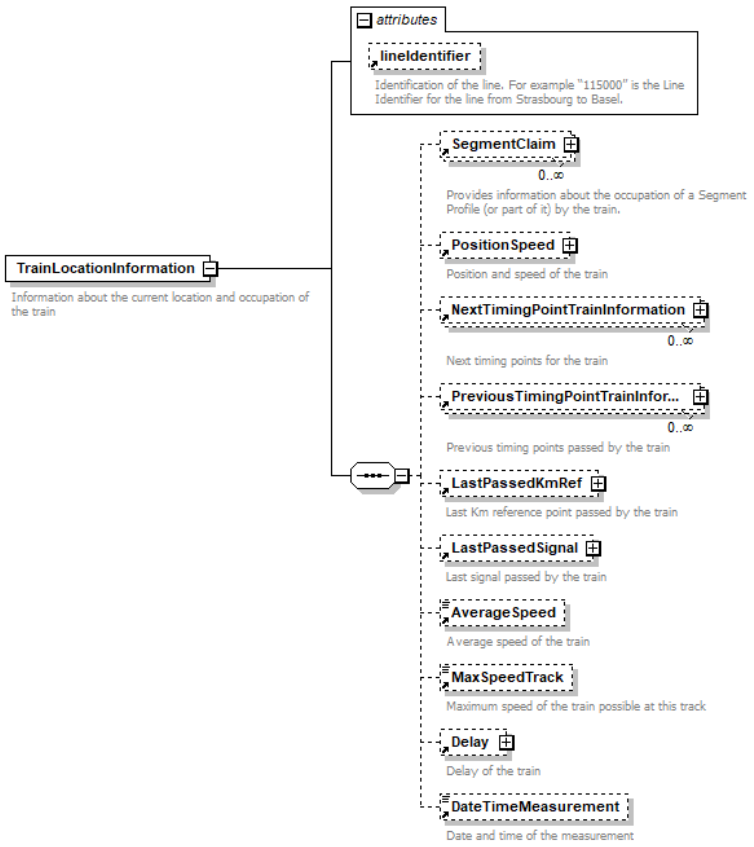
	support it.</xs:documentation> </xs:annotation> </xs:element>

element **TrainLocation**

diagram	
type	**teltsi_LocationIdent**
properties	content complex
children	**teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**
used by	elements **DelayNotice** **TrainReadyStatus**
source	<xs:element name="TrainLocation" type="teltsi_LocationIdent"/>

element **TrainLocationInformation**

diagram



properties

content complex

children

SegmentClaim PositionSpeed NextTimingPointTrainInformation PreviousTimingPointTrainInformation LastPassedKmRef LastPassedSignal AverageSpeed MaxSpeedTrack Delay DateTimeMeasurement

used by

element OwnTrain
complexType TrainType

attributes

Name	Type	Use	Default	Fixed	Annotation
lineIdentifier	**xs:string**				documentation Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.

annotation

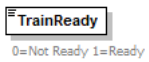
documentation
Information about the current location and occupation of the train

source

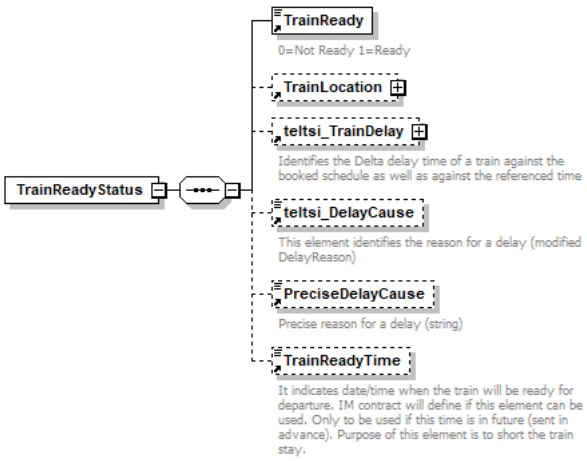
```
<xs:element name="TrainLocationInformation">
 <xs:annotation>
  <xs:documentation>Information about the current location and occupation of the train</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:sequence>
   <xs:element ref="SegmentClaim" minOccurs="0" maxOccurs="unbounded"/>
   <xs:element ref="PositionSpeed" minOccurs="0"/>
   <xs:element ref="NextTimingPointTrainInformation" minOccurs="0" maxOccurs="unbounded"/>
```

```
      <xs:element ref="PreviousTimingPointTrainInformation" minOccurs="0"
maxOccurs="unbounded"/>
      <xs:element ref="LastPassedKmRef" minOccurs="0"/>
      <xs:element ref="LastPassedSignal" minOccurs="0"/>
      <xs:element ref="AverageSpeed" minOccurs="0"/>
      <xs:element ref="MaxSpeedTrack" minOccurs="0"/>
      <xs:element ref="Delay" minOccurs="0"/>
      <xs:element ref="DateTimeMeasurement" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute ref="lineIdentifier"/>
  </xs:complexType>
</xs:element>
```

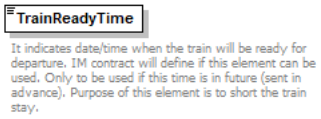
element **TrainReady**

diagram	
type	restriction of **xs:integer**
properties	content simple
used by	element **TrainReadyStatus**
facets	Kind Value Annotation enumeration 0 enumeration 1
annotation	documentation 0=Not Ready 1=Ready
source	<xs:element name="TrainReady"> <xs:annotation> <xs:documentation>0=Not Ready 1=Ready</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:integer"> <xs:enumeration value="0"/> <xs:enumeration value="1"/> </xs:restriction> </xs:simpleType> </xs:element>

element **TrainReadyStatus**

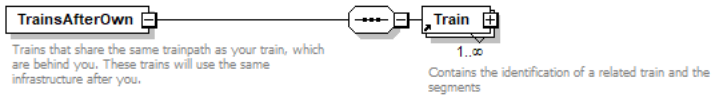
diagram	
properties	content complex
children	**TrainReady** **TrainLocation** **teltsi_TrainDelay** **teltsi_DelayCause** **PreciseDelayCause** **TrainReadyTime**
used by	element **B2G_StatusReport**
source	<xs:element name="TrainReadyStatus"> <xs:complexType> <xs:sequence> <xs:element ref="TrainReady"/> <xs:element ref="TrainLocation" minOccurs="0"/> <xs:element ref="teltsi_TrainDelay" minOccurs="0"/> <xs:element ref="teltsi_DelayCause" minOccurs="0"/> <xs:element ref="PreciseDelayCause" minOccurs="0"/> <xs:element ref="TrainReadyTime" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **TrainReadyTime**

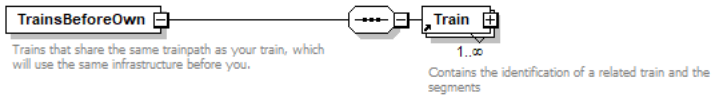
diagram	
type	**xs:dateTime**
properties	content simple
used by	element **TrainReadyStatus**
annotation	documentation It indicates date/time when the train will be ready for departure. IM contract will define if this element can be used. Only to be used if this time is in future (sent in advance). Purpose of this element is to short the train stay.
source	<xs:element name="TrainReadyTime" type="xs:dateTime"> <xs:annotation> <xs:documentation>It indicates date/time when the train will be ready for departure. IM contract will define if this element can be used. Only to be used if this time is in future (sent in advance). Purpose of this element is to short the train stay. </xs:documentation> </xs:annotation>

	</xs:element>

element **TrainsAfterOwn**

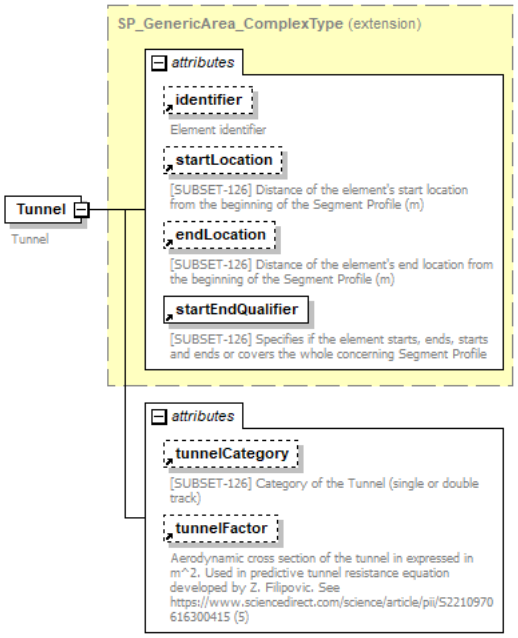
diagram	
properties	content complex
children	**Train**
used by	element **RelatedTrainInformation**
annotation	documentation Trains that share the same trainpath as your train, which are behind you. These trains will use the same infrastructure after you.
source	<xs:element name="TrainsAfterOwn"> <xs:annotation> <xs:documentation>Trains that share the same trainpath as your train, which are behind you. These trains will use the same infrastructure after you.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="Train" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **TrainsBeforeOwn**

diagram	
properties	content complex
children	**Train**
used by	element **RelatedTrainInformation**
annotation	documentation Trains that share the same trainpath as your train, which will use the same infrastructure before you.
source	<xs:element name="TrainsBeforeOwn"> <xs:annotation> <xs:documentation>Trains that share the same trainpath as your train, which will use the same infrastructure before you.</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="Train" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </xs:element>

element **Tunnel**

diagram



type: extension of **SP_GenericArea_ComplexType**

properties: content complex

used by: element **SP_Areas**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
tunnelCategory	**derived by: xs:string**				documentation [SUBSET-126] Category of the Tunnel (single or double track)
tunnelFactor	**xs:double**				documentation Aerodynamic cross section of the tunnel in expressed in m^2. Used in predictive tunnel resistance equation developed by Z. Filipovic. See https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)

annotation: documentation
Tunnel

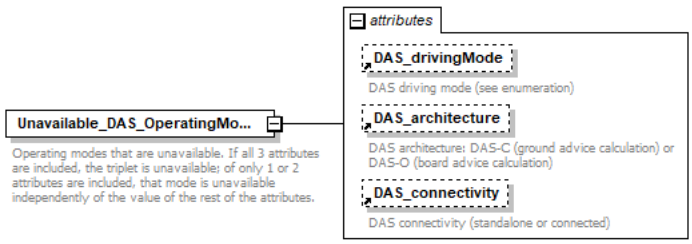
source

```
<xs:element name="Tunnel">
 <xs:annotation>
  <xs:documentation>Tunnel</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericArea_ComplexType">
```

```
      <xs:attribute ref="tunnelCategory"/>
      <xs:attribute ref="tunnelFactor"/>
    </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **Unavailable_DAS_OperatingModes**

diagram



properties: content complex

used by: elements **DAS_ModesUnavailabilityArea** **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
DAS_drivingMode	**derived by: xs:string**				documentation DAS driving mode (see enumeration)
DAS_architecture	**derived by: xs:string**				documentation DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation)
DAS_connectivity	**derived by: xs:string**				documentation DAS connectivity (standalone or connected)

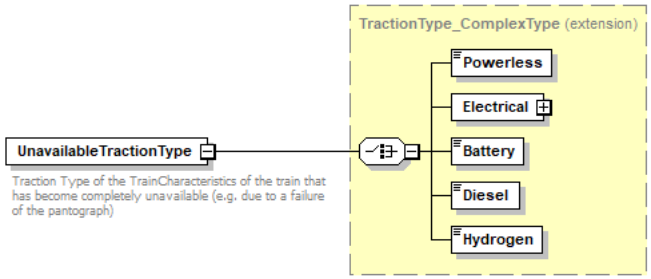
annotation

documentation
Operating modes that are unavailable. If all 3 attributes are included, the triplet is unavailable; of only 1 or 2 attributes are included, that mode is unavailable independently of the value of the rest of the attributes.

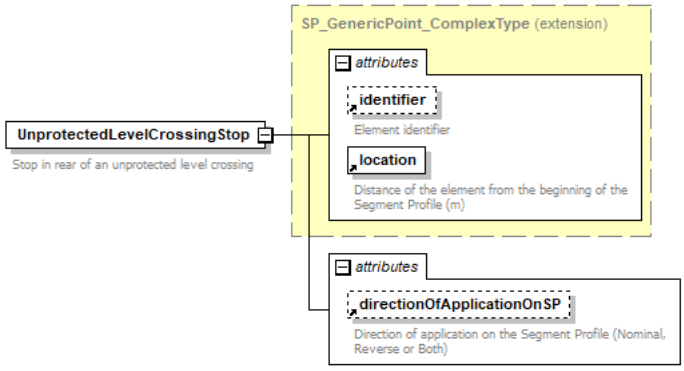
source

```
<xs:element name="Unavailable_DAS_OperatingModes">
 <xs:annotation>
  <xs:documentation>Operating modes that are unavailable. If all 3 attributes are included, the triplet is unavailable; of only 1 or 2 attributes are included, that mode is unavailable independently of the value of the rest of the attributes.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:attribute ref="DAS_drivingMode"/>
  <xs:attribute ref="DAS_architecture"/>
  <xs:attribute ref="DAS_connectivity"/>
 </xs:complexType>
</xs:element>
```

element **UnavailableTractionType**

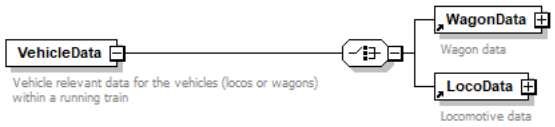
diagram	
type	extension of **TractionType_ComplexType**
properties	content complex
children	**Powerless** **Electrical** **Battery** **Diesel** **Hydrogen**
used by	element **TC_ChangeFeatures**
annotation	documentation Traction Type of the TrainCharacteristics of the train that has become completely unavailable (e.g. due to a failure of the pantograph)
source	<xs:element name="UnavailableTractionType"> <xs:annotation> <xs:documentation>Traction Type of the TrainCharacteristics of the train that has become completely unavailable (e.g. due to a failure of the pantograph)</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="TractionType_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

element **UnprotectedLevelCrossingStop**

diagram	
type	extension of **SP_GenericPoint_ComplexType**
properties	content complex
used by	element **SP_Points**
attributes	Name: identifier; Type: **xs:string**; Use: ; Default: ; Fixed: ; Annotation: documentation Element identifier

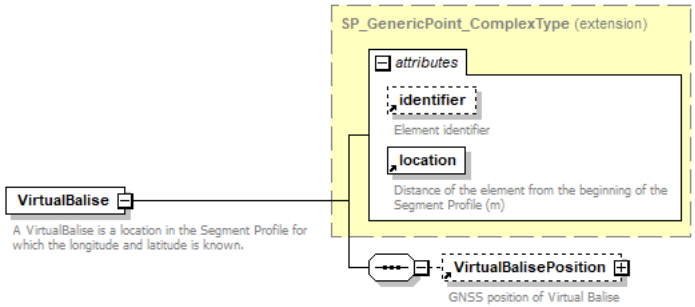
	location	**distanceType**	required	documentation Distance of the element from the beginning of the Segment Profile (m)
	directionOfApplicationOnSP	**derived by: xs:string**		documentation Direction of application on the Segment Profile (Nominal, Reverse or Both)
annotation	documentation Stop in rear of an unprotected level crossing			
source	<xs:element name="UnprotectedLevelCrossingStop"> <xs:annotation> <xs:documentation>Stop in rear of an unprotected level crossing</xs:documentation> </xs:annotation> <xs:complexType> <xs:complexContent> <xs:extension base="SP_GenericPoint_ComplexType"> <xs:attribute ref="directionOfApplicationOnSP"/> </xs:extension> </xs:complexContent> </xs:complexType> </xs:element>			

element **VehicleData**

diagram	
properties	content complex
children	**WagonData** **LocoData**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Vehicle relevant data for the vehicles (locos or wagons) within a running train
source	<xs:element name="VehicleData"> <xs:annotation> <xs:documentation>Vehicle relevant data for the vehicles (locos or wagons) within a running train</xs:documentation> </xs:annotation> <xs:complexType> <xs:choice> <xs:element ref="WagonData"/> <xs:element ref="LocoData"/> </xs:choice> </xs:complexType> </xs:element>

element **VirtualBalise**

diagram



type: extension of **SP_GenericPoint_ComplexType**

properties: content complex

children: **VirtualBalisePosition**

used by: element **SP_Points**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation: Element identifier
location	**distanceType**	required			documentation: Distance of the element from the beginning of the Segment Profile (m)

annotation

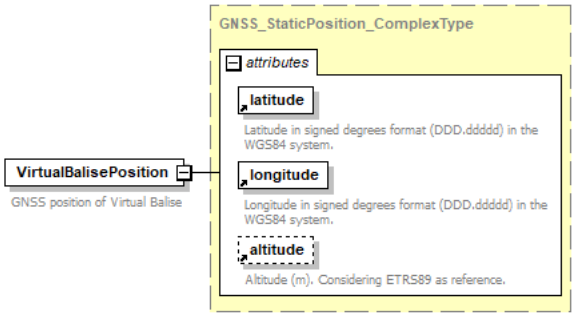
documentation
A VirtualBalise is a location in the Segment Profile for which the longitude and latitude is known.

source

```
<xs:element name="VirtualBalise">
 <xs:annotation>
  <xs:documentation>A VirtualBalise is a location in the Segment Profile for which the longitude and latitude is known.</xs:documentation>
 </xs:annotation>
 <xs:complexType>
  <xs:complexContent>
   <xs:extension base="SP_GenericPoint_ComplexType">
    <xs:sequence>
     <xs:element ref="VirtualBalisePosition" minOccurs="0"/>
    </xs:sequence>
   </xs:extension>
  </xs:complexContent>
 </xs:complexType>
</xs:element>
```

element **VirtualBalisePosition**

diagram



type **GNSS_StaticPosition_ComplexType**

properties
content complex

used by
element **VirtualBalise**

attributes

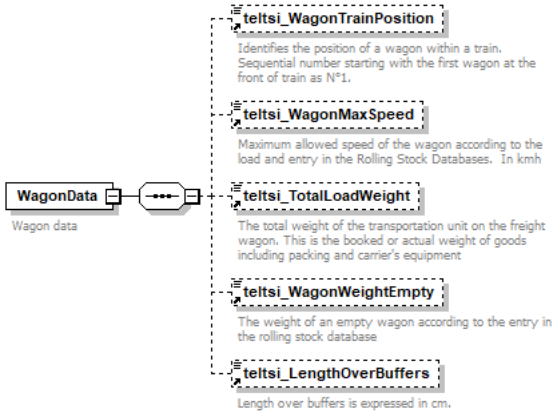
Name	Type	Use	Default	Fixed	Annotation
latitude	**xs:decimal**	required			documentation Latitude in signed degrees format (DDD.ddddd) in the WGS84 system.
longitude	**xs:decimal**	required			documentation Longitude in signed degrees format (DDD.ddddd) in the WGS84 system.
altitude	**xs:decimal**				documentation Altitude (m). Considering ETRS89 as reference.

annotation
documentation
GNSS position of Virtual Balise

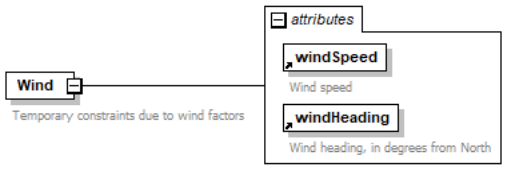
source

```
<xs:element name="VirtualBalisePosition" type="GNSS_StaticPosition_ComplexType">
  <xs:annotation>
    <xs:documentation>GNSS position of Virtual Balise</xs:documentation>
  </xs:annotation>
</xs:element>
```

element **WagonData**

diagram	
properties	content complex
children	**teltsi_WagonTrainPosition** **teltsi_WagonMaxSpeed** **teltsi_TotalLoadWeight** **teltsi_WagonWeightEmpty** **teltsi_LengthOverBuffers**
used by	element **VehicleData**
annotation	documentation Wagon data
source	<xs:element name="WagonData"> <xs:annotation> <xs:documentation>Wagon data</xs:documentation> </xs:annotation> <xs:complexType> <xs:sequence> <xs:element ref="teltsi_WagonTrainPosition" minOccurs="0"/> <xs:element ref="teltsi_WagonMaxSpeed" minOccurs="0"/> <xs:element ref="teltsi_TotalLoadWeight" minOccurs="0"/> <xs:element ref="teltsi_WagonWeightEmpty" minOccurs="0"/> <xs:element ref="teltsi_LengthOverBuffers" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element>

element **Wind**

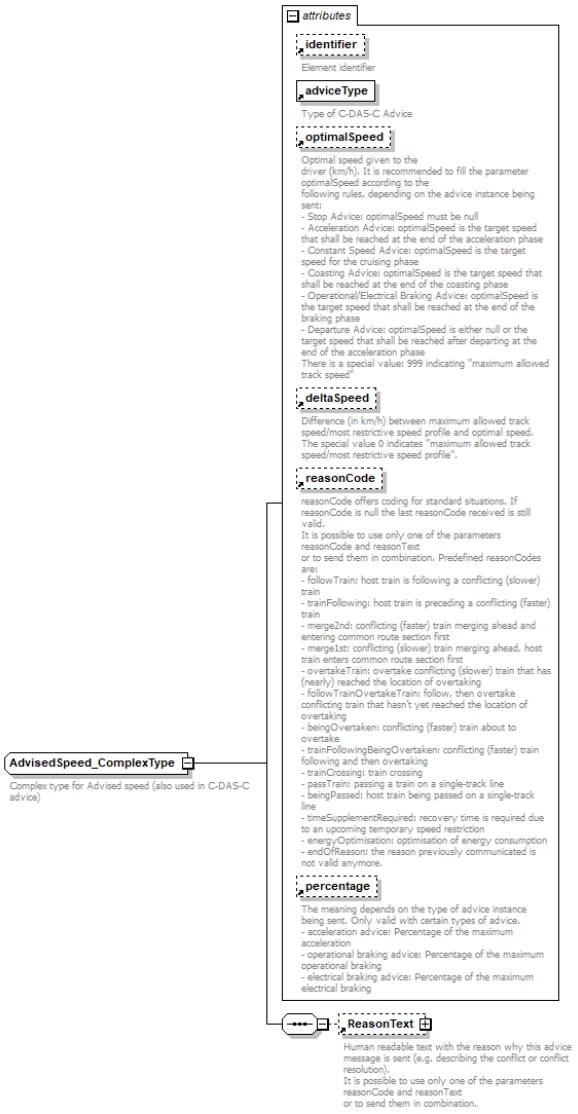
diagram	
properties	content complex
used by	element **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
windSpeed	**speedType**	required			documentation Wind speed
windHeading	**xs:unsignedShort**	required			documentation Wind heading, in degrees from North

annotation	documentation Temporary constraints due to wind factors
source	<xs:element name="Wind"> <xs:annotation> <xs:documentation>Temporary constraints due to wind factors</xs:documentation> </xs:annotation> <xs:complexType> <xs:attribute ref="windSpeed" use="required"/> <xs:attribute ref="windHeading" use="required"/> </xs:complexType> </xs:element>

complexType **AdvisedSpeed_ComplexType**

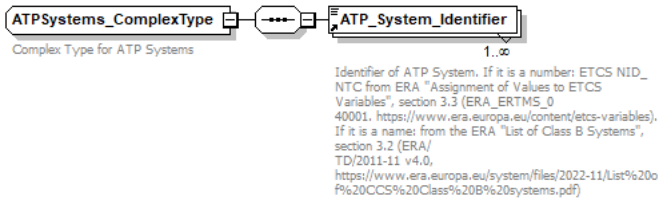
diagram	
children	**ReasonText**
used by	elements **AdvisedSpeed** **C_DAS_C_Advice**
attributes	Name Type Use Default Fixed Annotation

	identifier	**xs:string**		documentation Element identifier
	adviceType	**derived by:** **xs:string**	required	documentation Type of C-DAS-C Advice
	optimalSpeed	**speedType**		documentation Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on the advice instance being sent: - Stop Advice: optimalSpeed must be null - Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase - Constant Speed Advice: optimalSpeed is the target speed for the cruising phase - Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase - Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase - Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase There is a special value: 999 indicating "maximum allowed track speed"
	deltaSpeed	**speedType**		documentation Difference (in km/h) between maximum allowed track speed/most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed/most restrictive speed profile".
	reasonCode	**derived by:** **xs:string**		documentation reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid. It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are: - followTrain: host train is following a conflicting (slower) train - trainFollowing: host train is preceding a conflicting (faster) train - merge2nd: conflicting (faster) train merging ahead and entering common route

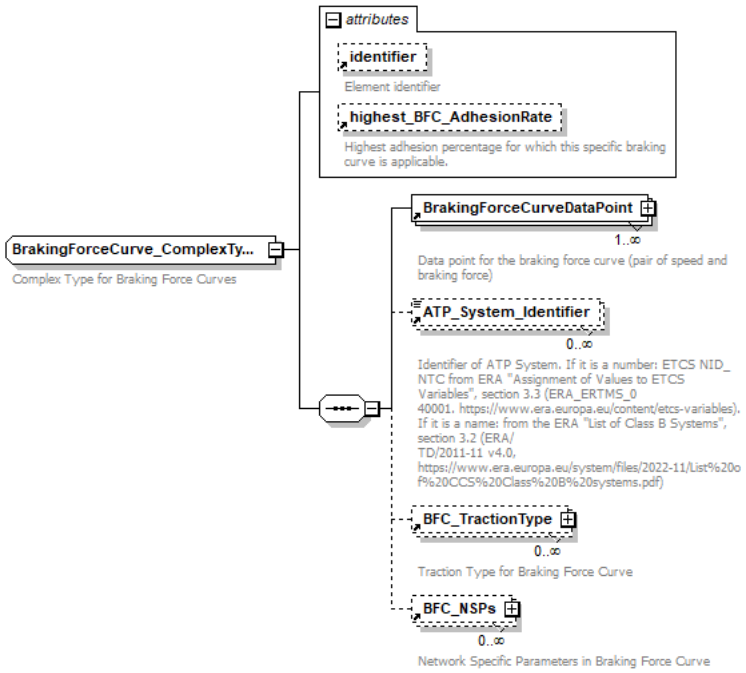
		section first - merge1st: conflicting (slower) train merging ahead, host train enters common route section first - overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking - followTrainOvertakeTrain: follow, then overtake conflicting train that hasn't yet reached the location of overtaking - beingOvertaken: conflicting (faster) train about to overtake - trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking - trainCrossing: train crossing - passTrain: passing a train on a single-track line - beingPassed: host train being passed on a single-track line - timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction - energyOptimisation: optimisation of energy consumption - endOfReason: the reason previously communicated is not valid anymore.
	percentage **percentageType**	documentation The meaning depends on the type of advice instance being sent. Only valid with certain types of advice. - acceleration advice: Percentage of the maximum acceleration - operational braking advice: Percentage of the maximum operational braking - electrical braking advice: Percentage of the maximum electrical braking
annotation	documentation Complex type for Advised speed (also used in C-DAS-C advice)	
source	<xs:complexType name="AdvisedSpeed_ComplexType"> <xs:annotation> <xs:documentation>Complex type for Advised speed (also used in C-DAS-C advice)</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="ReasonText" minOccurs="0"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="adviceType" use="required"/> <xs:attribute ref="optimalSpeed"/> <xs:attribute ref="deltaSpeed"/> <xs:attribute ref="reasonCode"/> <xs:attribute ref="percentage"/>	

	</xs:complexType>

complexType **ATPSystems_ComplexType**

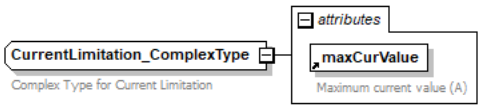
diagram	
children	ATP_System_Identifier
used by	elements ATP_SystemsChange ATP_SystemsStart Signal_ATP_Systems Train_ATP_Systems
annotation	documentation Complex Type for ATP Systems
source	<xs:complexType name="ATPSystems_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for ATP Systems</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="ATP_System_Identifier" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType>

complexType **BrakingForceCurve_ComplexType**

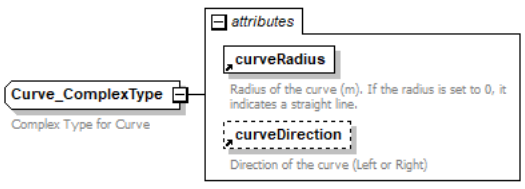
diagram	
children	BrakingForceCurveDataPoint ATP_System_Identifier BFC_TractionType BFC_NSPs
used by	element BrakingForceCurve
attributes	Name Type Use Default Fixed Annotation

	identifier	**xs:string**	documentation Element identifier
	highest_BFC_AdhesionRate	**percentageType**	documentation Highest adhesion percentage for which this specific braking curve is applicable.
annotation	documentation Complex Type for Braking Force Curves		
source	<xs:complexType name="BrakingForceCurve_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for Braking Force Curves</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="BrakingForceCurveDataPoint" maxOccurs="unbounded"/> <xs:element ref="ATP_System_Identifier" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="BFC_TractionType" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="BFC_NSPs" minOccurs="0" maxOccurs="unbounded"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="highest_BFC_AdhesionRate"/> </xs:complexType>		

complexType **CurrentLimitation_ComplexType**

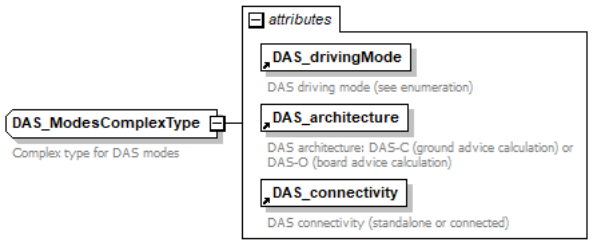
diagram						
used by	elements **CurrentLimitationChange** **CurrentLimitationStart**					
attributes	Name maxCurValue	Type **currentType**	Use required	Default	Fixed	Annotation documentation Maximum current value (A)
annotation	documentation Complex Type for Current Limitation					
source	<xs:complexType name="CurrentLimitation_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for Current Limitation</xs:documentation> </xs:annotation> <xs:attribute ref="maxCurValue" use="required"/> </xs:complexType>					

complexType **Curve_ComplexType**

diagram	

used by	elements **CurveChange CurveStart**
attributes	Name / Type / Use / Default / Fixed / Annotation curveRadius — **xs:unsignedShort** — required — documentation: Radius of the curve (m). If the radius is set to 0, it indicates a straight line. curveDirection — **derived by: xs:string** — documentation: Direction of the curve (Left or Right)
annotation	documentation Complex Type for Curve
source	<xs:complexType name="Curve_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for Curve</xs:documentation> </xs:annotation> <xs:attribute ref="curveRadius" use="required"/> <xs:attribute ref="curveDirection"/> </xs:complexType>

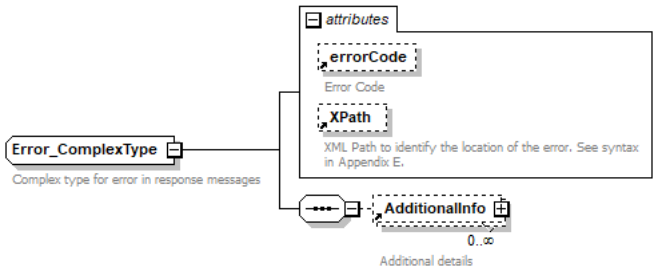
complexType **DAS_ModesComplexType**

diagram	
used by	elements **DAS_Mode DAS_OperatingModesSupported**
attributes	Name / Type / Use / Default / Fixed / Annotation DAS_drivingMode — **derived by: xs:string** — required — documentation: DAS driving mode (see enumeration) DAS_architecture — **derived by: xs:string** — required — documentation: DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation) DAS_connectivity — **derived by: xs:string** — required — documentation: DAS connectivity (standalone or connected)
annotation	documentation Complex type for DAS modes
source	<xs:complexType name="DAS_ModesComplexType"> <xs:annotation> <xs:documentation>Complex type for DAS modes</xs:documentation> </xs:annotation>

```
  <xs:attribute ref="DAS_drivingMode" use="required"/>
  <xs:attribute ref="DAS_architecture" use="required"/>
  <xs:attribute ref="DAS_connectivity" use="required"/>
</xs:complexType>
```

complexType **Error_ComplexType**

diagram



children **AdditionalInfo**

used by elements **B2G_Error** **G2B_Error**

attributes

Name	Type	Use	Default	Fixed	Annotation
errorCode	**xs:string**				documentation Error Code
XPath	**xs:string**				documentation XML Path to identify the location of the error. See syntax in Appendix E.

annotation

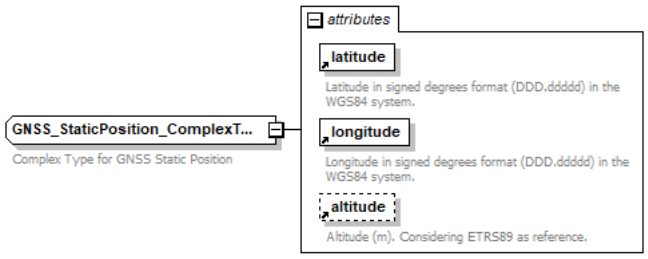
documentation
Complex type for error in response messages

source

```
<xs:complexType name="Error_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for error in response messages</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="AdditionalInfo" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="errorCode"/>
  <xs:attribute ref="XPath"/>
</xs:complexType>
```

complexType **GNSS_StaticPosition_ComplexType**

diagram



used by element **VirtualBalisePosition**

attributes

Name	Type	Use	Default	Fixed	Annotation
latitude	**xs:decimal**	required			documentation: Latitude in signed degrees format (DDD.ddddd) in the WGS84 system.
longitude	**xs:decimal**	required			documentation: Longitude in signed degrees format (DDD.ddddd) in the WGS84 system.
altitude	**xs:decimal**				documentation: Altitude (m). Considering ETRS89 as reference.

annotation

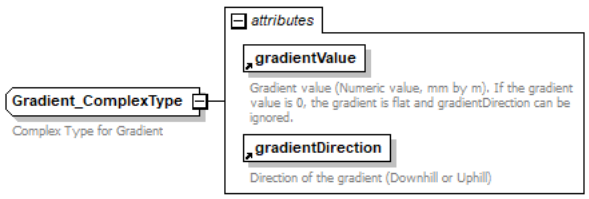
documentation
Complex Type for GNSS Static Position

source

```
<xs:complexType name="GNSS_StaticPosition_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for GNSS Static Position</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="latitude" use="required"/>
  <xs:attribute ref="longitude" use="required"/>
  <xs:attribute ref="altitude"/>
</xs:complexType>
```

complexType **Gradient_ComplexType**

diagram



used by

elements **GradientChange** **GradientStart** **GradientSteepestChange** **GradientSteepestStart**

attributes

Name	Type	Use	Default	Fixed	Annotation
gradientValue	**gradientValueType**	required			documentation: Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
gradientDirection	**gradientDirectionType**	required			documentation: Direction of the gradient (Downhill or Uphill)

annotation

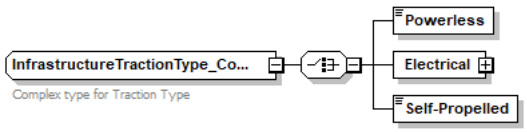
documentation
Complex Type for Gradient

source

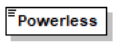
```
<xs:complexType name="Gradient_ComplexType">
  <xs:annotation>
```

	<xs:documentation>Complex Type for Gradient</xs:documentation> </xs:annotation> <xs:attribute ref="gradientValue" use="required"/> <xs:attribute ref="gradientDirection" use="required"/> </xs:complexType>

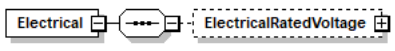
complexType **InfrastructureTractionType_ComplexType**

diagram	
children	**Powerless** **Electrical** **Self-Propelled**
used by	elements **TractionTypeTransitionBimodal** **TractionTypeTransitionElectric**
annotation	documentation Complex type for Traction Type
source	<xs:complexType name="InfrastructureTractionType_ComplexType"> <xs:annotation> <xs:documentation>Complex type for Traction Type</xs:documentation> </xs:annotation> <xs:choice> <xs:element name="Powerless"/> <xs:element name="Electrical"> <xs:complexType> <xs:sequence> <xs:element name="ElectricalRatedVoltage" type="Voltage_ComplexType" minOccurs="0"/> </xs:sequence> </xs:complexType> </xs:element> <xs:element name="Self-Propelled"/> </xs:choice> </xs:complexType>

element **InfrastructureTractionType_ComplexType/Powerless**

diagram	
source	<xs:element name="Powerless"/>

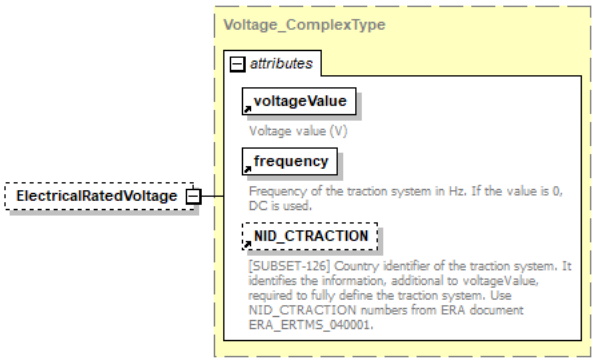
element **InfrastructureTractionType_ComplexType/Electrical**

diagram	
properties	content complex
children	**ElectricalRatedVoltage**
source	<xs:element name="Electrical"> <xs:complexType> <xs:sequence> <xs:element name="ElectricalRatedVoltage" type="Voltage_ComplexType" minOccurs="0"/>

```
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

element **InfrastructureTractionType_ComplexType/Electrical/ElectricalRatedVoltage**

diagram



type **Voltage_ComplexType**

properties

minOcc	0
maxOcc	1
content	complex

attributes

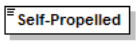
Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation: Voltage value (V)
frequency	**xs:decimal**	required			documentation: Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation: [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.

source

```
<xs:element name="ElectricalRatedVoltage" type="Voltage_ComplexType" minOccurs="0"/>
```

element **InfrastructureTractionType_ComplexType/Self-Propelled**

diagram

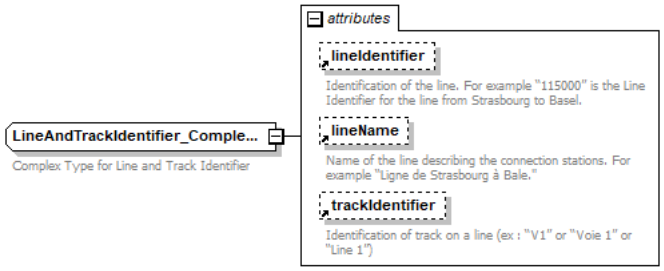


source

```
<xs:element name="Self-Propelled"/>
```

complexType **LineAndTrackIdentifier_ComplexType**

diagram



used by

elements **LineAndTrackIdentifierChange** **LineAndTrackIdentifierStart**

attributes

Name	Type	Use	Default	Fixed	Annotation
lineIdentifier	**xs:string**				documentation Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.
lineName					documentation Name of the line describing the connection stations. For example “Ligne de Strasbourg à Bale."
trackIdentifier					documentation Identification of track on a line (ex : “V1” or “Voie 1” or “Line 1”)

annotation

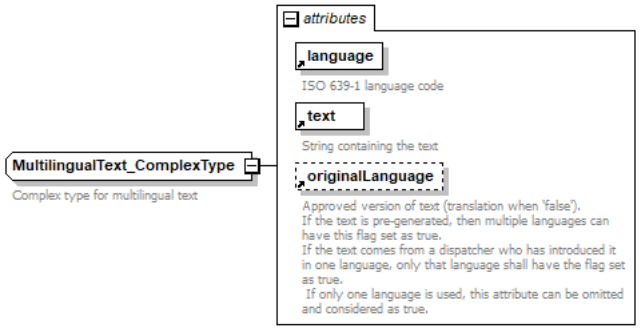
documentation
Complex Type for Line and Track Identifier

source

```
<xs:complexType name="LineAndTrackIdentifier_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for Line and Track Identifier</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="lineIdentifier"/>
  <xs:attribute ref="lineName"/>
  <xs:attribute ref="trackIdentifier"/>
</xs:complexType>
```

complexType **MultilingualText_ComplexType**

diagram



used by

elements **AdditionalInfo** **ChangedStoppingLocationAlert** **ConnectionTrackDescription** **Contact/ContactNotes** **JP_ChangeReason** **LevelCrossingDescription** **LocationText** **MessageText** **ReasonText** **RestrictionReason** **SSP_VisualReference** **StopTypeDetails** **TAF_TAP_LocationName** **TemporaryConstraintReason**

attributes

Name	Type	Use	Default	Fixed	Annotation
language	**derived by: xs:string**	required			documentation ISO 639-1 language code
text	**xs:string**	required			documentation String containing the text
originalLanguage	**xs:boolean**				documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.

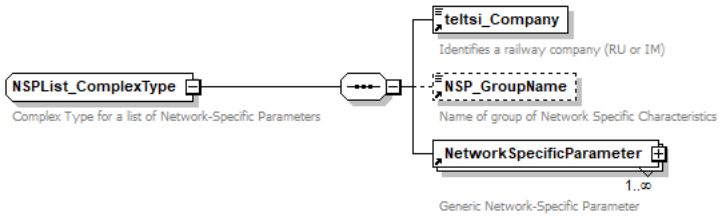
annotation

documentation
Complex type for multilingual text

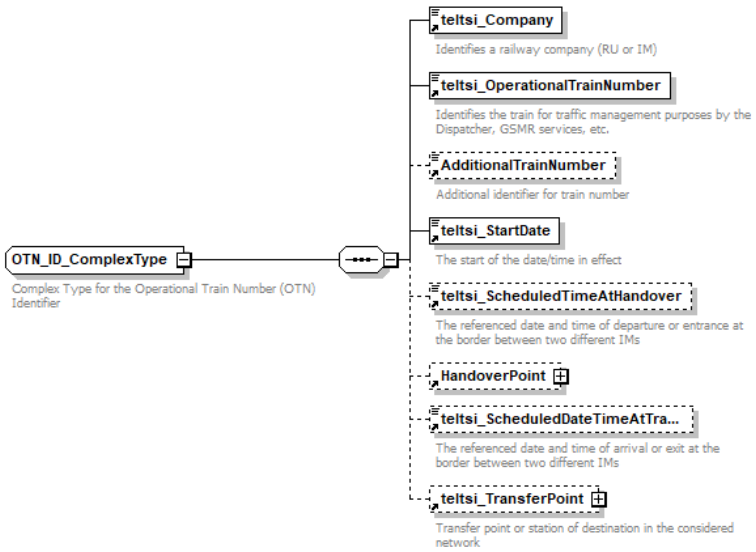
source

```
<xs:complexType name="MultilingualText_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for multilingual text</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="language" use="required"/>
  <xs:attribute ref="text" use="required"/>
  <xs:attribute ref="originalLanguage"/>
</xs:complexType>
```

complexType **NSPList_ComplexType**

diagram	
children	teltsi_Company NSP_GroupName NetworkSpecificParameter
used by	elements BFC_NSPs General_JP_Information_NSP JP_ContextInformation/JP_ContextInformation_NSPs NetworkSpecificConstraint NetworkSpecificEvent PhysicalCharacteristics_NSPs Signal_NSPs SignalInformation_NSPs StatusReport_NSPs TAF_TAP_Location_NSP TC_Features_NSPs TFC_NSPs TimingPoint_NSPs TimingPointConstraints_NSPs TrainCharacteristicsRef_NSPs
annotation	documentation Complex Type for a list of Network-Specific Parameters
source	<xs:complexType name="NSPList_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for a list of Network-Specific Parameters</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="teltsi_Company"/> <xs:element ref="NSP_GroupName" minOccurs="0"/> <xs:element ref="NetworkSpecificParameter" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType>

complexType **OTN_ID_ComplexType**

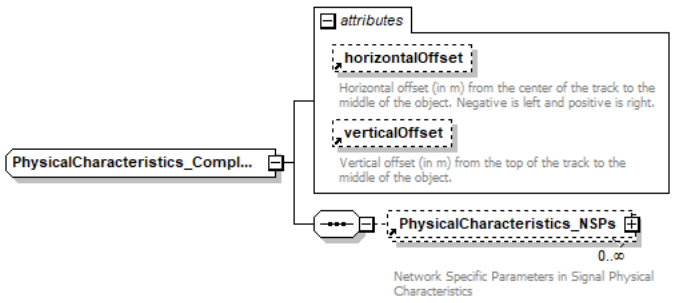
diagram	
children	teltsi_Company teltsi_OperationalTrainNumber AdditionalTrainNumber teltsi_StartDate teltsi_ScheduledTimeAtHandover HandoverPoint teltsi_ScheduledDateTimeAtTransfer teltsi_TransferPoint
used by	element OTN_ID
annotation	documentation Complex Type for the Operational Train Number (OTN) Identifier

source

```
<xs:complexType name="OTN_ID_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for the Operational Train Number (OTN)
Identifier</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="teltsi_Company"/>
    <xs:element ref="teltsi_OperationalTrainNumber"/>
    <xs:element ref="AdditionalTrainNumber" minOccurs="0"/>
    <xs:element ref="teltsi_StartDate"/>
    <xs:element ref="teltsi_ScheduledTimeAtHandover" minOccurs="0"/>
    <xs:element ref="HandoverPoint" minOccurs="0"/>
    <xs:element ref="teltsi_ScheduledDateTimeAtTransfer" minOccurs="0"/>
    <xs:element ref="teltsi_TransferPoint" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
```

complexType **PhysicalCharacteristics_ComplexType**

diagram



children **PhysicalCharacteristics_NSPs**

used by elements **KM_Reference** **SignalPhysicalCharacteristics**

attributes

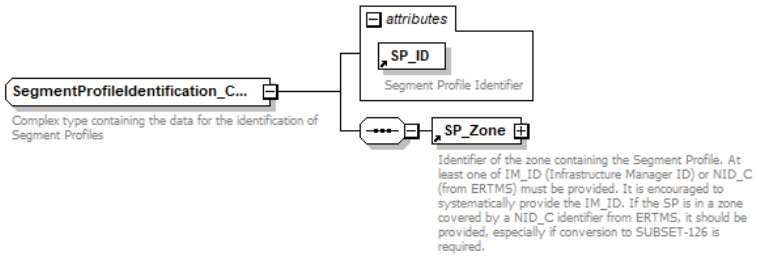
Name	Type	Use	Default	Fixed	Annotation
horizontalOffset	**distanceType**				documentation Horizontal offset (in m) from the center of the track to the middle of the object. Negative is left and positive is right.
verticalOffset	**distanceType**				documentation Vertical offset (in m) from the top of the track to the middle of the object.

source

```
<xs:complexType name="PhysicalCharacteristics_ComplexType">
  <xs:sequence>
    <xs:element ref="PhysicalCharacteristics_NSPs" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="horizontalOffset"/>
  <xs:attribute ref="verticalOffset"/>
```

	</xs:complexType>

complexType **SegmentProfileIdentification_ComplexType**

diagram	
children	**SP_Zone**
used by	elements **LastAbsoluteOdometryPoint** **PositionSpeed** **RequestFromSegmentProfile** **SegmentProfile** **SegmentProfileReference** **SP_EndDeviation** **SP_Request** **SP_StartDeviation** **StoppingPointApplicableIfPrecedingSP** complexType **SegmentProfileInfo_ComplexType**

attributes	Name	Type	Use	Default	Fixed	Annotation
	SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier

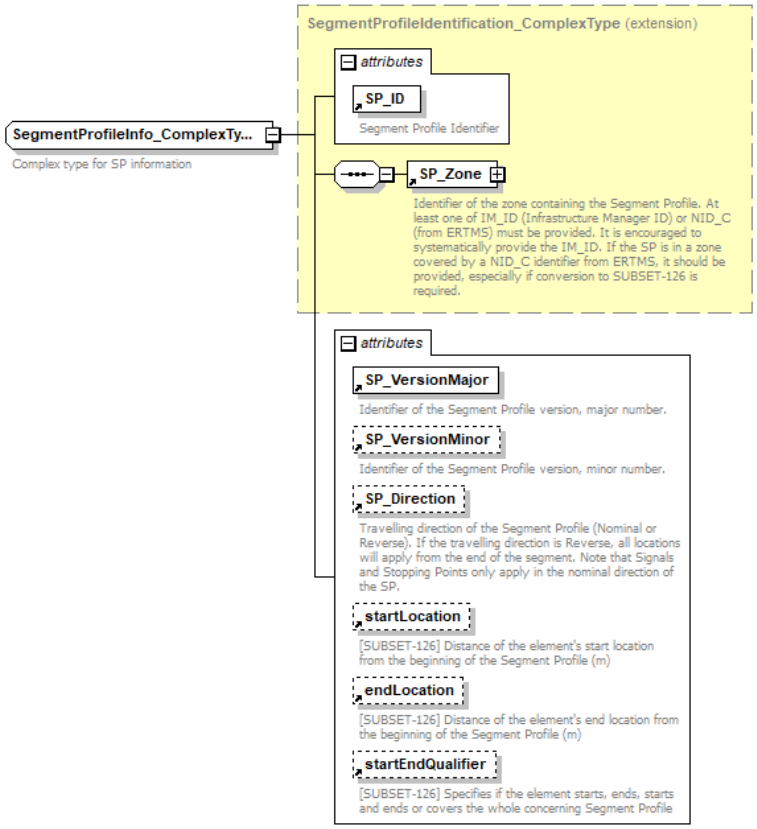
annotation	documentation Complex type containing the data for the identification of Segment Profiles

source

```
<xs:complexType name="SegmentProfileIdentification_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type containing the data for the identification of Segment Profiles</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="SP_Zone"/>
  </xs:sequence>
  <xs:attribute ref="SP_ID" use="required"/>
</xs:complexType>
```

complexType **SegmentProfileInfo_ComplexType**

diagram



type: extension of **SegmentProfileIdentification_ComplexType**

properties: base SegmentProfileIdentification_ComplexType

children: **SP_Zone**

used by: elements **SegmentClaim** **SegmentProfileInfo**

attributes

Name	Type	Use	Default	Fixed	Annotation
SP_ID	**derived by: xs:string**	required			documentation Segment Profile Identifier
SP_VersionMajor	**xs:string**	required			documentation Identifier of the Segment Profile version, major number.
SP_VersionMinor	**xs:string**				documentation Identifier of the Segment Profile version, minor number.
SP_Direction	**derived by: xs:string**				documentation Travelling direction of the Segment Profile (Nominal or Reverse). If the travelling direction is

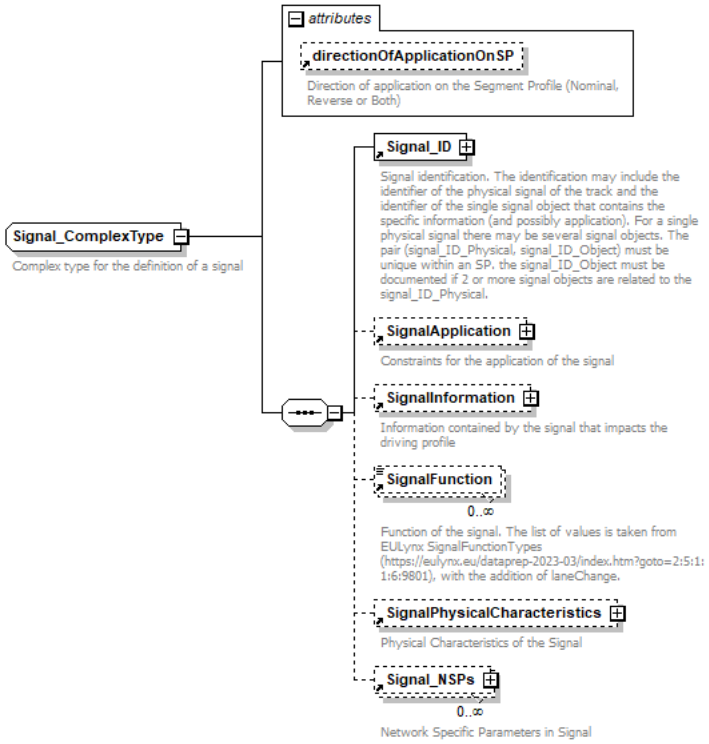
	startLocation	**distanceType**	Reverse, all locations will apply from the end of the segment. Note that Signals and Stopping Points only apply in the nominal direction of the SP. documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
	endLocation	**distanceType**	documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
	startEndQualifier	**derived by:** **xs:string**	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
annotation	documentation Complex type for SP information		

source

```
<xs:complexType name="SegmentProfileInfo_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for SP information</xs:documentation>
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="SegmentProfileIdentification_ComplexType">
      <xs:attribute ref="SP_VersionMajor" use="required"/>
      <xs:attribute ref="SP_VersionMinor"/>
      <xs:attribute ref="SP_Direction"/>
      <xs:attribute ref="startLocation"/>
      <xs:attribute ref="endLocation"/>
      <xs:attribute ref="startEndQualifier"/>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
```

complexType **Signal_ComplexType**

diagram



children **Signal_ID SignalApplication SignalInformation SignalFunction SignalPhysicalCharacteristics Signal_NSPs**

used by elements **Signal TemporarySignalConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
directionOfApplicationOnSP	**derived by: xs:string**				documentation: Direction of application on the Segment Profile (Nominal, Reverse or Both)

annotation

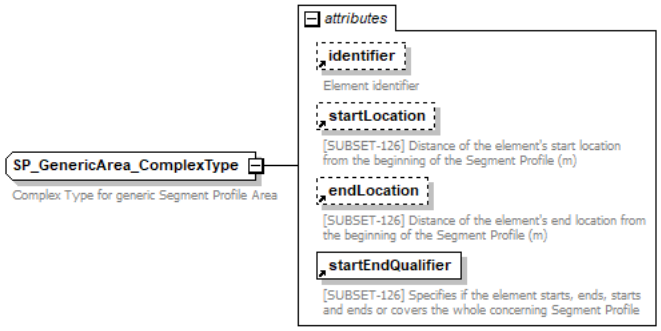
documentation
Complex type for the definition of a signal

source

```
<xs:complexType name="Signal_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for the definition of a signal</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="Signal_ID"/>
    <xs:element ref="SignalApplication" minOccurs="0"/>
    <xs:element ref="SignalInformation" minOccurs="0"/>
    <xs:element ref="SignalFunction" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element ref="SignalPhysicalCharacteristics" minOccurs="0"/>
    <xs:element ref="Signal_NSPs" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="directionOfApplicationOnSP"/>
</xs:complexType>
```

complexType **SP_GenericArea_ComplexType**

diagram



used by

elements **AxleLoadSpeedProfile BatteryChargingArea CommunicationNetwork ContactList DAS_ModesUnavailabilityArea DecisiveGradientArea DynamicBrakeForceInhibitionArea GuardRails LevelCrossingArea LimitedDynamicBrakeForceArea NetworkSpecificArea NoAdvicePresentedToDriver NoStoppingArea PermittedBrakingDistanceArea Platform SwitchOffEddyCurrentBrakeArea SwitchOffEddyCurrentEmergencyBrakeArea SwitchOffMagneticShoeBrakeArea SwitchOffRegenerativeBrakeArea TAF_TAP_Location Tunnel**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile

annotation

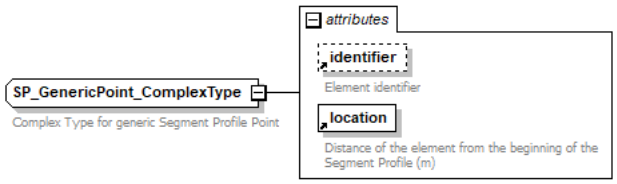
documentation
Complex Type for generic Segment Profile Area

source

```
<xs:complexType name="SP_GenericArea_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for generic Segment Profile Area</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="identifier"/>
  <xs:attribute ref="startLocation"/>
  <xs:attribute ref="endLocation"/>
  <xs:attribute ref="startEndQualifier" use="required"/>
</xs:complexType>
```

complexType **SP_GenericPoint_ComplexType**

diagram



used by

elements **Balise** **ConnectionTrack** **KilometreReferencePoint** **NetworkSpecificPoint** **UnprotectedLevelCrossingStop** **VirtualBalise**

attributes

Name	Type	Use	Default	Fixed	Annotation
identifier	**xs:string**				documentation Element identifier
location	**distanceType**	required			documentation Distance of the element from the beginning of the Segment Profile (m)

annotation

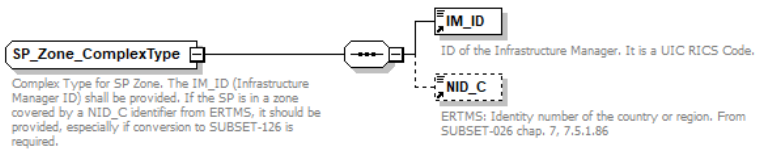
documentation
Complex Type for generic Segment Profile Point

source

```
<xs:complexType name="SP_GenericPoint_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for generic Segment Profile Point</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="identifier"/>
  <xs:attribute ref="location" use="required"/>
</xs:complexType>
```

complexType **SP_Zone_ComplexType**

diagram



children **IM_ID** **NID_C**

used by

element **SP_Zone**

annotation

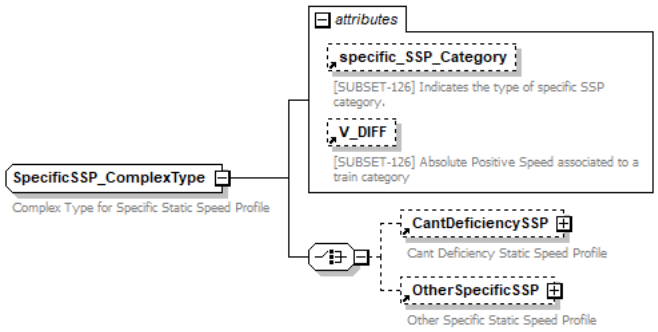
documentation
Complex Type for SP Zone. The IM_ID (Infrastructure Manager ID) shall be provided. If the SP is in a zone covered by a NID_C identifier from ERTMS, it should be provided, especially if conversion to SUBSET-126 is required.

source

```
<xs:complexType name="SP_Zone_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for SP Zone. The IM_ID (Infrastructure Manager ID) shall be provided. If the SP is in a zone covered by a NID_C identifier from ERTMS, it should be provided, especially if conversion to SUBSET-126 is required.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="IM_ID"/>
    <xs:element ref="NID_C" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
```

complexType **SpecificSSP_ComplexType**

diagram



children **CantDeficiencySSP** **OtherSpecificSSP**

used by element **SpecificSSP**

attributes

Name	Type	Use	Default	Fixed	Annotation
specific_SSP_Category	**derived by: xs:string**				documentation [SUBSET-126] Indicates the type of specific SSP category.
V_DIFF	**speedType**				documentation [SUBSET-126] Absolute Positive Speed associated to a train category

annotation

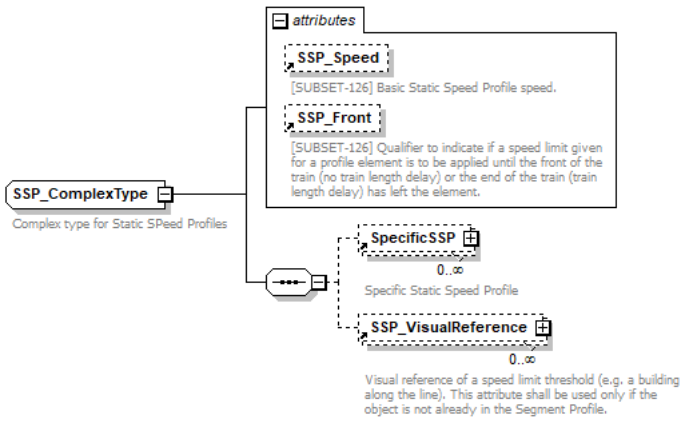
documentation
Complex Type for Specific Static Speed Profile

source

```
<xs:complexType name="SpecificSSP_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for Specific Static Speed Profile</xs:documentation>
  </xs:annotation>
  <xs:choice>
    <xs:element ref="CantDeficiencySSP" minOccurs="0"/>
    <xs:element ref="OtherSpecificSSP" minOccurs="0"/>
  </xs:choice>
  <xs:attribute ref="specific_SSP_Category"/>
  <xs:attribute ref="V_DIFF"/>
</xs:complexType>
```

complexType **SSP_ComplexType**

diagram



children **SpecificSSP** **SSP_VisualReference**

used by elements **StaticSpeedProfileChange** **StaticSpeedProfileStart**

attributes

Name	Type	Use	Default	Fixed	Annotation
SSP_Speed	**speedType**				documentation [SUBSET-126] Basic Static Speed Profile speed.
SSP_Front	**xs:boolean**				documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (no train length delay) or the end of the train (train length delay) has left the element.

annotation

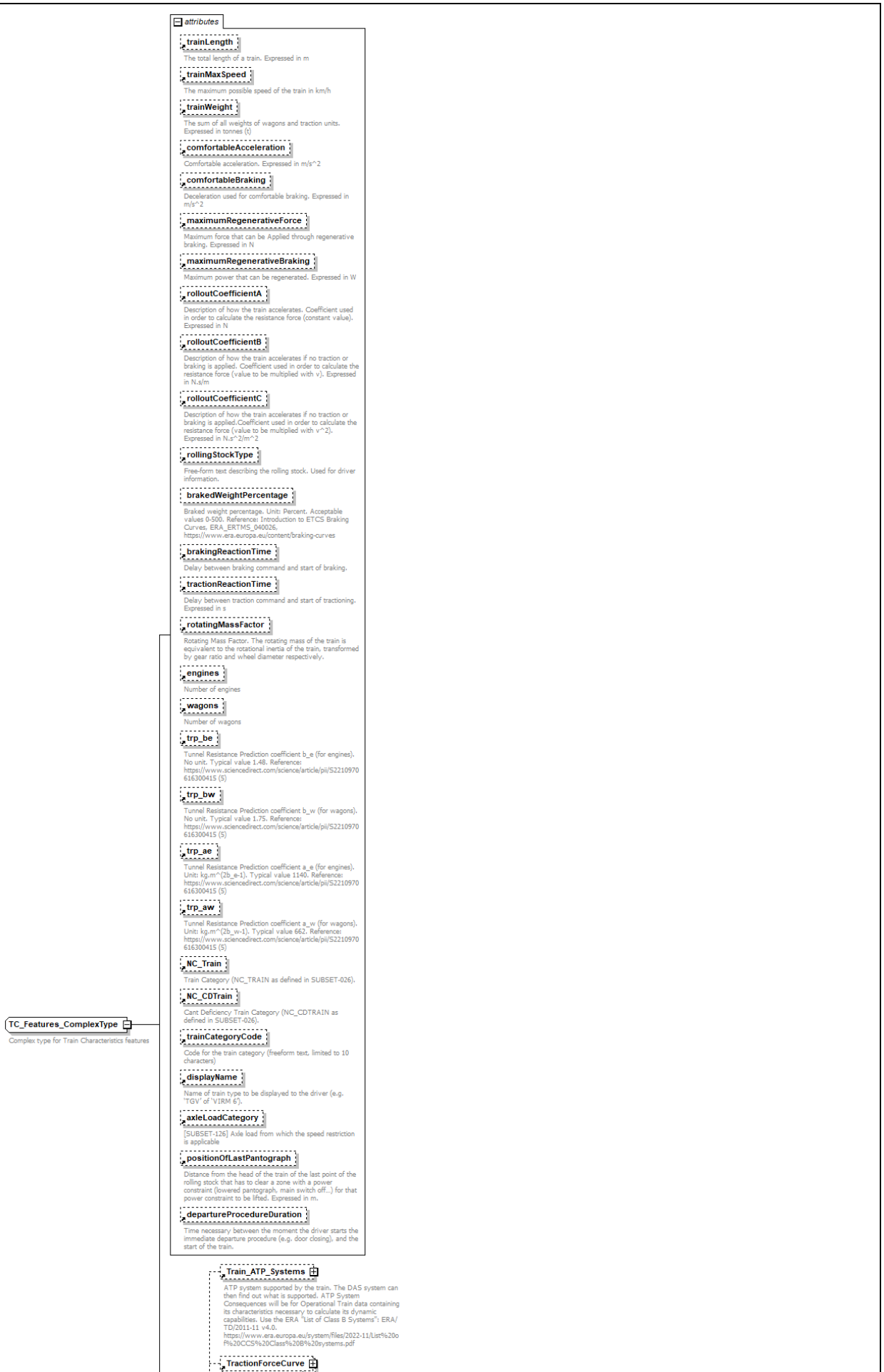
documentation
Complex type for Static SPeed Profiles

source

```
<xs:complexType name="SSP_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for Static SPeed Profiles</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="SpecificSSP" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element ref="SSP_VisualReference" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="SSP_Speed"/>
  <xs:attribute ref="SSP_Front"/>
</xs:complexType>
```

complexType **TC_Features_ComplexType**

diagram


attributes
trainLength
The total length of a train. Expressed in m
trainMaxSpeed
The maximum possible speed of the train in km/h
trainWeight
The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration
Comfortable acceleration. Expressed in m/s^2
comfortableBraking
Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce
Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking
Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA
Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB
Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC
Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType
Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage
Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026. https://www.era.europa.eu/content/braking-curves
brakingReactionTime
Delay between braking command and start of braking.
tractionReactionTime
Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor
Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines
Number of engines
wagons
Number of wagons
trp_be
Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_bw
Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_ae
Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_aw
Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
NC_Train
Train Category (NC_TRAIN as defined in SUBSET-026).
NC_CDTrain
Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
trainCategoryCode
Code for the train category (freeform text, limited to 10 characters)
displayName
Name of train type to be displayed to the driver (e.g. "TGV" of "VIRM 6").
axleLoadCategory
[SUBSET-126] Axle load from which the speed restriction is applicable
positionOfLastPantograph
Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off...) for that power constraint to be lifted. Expressed in m.
departureProcedureDuration
Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
TC_Features_ComplexType
Complex type for Train Characteristics features
Train_ATP_Systems
ATP system supported by the train. The DAS system can then find out what is supported. ATP System Consequences will be for Operational Train data containing its characteristics necessary to calculate its dynamic capabilities. Use the ERA "List of Class B Systems": ERA/TD/2011-11 v4.0. https://www.era.europa.eu/system/files/2022-11/List%20of%20CCS%20Class%20B%20systems.pdf
TractionForceCurve

children	**Train_ATP_Systems TractionForceCurve BrakingForceCurve Engine VehicleData TC_Features_NSPs**
used by	elements **TC_ChangeFeatures TC_Features**

attributes

Name	Type	Use	Default	Fixed	Annotation
trainLength	**xs:integer**	optional			documentation The total length of a train. Expressed in m
trainMaxSpeed	**xs:integer**	optional			documentation The maximum possible speed of the train in km/h
trainWeight	**xs:int**	optional			documentation The sum of all weights of wagons and traction units. Expressed in tonnes (t)
comfortableAcceleration	**xs:double**	optional			documentation Comfortable acceleration. Expressed in m/s^2
comfortableBraking	**xs:double**	optional			documentation Deceleration used for comfortable braking. Expressed in m/s^2
maximumRegenerativeForce	**xs:double**	optional			documentation Maximum force that can be Applied through regenerative braking. Expressed in N
maximumRegenerativeBraking	**xs:double**	optional			documentation Maximum power that can be regenerated. Expressed in W
rolloutCoefficientA	**xs:double**	optional			documentation Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
rolloutCoefficientB	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
rolloutCoefficientC	**xs:double**	optional			documentation Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
rollingStockType	**xs:string**	optional			documentation Free-form text describing the rolling stock. Used for driver information.
brakedWeightPercentage	**xs:decimal**				documentation Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves
brakingReactionTime	**xs:duration**	optional			documentation Delay between braking command and start of braking.
tractionReactionTime	**xs:int**	optional			documentation Delay between traction command and start of tractioning. Expressed in s
rotatingMassFactor	**xs:decimal**				documentation Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
engines	**xs:int**				documentation Number of engines
wagons	**xs:int**				documentation Number of wagons
trp_be	**xs:decimal**				documentation Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
trp_bw	**xs:decimal**				documentation Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)

	trp_ae	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	trp_aw	**xs:decimal**	documentation Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
	NC_Train	**NC_TrainType**	documentation Train Category (NC_TRAIN as defined in SUBSET-026).
	NC_CDTrain	**NC_CDTrainType**	documentation Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
	trainCategoryCode	**derived by: xs:string**	documentation Code for the train category (freeform text, limited to 10 characters)
	displayName	**xs:string**	documentation Name of train type to be displayed to the driver (e.g. 'TGV' of 'VIRM 6').
	axleLoadCategory	**derived by: xs:string**	documentation [SUBSET-126] Axle load from which the speed restriction is applicable
	positionOfLastPantograph	**distanceType**	documentation Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off…) for that power constraint to be lifted. Expressed in m.
	departureProcedureDuration	**xs:duration**	documentation Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
annotation	documentation Complex type for Train Characteristics features		

source

```
<xs:complexType name="TC_Features_ComplexType">
 <xs:annotation>
  <xs:documentation>Complex type for Train Characteristics features</xs:documentation>
 </xs:annotation>
 <xs:sequence>
  <xs:element ref="Train_ATP_Systems" minOccurs="0"/>
  <xs:element ref="TractionForceCurve" minOccurs="0" maxOccurs="unbounded"/>
  <xs:element ref="BrakingForceCurve" minOccurs="0" maxOccurs="unbounded"/>
  <xs:element ref="Engine" minOccurs="0"/>
  <xs:element ref="VehicleData" minOccurs="0" maxOccurs="unbounded"/>
  <xs:element ref="TC_Features_NSPs" minOccurs="0" maxOccurs="unbounded"/>
 </xs:sequence>
 <xs:attribute ref="trainLength" use="optional"/>
 <xs:attribute ref="trainMaxSpeed" use="optional"/>
 <xs:attribute ref="trainWeight" use="optional"/>
 <xs:attribute ref="comfortableAcceleration" use="optional"/>
 <xs:attribute ref="comfortableBraking" use="optional"/>
 <xs:attribute ref="maximumRegenerativeForce" use="optional"/>
 <xs:attribute ref="maximumRegenerativeBraking" use="optional"/>
 <xs:attribute ref="rolloutCoefficientA" use="optional"/>
 <xs:attribute ref="rolloutCoefficientB" use="optional"/>
 <xs:attribute ref="rolloutCoefficientC" use="optional"/>
 <xs:attribute ref="rollingStockType" use="optional"/>
 <xs:attribute name="brakedWeightPercentage" type="xs:decimal">
```

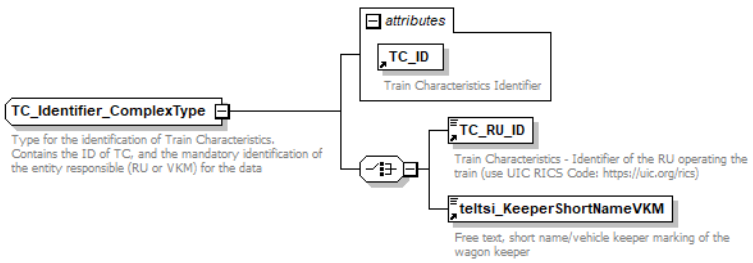
```
    <xs:annotation>
      <xs:documentation>Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference:
Introduction to ETCS Braking Curves, ERA_ERTMS_040026,
https://www.era.europa.eu/content/braking-curves</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute ref="brakingReactionTime" use="optional"/>
  <xs:attribute ref="tractionReactionTime" use="optional"/>
  <xs:attribute ref="rotatingMassFactor"/>
  <xs:attribute ref="engines"/>
  <xs:attribute ref="wagons"/>
  <xs:attribute ref="trp_be"/>
  <xs:attribute ref="trp_bw"/>
  <xs:attribute ref="trp_ae"/>
  <xs:attribute ref="trp_aw"/>
  <xs:attribute ref="NC_Train"/>
  <xs:attribute ref="NC_CDTrain"/>
  <xs:attribute ref="trainCategoryCode"/>
  <xs:attribute ref="displayName"/>
  <xs:attribute ref="axleLoadCategory"/>
  <xs:attribute ref="positionOfLastPantograph"/>
  <xs:attribute ref="departureProcedureDuration"/>
</xs:complexType>
```

attribute **TC_Features_ComplexType/@brakedWeightPercentage**

type	**xs:decimal**
annotation	documentation Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves

source	`<xs:attribute name="brakedWeightPercentage" type="xs:decimal">`
	`  <xs:annotation>`
	`  <xs:documentation>Braked weight percentage. Unit: Percent. Acceptable values 0-500. Reference: Introduction to ETCS Braking Curves, ERA_ERTMS_040026, https://www.era.europa.eu/content/braking-curves</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

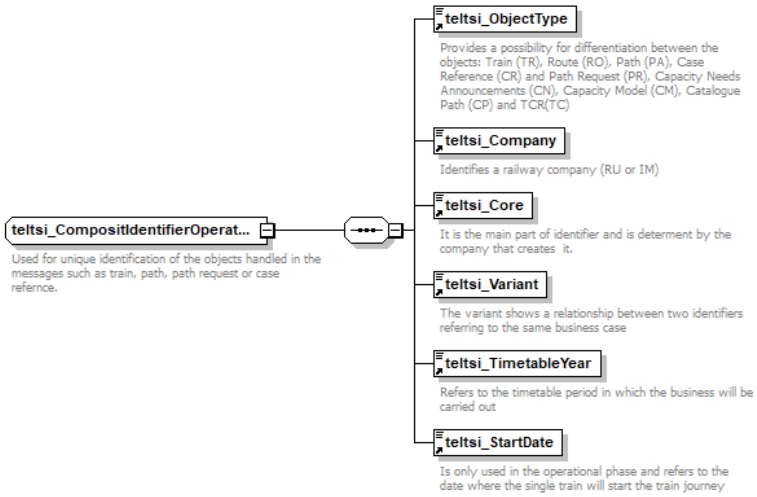
complexType **TC_Identifier_ComplexType**

diagram	
children	**TC_RU_ID teltsi_KeeperShortNameVKM**
used by	elements **TC_Request TrainCharacteristics TrainCharacteristicsChange TrainCharacteristicsRef**

attributes	Name	Type	Use	Default	Fixed	Annotation
	TC_ID	**xs:string**	required			documentation

	Train Characteristics Identifier
annotation	documentation Type for the identification of Train Characteristics. Contains the ID of TC, and the mandatory identification of the entity responsible (RU or VKM) for the data
source	<xs:complexType name="TC_Identifier_ComplexType"> <xs:annotation> <xs:documentation>Type for the identification of Train Characteristics. Contains the ID of TC, and the mandatory identification of the entity responsible (RU or VKM) for the data</xs:documentation> </xs:annotation> <xs:choice> <xs:element ref="TC_RU_ID"/> <xs:element ref="teltsi_KeeperShortNameVKM"/> </xs:choice> <xs:attribute ref="TC_ID" use="required"/> </xs:complexType>

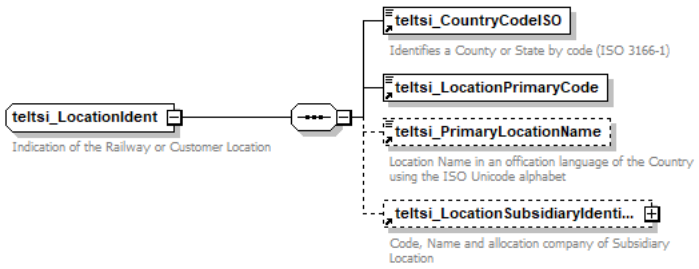
complexType **teltsi_CompositIdentifierOperationalType**

diagram	
children	**teltsi_ObjectType** **teltsi_Company** **teltsi_Core** **teltsi_Variant** **teltsi_TimetableYear** **teltsi_StartDate**
used by	element **TAF_TAP_ID**
annotation	documentation Used for unique identification of the objects handled in the messages such as train, path, path request or case refernce.
source	<xs:complexType name="teltsi_CompositIdentifierOperationalType"> <xs:annotation> <xs:documentation>Used for unique identification of the objects handled in the messages such as train, path, path request or case refernce.</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="teltsi_ObjectType"/> <xs:element ref="teltsi_Company"/> <xs:element ref="teltsi_Core"/> <xs:element ref="teltsi_Variant"/> <xs:element ref="teltsi_TimetableYear"/> <xs:element ref="teltsi_StartDate"> <xs:annotation>

```
      <xs:documentation>Is only used in the operational phase and refers to the date where the single train will start the train journey</xs:documentation>
    </xs:annotation>
  </xs:element>
 </xs:sequence>
</xs:complexType>
```

complexType **teltsi_LocationIdent**

diagram



children **teltsi_CountryCodeISO** **teltsi_LocationPrimaryCode** **teltsi_PrimaryLocationName** **teltsi_LocationSubsidiaryIdentification**

used by

elements **HandoverPoint** **TAF_TAP_LocationIdent** **teltsi_Location** **teltsi_NetworkProjectedLocation/teltsi_NextLocation** **teltsi_TransferPoint** **To_TAF_TAP_Location** **TrainLocation**

annotation

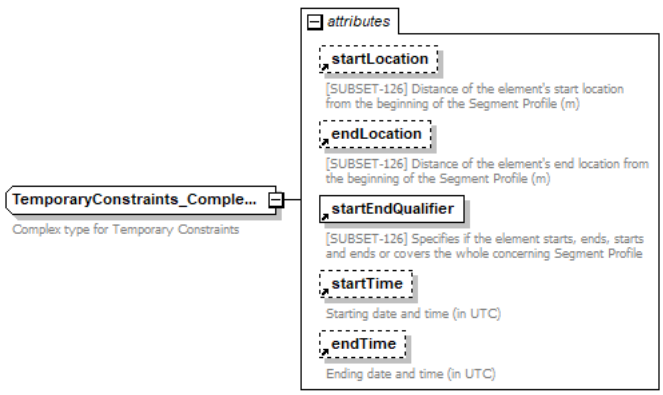
documentation
Indication of the Railway or Customer Location

source

```
<xs:complexType name="teltsi_LocationIdent">
 <xs:annotation>
  <xs:documentation>Indication of the Railway or Customer Location</xs:documentation>
 </xs:annotation>
 <xs:sequence>
  <xs:element ref="teltsi_CountryCodeISO"/>
  <xs:element ref="teltsi_LocationPrimaryCode"/>
  <xs:element ref="teltsi_PrimaryLocationName" minOccurs="0"/>
  <xs:element ref="teltsi_LocationSubsidiaryIdentification" minOccurs="0"/>
 </xs:sequence>
</xs:complexType>
```

complexType **TemporaryConstraints_ComplexType**

diagram



used by

elements **Applicable_OTN** **JP_ContextInformation_NSP_Constraints** **TemporaryConstraints**

attributes

Name	Type	Use	Default	Fixed	Annotation
startLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
endLocation	**distanceType**				documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
startEndQualifier	**derived by: xs:string**	required			documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
startTime	**UTCDateTimeType**				documentation Starting date and time (in UTC)
endTime	**UTCDateTimeType**				documentation Ending date and time (in UTC)

annotation

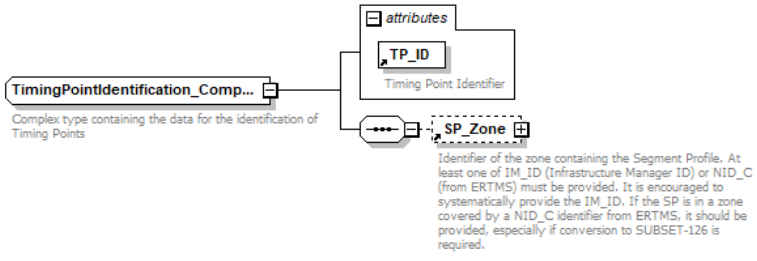
documentation
Complex type for Temporary Constraints

source

```
<xs:complexType name="TemporaryConstraints_ComplexType">
 <xs:annotation>
  <xs:documentation>Complex type for Temporary Constraints</xs:documentation>
 </xs:annotation>
 <xs:attribute ref="startLocation"/>
 <xs:attribute ref="endLocation"/>
 <xs:attribute ref="startEndQualifier" use="required"/>
 <xs:attribute ref="startTime"/>
 <xs:attribute ref="endTime"/>
</xs:complexType>
```

complexType **TimingPointIdentification_ComplexType**

diagram



children	**SP_Zone**
used by	elements **PreviousTimingPoint StationAheadSign TP_ID_Reference**

attributes	Name	Type	Use	Default	Fixed	Annotation
	TP_ID	**xs:string**	required			documentation Timing Point Identifier

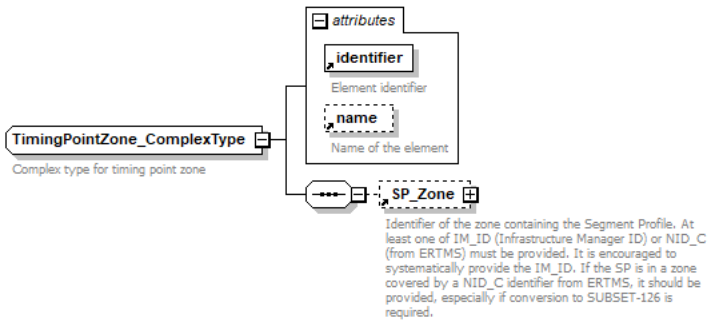
annotation	documentation Complex type containing the data for the identification of Timing Points

source

```
<xs:complexType name="TimingPointIdentification_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type containing the data for the identification of Timing Points</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="SP_Zone" minOccurs="0"/>
  </xs:sequence>
  <xs:attribute ref="TP_ID" use="required"/>
</xs:complexType>
```

complexType **TimingPointZone_ComplexType**

diagram



children	**SP_Zone**
used by	elements **TimingPointZone TimingPointZoneReference**

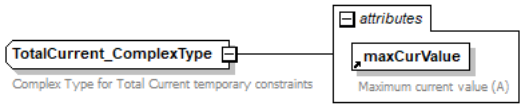
attributes	Name	Type	Use	Default	Fixed	Annotation
	identifier	**xs:string**	required			documentation Element identifier
	name	**xs:string**				documentation Name of the element

annotation	documentation Complex type for timing point zone

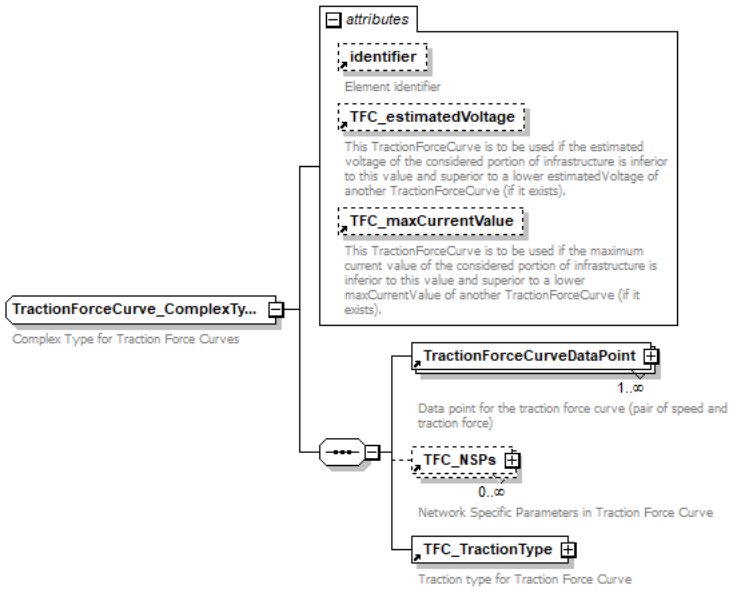
source

```
<xs:complexType name="TimingPointZone_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex type for timing point zone</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="SP_Zone" minOccurs="0"/>
  </xs:sequence>
  <xs:attribute ref="identifier" use="required"/>
  <xs:attribute ref="name"/>
</xs:complexType>
```

complexType **TotalCurrent_ComplexType**

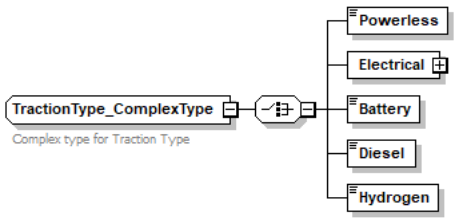
diagram	
used by	elements **RegenerationTotalCurrent** **TractionTotalCurrent**
attributes	Name: maxCurValue; Type: **currentType**; Use: required; Default: ; Fixed: ; Annotation: documentation: Maximum current value (A)
annotation	documentation Complex Type for Total Current temporary constraints
source	<xs:complexType name="TotalCurrent_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for Total Current temporary constraints</xs:documentation> </xs:annotation> <xs:attribute ref="maxCurValue" use="required"/> </xs:complexType>

complexType **TractionForceCurve_ComplexType**

diagram	
children	**TractionForceCurveDataPoint** **TFC_NSPs** **TFC_TractionType**
used by	element **TractionForceCurve**
attributes	Name: identifier; Type: **xs:string**; Use: ; Default: ; Fixed: ; Annotation: documentation: Element identifier Name: TFC_estimatedVoltage; Type: **voltageType**; Use: ; Default: ; Fixed: ; Annotation: documentation: This TractionForceCurve is to be used if the estimated voltage of the considered portion of infrastructure is

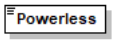
	TFC_maxCurrentValue **currentType**	inferior to this value and superior to a lower estimatedVoltage of another TractionForceCurve (if it exists). documentation This TractionForceCurve is to be used if the maximum current value of the considered portion of infrastructure is inferior to this value and superior to a lower maxCurrentValue of another TractionForceCurve (if it exists).
annotation	documentation Complex Type for Traction Force Curves	
source	<xs:complexType name="TractionForceCurve_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for Traction Force Curves</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="TractionForceCurveDataPoint" maxOccurs="unbounded"/> <xs:element ref="TFC_NSPs" minOccurs="0" maxOccurs="unbounded"/> <xs:element ref="TFC_TractionType"/> </xs:sequence> <xs:attribute ref="identifier"/> <xs:attribute ref="TFC_estimatedVoltage"/> <xs:attribute ref="TFC_maxCurrentValue"/> </xs:complexType>	

complexType **TractionType_ComplexType**

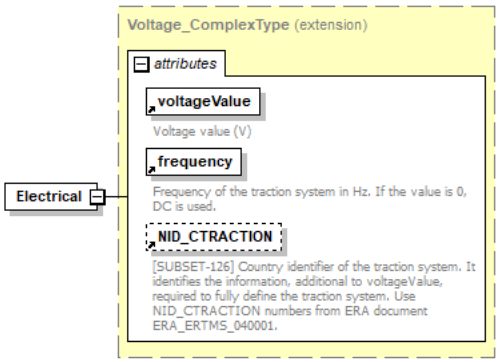
diagram	
children	**Powerless** **Electrical** **Battery** **Diesel** **Hydrogen**
used by	elements **BFC_TractionType** **EngineTractionType** **TFC_TractionType** **UnavailableTractionType**
annotation	documentation Complex type for Traction Type
source	<xs:complexType name="TractionType_ComplexType"> <xs:annotation> <xs:documentation>Complex type for Traction Type</xs:documentation> </xs:annotation> <xs:choice> <xs:element name="Powerless"/>

```
    <xs:element name="Electrical">
     <xs:complexType>
      <xs:complexContent>
       <xs:extension base="Voltage_ComplexType"/>
      </xs:complexContent>
     </xs:complexType>
    </xs:element>
    <xs:element name="Battery"/>
    <xs:element name="Diesel"/>
    <xs:element name="Hydrogen"/>
   </xs:choice>
</xs:complexType>
```

element **TractionType_ComplexType/Powerless**

diagram	
source	`<xs:element name="Powerless"/>`

element **TractionType_ComplexType/Electrical**

diagram	
type	extension of **Voltage_ComplexType**
properties	content complex

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation Voltage value (V)
frequency	**xs:decimal**	required			documentation Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.

source	<xs:element name="Electrical"> <xs:complexType> <xs:complexContent> <xs:extension base="Voltage_ComplexType"/> </xs:complexContent> </xs:complexType> </xs:element>

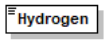
element **TractionType_ComplexType/Battery**

diagram	
source	<xs:element name="Battery"/>

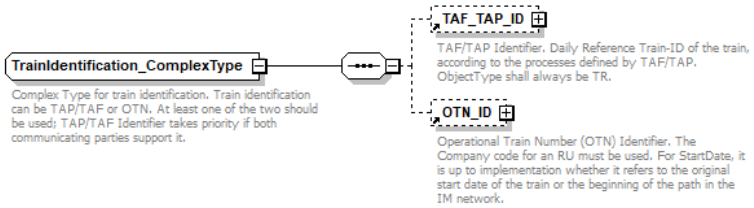
element **TractionType_ComplexType/Diesel**

diagram	
source	<xs:element name="Diesel"/>

element **TractionType_ComplexType/Hydrogen**

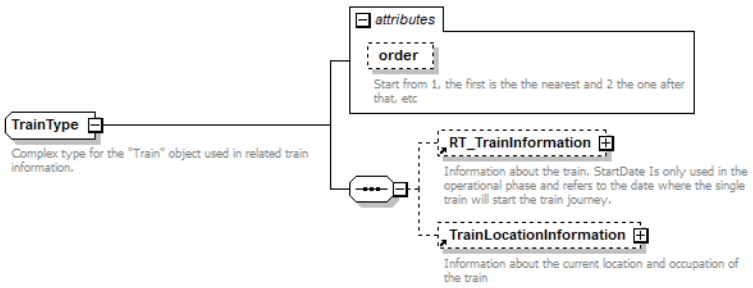
diagram	
source	<xs:element name="Hydrogen"/>

complexType **TrainIdentification_ComplexType**

diagram	
children	**TAF_TAP_ID** **OTN_ID**
used by	element **TrainIdentification**
annotation	documentation Complex Type for train identification. Train identification can be TAP/TAF or OTN. At least one of the two should be used; TAP/TAF Identifier takes priority if both communicating parties support it.
source	<xs:complexType name="TrainIdentification_ComplexType"> <xs:annotation> <xs:documentation>Complex Type for train identification. Train identification can be TAP/TAF or OTN. At least one of the two should be used; TAP/TAF Identifier takes priority if both communicating parties support it.</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="TAF_TAP_ID" minOccurs="0"/> <xs:element ref="OTN_ID" minOccurs="0"/>

	</xs:sequence> </xs:complexType>

complexType **TrainType**

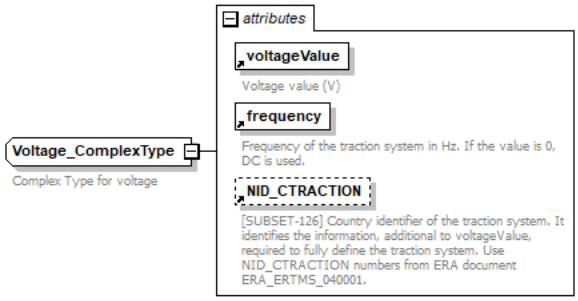
diagram	
children	**RT_TrainInformation** **TrainLocationInformation**
used by	element **Train**
attributes	Name: order; Type: ; Use: ; Default: ; Fixed: ; Annotation: documentation: Start from 1, the first is the the nearest and 2 the one after that, etc
annotation	documentation Complex type for the "Train" object used in related train information.
source	<xs:complexType name="TrainType"> <xs:annotation> <xs:documentation>Complex type for the "Train" object used in related train information.</xs:documentation> </xs:annotation> <xs:sequence> <xs:element ref="RT_TrainInformation" minOccurs="0"/> <xs:element ref="TrainLocationInformation" minOccurs="0"/> </xs:sequence> <xs:attribute name="order"> <xs:annotation> <xs:documentation>Start from 1, the first is the the nearest and 2 the one after that, etc</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType>

attribute **TrainType/@order**

annotation	documentation Start from 1, the first is the the nearest and 2 the one after that, etc
source	<xs:attribute name="order"> <xs:annotation> <xs:documentation>Start from 1, the first is the the nearest and 2 the one after that, etc</xs:documentation> </xs:annotation> </xs:attribute>

complexType **Voltage_ComplexType**

diagram



used by

elements **TractionType_ComplexType/Electrical** **InfrastructureTractionType_ComplexType/Electrical/ElectricalRatedVoltage** **EstimatedVoltageChange** **EstimatedVoltageStart** **RatedVoltageChange** **RatedVoltageStart**

attributes

Name	Type	Use	Default	Fixed	Annotation
voltageValue	**voltageType**	required			documentation: Voltage value (V)
frequency	**xs:decimal**	required			documentation: Frequency of the traction system in Hz. If the value is 0, DC is used.
NID_CTRACTION	**xs:unsignedByte**				documentation: [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.

annotation

documentation
Complex Type for voltage

source

```
<xs:complexType name="Voltage_ComplexType">
  <xs:annotation>
    <xs:documentation>Complex Type for voltage</xs:documentation>
  </xs:annotation>
  <xs:attribute ref="voltageValue" use="required"/>
  <xs:attribute ref="frequency" use="required"/>
  <xs:attribute ref="NID_CTRACTION"/>
</xs:complexType>
```

simpleType **adhesionCategoryType**

type

restriction of **xs:string**

properties

base xs:string

used by

attribute **lowAdhesionCategory**

facets

Kind	Value	Annotation
enumeration	Dry Rail	
enumeration	Dry Rail (Medium)	
enumeration	Dry Rail (Low)	
enumeration	Low Adhesion	

	enumeration Very Low Adhesion enumeration Extremely Low Adhesion
annotation	documentation Simple Type for adhesion category
source	<xs:simpleType name="adhesionCategoryType"> <xs:annotation> <xs:documentation>Simple Type for adhesion category</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="Dry Rail"/> <xs:enumeration value="Dry Rail (Medium)"/> <xs:enumeration value="Dry Rail (Low)"/> <xs:enumeration value="Low Adhesion"/> <xs:enumeration value="Very Low Adhesion"/> <xs:enumeration value="Extremely Low Adhesion"/> </xs:restriction> </xs:simpleType>

simpleType **ClaimTypeType**

type	restriction of **xs:string**
properties	base xs:string
used by	element **ClaimType**
facets	Kind Value Annotation enumeration Planned enumeration Reserved enumeration ReservedImplicit enumeration Occupied enumeration Unknown
source	<xs:simpleType name="ClaimTypeType"> <xs:restriction base="xs:string"> <xs:enumeration value="Planned"/> <xs:enumeration value="Reserved"/> <xs:enumeration value="ReservedImplicit"/> <xs:enumeration value="Occupied"/> <xs:enumeration value="Unknown"/> </xs:restriction> </xs:simpleType>

simpleType **currentType**

type	**xs:decimal**
properties	base xs:decimal
used by	attributes **expectedConsumptionCurrent** **expectedRegenerativeCurrent** **maxChargingCurrent** **maxCurValue** **maxDischargingCurrent** **measuredCurrent** **TFC_maxCurrentValue**
annotation	documentation Simple Type for current (A)
source	<xs:simpleType name="currentType"> <xs:annotation> <xs:documentation>Simple Type for current (A)</xs:documentation>

	</xs:annotation> <xs:restriction base="xs:decimal"/> </xs:simpleType>

simpleType **deviceType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **destinationDevice** **sourceDevice**
facets	Kind Value Annotation maxLength 128
annotation	documentation Simple type for DAS devices
source	<xs:simpleType name="deviceType"> <xs:annotation> <xs:documentation>Simple type for DAS devices</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:maxLength value="128"/> </xs:restriction> </xs:simpleType>

simpleType **distanceType**

type	**xs:double**
properties	base xs:double
used by	attributes **applicationDistance** **distance_EoA_Offset** **distance_STP_Reached** **distanceBeforeSignal** **distanceToPowerConstraint** **distanceToTractionTypeTransition** **endLocation** **endPointOfDownLineSpeedApplication** **horizontalOffset** **location** **location_NegativeErrorMargin** **location_PositiveErrorMargin** **permittedBrakingDistance** **positionOfLastPantograph** **SP_Length** **startLocation** **startPointOfDownLineSpeedApplication** **stopLocationTolerance** **stoppingPointOffset** **verticalOffset**
annotation	documentation Simple Type for distance attributes (m)
source	<xs:simpleType name="distanceType"> <xs:annotation> <xs:documentation>Simple Type for distance attributes (m)</xs:documentation> </xs:annotation> <xs:restriction base="xs:double"/> </xs:simpleType>

simpleType **energyType**

type	**xs:decimal**
properties	base xs:decimal
used by	attribute **autonomy**
annotation	documentation Simple Type for energy (kWh)
source	<xs:simpleType name="energyType"> <xs:annotation> <xs:documentation>Simple Type for energy (kWh)</xs:documentation>

	</xs:annotation> <xs:restriction base="xs:decimal"/> </xs:simpleType>

simpleType **forceType**

type	**xs:decimal**
properties	base xs:decimal
used by	attributes **brakingForce** **tractionForce**
annotation	documentation Force Type
source	<xs:simpleType name="forceType"> <xs:annotation> <xs:documentation>Force Type</xs:documentation> </xs:annotation> <xs:restriction base="xs:decimal"/> </xs:simpleType>

simpleType **gradientDirectionType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **gradientDirection** **gradientDirection_PBD**
facets	Kind Value Annotation enumeration Downhill enumeration Uphill
annotation	documentation Type for gradient direction (Downhill or Uphill)
source	<xs:simpleType name="gradientDirectionType"> <xs:annotation> <xs:documentation>Type for gradient direction (Downhill or Uphill)</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="Downhill"/> <xs:enumeration value="Uphill"/> </xs:restriction> </xs:simpleType>

simpleType **gradientValueType**

type	**xs:decimal**
properties	base xs:decimal
used by	attributes **gradient_PBD** **gradientValue**
annotation	documentation Type for gradient values (Numeric value, mm by m)
source	<xs:simpleType name="gradientValueType"> <xs:annotation> <xs:documentation>Type for gradient values (Numeric value, mm by m)</xs:documentation> </xs:annotation>

	<xs:restriction base="xs:decimal"/> </xs:simpleType>

simpleType **LeavingEnteringStatusType**

type	restriction of **xs:string**
properties	base xs:string
used by	element **LeavingEnteringStatus**
facets	Kind / Value / Annotation enumeration Entering enumeration Leaving enumeration Crossing
source	<xs:simpleType name="LeavingEnteringStatusType"> <xs:restriction base="xs:string"> <xs:enumeration value="Entering"/> <xs:enumeration value="Leaving"/> <xs:enumeration value="Crossing"/> </xs:restriction> </xs:simpleType>

simpleType **NC_CDTrainType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **NC_CDTrain** **SSP_NC_CDDIFF**
facets	Kind / Value / Annotation enumeration 80mm enumeration 100mm enumeration 130mm enumeration 150mm enumeration 165mm enumeration 180mm enumeration 210mm enumeration 225mm enumeration 245mm enumeration 275mm enumeration 300mm
annotation	documentation Simple Type for Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
source	<xs:simpleType name="NC_CDTrainType"> <xs:annotation> <xs:documentation>Simple Type for Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="80mm"/> <xs:enumeration value="100mm"/> <xs:enumeration value="130mm"/>

```
    <xs:enumeration value="150mm"/>
    <xs:enumeration value="165mm"/>
    <xs:enumeration value="180mm"/>
    <xs:enumeration value="210mm"/>
    <xs:enumeration value="225mm"/>
    <xs:enumeration value="245mm"/>
    <xs:enumeration value="275mm"/>
    <xs:enumeration value="300mm"/>
  </xs:restriction>
</xs:simpleType>
```

simpleType **NC_TrainType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **NC_Train** **SSP_NC_DIFF**
facets	Kind / Value / Annotation enumeration Freight_Train_P_Position enumeration Freight_Train_G_Position enumeration Passenger_Train
annotation	documentation Simple Type for Train Category (NC_TRAIN as defined in SUBSET-026).
source	<xs:simpleType name="NC_TrainType"> <xs:annotation> <xs:documentation>Simple Type for Train Category (NC_TRAIN as defined in SUBSET-026).</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="Freight_Train_P_Position"/> <xs:enumeration value="Freight_Train_G_Position"/> <xs:enumeration value="Passenger_Train"/> </xs:restriction> </xs:simpleType>

simpleType **NID_C_Type**

type	restriction of **xs:short**
properties	base xs:short
used by	element **NID_C**
facets	Kind / Value / Annotation minInclusive 0 maxInclusive 1023
annotation	documentation Simple type for NID_C
source	<xs:simpleType name="NID_C_Type"> <xs:annotation> <xs:documentation>Simple type for NID_C</xs:documentation> </xs:annotation> <xs:restriction base="xs:short"> <xs:minInclusive value="0"/> <xs:maxInclusive value="1023"/>

```
  </xs:restriction>
</xs:simpleType>
```

simpleType **percentageType**

type	restriction of **xs:decimal**
properties	base xs:decimal
used by	attributes **alternativeItineraryProbability** **highest_BFC_AdhesionRate** **percentage** **remainingEngineCharge** **remainingEnginePower**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1
annotation	documentation Percentage Type
source	<xs:simpleType name="percentageType"> <xs:annotation> <xs:documentation>Percentage Type</xs:documentation> </xs:annotation> <xs:restriction base="xs:decimal"> <xs:minInclusive value="0"/> <xs:maxInclusive value="1"/> </xs:restriction> </xs:simpleType>

simpleType **powerTypekW**

type	**xs:double**
properties	base xs:double
used by	attribute **maxPower**
annotation	documentation Simple Type for power (kW)
source	<xs:simpleType name="powerTypekW"> <xs:annotation> <xs:documentation>Simple Type for power (kW)</xs:documentation> </xs:annotation> <xs:restriction base="xs:double"/> </xs:simpleType>

simpleType **powerTypeW**

type	**xs:double**
properties	base xs:double
used by	attribute **enginePower**
annotation	documentation Simple Type for power (W)
source	<xs:simpleType name="powerTypeW"> <xs:annotation> <xs:documentation>Simple Type for power (W)</xs:documentation> </xs:annotation> <xs:restriction base="xs:double"/>

	</xs:simpleType>

simpleType **relatedTrainRequestType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **relatedTrainRequest relatedTrainSupported**
facets	Kind / Value / Annotation enumeration None enumeration OwnTrain enumeration RelatedTrains enumeration OwnTrainAndRelatedTrains enumeration OwnTrainAndOrRelatedTrains
source	<xs:simpleType name="relatedTrainRequestType"> <xs:restriction base="xs:string"> <xs:enumeration value="None"/> <xs:enumeration value="OwnTrain"/> <xs:enumeration value="RelatedTrains"/> <xs:enumeration value="OwnTrainAndRelatedTrains"/> <xs:enumeration value="OwnTrainAndOrRelatedTrains"/> </xs:restriction> </xs:simpleType>

simpleType **speedType**

type	**xs:decimal**
properties	base xs:decimal
used by	attributes **ASR_Speed deltaSpeed maxAdvisedSpeed minAdvisedSpeed newSpeedLevel optimalSpeed rangeMaxSpeed rangeMinSpeed speed SSP_Speed V_DIFF windSpeed**
annotation	documentation Speed Type
source	<xs:simpleType name="speedType"> <xs:annotation> <xs:documentation>Speed Type</xs:documentation> </xs:annotation> <xs:restriction base="xs:decimal"/> </xs:simpleType>

simpleType **teltsi_CompanyCode**

type	restriction of **teltsi_String4-4**
properties	base teltsi_String4-4
used by	elements **IM_ID Recipient Sender TC_RU_ID teltsi_AllocationCompany teltsi_Company** attributes **destinationCompany sourceCompany**
facets	Kind / Value / Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation

	Identifies the RU, IM or other company involved in the Rail Transport Chain
source	<xs:simpleType name="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>Identifies the RU, IM or other company involved in the Rail Transport Chain</xs:documentation> </xs:annotation> <xs:restriction base="teltsi_String4-4"> <xs:pattern value="[0-9A-Z]{4}"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_CountryIdentISO**

type	restriction of **xs:string**
properties	base xs:string
used by	element **teltsi_CountryCodeISO**
facets	Kind Value Annotation minLength 2 maxLength 2
annotation	documentation ISO 3166-1 alpha code (2 positions)
source	<xs:simpleType name="teltsi_CountryIdentISO"> <xs:annotation> <xs:documentation>ISO 3166-1 alpha code (2 positions)</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:minLength value="2"/> <xs:maxLength value="2"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_DelayCode**

type	restriction of **xs:token**
properties	base xs:token
used by	elements **teltsi_DelayCause** **teltsi_InterruptionReason**
facets	Kind Value Annotation enumeration 11 documentation Formation of trains if managed by infrastructure manager enumeration 10 documentation Timetable compilation enumeration 12 documentation Mistakes in operational procedures enumeration 13 documentation Wrong application of priority rules enumeration 14 enumeration 18 documentation Staff enumeration 19 documentation Other causes related to the operational planning and management enumeration 20 documentation Signalling installations enumeration 21 documentation Signalling installations at level crossings

	enumeration	22	documentation Telecommunication installations
	enumeration	23	documentation Power supply equipment
	enumeration	24	documentation Track
	enumeration	25	documentation Structures
	enumeration	28	documentation Staff
	enumeration	29	documentation Other causes related to infrastructure installations
	enumeration	31	documentation Irregularities in execution of construction work
	enumeration	30	documentation Planned construction work
	enumeration	32	documentation Speed restrictions due to defective track
	enumeration	39	documentation Other causes related to Civil engineering
	enumeration	40	documentation Delay caused by next IM
	enumeration	41	documentation Delay caused by previous IM
	enumeration	50	documentation exceeding the stop time
	enumeration	51	documentation Request of the RU
	enumeration	52	documentation Loading operations
	enumeration	53	documentation Loading irregularities
	enumeration	54	documentation Commercial preparation of the train
	enumeration	58	documentation Staff
	enumeration	59	documentation Other causes related to commercial causes
	enumeration	60	documentation Roster planning/re-rostering
	enumeration	61	documentation Formation of trains by the RU
	enumeration	62	documentation Problems affecting coaches
	enumeration	63	documentation Problems affecting wagons
	enumeration	64	documentation Problems affecting traction units
	enumeration	68	documentation Staff
	enumeration	69	documentation Other causes related to Rolling Stock
	enumeration	70	documentation Delay caused by next RU
	enumeration	71	documentation Delay caused by previous RU
	enumeration	80	documentation Strike
	enumeration	81	documentation Administrative formalites
	enumeration	82	documentation Outside influence
	enumeration	83	documentation Effects of weather and natural causes
	enumeration	84	documentation Delay caused by external reasons on the next network
	enumeration	89	documentation Other external causes
	enumeration	90	documentation Dangerous incidents, accidents and hazards

	enumeration 91 documentation Track occupation caused by the lateness of the same train enumeration 92 documentation Track occupation caused by the lateness of another train enumeration 93 documentation Turn round enumeration 94 documentation Connection enumeration 95 documentation Further investigation needed
annotation	documentation Reason for a delay or interruption. UIC Leaflet 450-2, Appendix C.
source	<xs:simpleType name="teltsi_DelayCode"> <xs:annotation> <xs:documentation>Reason for a delay or interruption. UIC Leaflet 450-2, Appendix C.</xs:documentation> </xs:annotation> <xs:restriction base="xs:token"> <xs:enumeration value="11"> <xs:annotation> <xs:documentation>Formation of trains if managed by infrastructure manager</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="10"> <xs:annotation> <xs:documentation>Timetable compilation</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="12"> <xs:annotation> <xs:documentation>Mistakes in operational procedures</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="13"> <xs:annotation> <xs:documentation>Wrong application of priority rules</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="14"/> <xs:enumeration value="18"> <xs:annotation> <xs:documentation>Staff</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="19"> <xs:annotation> <xs:documentation>Other causes related to the operational planning and management</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="20"> <xs:annotation> <xs:documentation>Signalling installations</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="21"> <xs:annotation>

```
      <xs:documentation>Signalling installations at level crossings</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="22">
    <xs:annotation>
      <xs:documentation>Telecommunication installations</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="23">
    <xs:annotation>
      <xs:documentation>Power supply equipment</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="24">
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      <xs:documentation>Track</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="25">
    <xs:annotation>
      <xs:documentation>Structures</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="28">
    <xs:annotation>
      <xs:documentation>Staff</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="29">
    <xs:annotation>
      <xs:documentation>Other causes related to infrastructure installations</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="31">
    <xs:annotation>
      <xs:documentation>Irregularities in execution of construction work</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="30">
    <xs:annotation>
      <xs:documentation>Planned construction work</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="32">
    <xs:annotation>
      <xs:documentation>Speed restrictions due to defective track</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="39">
    <xs:annotation>
      <xs:documentation>Other causes related to Civil engineering</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="40">
    <xs:annotation>
      <xs:documentation>Delay caused by next IM</xs:documentation>
```

```
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="41">
    <xs:annotation>
      <xs:documentation>Delay caused by previous IM</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
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    <xs:annotation>
      <xs:documentation>exceeding the stop time</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="51">
    <xs:annotation>
      <xs:documentation>Request of the RU</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="52">
    <xs:annotation>
      <xs:documentation>Loading operations</xs:documentation>
    </xs:annotation>
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      <xs:documentation>Loading irregularities</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="54">
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    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="58">
    <xs:annotation>
      <xs:documentation>Staff</xs:documentation>
    </xs:annotation>
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  <xs:enumeration value="59">
    <xs:annotation>
      <xs:documentation>Other causes related to commercial causes</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="60">
    <xs:annotation>
      <xs:documentation>Roster planning/re-rostering</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="61">
    <xs:annotation>
      <xs:documentation>Formation of trains by the RU</xs:documentation>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="62">
    <xs:annotation>
      <xs:documentation>Problems affecting coaches</xs:documentation>
    </xs:annotation>
```

```
    </xs:enumeration>
    <xs:enumeration value="63">
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        <xs:documentation>Problems affecting wagons</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="64">
      <xs:annotation>
        <xs:documentation>Problems affecting traction units</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="68">
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        <xs:documentation>Staff</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="69">
      <xs:annotation>
        <xs:documentation>Other causes related to Rolling Stock</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="70">
      <xs:annotation>
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      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="71">
      <xs:annotation>
        <xs:documentation>Delay caused by previous RU</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="80">
      <xs:annotation>
        <xs:documentation>Strike</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="81">
      <xs:annotation>
        <xs:documentation>Administrative formalites</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="82">
      <xs:annotation>
        <xs:documentation>Outside influence</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="83">
      <xs:annotation>
        <xs:documentation>Effects of weather and natural causes</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="84">
      <xs:annotation>
        <xs:documentation>Delay caused by external reasons on the next
network</xs:documentation>
      </xs:annotation>
```

```
    </xs:enumeration>
    <xs:enumeration value="89">
      <xs:annotation>
        <xs:documentation>Other external causes</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="90">
      <xs:annotation>
        <xs:documentation>Dangerous incidents, accidents and hazards</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="91">
      <xs:annotation>
        <xs:documentation>Track occupation caused by the lateness of the same
train</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="92">
      <xs:annotation>
        <xs:documentation>Track occupation caused by the lateness of another
train</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="93">
      <xs:annotation>
        <xs:documentation>Turn round</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="94">
      <xs:annotation>
        <xs:documentation>Connection</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="95">
      <xs:annotation>
        <xs:documentation>Further investigation needed</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
  </xs:restriction>
</xs:simpleType>
```

simpleType **teltsi_DeltaTime**

type	restriction of **xs:string**
properties	base xs:string
used by	elements **teltsi_AgainstBooked** **teltsi_AgainstReferenced**
facets	Kind Value Annotation length 5
annotation	documentation Time difference delay (+) or ahead of schedule (-) this shall be 1character + 4 Numeric
source	<xs:simpleType name="teltsi_DeltaTime"> <xs:annotation> <xs:documentation>Time difference delay (+) or ahead of schedule (-) this shall be 1character +

	4 Numeric</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:length value="5"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_FreeText**

type	restriction of **xs:string**
properties	base xs:string
used by	elements **DetailedDescriptionOfLocation** **MessageIdentifier** **teltsi_InterruptionPoint/teltsi_DetailedDescriptionOfLocation** **teltsi_InternalReferenceIdentifier** **teltsi_InterruptionDescription** **teltsi_LocationSubsidiaryName** **teltsi_PrimaryLocationName** **teltsi_Remarks**
facets	Kind Value Annotation minLength 1 maxLength 255
annotation	documentation Clear Text in ISO Unicode character set
source	<xs:simpleType name="teltsi_FreeText"> <xs:annotation> <xs:documentation>Clear Text in ISO Unicode character set</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:maxLength value="255"/> <xs:minLength value="1"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_Numeric1-5**

type	restriction of **xs:positiveInteger**
properties	base xs:positiveInteger
used by	element **teltsi_LocationPrimaryCode**
facets	Kind Value Annotation minInclusive 1 maxInclusive 99999
source	<xs:simpleType name="teltsi_Numeric1-5"> <xs:restriction base="xs:positiveInteger"> <xs:minInclusive value="1"/> <xs:maxInclusive value="99999"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_Numeric2-2**

type	restriction of **xs:integer**
properties	base xs:integer

used by	attribute **CI_InstanceNumber**
facets	Kind Value Annotation minInclusive 01 maxInclusive 99
source	<xs:simpleType name="teltsi_Numeric2-2"> <xs:restriction base="xs:integer"> <xs:minInclusive value="01"/> <xs:maxInclusive value="99"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_Numeric3-3**

type	restriction of **xs:integer**
properties	base xs:integer
used by	simpleType **teltsi_Speed**
facets	Kind Value Annotation minInclusive 001 maxInclusive 999
source	<xs:simpleType name="teltsi_Numeric3-3"> <xs:restriction base="xs:integer"> <xs:minInclusive value="001"/> <xs:maxInclusive value="999"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_Percentage**

type	restriction of **xs:float**
properties	base xs:float
used by	element **teltsi_NetworkProjectedLocation/teltsi_ProportionOfDistanceBetweenLocations**
facets	Kind Value Annotation minInclusive 0 maxInclusive 100
annotation	documentation decimal value between 0 and 100
source	<xs:simpleType name="teltsi_Percentage"> <xs:annotation> <xs:documentation>decimal value between 0 and 100</xs:documentation> </xs:annotation> <xs:restriction base="xs:float"> <xs:minInclusive value="0"/> <xs:maxInclusive value="100"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_Speed**

type	**teltsi_Numeric3-3**
properties	base teltsi_Numeric3-3
used by	element **teltsi_GNSS_DynamicPosition/teltsi_CurrentSpeed**
facets	Kind Value Annotation minInclusive 001 maxInclusive 999
annotation	documentation Shown in Km/h
source	<xs:simpleType name="teltsi_Speed"> <xs:annotation> <xs:documentation>Shown in Km/h</xs:documentation> </xs:annotation> <xs:restriction base="teltsi_Numeric3-3"/> </xs:simpleType>

simpleType **teltsi_String1-10**

type	restriction of **xs:string**
properties	base xs:string
used by	element **teltsi_LocationSubsidiaryCode**
facets	Kind Value Annotation minLength 1 maxLength 10
source	<xs:simpleType name="teltsi_String1-10"> <xs:restriction base="xs:string"> <xs:minLength value="1"/> <xs:maxLength value="10" fixed="false"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_String1-8**

type	restriction of **xs:string**
properties	base xs:string
used by	element **teltsi_OperationalTrainNumber**
facets	Kind Value Annotation minLength 1 maxLength 8
source	<xs:simpleType name="teltsi_String1-8"> <xs:restriction base="xs:string"> <xs:minLength value="1"/> <xs:maxLength value="8"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_String4-4**

type	restriction of **xs:string**
properties	base xs:string
used by	simpleType **teltsi_CompanyCode**
facets	Kind / Value / Annotation minLength 4 maxLength 4
source	<xs:simpleType name="teltsi_String4-4"> <xs:restriction base="xs:string"> <xs:minLength value="4"/> <xs:maxLength value="4"/> </xs:restriction> </xs:simpleType>

simpleType **teltsi_WeightValueKilo**

type	restriction of **xs:integer**
properties	base xs:integer
used by	elements **LocoWeight** **teltsi_TotalLoadWeight** **teltsi_WagonWeightEmpty**
facets	Kind / Value / Annotation minInclusive 0 maxInclusive 999999 whiteSpace collapse
annotation	documentation In Kilograms
source	<xs:simpleType name="teltsi_WeightValueKilo"> <xs:annotation> <xs:documentation>In Kilograms</xs:documentation> </xs:annotation> <xs:restriction base="xs:integer"> <xs:minInclusive value="0"/> <xs:maxInclusive value="999999"/> <xs:whiteSpace value="collapse"/> </xs:restriction> </xs:simpleType>

simpleType **tractionTypeType**

type	restriction of **xs:string**
properties	base xs:string
facets	Kind / Value / Annotation enumeration 600V DC enumeration 750V DC enumeration 1500V DC enumeration 3000V DC enumeration 25000V AC 50Hz enumeration 15000V AC 16Hz2/3 enumeration Self-propelled

	enumeration Powerless
annotation	documentation Simple Type for TractionTypes
source	<xs:simpleType name="tractionTypeType"> <xs:annotation> <xs:documentation>Simple Type for TractionTypes</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="600V DC"/> <xs:enumeration value="750V DC"/> <xs:enumeration value="1500V DC"/> <xs:enumeration value="3000V DC"/> <xs:enumeration value="25000V AC 50Hz"/> <xs:enumeration value="15000V AC 16Hz2/3"/> <xs:enumeration value="Self-propelled"/> <xs:enumeration value="Powerless"/> </xs:restriction> </xs:simpleType>

simpleType **trainPartType**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **endSignalApplicationMustBeClearedBy** **startSignalApplicationMustBeClearedBy**
facets	Kind Value Annotation enumeration TrainHead enumeration TrainEnd enumeration TrainLastPantograph
annotation	documentation Simple type for train part (head, end or last pantograph). To be used for defining which part of the train has to clear a signal for application.
source	<xs:simpleType name="trainPartType"> <xs:annotation> <xs:documentation>Simple type for train part (head, end or last pantograph). To be used for defining which part of the train has to clear a signal for application.</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:enumeration value="TrainHead"/> <xs:enumeration value="TrainEnd"/> <xs:enumeration value="TrainLastPantograph"/> </xs:restriction> </xs:simpleType>

simpleType **UTCDateTimeType**

type	restriction of **xs:dateTime**
properties	base xs:dateTime
used by	elements **DateTimeMeasurement** **GeoLocalisationTimestamp** attributes **arrivalTime** **dataFirstAvailable** **dateEndApplication** **datePublished** **dateStartApplication** **departureTime** **earliestDepartureAdviceTime** **earliestDepartureTime** **endTime** **endValidity** **endValidityAbs** **forecastedArrival** **forecastedDeparture** **latestDepartureAdviceTime** **plannedDepartureTime** **startTime** **startValidity** **timestamp** **TP_latestArrivalTime** **TP_PlannedLatestArrivalTime**

facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Simple Type for date and time in UTC
source	<xs:simpleType name="UTCDateTimeType"> <xs:annotation> <xs:documentation>Simple Type for date and time in UTC</xs:documentation> </xs:annotation> <xs:restriction base="xs:dateTime"> <xs:pattern value="\d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z"/> </xs:restriction> </xs:simpleType>

simpleType **UUIDv4_Type**

type	restriction of **xs:string**
properties	base xs:string
used by	attributes **correlation_ID** **message_ID** **referenceIDAbs**
facets	Kind Value Annotation length 36 pattern [0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[8-9a-bA-B][0-9a-fA-F]{3}-[0-9a-fA-F]{12}
annotation	documentation UUIDv4 (random) simpleType as defined in RFC4122
source	<xs:simpleType name="UUIDv4_Type"> <xs:annotation> <xs:documentation>UUIDv4 (random) simpleType as defined in RFC4122</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:length value="36" fixed="true"/> <xs:pattern value="[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[8-9a-bA-B][0-9a-fA-F]{3}-[0-9a-fA-F]{12}"/> </xs:restriction> </xs:simpleType>

simpleType **voltageType**

type	**xs:int**
properties	base xs:int
used by	attributes **measuredVoltage** **TFC_estimatedVoltage** **voltageValue**
annotation	documentation Simple Type for voltage (V)
source	<xs:simpleType name="voltageType"> <xs:annotation> <xs:documentation>Simple Type for voltage (V)</xs:documentation> </xs:annotation> <xs:restriction base="xs:int"/> </xs:simpleType>

attribute **adviceType**

type	restriction of **xs:string**
used by	complexType **AdvisedSpeed_ComplexType**
facets	Kind / Value / Annotation enumeration StopAdvice enumeration AccelerationAdvice enumeration ConstantspeedAdvice enumeration CoastingAdvice enumeration OperationalBrakingAdvice enumeration ElectricalBrakingAdvice enumeration EndOfAdvice enumeration DepartureAdvice enumeration TextAdvice enumeration DeleteAdvice
annotation	documentation Type of C-DAS-C Advice
source	<xs:attribute name="adviceType"> <xs:annotation> <xs:documentation>Type of C-DAS-C Advice</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="StopAdvice"/> <xs:enumeration value="AccelerationAdvice"/> <xs:enumeration value="ConstantspeedAdvice"/> <xs:enumeration value="CoastingAdvice"/> <xs:enumeration value="OperationalBrakingAdvice"/> <xs:enumeration value="ElectricalBrakingAdvice"/> <xs:enumeration value="EndOfAdvice"/> <xs:enumeration value="DepartureAdvice"/> <xs:enumeration value="TextAdvice"/> <xs:enumeration value="DeleteAdvice"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **aimEarlyArrival**

type	**xs:boolean**
used by	element **TimingPointConstraints**
annotation	documentation Aim to be as near as possible to the beginning of the arrival window
source	<xs:attribute name="aimEarlyArrival" type="xs:boolean"> <xs:annotation> <xs:documentation>Aim to be as near as possible to the beginning of the arrival window</xs:documentation> </xs:annotation> </xs:attribute>

attribute **ALSP_Front**

type	**xs:boolean**
used by	element **AxleLoadSpeedProfile**
annotation	documentation [SUBSET-126] Qualifier to indicate if the ALSP is to be applied to the front of the train (no train length delay) or to the end of the train (train length delay).
source	<xs:attribute name="ALSP_Front" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] Qualifier to indicate if the ALSP is to be applied to the front of the train (no train length delay) or to the end of the train (train length delay).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **alternativeItineraryProbability**

type	**percentageType**
used by	element **AlternativeItinerary**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1
annotation	documentation Probability that the alternative itinerary is taken
source	<xs:attribute name="alternativeItineraryProbability" type="percentageType"> <xs:annotation> <xs:documentation>Probability that the alternative itinerary is taken</xs:documentation> </xs:annotation> </xs:attribute>

attribute **altitude**

type	**xs:decimal**
used by	complexType **GNSS_StaticPosition_ComplexType**
annotation	documentation Altitude (m). Considering ETRS89 as reference.
source	<xs:attribute name="altitude" type="xs:decimal"> <xs:annotation> <xs:documentation>Altitude (m). Considering ETRS89 as reference.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **amountLevelCrossings**

type	**xs:unsignedByte**
used by	element **Balise**
annotation	documentation Amount of guarded level crossings (to be used only if levelCrossingGuarding is true)
source	<xs:attribute name="amountLevelCrossings" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>Amount of guarded level crossings (to be used only if levelCrossingGuarding

	is true)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **applicationDistance**

type	**distanceType**
annotation	documentation How far (in m) after the signal the speed must be maintained.
source	<xs:attribute name="applicationDistance" type="distanceType"> <xs:annotation> <xs:documentation>How far (in m) after the signal the speed must be maintained.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **arrivalTime**

type	**UTCDateTimeType**
used by	element **PreviousTimingPointTrainInformation**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Arrival time at the Timing Point
source	<xs:attribute name="arrivalTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Arrival time at the Timing Point</xs:documentation> </xs:annotation> </xs:attribute>

attribute **arrivalWindow**

type	**xs:duration**
used by	element **TimingPointConstraints**
annotation	documentation Acceptable time allowance to be earlier at the TP
source	<xs:attribute name="arrivalWindow" type="xs:duration"> <xs:annotation> <xs:documentation>Acceptable time allowance to be earlier at the TP</xs:documentation> </xs:annotation> </xs:attribute>

attribute **ascending**

type	restriction of **xs:string**
used by	element **KM_Reference**
facets	Kind Value Annotation enumeration Ascending enumeration Descending

annotation	documentation Ascending or Descending
source	<xs:attribute name="ascending"> <xs:annotation> <xs:documentation>Ascending or Descending</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Ascending"/> <xs:enumeration value="Descending"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **ASR_Front**

type	**xs:boolean**
used by	element **AdditionalSpeedRestriction**
annotation	documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (1, no train length delay) or the end of the train (0, train length delay) has left the element
source	<xs:attribute name="ASR_Front" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (1, no train length delay) or the end of the train (0, train length delay) has left the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **ASR_Speed**

type	**speedType**
used by	element **AdditionalSpeedRestriction**
annotation	documentation [SUBSET-126] Value of the speed level restriction (km/h)
source	<xs:attribute name="ASR_Speed" type="speedType"> <xs:annotation> <xs:documentation>[SUBSET-126] Value of the speed level restriction (km/h)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **ATO_HandingOver**

type	**xs:boolean**
used by	element **HandshakeRequest**
annotation	documentation [SUBSET-126] This attribute allows the ATO-TS to know if the handshake request is due to a handing over ATO-TS.
source	<xs:attribute name="ATO_HandingOver" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] This attribute allows the ATO-TS to know if the handshake

	request is due to a handing over ATO-TS.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **ATO_State**

type	restriction of **xs:string**
used by	element **B2G_StatusReport**
facets	Kind / Value / Annotation enumeration Unknown enumeration CO enumeration NA enumeration AV enumeration RE enumeration EG enumeration DE enumeration FA
annotation	documentation [SUBSET-126] ATO State, from Subset-126. Only relevant when in ATO mode.
source	<xs:attribute name="ATO_State"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO State, from Subset-126. Only relevant when in ATO mode.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Unknown"/> <xs:enumeration value="CO"/> <xs:enumeration value="NA"/> <xs:enumeration value="AV"/> <xs:enumeration value="RE"/> <xs:enumeration value="EG"/> <xs:enumeration value="DE"/> <xs:enumeration value="FA"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **ATOTS_ID**

type	**xs:string**
used by	elements **ATOTS_ContactInfo** **HandshakeAcknowledgement** **HandshakeReject**
annotation	documentation [SUBSET-126] Identifier of the ATO-TS
source	<xs:attribute name="ATOTS_ID" type="xs:string"> <xs:annotation> <xs:documentation>[SUBSET-126] Identifier of the ATO-TS</xs:documentation> </xs:annotation> </xs:attribute>

attribute **automaticClosing**

type	**xs:boolean**
used by	element **StoppingPointDepartureDetails**
annotation	documentation [SUBSET-126] This variable defines if the ATO-OB has to manage the train doors closing (1) or not (0)
source	<xs:attribute name="automaticClosing" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] This variable defines if the ATO-OB has to manage the train doors closing (1) or not (0)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **autonomy**

type	**energyType**
used by	element **EnergyData**
annotation	documentation Amount of energy still available on-board for the self-propelled energy source (in kWh)
source	<xs:attribute name="autonomy" type="energyType"> <xs:annotation> <xs:documentation>Amount of energy still available on-board for the self-propelled energy source (in kWh)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **axleLoadCategory**

type	restriction of **xs:string**
used by	element **AxleLoadSpeedProfile** complexType **TC_Features_ComplexType**
facets	Kind / Value / Annotation enumeration A enumeration HS17 enumeration B1 enumeration B2 enumeration C2 enumeration C3 enumeration C4 enumeration D2 enumeration D3 enumeration D4 enumeration D4XL enumeration E4 enumeration E5
annotation	documentation [SUBSET-126] Axle load from which the speed restriction is applicable
source	<xs:attribute name="axleLoadCategory"> <xs:annotation>

```
    <xs:documentation>[SUBSET-126] Axle load from which the speed restriction is
applicable</xs:documentation>
  </xs:annotation>
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:enumeration value="A"/>
      <xs:enumeration value="HS17"/>
      <xs:enumeration value="B1"/>
      <xs:enumeration value="B2"/>
      <xs:enumeration value="C2"/>
      <xs:enumeration value="C3"/>
      <xs:enumeration value="C4"/>
      <xs:enumeration value="D2"/>
      <xs:enumeration value="D3"/>
      <xs:enumeration value="D4"/>
      <xs:enumeration value="D4XL"/>
      <xs:enumeration value="E4"/>
      <xs:enumeration value="E5"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **baliseGroupIdentifier**

type	**xs:unsignedShort**
used by	element **BaliseGroup**
annotation	documentation [SUBSET-126] Identifier of the balise group
source	<xs:attribute name="baliseGroupIdentifier" type="xs:unsignedShort"> <xs:annotation> <xs:documentation>[SUBSET-126] Identifier of the balise group</xs:documentation> </xs:annotation> </xs:attribute>

attribute **baliseRelativePosition**

type	**xs:unsignedByte**
used by	element **Balise**
annotation	documentation [SUBSET-126] Defines the relative position in a balise group
source	<xs:attribute name="baliseRelativePosition" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>[SUBSET-126] Defines the relative position in a balise group</xs:documentation> </xs:annotation> </xs:attribute>

attribute **boardingPassengers**

type	**xs:unsignedInt**

used by	element **PassengerMovement**
annotation	documentation Number of passengers boarding the train
source	<xs:attribute name="boardingPassengers" type="xs:unsignedInt"> <xs:annotation> <xs:documentation>Number of passengers boarding the train</xs:documentation> </xs:annotation> </xs:attribute>

attribute **brakingForce**

type	**forceType**
used by	element **BrakingForceCurveDataPoint**
annotation	documentation Braking Force (in N)
source	<xs:attribute name="brakingForce" type="forceType"> <xs:annotation> <xs:documentation>Braking Force (in N)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **brakingReactionTime**

type	**xs:duration**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Delay between braking command and start of braking.
source	<xs:attribute name="brakingReactionTime" type="xs:duration"> <xs:annotation> <xs:documentation>Delay between braking command and start of braking.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **C_DAS_C_Able**

type	**xs:boolean**
annotation	documentation 1 = The device is able to receive C-DAS-C messages. 0 (Default) = The device is unable to receive C-DAS-C messages
source	<xs:attribute name="C_DAS_C_Able" type="xs:boolean"> <xs:annotation> <xs:documentation>1 = The device is able to receive C-DAS-C messages. 0 (Default) = The device is unable to receive C-DAS-C messages</xs:documentation> </xs:annotation> </xs:attribute>

attribute **centralisedOpening**

type	**xs:boolean**
used by	element **StoppingPointInformation**

annotation	documentation [SUBSET-126] This variable defines if the doors are to be opened centralised (1) or by the passengers (0)
source	<xs:attribute name="centralisedOpening" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] This variable defines if the doors are to be opened centralised (1) or by the passengers (0)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **CI_InstanceNumber**

type	**teltsi_Numeric2-2**
used by	elements **Recipient** **Sender**
facets	Kind Value Annotation minInclusive 01 maxInclusive 99
annotation	documentation Number of a Common Interface Instance for the same Company
source	<xs:attribute name="CI_InstanceNumber" type="teltsi_Numeric2-2"> <xs:annotation> <xs:documentation>Number of a Common Interface Instance for the same Company</xs:documentation> </xs:annotation> </xs:attribute>

attribute **comfortableAcceleration**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Comfortable acceleration. Expressed in m/s^2
source	<xs:attribute name="comfortableAcceleration" type="xs:double"> <xs:annotation> <xs:documentation>Comfortable acceleration. Expressed in m/s^2</xs:documentation> </xs:annotation> </xs:attribute>

attribute **comfortableBraking**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Deceleration used for comfortable braking. Expressed in m/s^2
source	<xs:attribute name="comfortableBraking" type="xs:double"> <xs:annotation> <xs:documentation>Deceleration used for comfortable braking. Expressed in m/s^2</xs:documentation> </xs:annotation> </xs:attribute>

attribute **communicationNetworkChannel**

type	**xs:string**
used by	element **CommunicationNetwork**
annotation	documentation Channel for communication network
source	<xs:attribute name="communicationNetworkChannel" type="xs:string"> <xs:annotation> <xs:documentation>Channel for communication network</xs:documentation> </xs:annotation> </xs:attribute>

attribute **communicationNetworkType**

type	**xs:string**
used by	element **CommunicationNetwork**
annotation	documentation Type of communication network. Standard possibilities are: "GSM-R", "FRMCS", "Public GSM", "Public WiFi" and "Private WiFi".
source	<xs:attribute name="communicationNetworkType" type="xs:string"> <xs:annotation> <xs:documentation>Type of communication network. Standard possibilities are: "GSM-R", "FRMCS", "Public GSM", "Public WiFi" and "Private WiFi".</xs:documentation> </xs:annotation> </xs:attribute>

attribute **connectionTrackType**

type	restriction of **xs:string**		
used by	element **ConnectionTrack**		
facets	Kind	Value	Annotation
	enumeration	ConvergenceFromRight	
	enumeration	ConvergingIntoRight	
	enumeration	ConvergenceFromLeft	
	enumeration	ConvergingIntoLeft	
	enumeration	CrossingFromRightToLeft	
	enumeration	CrossingFromLeftToRight	
	enumeration	CrossingALineOnLeft	
	enumeration	CrossingALineOnRight	
	enumeration	DivergenceIntoRight	
	enumeration	DivergenceAtRight	
	enumeration	DivergenceIntoLeft	
	enumeration	DivergenceAtLeft	
	enumeration	NominalDirectionBufferStop	
	enumeration	ReverseDirectionBufferStop	
	enumeration	Unknown	
annotation	documentation Type of connection track		

source

```
<xs:attribute name="connectionTrackType">
  <xs:annotation>
    <xs:documentation>Type of connection track</xs:documentation>
  </xs:annotation>
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:enumeration value="ConvergenceFromRight"/>
      <xs:enumeration value="ConvergingIntoRight"/>
      <xs:enumeration value="ConvergenceFromLeft"/>
      <xs:enumeration value="ConvergingIntoLeft"/>
      <xs:enumeration value="CrossingFromRightToLeft"/>
      <xs:enumeration value="CrossingFromLeftToRight"/>
      <xs:enumeration value="CrossingALineOnLeft"/>
      <xs:enumeration value="CrossingALineOnRight"/>
      <xs:enumeration value="DivergenceIntoRight"/>
      <xs:enumeration value="DivergenceAtRight"/>
      <xs:enumeration value="DivergenceIntoLeft"/>
      <xs:enumeration value="DivergenceAtLeft"/>
      <xs:enumeration value="NominalDirectionBufferStop"/>
      <xs:enumeration value="ReverseDirectionBufferStop"/>
      <xs:enumeration value="Unknown"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **contactIdentifier**

type	**xs:string**
used by	elements **Contact/OtherContactType** **Contact/SFERA_PlainTextMessaging**
annotation	documentation Identifier to be used to reach the contact on the communication platform defined in ContactType (e.g. telephone number, GSM-R number, pseudo...)

source

```
<xs:attribute name="contactIdentifier" type="xs:string">
  <xs:annotation>
    <xs:documentation>Identifier to be used to reach the contact on the communication platform defined in ContactType (e.g. telephone number, GSM-R number, pseudo...)</xs:documentation>
  </xs:annotation>
</xs:attribute>
```

attribute **contactNotes**

type	**xs:string**
annotation	documentation Notes related to the contact

source

```
<xs:attribute name="contactNotes" type="xs:string">
  <xs:annotation>
    <xs:documentation>Notes related to the contact</xs:documentation>
  </xs:annotation>
</xs:attribute>
```

attribute **contactRole**

type	**xs:string**
used by	element **Contact**
annotation	documentation Role of the contact
source	<xs:attribute name="contactRole" type="xs:string"> <xs:annotation> <xs:documentation>Role of the contact</xs:documentation> </xs:annotation> </xs:attribute>

attribute **contactType**

type	**xs:string**
annotation	documentation Type of contact. Communication platform to be used to contact this person: e.g. telephone, GSM-R, social network, messaging app, etc.
source	<xs:attribute name="contactType" type="xs:string"> <xs:annotation> <xs:documentation>Type of contact. Communication platform to be used to contact this person: e.g. telephone, GSM-R, social network, messaging app, etc.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **correlation_ID**

type	**UUIDv4_Type**
used by	element **MessageHeader**
facets	Kind Value Annotation length 36 pattern [0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[8-9a-bA-B][0-9a-fA-F]{3}-[0-9a-fA-F]{12}
annotation	documentation ID of the message for which the current message is a reply
source	<xs:attribute name="correlation_ID" type="UUIDv4_Type"> <xs:annotation> <xs:documentation>ID of the message for which the current message is a reply</xs:documentation> </xs:annotation> </xs:attribute>

attribute **currentPassengers**

type	**xs:unsignedInt**
used by	element **PassengerData**
annotation	documentation Current number of passengers
source	<xs:attribute name="currentPassengers" type="xs:unsignedInt"> <xs:annotation> <xs:documentation>Current number of passengers</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **curveDirection**

type	restriction of **xs:string**
used by	complexType **Curve_ComplexType**
facets	Kind / Value / Annotation enumeration Unknown enumeration Left enumeration Right
annotation	documentation Direction of the curve (Left or Right)
source	<xs:attribute name="curveDirection"> <xs:annotation> <xs:documentation>Direction of the curve (Left or Right)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Unknown"/> <xs:enumeration value="Left"/> <xs:enumeration value="Right"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **curveRadius**

type	**xs:unsignedShort**
used by	complexType **Curve_ComplexType**
annotation	documentation Radius of the curve (m). If the radius is set to 0, it indicates a straight line.
source	<xs:attribute name="curveRadius" type="xs:unsignedShort"> <xs:annotation> <xs:documentation>Radius of the curve (m). If the radius is set to 0, it indicates a straight line.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **DAS_architecture**

type	restriction of **xs:string**
used by	elements **DAS_OperatingModeSelected** **Unavailable_DAS_OperatingModes** complexType **DAS_ModesComplexType**
facets	Kind / Value / Annotation enumeration GroundAdviceCalculation enumeration BoardAdviceCalculation
annotation	documentation DAS architecture: DAS-C (ground advice calculation) or DAS-O (board advice calculation)
source	<xs:attribute name="DAS_architecture"> <xs:annotation> <xs:documentation>DAS architecture: DAS-C (ground advice calculation) or DAS-O (board

```
advice calculation)</xs:documentation>
  </xs:annotation>
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:enumeration value="GroundAdviceCalculation"/>
      <xs:enumeration value="BoardAdviceCalculation"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **DAS_Behaviour**

type	restriction of **xs:string**
facets	Kind / Value / Annotation enumeration Standalone enumeration Connected enumeration C-DAS-C_Only
annotation	documentation Behaviour of DAS (Standalone or Connected)
source	<xs:attribute name="DAS_Behaviour"> <xs:annotation> <xs:documentation>Behaviour of DAS (Standalone or Connected)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Standalone"/> <xs:enumeration value="Connected"/> <xs:enumeration value="C-DAS-C_Only"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **DAS_connectivity**

type	restriction of **xs:string**
used by	elements **DAS_OperatingModeSelected** **Unavailable_DAS_OperatingModes** complexType **DAS_ModesComplexType**
facets	Kind / Value / Annotation enumeration Standalone enumeration Connected
annotation	documentation DAS connectivity (standalone or connected)
source	<xs:attribute name="DAS_connectivity"> <xs:annotation> <xs:documentation>DAS connectivity (standalone or connected)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Standalone"/> <xs:enumeration value="Connected"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **DAS_drivingMode**

type	restriction of **xs:string**
used by	elements **ForceDrivingModeChange** **Reported_DAS_DrivingMode** **Unavailable_DAS_OperatingModes** complexType **DAS_ModesComplexType**
facets	Kind / Value / Annotation enumeration Inactive enumeration Timetable enumeration Read-Only enumeration DAS not connected to ATP enumeration GoA1 enumeration GoA2 enumeration GoA3 enumeration GoA4
annotation	documentation DAS driving mode (see enumeration)
source	<xs:attribute name="DAS_drivingMode"> <xs:annotation> <xs:documentation>DAS driving mode (see enumeration)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Inactive"/> <xs:enumeration value="Timetable"/> <xs:enumeration value="Read-Only"/> <xs:enumeration value="DAS not connected to ATP"/> <xs:enumeration value="GoA1"/> <xs:enumeration value="GoA2"/> <xs:enumeration value="GoA3"/> <xs:enumeration value="GoA4"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **dataFirstAvailable**

type	**UTCDateTimeType**
used by	element **G2B_Error**
facets	Kind / Value / Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Date and time of first availability of the requested data
source	<xs:attribute name="dataFirstAvailable" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Date and time of first availability of the requested data</xs:documentation> </xs:annotation> </xs:attribute>

attribute **dateEndApplication**

type	**UTCDateTimeType**
used by	elements **SegmentProfile** **TrainCharacteristics**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation date of end of application of the element
source	<xs:attribute name="dateEndApplication" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>date of end of application of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **datePublished**

type	**UTCDateTimeType**
used by	elements **SegmentProfile** **TrainCharacteristics**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation date of publication of the element
source	<xs:attribute name="datePublished" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>date of publication of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **dateStartApplication**

type	**UTCDateTimeType**
used by	elements **SegmentProfile** **TrainCharacteristics**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation date of start of application of the element
source	<xs:attribute name="dateStartApplication" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>date of start of application of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **daylightSaving**

type	**xs:boolean**
used by	element **TimingPointConstraints**
annotation	documentation [SUBSET-126] Specifies some information specific for the TPThis variable defines if the daylight saving hour is applicable to calculate the local time
source	<xs:attribute name="daylightSaving" type="xs:boolean"> <xs:annotation>

	<xs:documentation>[SUBSET-126] Specifies some information specific for the TPThis variable defines if the daylight saving hour is applicable to calculate the local time</xs:documentation> </xs:annotation> </xs:attribute>

attribute **Delay**

type	**xs:duration**
used by	element **Delay**
annotation	documentation Delay of the train
source	<xs:attribute name="Delay" type="xs:duration"> <xs:annotation> <xs:documentation>Delay of the train</xs:documentation> </xs:annotation> </xs:attribute>

attribute **DelayCategory**

type	restriction of **xs:string**
used by	element **Delay**
facets	Kind / Value / Annotation enumeration Measured enumeration Expected enumeration Upcoming enumeration Unknown
annotation	documentation Category of the delay. Delay could be measured, delay can be expected (for example before departure), announcement that the delay will be published (upcoming) or can be unkown
source	<xs:attribute name="DelayCategory"> <xs:annotation> <xs:documentation>Category of the delay. Delay could be measured, delay can be expected (for example before departure), announcement that the delay will be published (upcoming) or can be unkown </xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Measured"/> <xs:enumeration value="Expected"/> <xs:enumeration value="Upcoming"/> <xs:enumeration value="Unknown"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **deltaSpeed**

type	**speedType**

used by	complexType **AdvisedSpeed_ComplexType**
annotation	documentation Difference (in km/h) between maximum allowed track speed/most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed/most restrictive speed profile".
source	<xs:attribute name="deltaSpeed" type="speedType"> <xs:annotation> <xs:documentation>Difference (in km/h) between maximum allowed track speed/most restrictive speed profile and optimal speed. The special value 0 indicates "maximum allowed track speed/most restrictive speed profile".</xs:documentation> </xs:annotation> </xs:attribute>

attribute **departureAuthorisationType**

type	**xs:string**
used by	element **StoppingPointDepartureDetails**
annotation	documentation Type of authorisation for departure (e.g. SMS)
source	<xs:attribute name="departureAuthorisationType" type="xs:string"> <xs:annotation> <xs:documentation>Type of authorisation for departure (e.g. SMS)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **departureProcedureDuration**

type	**xs:duration**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.
source	<xs:attribute name="departureProcedureDuration" type="xs:duration"> <xs:annotation> <xs:documentation>Time necessary between the moment the driver starts the immediate departure procedure (e.g. door closing), and the start of the train.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **departureTime**

type	**UTCDateTimeType**
used by	elements **PreviousTimingPointTrainInformation** **StoppingPointDepartureDetails**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Expected time of departure at Timing Point expressed in UTC
source	<xs:attribute name="departureTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Expected time of departure at Timing Point expressed in

	UTC</xs:documentation> </xs:annotation> </xs:attribute>

attribute **description**

type	**xs:string**
used by	element **NetworkSpecificParameter**
annotation	documentation Description of the element
source	<xs:attribute name="description" type="xs:string"> <xs:annotation> <xs:documentation>Description of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **destinationCompany**

type	**teltsi_CompanyCode**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation ID of the company to which the message is addressed. Must be a UIC RICS Code: : https://uic.org/rics
source	<xs:attribute name="destinationCompany" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>ID of the company to which the message is addressed. Must be a UIC RICS Code: : https://uic.org/rics</xs:documentation> </xs:annotation> </xs:attribute>

attribute **destinationDevice**

type	**deviceType**
used by	element **MessageHeader**
facets	Kind Value Annotation maxLength 128
annotation	documentation ID of the device to which the message is addressed. For any given company, this ID must be unique. Maximum 128 characters. For G2B messages, if this attribute is empty, the message is directed to all devices in the train. For B2G messages, this attribute can be ignored.
source	<xs:attribute name="destinationDevice" type="deviceType"> <xs:annotation> <xs:documentation>ID of the device to which the message is addressed. For any given company, this ID must be unique. Maximum 128 characters. For G2B messages, if this attribute is empty, the message is directed to all devices in the train. For B2G messages, this attribute can be ignored.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **directionOfApplicationOnSP**

type	restriction of **xs:string**
used by	elements **DecisiveGradientArea** **SP_NextSP** **UnprotectedLevelCrossingStop** complexType **Signal_ComplexType**
facets	Kind Value Annotation enumeration Nominal enumeration Reverse enumeration Both
annotation	documentation Direction of application on the Segment Profile (Nominal, Reverse or Both)
source	<xs:attribute name="directionOfApplicationOnSP"> <xs:annotation> <xs:documentation>Direction of application on the Segment Profile (Nominal, Reverse or Both)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Nominal"/> <xs:enumeration value="Reverse"/> <xs:enumeration value="Both"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **disembarkingPassengers**

type	**xs:unsignedInt**
used by	element **PassengerMovement**
annotation	documentation Number of passengers disembarking the train
source	<xs:attribute name="disembarkingPassengers" type="xs:unsignedInt"> <xs:annotation> <xs:documentation>Number of passengers disembarking the train</xs:documentation> </xs:annotation> </xs:attribute>

attribute **displayName**

type	**xs:string**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Name of train type to be displayed to the driver (e.g. ‘TGV’ of ‘VIRM 6’).
source	<xs:attribute name="displayName" type="xs:string"> <xs:annotation> <xs:documentation>Name of train type to be displayed to the driver (e.g. ‘TGV’ of ‘VIRM 6’).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **distance_EoA_Offset**

type	**distanceType**
used by	element **SegmentProfile**
annotation	documentation [SUBSET-126] Distance to stop the train in rear of the End of Authority (in m)
source	<xs:attribute name="distance_EoA_Offset" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Distance to stop the train in rear of the End of Authority (in m)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **distance_STP_Reached**

type	**distanceType**
used by	element **StoppingPointLocation**
annotation	documentation [SUBSET-126] Distance (in m) from a Stopping Point to consider it as reached
source	<xs:attribute name="distance_STP_Reached" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Distance (in m) from a Stopping Point to consider it as reached</xs:documentation> </xs:annotation> </xs:attribute>

attribute **distanceBeforeSignal**

type	**distanceType**
used by	element **CancelPreviousSignal**
annotation	documentation The distanceBeforeSignal attribute gives the point from which to restart accelerating. Depending on the type of signalling, the position of this point will be defined by the visual identification by the driver, the position of a balise, or the time at which the driver will receive an updated information for in-cab signalling. This distance applies to the front of the train.
source	<xs:attribute name="distanceBeforeSignal" type="distanceType"> <xs:annotation> <xs:documentation>The distanceBeforeSignal attribute gives the point from which to restart accelerating. Depending on the type of signalling, the position of this point will be defined by the visual identification by the driver, the position of a balise, or the time at which the driver will receive an updated information for in-cab signalling. This distance applies to the front of the train.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **distanceToPowerConstraint**

type	**distanceType**
annotation	documentation Distance (in m) until the Power Constraint is to be applied
source	<xs:attribute name="distanceToPowerConstraint" type="distanceType">

	<xs:annotation> <xs:documentation>Distance (in m) until the Power Constraint is to be applied</xs:documentation> </xs:annotation> </xs:attribute>

attribute **distanceToTractionTypeTransition**

type	**distanceType**
annotation	documentation Distance (in m) from the actual Transition Point
source	<xs:attribute name="distanceToTractionTypeTransition" type="distanceType"> <xs:annotation> <xs:documentation>Distance (in m) from the actual Transition Point</xs:documentation> </xs:annotation> </xs:attribute>

attribute **driverID**

type	**xs:string**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Driver identifier number as defined in SUBSET-027 v300 §4.2.3.7.
source	<xs:attribute name="driverID" type="xs:string"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Driver identifier number as defined in SUBSET-027 v300 §4.2.3.7.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **dynamicBrakeForceLimit**

type	**xs:unsignedShort**
used by	element **LimitedDynamicBrakeForceArea**
annotation	documentation [SUBSET-126] Maximum Brake Force (kN) to be applied by the Dynamic Brake
source	<xs:attribute name="dynamicBrakeForceLimit" type="xs:unsignedShort"> <xs:annotation> <xs:documentation>[SUBSET-126] Maximum Brake Force (kN) to be applied by the Dynamic Brake</xs:documentation> </xs:annotation> </xs:attribute>

attribute **earliestDepartureAdviceTime**

type	**UTCDateTimeType**		
used by	element **C_DAS_C_Advice**		
facets	Kind	Value	Annotation
	pattern	\d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z	

annotation	documentation Lower bound of time window for train departure. Only to be used if departure advice is sent. If earliestDepartureTime is not specified, the advice should be obeyed at any time earlier than latestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
source	<xs:attribute name="earliestDepartureAdviceTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Lower bound of time window for train departure. Only to be used if departure advice is sent. If earliestDepartureTime is not specified, the advice should be obeyed at any time earlier than latestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **earliestDepartureTime**

type	**UTCDateTimeType**
used by	element **StoppingPointDepartureDetails**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Earliest departure time desired, expressed in UTC.
source	<xs:attribute name="earliestDepartureTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Earliest departure time desired, expressed in UTC.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endKm**

type	**xs:double**
used by	element **C_DAS_C_Advice**
annotation	documentation References the kilometre at the end of the message, preferably linked with a kilometer sign near the track
source	<xs:attribute name="endKm" type="xs:double"> <xs:annotation> <xs:documentation>References the kilometre at the end of the message, preferably linked with a kilometer sign near the track</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endKmAbs**

type	**xs:double**
used by	element **C_DAS_C_Advice**
annotation	documentation References the kilometre where the optimalSpeed changes next, preferably linked with a kilometer sign near the track
source	<xs:attribute name="endKmAbs" type="xs:double"> <xs:annotation> <xs:documentation>References the kilometre where the optimalSpeed changes next, preferably linked with a kilometer sign near the track</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **endLine**

type	**xs:int**
used by	element **C_DAS_C_Advice**
annotation	documentation References the (national) line number at the end of the message
source	<xs:attribute name="endLine" type="xs:int"> <xs:annotation> <xs:documentation>References the (national) line number at the end of the message</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endLineAbs**

type	**xs:int**
used by	element **C_DAS_C_Advice**
annotation	documentation References the (national) line number where the optimalSpeed changes next
source	<xs:attribute name="endLineAbs" type="xs:int"> <xs:annotation> <xs:documentation>References the (national) line number where the optimalSpeed changes next</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endLocation**

type	**distanceType**
used by	elements **ChangeReason** **PlatformChange/PlatformLengthChange** complexTypes **SegmentProfileInfo_ComplexType** **SP_GenericArea_ComplexType** **TemporaryConstraints_ComplexType**
annotation	documentation [SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)
source	<xs:attribute name="endLocation" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Distance of the element's end location from the beginning of the Segment Profile (m)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endPointOfDownLineSpeedApplication**

type	**distanceType**
annotation	documentation How far (in m) after the considered signal the speed advice must cease to be respected.
source	<xs:attribute name="endPointOfDownLineSpeedApplication" type="distanceType"> <xs:annotation>

	<xs:documentation>How far (in m) after the considered signal the speed advice must cease to be respected.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endSignal**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the name of the end signal of the message, if the end of the message is at or near a signal
source	<xs:attribute name="endSignal" type="xs:string"> <xs:annotation> <xs:documentation>References the name of the end signal of the message, if the end of the message is at or near a signal</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endSignalAbs**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the name of the signal where the optimalSpeed changes next, if this is at or near a signal
source	<xs:attribute name="endSignalAbs" type="xs:string"> <xs:annotation> <xs:documentation>References the name of the signal where the optimalSpeed changes next, if this is at or near a signal</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endSignalApplicationMustBeClearedBy**

type	**trainPartType**
used by	element **EndSignalApplication**
facets	Kind / Value / Annotation enumeration TrainHead enumeration TrainEnd enumeration TrainLastPantograph
annotation	documentation What part of the train must clear the signal application before the signal stops being applied
source	<xs:attribute name="endSignalApplicationMustBeClearedBy" type="trainPartType"> <xs:annotation> <xs:documentation>What part of the train must clear the signal application before the signal stops being applied</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endStationLong**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the start station at the end of the message (long form)
source	<xs:attribute name="endStationLong" type="xs:string"> <xs:annotation> <xs:documentation>References the start station at the end of the message (long form)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endStationLongAbs**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the station where the optimalSpeed changes next (long form)
source	<xs:attribute name="endStationLongAbs" type="xs:string"> <xs:annotation> <xs:documentation>References the station where the optimalSpeed changes next (long form)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endStationShort**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the start station at the end of the message (short form)
source	<xs:attribute name="endStationShort" type="xs:string"> <xs:annotation> <xs:documentation>References the start station at the end of the message (short form)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endStationShortAbs**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the station where the optimalSpeed changes next (short form)
source	<xs:attribute name="endStationShortAbs" type="xs:string"> <xs:annotation> <xs:documentation>References the station where the optimalSpeed changes next (short form)</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **endTime**

type	**UTCDateTimeType**
used by	elements **ChangeReason** **ExpectedCurrentData** **TemporarySignalConstraints** complexType **TemporaryConstraints_ComplexType**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Ending date and time (in UTC)
source	<xs:attribute name="endTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Ending date and time (in UTC)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endValidity**

type	**UTCDateTimeType**
used by	element **C_DAS_C_Advice**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation End of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
source	<xs:attribute name="endValidity" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>End of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **endValidityAbs**

type	**UTCDateTimeType**
used by	element **C_DAS_C_Advice**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Time at the end of the optimalSpeed
source	<xs:attribute name="endValidityAbs" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Time at the end of the optimalSpeed</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **enginePower**

type	**powerTypeW**
used by	element **Engine**
annotation	documentation Maximum power of the engine. Expressed in W
source	<xs:attribute name="enginePower" type="powerTypeW"> <xs:annotation> <xs:documentation>Maximum power of the engine. Expressed in W</xs:documentation> </xs:annotation> </xs:attribute>

attribute **engines**

type	**xs:int**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Number of engines
source	<xs:attribute name="engines" type="xs:int"> <xs:annotation> <xs:documentation>Number of engines</xs:documentation> </xs:annotation> </xs:attribute>

attribute **errorCode**

type	**xs:string**
used by	complexType **Error_ComplexType**
annotation	documentation Error Code
source	<xs:attribute name="errorCode" type="xs:string"> <xs:annotation> <xs:documentation>Error Code</xs:documentation> </xs:annotation> </xs:attribute>

attribute **EuropeanVehicleNumber**

type	restriction of **xs:string**
used by	element **MessageHeader**
facets	Kind Value Annotation minLength 12 maxLength 12
annotation	documentation Number on 12 digits allocated to each rail vehicle to uniquely identify it from any other rail vehicle
source	<xs:attribute name="EuropeanVehicleNumber">

	<xs:annotation> <xs:documentation>Number on 12 digits allocated to each rail vehicle to uniquely identify it from any other rail vehicle</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:minLength value="12"/> <xs:maxLength value="12"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **expectedConsumptionCurrent**

type	**currentType**
used by	element **ExpectedCurrentData**
annotation	documentation Expected consumption current (A)
source	<xs:attribute name="expectedConsumptionCurrent" type="currentType"> <xs:annotation> <xs:documentation>Expected consumption current (A)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **expectedRegenerativeCurrent**

type	**currentType**
used by	element **ExpectedCurrentData**
annotation	documentation Expected regenerative current (A)
source	<xs:attribute name="expectedRegenerativeCurrent" type="currentType"> <xs:annotation> <xs:documentation>Expected regenerative current (A)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **forecastedArrival**

type	**UTCDateTimeType**
used by	elements **NextTimingPointTrainInformation** **TimingPointEstimation**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Forecasted arrival at the Timing Point
source	<xs:attribute name="forecastedArrival" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Forecasted arrival at the Timing Point</xs:documentation> </xs:annotation> </xs:attribute>

attribute **forecastedDeparture**

type	**UTCDateTimeType**
used by	element **NextTimingPointTrainInformation**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Forecasted departure from the Timing Point
source	<xs:attribute name="forecastedDeparture" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Forecasted departure from the Timing Point</xs:documentation> </xs:annotation> </xs:attribute>

attribute **frequency**

type	**xs:decimal**
used by	complexType **Voltage_ComplexType**
annotation	documentation Frequency of the traction system in Hz. If the value is 0, DC is used.
source	<xs:attribute name="frequency" type="xs:decimal"> <xs:annotation> <xs:documentation>Frequency of the traction system in Hz. If the value is 0, DC is used.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **gradient_PBD**

type	**gradientValueType**
used by	element **PermittedBrakingDistanceArea**
annotation	documentation [SUBSET-126] A single gradient value applicable for the calculation of the Permitted Braking Distance
source	<xs:attribute name="gradient_PBD" type="gradientValueType"> <xs:annotation> <xs:documentation>[SUBSET-126] A single gradient value applicable for the calculation of the Permitted Braking Distance</xs:documentation> </xs:annotation> </xs:attribute>

attribute **gradientDirection**

type	**gradientDirectionType**
used by	element **DecisiveGradientArea** complexType **Gradient_ComplexType**
facets	Kind Value Annotation enumeration Downhill enumeration Uphill
annotation	documentation Direction of the gradient (Downhill or Uphill)

source	<xs:attribute name="gradientDirection" type="gradientDirectionType"> <xs:annotation> <xs:documentation>Direction of the gradient (Downhill or Uphill)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **gradientDirection_PBD**

type	**gradientDirectionType**
used by	element **PermittedBrakingDistanceArea**
facets	Kind Value Annotation enumeration Downhill enumeration Uphill
annotation	documentation [SUBSET-126] Direction of the gradient for the Permitted Braking Distance
source	<xs:attribute name="gradientDirection_PBD" type="gradientDirectionType"> <xs:annotation> <xs:documentation>[SUBSET-126] Direction of the gradient for the Permitted Braking Distance</xs:documentation> </xs:annotation> </xs:attribute>

attribute **gradientValue**

type	**gradientValueType**
used by	element **DecisiveGradientArea** complexType **Gradient_ComplexType**
annotation	documentation Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.
source	<xs:attribute name="gradientValue" type="gradientValueType"> <xs:annotation> <xs:documentation>Gradient value (Numeric value, mm by m). If the gradient value is 0, the gradient is flat and gradientDirection can be ignored.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **groundContactIdentifier**

type	**xs:string**
used by	element **PlaintextMessage**
annotation	documentation References the contactIdentifier under Contact
source	<xs:attribute name="groundContactIdentifier" type="xs:string"> <xs:annotation> <xs:documentation>References the contactIdentifier under Contact</xs:documentation> </xs:annotation> </xs:attribute>

attribute **highest_BFC_AdhesionRate**

type	**percentageType**
used by	complexType **BrakingForceCurve_ComplexType**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1
annotation	documentation Highest adhesion percentage for which this specific braking curve is applicable.
source	<xs:attribute name="highest_BFC_AdhesionRate" type="percentageType"> <xs:annotation> <xs:documentation>Highest adhesion percentage for which this specific braking curve is applicable.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **horizontalOffset**

type	**distanceType**
used by	complexType **PhysicalCharacteristics_ComplexType**
annotation	documentation Horizontal offset (in m) from the center of the track to the middle of the object. Negative is left and positive is right.
source	<xs:attribute name="horizontalOffset" type="distanceType"> <xs:annotation> <xs:documentation>Horizontal offset (in m) from the center of the track to the middle of the object. Negative is left and positive is right.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **identifier**

type	**xs:string**
used by	elements **B2G_StatusReport** **C_DAS_C_Shown** **ChangeReason** **ForceDrivingModeChange** **LastForceStatusChangeMessage** **LastNetworkSpecificEvent** **LastPlaintextMessage** **LastStatusReportMessage** **PlaintextMessage** **PlatformChange/PlatformLengthChange** **StoppingPointLocation** complexTypes **AdvisedSpeed_ComplexType** **BrakingForceCurve_ComplexType** **SP_GenericArea_ComplexType** **SP_GenericPoint_ComplexType** **TimingPointZone_ComplexType** **TractionForceCurve_ComplexType**
annotation	documentation Element identifier
source	<xs:attribute name="identifier" type="xs:string"> <xs:annotation> <xs:documentation>Element identifier</xs:documentation> </xs:annotation> </xs:attribute>

attribute **JP_SP_ConsistencyError**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation

	[SUBSET-126] ATO: JP SP Consistency Error, from Subset-126 (boolean)
source	<xs:attribute name="JP_SP_ConsistencyError" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: JP SP Consistency Error, from Subset-126 (boolean)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **JP_Status**

type	restriction of **xs:string**
used by	element **JourneyProfile**
facets	Kind / Value / Annotation enumeration Valid enumeration Invalid enumeration Unavailable enumeration Update enumeration Overwrite
annotation	documentation [SUBSET-126] Status of the Journey Profile: "Valid": JP containing the data requested. "Unavailable": JP specifies that the requested part of the Journey Profile is currently not available yet (but all the previously sent JPs are still applicable). "Invalid": JP specifies that the TP identifier asserted in the JPReq does not belong to the preceding JP already sent to the ATO-OB. "Update": JP specifies that the Journey Profile has been updated by the TMS within the current visibility of the ATO-OB. "Overwrite": JP specifies that the previously sent JPs shall be completely overwritten by this data.
source	<xs:attribute name="JP_Status"> <xs:annotation> <xs:documentation>[SUBSET-126] Status of the Journey Profile: "Valid": JP containing the data requested. "Unavailable": JP specifies that the requested part of the Journey Profile is currently not available yet (but all the previously sent JPs are still applicable). "Invalid": JP specifies that the TP identifier asserted in the JPReq does not belong to the preceding JP already sent to the ATO-OB. "Update": JP specifies that the Journey Profile has been updated by the TMS within the current visibility of the ATO-OB. "Overwrite": JP specifies that the previously sent JPs shall be completely overwritten by this data.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Valid"/> <xs:enumeration value="Invalid"/> <xs:enumeration value="Unavailable"/> <xs:enumeration value="Update"/> <xs:enumeration value="Overwrite"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **JP_Version**

type	**xs:unsignedInt**
used by	elements **JourneyProfile** **JP_InUse**
annotation	documentation Version of the Journey Profile. Newer JPs shall have JP_Version higher than older JPs.
source	<xs:attribute name="JP_Version" type="xs:unsignedInt">

```
  <xs:annotation>
    <xs:documentation>Version of the Journey Profile. Newer JPs shall have JP_Version higher than older JPs.</xs:documentation>
  </xs:annotation>
</xs:attribute>
```

attribute **kmRef**

type	**xs:double**
used by	elements **KM_Reference** **LastPassedKmRef**
annotation	documentation KM reference point (in km)
source	<xs:attribute name="kmRef" type="xs:double"> <xs:annotation> <xs:documentation>KM reference point (in km)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **language**

type	restriction of **xs:string**
used by	elements **Contact/LanguagesSupported** **TP_Name** complexType **MultilingualText_ComplexType**
facets	Kind Value Annotation length 2
annotation	documentation ISO 639-1 language code
source	<xs:attribute name="language"> <xs:annotation> <xs:documentation>ISO 639-1 language code</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:length value="2"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **latestDepartureAdviceTime**

type	**UTCDateTimeType**
used by	element **C_DAS_C_Advice**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Upper bound of time window for train departure. Only to be used if departure advice is sent. If latestDepartureTime is not specified, the advice should be obeyed at any time later than earliestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.
source	<xs:attribute name="latestDepartureAdviceTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Upper bound of time window for train departure. Only to be used if departure advice is sent. If latestDepartureTime is not specified, the advice should be obeyed at any time later

	than earliestDepartureTime. If both attributes are not specified, the advice should be obeyed immediately.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **latitude**

type	**xs:decimal**
used by	complexType **GNSS_StaticPosition_ComplexType**
annotation	documentation Latitude in signed degrees format (DDD.ddddd) in the WGS84 system.
source	<xs:attribute name="latitude" type="xs:decimal"> <xs:annotation> <xs:documentation>Latitude in signed degrees format (DDD.ddddd) in the WGS84 system.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **leaveASAP**

type	**xs:boolean**
used by	element **StoppingPointDepartureDetails**
annotation	documentation Leave as soon as possible, taking into account also the other variables (boolean).
source	<xs:attribute name="leaveASAP" type="xs:boolean"> <xs:annotation> <xs:documentation>Leave as soon as possible, taking into account also the other variables (boolean).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **levelCrossingGuarding**

type	**xs:boolean**
used by	element **Balise**
annotation	documentation Determines if the balise is guarding any level crossing
source	<xs:attribute name="levelCrossingGuarding" type="xs:boolean"> <xs:annotation> <xs:documentation>Determines if the balise is guarding any level crossing</xs:documentation> </xs:annotation> </xs:attribute>

attribute **levelCrossingType**

type	**xs:string**
used by	element **LevelCrossingArea**
annotation	documentation Type of level crossing

source	<xs:attribute name="levelCrossingType" type="xs:string"> <xs:annotation> <xs:documentation>Type of level crossing</xs:documentation> </xs:annotation> </xs:attribute>

attribute **lineIdentifier**

type	**xs:string**
used by	element **TrainLocationInformation** complexType **LineAndTrackIdentifier_ComplexType**
annotation	documentation Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.
source	<xs:attribute name="lineIdentifier" type="xs:string"> <xs:annotation> <xs:documentation>Identification of the line. For example “115000” is the Line Identifier for the line from Strasbourg to Basel.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **lineName**

used by	complexType **LineAndTrackIdentifier_ComplexType**
annotation	documentation Name of the line describing the connection stations. For example “Ligne de Strasbourg à Bale."
source	<xs:attribute name="lineName"> <xs:annotation> <xs:documentation>Name of the line describing the connection stations. For example “Ligne de Strasbourg à Bale."</xs:documentation> </xs:annotation> </xs:attribute>

attribute **location**

type	**distanceType**
used by	elements **ATP_SystemsChange** **CurrentLimitationChange** **CurveChange** **EndSignalApplication** **EstimatedVoltageChange** **GradientChange** **GradientSteepestChange** **LastAbsoluteOdometryPoint** **LineAndTrackIdentifierChange** **NSP_Change** **PositionSpeed** **RatedVoltageChange** **Signal_ID** **StartSignalApplication** **StaticSpeedProfileChange** **SuperElevationChange** **TimingPoint** **TrainCharacteristicsRef** complexType **SP_GenericPoint_ComplexType**
annotation	documentation Distance of the element from the beginning of the Segment Profile (m)
source	<xs:attribute name="location" type="distanceType"> <xs:annotation> <xs:documentation>Distance of the element from the beginning of the Segment Profile (m)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **location_NegativeErrorMargin**

type	**distanceType**
used by	element **PositionSpeed**
annotation	documentation Error margin of the position of the head of the train. This margin subtracted from the location attribute gives the least downline position of the train possible at the moment of the measure. Expressed in m
source	<xs:attribute name="location_NegativeErrorMargin" type="distanceType"> <xs:annotation> <xs:documentation>Error margin of the position of the head of the train. This margin subtracted from the location attribute gives the least downline position of the train possible at the moment of the measure. Expressed in m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **location_PositiveErrorMargin**

type	**distanceType**
used by	element **PositionSpeed**
annotation	documentation Error margin of the position of the head of the train. This margin added to the location attribute gives the most downline position of the train possible at the moment of the measure. Expressed in m
source	<xs:attribute name="location_PositiveErrorMargin" type="distanceType"> <xs:annotation> <xs:documentation>Error margin of the position of the head of the train. This margin added to the location attribute gives the most downline position of the train possible at the moment of the measure. Expressed in m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **longitude**

type	**xs:decimal**
used by	complexType **GNSS_StaticPosition_ComplexType**
annotation	documentation Longitude in signed degrees format (DDD.ddddd) in the WGS84 system.
source	<xs:attribute name="longitude" type="xs:decimal"> <xs:annotation> <xs:documentation>Longitude in signed degrees format (DDD.ddddd) in the WGS84 system.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **lowAdhesionCategory**

type	**adhesionCategoryType**		
used by	elements **AdhesionConditionsChange** **LowAdhesion**		
facets	Kind	Value	Annotation
	enumeration	Dry Rail	
	enumeration	Dry Rail (Medium)	

	enumeration Dry Rail (Low) enumeration Low Adhesion enumeration Very Low Adhesion enumeration Extremely Low Adhesion
annotation	documentation Adhesion category aligned with the definition in EN15595 of the wheel-rail interface for the lower boundaries (definition and values taken from SUBSET-126).
source	<xs:attribute name="lowAdhesionCategory" type="adhesionCategoryType"> <xs:annotation> <xs:documentation>Adhesion category aligned with the definition in EN15595 of the wheel-rail interface for the lower boundaries (definition and values taken from SUBSET-126).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **mainContact**

type	**xs:boolean**
used by	element **Contact**
annotation	documentation Specifies if this is the main contact for the area
source	<xs:attribute name="mainContact" type="xs:boolean"> <xs:annotation> <xs:documentation>Specifies if this is the main contact for the area</xs:documentation> </xs:annotation> </xs:attribute>

attribute **mandatoryStop**

type	**xs:boolean**
used by	element **StopType**
annotation	documentation Flag indicating whether the stop is mandatory (1) or optional (0)
source	<xs:attribute name="mandatoryStop" type="xs:boolean"> <xs:annotation> <xs:documentation>Flag indicating whether the stop is mandatory (1) or optional (0)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maxAdvisedSpeed**

type	**speedType**
used by	element **AdvisedSpeed**
annotation	documentation Maximum advised speed. Must be higher than "speed".
source	<xs:attribute name="maxAdvisedSpeed" type="speedType"> <xs:annotation> <xs:documentation>Maximum advised speed. Must be higher than "speed".</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **maxChargingCurrent**

type	**currentType**
used by	element **BatteryChargingArea**
annotation	documentation Maximum charging current to be applied to the train. Expressed in A.
source	<xs:attribute name="maxChargingCurrent" type="currentType"> <xs:annotation> <xs:documentation>Maximum charging current to be applied to the train. Expressed in A.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maxCurValue**

type	**currentType**
used by	complexTypes **CurrentLimitation_ComplexType** **TotalCurrent_ComplexType**
annotation	documentation Maximum current value (A)
source	<xs:attribute name="maxCurValue" type="currentType"> <xs:annotation> <xs:documentation>Maximum current value (A)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maxDischargingCurrent**

type	**currentType**
used by	element **BatteryChargingArea**
annotation	documentation Maximum current that may be returned to charging infrastructure. Expressed in A.
source	<xs:attribute name="maxDischargingCurrent" type="currentType"> <xs:annotation> <xs:documentation>Maximum current that may be returned to charging infrastructure. Expressed in A.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maximumCantDeficiency**

type	**xs:int**
annotation	documentation Maximum Cant Deficiency supported by the train (in mm)
source	<xs:attribute name="maximumCantDeficiency" type="xs:int"> <xs:annotation> <xs:documentation>Maximum Cant Deficiency supported by the train (in mm)</xs:documentation>

	</xs:annotation> </xs:attribute>

attribute **maximumNumberOfCars**

type	**xs:short**
used by	element **StoppingPointApplicability**
annotation	documentation Maximum number of cars
source	<xs:attribute name="maximumNumberOfCars" type="xs:short"> <xs:annotation> <xs:documentation>Maximum number of cars</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maximumNumberOfRSType**

type	**xs:unsignedByte**
used by	element **RollingStockApplicability**
annotation	documentation Maximum number of a certain type of rolling stock
source	<xs:attribute name="maximumNumberOfRSType" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>Maximum number of a certain type of rolling stock</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maximumRegenerativeBraking**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Maximum power that can be regenerated. Expressed in W
source	<xs:attribute name="maximumRegenerativeBraking" type="xs:double"> <xs:annotation> <xs:documentation>Maximum power that can be regenerated. Expressed in W</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maximumRegenerativeForce**

type	**xs:double**
used by	element **BrakingForceCurveDataPoint** complexType **TC_Features_ComplexType**
annotation	documentation Maximum force that can be Applied through regenerative braking. Expressed in N
source	<xs:attribute name="maximumRegenerativeForce" type="xs:double"> <xs:annotation> <xs:documentation>Maximum force that can be Applied through regenerative braking. Expressed

	in N</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maximumTrainLength**

type	**xs:double**
used by	elements **SignalTrainLength** **StoppingPointApplicability**
annotation	documentation Maximum Train Length for which stopping point is valid. Expressed in m
source	<xs:attribute name="maximumTrainLength" type="xs:double"> <xs:annotation> <xs:documentation>Maximum Train Length for which stopping point is valid. Expressed in m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **maxPower**

type	**powerTypekW**
used by	element **EnergyData**
annotation	documentation Maximal power that can be delivered form energy source available on-board (in kW)
source	<xs:attribute name="maxPower" type="powerTypekW"> <xs:annotation> <xs:documentation>Maximal power that can be delivered form energy source available on-board (in kW)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **measuredCurrent**

type	**currentType**
used by	element **EnergyData**
annotation	documentation Measured current taken from overhead contact line, negative during regeneration (in A)
source	<xs:attribute name="measuredCurrent" type="currentType"> <xs:annotation> <xs:documentation>Measured current taken from overhead contact line, negative during regeneration (in A)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **measuredVoltage**

type	**voltageType**
used by	element **EnergyData**
annotation	documentation Measured voltage at pantograph (in V)

source	<xs:attribute name="measuredVoltage" type="voltageType"> <xs:annotation> <xs:documentation>Measured voltage at pantograph (in V)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **message_ID**

type	**UUIDv4_Type**
used by	elements **C_DAS_C_Shown** **MessageHeader**
facets	Kind Value Annotation length 36 pattern [0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[8-9a-bA-B][0-9a-fA-F]{3}-[0-9a-fA-F]{12}
annotation	documentation Identifier of the SFERA message
source	<xs:attribute name="message_ID" type="UUIDv4_Type"> <xs:annotation> <xs:documentation>Identifier of the SFERA message</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minAdvisedSpeed**

type	**speedType**
used by	element **AdvisedSpeed**
annotation	documentation Minimum advised speed. Must be lower than "speed".
source	<xs:attribute name="minAdvisedSpeed" type="speedType"> <xs:annotation> <xs:documentation>Minimum advised speed. Must be lower than "speed".</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minimumDwellTime**

type	**xs:duration**
used by	element **StoppingPointDepartureDetails**
annotation	documentation [SUBSET-126] Minimum dwell time at given Stopping Point (in seconds)
source	<xs:attribute name="minimumDwellTime" type="xs:duration"> <xs:annotation> <xs:documentation>[SUBSET-126] Minimum dwell time at given Stopping Point (in seconds)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minimumNumberOfCars**

type	**xs:short**

used by	element **StoppingPointApplicability**
annotation	documentation Minimum number of cars
source	<xs:attribute name="minimumNumberOfCars" type="xs:short"> <xs:annotation> <xs:documentation>Minimum number of cars</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minimumNumberOfRSType**

type	**xs:unsignedByte**
used by	element **RollingStockApplicability**
annotation	documentation Minimum number of a certain type of rolling stock
source	<xs:attribute name="minimumNumberOfRSType" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>Minimum number of a certain type of rolling stock</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minimumPositionUpdateRequestPeriod**

type	**xs:duration**
used by	element **HandshakeAcknowledgement**
annotation	documentation Minimum time to be waited between two position requests by board
source	<xs:attribute name="minimumPositionUpdateRequestPeriod" type="xs:duration"> <xs:annotation> <xs:documentation>Minimum time to be waited between two position requests by board</xs:documentation> </xs:annotation> </xs:attribute>

attribute **minimumTrainLength**

type	**xs:double**
used by	elements **SignalTrainLength** **StoppingPointApplicability**
annotation	documentation Minimum Train Length for which stopping point is valid. Expressed in m
source	<xs:attribute name="minimumTrainLength" type="xs:double"> <xs:annotation> <xs:documentation>Minimum Train Length for which stopping point is valid. Expressed in m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **name**

type	**xs:string**
used by	elements **AlternativeItinerary** **NetworkSpecificParameter** **TP_Name** complexType **TimingPointZone_ComplexType**
annotation	documentation Name of the element
source	<xs:attribute name="name" type="xs:string"> <xs:annotation> <xs:documentation>Name of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **NC_CDTrain**

type	**NC_CDTrainType**
used by	element **SignalTrainCategory/Signal_CantDeficiencyTrainCategory** complexType **TC_Features_ComplexType**
facets	Kind Value Annotation enumeration 80mm enumeration 100mm enumeration 130mm enumeration 150mm enumeration 165mm enumeration 180mm enumeration 210mm enumeration 225mm enumeration 245mm enumeration 275mm enumeration 300mm
annotation	documentation Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).
source	<xs:attribute name="NC_CDTrain" type="NC_CDTrainType"> <xs:annotation> <xs:documentation>Cant Deficiency Train Category (NC_CDTRAIN as defined in SUBSET-026).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **NC_Train**

type	**NC_TrainType**
used by	element **SignalTrainCategory/Signal_OtherTrainCategory** complexType **TC_Features_ComplexType**
facets	Kind Value Annotation enumeration Freight_Train_P_Position enumeration Freight_Train_G_Position enumeration Passenger_Train
annotation	documentation Train Category (NC_TRAIN as defined in SUBSET-026).
source	<xs:attribute name="NC_Train" type="NC_TrainType"> <xs:annotation>

	<xs:documentation>Train Category (NC_TRAIN as defined in SUBSET-026).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **newAdhesionConditions**

type	restriction of **xs:string**
used by	element **AdhesionConditionsChange**
facets	Kind Value Annotation enumeration Better enumeration Worse
annotation	documentation New adhesion conditions
source	<xs:attribute name="newAdhesionConditions"> <xs:annotation> <xs:documentation>New adhesion conditions</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Better"/> <xs:enumeration value="Worse"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **newSpeedLevel**

type	**speedType**
used by	element **AxleLoadSpeedProfile**
annotation	documentation [SUBSET-126] Speed restriction to be applied if the axle load of the train ≥ axleLoadCategory
source	<xs:attribute name="newSpeedLevel" type="speedType"> <xs:annotation> <xs:documentation>[SUBSET-126] Speed restriction to be applied if the axle load of the train ≥ axleLoadCategory</xs:documentation> </xs:annotation> </xs:attribute>

attribute **nextStoppingPointSkip**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: skip next stopping point, from Subset-126 (boolean)
source	<xs:attribute name="nextStoppingPointSkip" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: skip next stopping point, from Subset-126 (boolean)</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **NID_CTRACTION**

type	**xs:unsignedByte**
used by	complexType **Voltage_ComplexType**
annotation	documentation [SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.
source	<xs:attribute name="NID_CTRACTION" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>[SUBSET-126] Country identifier of the traction system. It identifies the information, additional to voltageValue, required to fully define the traction system. Use NID_CTRACTION numbers from ERA document ERA_ERTMS_040001.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **NID_ENGINE**

type	restriction of **xs:unsignedInt**
used by	element **MessageHeader**
facets	Kind Value Annotation minInclusive 0 maxInclusive 16777216
annotation	documentation [SUBSET-126] ETCS identity number uniquely defined for ERTMS/ETCS purposes. Values to be assigned.
source	<xs:attribute name="NID_ENGINE"> <xs:annotation> <xs:documentation>[SUBSET-126] ETCS identity number uniquely defined for ERTMS/ETCS purposes. Values to be assigned.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:unsignedInt"> <xs:minInclusive value="0"/> <xs:maxInclusive value="16777216"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **NID_OPERATIONAL**

type	**xs:unsignedInt**
used by	element **MessageHeader**
annotation	documentation [SUBSET-126] Operational train running number. In SUBSET-126, NID_OPERATIONAL consists of up to 8 digits which are entered left adjusted into the data field, the leftmost digit is the digit to be entered first. In case the NID_OPERATIONAL is shorter than 8 digits, the remaining space is to be filled with special character “F”. In SFERA, this is converted to an integer.
source	<xs:attribute name="NID_OPERATIONAL" type="xs:unsignedInt"> <xs:annotation> <xs:documentation>[SUBSET-126] Operational train running number. In SUBSET-126,

	NID_OPERATIONAL consists of up to 8 digits which are entered left adjusted into the data field, the leftmost digit is the digit to be entered first. In case the NID_OPERATIONAL is shorter than 8 digits, the remaining space is to be filled with special character “F”. In SFERA, this is converted to an integer.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **openingDoorSide**

type	restriction of **xs:string**
used by	element **StoppingPointInformation**
facets	Kind / Value / Annotation enumeration None enumeration Right enumeration Left enumeration Both
annotation	documentation [SUBSET-126] Specifies if the ATO-OB has to manage the train doors opening and on which side the passenger exchange doors have to be opened
source	<xs:attribute name="openingDoorSide"> <xs:annotation> <xs:documentation>[SUBSET-126] Specifies if the ATO-OB has to manage the train doors opening and on which side the passenger exchange doors have to be opened</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="None"/> <xs:enumeration value="Right"/> <xs:enumeration value="Left"/> <xs:enumeration value="Both"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **operationalConditionsFulfillment**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Operational conditions fulfilment, from Subset-126 (boolean)
source	<xs:attribute name="operationalConditionsFulfillment" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Operational conditions fulfilment, from Subset-126 (boolean)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **optimalSpeed**

type	**speedType**

used by	complexType **AdvisedSpeed_ComplexType**
annotation	documentation Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on the advice instance being sent: - Stop Advice: optimalSpeed must be null - Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase - Constant Speed Advice: optimalSpeed is the target speed for the cruising phase - Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase - Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase - Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase There is a special value: 999 indicating "maximum allowed track speed"
source	<xs:attribute name="optimalSpeed" type="speedType"> <xs:annotation> <xs:documentation>Optimal speed given to the driver (km/h). It is recommended to fill the parameter optimalSpeed according to the following rules, depending on the advice instance being sent: - Stop Advice: optimalSpeed must be null - Acceleration Advice: optimalSpeed is the target speed that shall be reached at the end of the acceleration phase - Constant Speed Advice: optimalSpeed is the target speed for the cruising phase - Coasting Advice: optimalSpeed is the target speed that shall be reached at the end of the coasting phase - Operational/Electrical Braking Advice: optimalSpeed is the target speed that shall be reached at the end of the braking phase - Departure Advice: optimalSpeed is either null or the target speed that shall be reached after departing at the end of the acceleration phase There is a special value: 999 indicating "maximum allowed track speed"</xs:documentation> </xs:annotation> </xs:attribute>

attribute **originalLanguage**

type	**xs:boolean**
used by	elements **Contact/LanguagesSupported** **TP_Name** complexType **MultilingualText_ComplexType**
annotation	documentation Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.
source	<xs:attribute name="originalLanguage" type="xs:boolean"> <xs:annotation> <xs:documentation>Approved version of text (translation when 'false'). If the text is pre-generated, then multiple languages can have this flag set as true. If the text comes from a dispatcher who has introduced it in one language, only that language shall have the flag set as true. If only one language is used, this attribute can be omitted and considered as true.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **OTN**

type	**xs:string**
used by	element **Applicable_OTN**
annotation	documentation Operational Train Number (free text). For driver information.
source	<xs:attribute name="OTN" type="xs:string"> <xs:annotation> <xs:documentation>Operational Train Number (free text). For driver information.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **PBD_Brake**

type	restriction of **xs:string**
used by	element **PermittedBrakingDistanceArea**
facets	Kind Value Annotation enumeration Service_Brake enumeration Emergency_Brake
annotation	documentation [SUBSET-126] Whether the permitted braking distance is to be achieved with the Service Brake or Emergency Brake
source	<xs:attribute name="PBD_Brake"> <xs:annotation> <xs:documentation>[SUBSET-126] Whether the permitted braking distance is to be achieved with the Service Brake or Emergency Brake</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Service_Brake"/> <xs:enumeration value="Emergency_Brake"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **percentage**

type	**percentageType**
used by	complexType **AdvisedSpeed_ComplexType**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1
annotation	documentation The meaning depends on the type of advice instance being sent. Only valid with certain types of advice. - acceleration advice: Percentage of the maximum acceleration - operational braking advice: Percentage of the maximum operational braking - electrical braking advice: Percentage of the maximum electrical braking
source	<xs:attribute name="percentage" type="percentageType"> <xs:annotation> <xs:documentation>The meaning depends on the type of advice instance being sent. Only valid with certain types of advice. - acceleration advice: Percentage of the maximum acceleration

	- operational braking advice: Percentage of the maximum operational braking - electrical braking advice: Percentage of the maximum electrical braking</xs:documentation> </xs:annotation> </xs:attribute>

attribute **permittedBrakingDistance**

type	**distanceType**
used by	element **PermittedBrakingDistanceArea**
annotation	documentation [SUBSET-126] Permitted Braking Distance value (in m)
source	<xs:attribute name="permittedBrakingDistance" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Permitted Braking Distance value (in m)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **plaintextMessageDirection**

type	restriction of **xs:string**
used by	element **PlaintextMessage**
facets	Kind Value Annotation enumeration B2G enumeration G2B
annotation	documentation Direction of the plain text message (ground to board or board to ground).
source	<xs:attribute name="plaintextMessageDirection"> <xs:annotation> <xs:documentation>Direction of the plain text message (ground to board or board to ground).</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="B2G"/> <xs:enumeration value="G2B"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **plannedDepartureTime**

type	**UTCDateTimeType**
used by	element **StoppingPointDepartureDetails**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation The original planned departure time from the Timing Point expressed in UTC. Useful for driver information.
source	<xs:attribute name="plannedDepartureTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>The original planned departure time from the Timing Point expressed in

	UTC. Useful for driver information.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **plannedStop**

type	**xs:boolean**
used by	element **StopType**
annotation	documentation Flag indicating whether the stop is planned (1) or unplanned (0)
source	<xs:attribute name="plannedStop" type="xs:boolean"> <xs:annotation> <xs:documentation>Flag indicating whether the stop is planned (1) or unplanned (0)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **platformName**

type	**xs:string**
used by	element **Platform**
annotation	documentation Name of the platform as known by driver and passengers
source	<xs:attribute name="platformName" type="xs:string"> <xs:annotation> <xs:documentation>Name of the platform as known by driver and passengers</xs:documentation> </xs:annotation> </xs:attribute>

attribute **platformSide**

type	restriction of **xs:string**
used by	element **Platform**
facets	Kind / Value / Annotation enumeration Left enumeration Right
annotation	documentation Side of the platform (relative to the nominal direction of the SP)
source	<xs:attribute name="platformSide"> <xs:annotation> <xs:documentation>Side of the platform (relative to the nominal direction of the SP)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Left"/> <xs:enumeration value="Right"/> </xs:restriction> </xs:simpleType>

	</xs:attribute>

attribute **platformUsage**

type	restriction of **xs:string**
used by	elements **Platform** **PlatformChange/PlatformUnavailability/PlatformUnavailableUsage**
facets	Kind / Value / Annotation enumeration Passenger enumeration PassengerIncidentOnly enumeration Freight enumeration Service enumeration Maintenance enumeration DriverAccess enumeration PassengerTrainServicePreparation enumeration PassengerReducedMobility
annotation	documentation Usage of the platform
source	<xs:attribute name="platformUsage"> <xs:annotation> <xs:documentation>Usage of the platform</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Passenger"/> <xs:enumeration value="PassengerIncidentOnly"/> <xs:enumeration value="Freight"/> <xs:enumeration value="Service"/> <xs:enumeration value="Maintenance"/> <xs:enumeration value="DriverAccess"/> <xs:enumeration value="PassengerTrainServicePreparation"/> <xs:enumeration value="PassengerReducedMobility"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **positionOfLastPantograph**

type	**distanceType**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off…) for that power constraint to be lifted. Expressed in m.
source	<xs:attribute name="positionOfLastPantograph" type="distanceType"> <xs:annotation> <xs:documentation>Distance from the head of the train of the last point of the rolling stock that has to clear a zone with a power constraint (lowered pantograph, main switch off…) for that power constraint to be lifted. Expressed in m.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **positionSpeedRequest**

type	**xs:boolean**
annotation	documentation Request for position/speed update by board
source	<xs:attribute name="positionSpeedRequest" type="xs:boolean"> <xs:annotation> <xs:documentation>Request for position/speed update by board</xs:documentation> </xs:annotation> </xs:attribute>

attribute **powerConstraintInstruction**

type	restriction of **xs:string**
used by	element **PowerConstraint**
facets	Kind / Value / Annotation enumeration Pantograph_Down enumeration Pantograph_Up enumeration Main_Switch_Off enumeration Main_Switch_On enumeration Traction_Off enumeration Traction_On enumeration HVAC_Off enumeration HVAC_On
annotation	documentation Instruction for Power Constraint
source	<xs:attribute name="powerConstraintInstruction"> <xs:annotation> <xs:documentation>Instruction for Power Constraint</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Pantograph_Down"/> <xs:enumeration value="Pantograph_Up"/> <xs:enumeration value="Main_Switch_Off"/> <xs:enumeration value="Main_Switch_On"/> <xs:enumeration value="Traction_Off"/> <xs:enumeration value="Traction_On"/> <xs:enumeration value="HVAC_Off"/> <xs:enumeration value="HVAC_On"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **powerConsumptionChange**

type	restriction of **xs:string**
used by	element **PowerAdvice**
facets	Kind / Value / Annotation enumeration Maximise

	enumeration Minimise enumeration No Advice
annotation	documentation Change in power consumption: maximise, minimise or "no advice" ("no advice" cancels a previous advice)
source	<xs:attribute name="powerConsumptionChange"> <xs:annotation> <xs:documentation>Change in power consumption: maximise, minimise or "no advice" ("no advice" cancels a previous advice)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Maximise"/> <xs:enumeration value="Minimise"/> <xs:enumeration value="No Advice"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **preciseDelayReason**

type	**xs:string**
annotation	documentation Precise reason of the delay
source	<xs:attribute name="preciseDelayReason" type="xs:string"> <xs:annotation> <xs:documentation>Precise reason of the delay</xs:documentation> </xs:annotation> </xs:attribute>

attribute **previousPlaintextMessageIdentifier**

used by	element **PlaintextMessage**
annotation	documentation References the identifier of the plaintext message to which this message is an answer.
source	<xs:attribute name="previousPlaintextMessageIdentifier"> <xs:annotation> <xs:documentation>References the identifier of the plaintext message to which this message is an answer.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **protectedLevelCrossing**

type	**xs:boolean**
used by	element **LevelCrossingArea**
annotation	documentation Is the level crossing protected
source	<xs:attribute name="protectedLevelCrossing" type="xs:boolean"> <xs:annotation> <xs:documentation>Is the level crossing protected</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **rangeMaxSpeed**

type	**speedType**
used by	element **AvoidSpeedRange**
annotation	documentation Maximum speed in the range of speed to be avoided
source	<xs:attribute name="rangeMaxSpeed" type="speedType"> <xs:annotation> <xs:documentation>Maximum speed in the range of speed to be avoided</xs:documentation> </xs:annotation> </xs:attribute>

attribute **rangeMinSpeed**

type	**speedType**
used by	element **AvoidSpeedRange**
annotation	documentation Minimum speed in the range of speed to be avoided
source	<xs:attribute name="rangeMinSpeed" type="speedType"> <xs:annotation> <xs:documentation>Minimum speed in the range of speed to be avoided</xs:documentation> </xs:annotation> </xs:attribute>

attribute **reasonCode**

type	restriction of **xs:string**
used by	complexType **AdvisedSpeed_ComplexType**
facets	Kind / Value / Annotation enumeration followTrain enumeration trainFollowing enumeration merge2nd enumeration merge1st enumeration overtakeTrain enumeration followTrainOvertakeTrain enumeration beingOvertaken enumeration trainFollowingBeingOvertaken enumeration trainCrossing enumeration passTrain enumeration beingPassed enumeration timeSupplementRequired enumeration energyOptimisation enumeration endOfReason enumeration nationalUse1 enumeration nationalUse2

	enumeration nationalUse3 enumeration nationalUse4 enumeration nationalUse5 enumeration nationalUse6 enumeration nationalUse7 enumeration nationalUse8 enumeration nationalUse9 enumeration nationalUse10
annotation	documentation reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid. It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are: - followTrain: host train is following a conflicting (slower) train - trainFollowing: host train is preceding a conflicting (faster) train - merge2nd: conflicting (faster) train merging ahead and entering common route section first - merge1st: conflicting (slower) train merging ahead, host train enters common route section first - overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking - followTrainOvertakeTrain: follow, then overtake conflicting train that hasn't yet reached the location of overtaking - beingOvertaken: conflicting (faster) train about to overtake - trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking - trainCrossing: train crossing - passTrain: passing a train on a single-track line - beingPassed: host train being passed on a single-track line - timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction - energyOptimisation: optimisation of energy consumption - endOfReason: the reason previously communicated is not valid anymore.
source	<xs:attribute name="reasonCode"> <xs:annotation> <xs:documentation>reasonCode offers coding for standard situations. If reasonCode is null the last reasonCode received is still valid. It is possible to use only one of the parameters reasonCode and reasonText or to send them in combination. Predefined reasonCodes are: - followTrain: host train is following a conflicting (slower) train - trainFollowing: host train is preceding a conflicting (faster) train - merge2nd: conflicting (faster) train merging ahead and entering common route section first - merge1st: conflicting (slower) train merging ahead, host train enters common route section first - overtakeTrain: overtake conflicting (slower) train that has (nearly) reached the location of overtaking - followTrainOvertakeTrain: follow, then overtake conflicting train that hasn't yet reached the location of overtaking - beingOvertaken: conflicting (faster) train about to overtake - trainFollowingBeingOvertaken: conflicting (faster) train following and then overtaking - trainCrossing: train crossing - passTrain: passing a train on a single-track line - beingPassed: host train being passed on a single-track line - timeSupplementRequired: recovery time is required due to an upcoming temporary speed restriction - energyOptimisation: optimisation of energy consumption - endOfReason: the reason previously communicated is not valid anymore.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="followTrain"/> <xs:enumeration value="trainFollowing"/> <xs:enumeration value="merge2nd"/> <xs:enumeration value="merge1st"/> <xs:enumeration value="overtakeTrain"/>

```
      <xs:enumeration value="followTrainOvertakeTrain"/>
      <xs:enumeration value="beingOvertaken"/>
      <xs:enumeration value="trainFollowingBeingOvertaken"/>
      <xs:enumeration value="trainCrossing"/>
      <xs:enumeration value="passTrain"/>
      <xs:enumeration value="beingPassed"/>
      <xs:enumeration value="timeSupplementRequired"/>
      <xs:enumeration value="energyOptimisation"/>
      <xs:enumeration value="endOfReason"/>
      <xs:enumeration value="nationalUse1"/>
      <xs:enumeration value="nationalUse2"/>
      <xs:enumeration value="nationalUse3"/>
      <xs:enumeration value="nationalUse4"/>
      <xs:enumeration value="nationalUse5"/>
      <xs:enumeration value="nationalUse6"/>
      <xs:enumeration value="nationalUse7"/>
      <xs:enumeration value="nationalUse8"/>
      <xs:enumeration value="nationalUse9"/>
      <xs:enumeration value="nationalUse10"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **referenceIDAbs**

type	**UUIDv4_Type**
used by	element **C_DAS_C_Advice**
facets	Kind: length — Value: 36 Kind: pattern — Value: [0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[8-9a-bA-B][0-9a-fA-F]{3}-[0-9a-fA-F]{12}
annotation	documentation References the ID of the message, that defined the absolute speed, which is still valid

source	`<xs:attribute name="referenceIDAbs" type="UUIDv4_Type">`
	`  <xs:annotation>`
	`  <xs:documentation>References the ID of the message, that defined the absolute speed, which is still valid</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

attribute **relatedTrainRequest**

type	**relatedTrainRequestType**
used by	element **HandshakeRequest**
facets	Kind: enumeration — Value: None Kind: enumeration — Value: OwnTrain Kind: enumeration — Value: RelatedTrains Kind: enumeration — Value: OwnTrainAndRelatedTrains Kind: enumeration — Value: OwnTrainAndOrRelatedTrains
annotation	documentation Request for related train information

source	<xs:attribute name="relatedTrainRequest" type="relatedTrainRequestType"> <xs:annotation> <xs:documentation>Request for related train information</xs:documentation> </xs:annotation> </xs:attribute>

attribute **relatedTrainSupported**

type	**relatedTrainRequestType**
used by	element **HandshakeAcknowledgement**
facets	Kind / Value / Annotation enumeration None enumeration OwnTrain enumeration RelatedTrains enumeration OwnTrainAndRelatedTrains enumeration OwnTrainAndOrRelatedTrains
annotation	documentation Information on whether related train information is supported
source	<xs:attribute name="relatedTrainSupported" type="relatedTrainRequestType"> <xs:annotation> <xs:documentation>Information on whether related train information is supported</xs:documentation> </xs:annotation> </xs:attribute>

attribute **relaxedCoupler**

type	**xs:boolean**
used by	element **StoppingPointInformation**
annotation	documentation [SUBSET-126] request to the train to stop with couplers relaxed
source	<xs:attribute name="relaxedCoupler" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] request to the train to stop with couplers relaxed</xs:documentation> </xs:annotation> </xs:attribute>

attribute **remainingEngineCharge**

type	**percentageType**
used by	element **TFC_TractionType**
facets	Kind / Value / Annotation minInclusive 0 maxInclusive 1
annotation	documentation Remaining charge available in percentage
source	<xs:attribute name="remainingEngineCharge" type="percentageType"> <xs:annotation>

	<xs:documentation>Remaining charge available in percentage</xs:documentation> </xs:annotation> </xs:attribute>

attribute **remainingEnginePower**

type	**percentageType**
used by	elements **TC_ChangeFeatures** **TFC_TractionType**
facets	Kind Value Annotation minInclusive 0 maxInclusive 1
annotation	documentation Percentage of power available due to engine problems
source	<xs:attribute name="remainingEnginePower" type="percentageType"> <xs:annotation> <xs:documentation>Percentage of power available due to engine problems</xs:documentation> </xs:annotation> </xs:attribute>

attribute **reportingTime**

type	**xs:duration**
used by	element **HandshakeAcknowledgement**
annotation	documentation [SUBSET-126] Reporting time cycle for triggering a Status Report for data reporting purposes
source	<xs:attribute name="reportingTime" type="xs:duration"> <xs:annotation> <xs:documentation>[SUBSET-126] Reporting time cycle for triggering a Status Report for data reporting purposes</xs:documentation> </xs:annotation> </xs:attribute>

attribute **result**

type	restriction of **xs:string**
used by	elements **B2G_MessageResponse** **G2B_MessageResponse**
facets	Kind Value Annotation enumeration OK enumeration ERROR
annotation	documentation Result of the message
source	<xs:attribute name="result"> <xs:annotation> <xs:documentation>Result of the message</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="OK"/> <xs:enumeration value="ERROR"/> </xs:restriction>

	</xs:simpleType> </xs:attribute>

attribute **rollingStockType**

type	**xs:string**
used by	element **RollingStockApplicability** complexType **TC_Features_ComplexType**
annotation	documentation Free-form text describing the rolling stock. Used for driver information.
source	<xs:attribute name="rollingStockType" type="xs:string"> <xs:annotation> <xs:documentation>Free-form text describing the rolling stock. Used for driver information.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **rolloutCoefficientA**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N
source	<xs:attribute name="rolloutCoefficientA" type="xs:double"> <xs:annotation> <xs:documentation>Description of how the train accelerates. Coefficient used in order to calculate the resistance force (constant value). Expressed in N</xs:documentation> </xs:annotation> </xs:attribute>

attribute **rolloutCoefficientB**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m
source	<xs:attribute name="rolloutCoefficientB" type="xs:double"> <xs:annotation> <xs:documentation>Description of how the train accelerates if no traction or braking is applied. Coefficient used in order to calculate the resistance force (value to be multiplied with v). Expressed in N.s/m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **rolloutCoefficientC**

type	**xs:double**
used by	complexType **TC_Features_ComplexType**

annotation	documentation Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2
source	<xs:attribute name="rolloutCoefficientC" type="xs:double"> <xs:annotation> <xs:documentation>Description of how the train accelerates if no traction or braking is applied.Coefficient used in order to calculate the resistance force (value to be multiplied with v^2). Expressed in N.s^2/m^2</xs:documentation> </xs:annotation> </xs:attribute>

attribute **rotatingMassFactor**

type	**xs:decimal**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.
source	<xs:attribute name="rotatingMassFactor" type="xs:decimal"> <xs:annotation> <xs:documentation>Rotating Mass Factor. The rotating mass of the train is equivalent to the rotational inertia of the train, transformed by gear ratio and wheel diameter respectively.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **routingError**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Routing Error, from Subset-126 (boolean). This parameter can also be used by the DAS-OB to notify the DAS-TS that the train is not on the expected route.
source	<xs:attribute name="routingError" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Routing Error, from Subset-126 (boolean). This parameter can also be used by the DAS-OB to notify the DAS-TS that the train is not on the expected route.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **sessionTerminationReason**

type	restriction of **xs:string**		
used by	element **SessionTermination**		
facets	Kind	Value	Annotation
	enumeration	End of Journey reached	
	enumeration	DAS-TS/ATO-TS Termination Request	
	enumeration	Last SP left	
	enumeration	Cab inactive	

	enumeration TRN or train length not valid enumeration ETCS-OB in NL enumeration ETCS-OB in SH enumeration ATO-TS Handover cancelled
annotation	documentation Reason for session termination
source	<xs:attribute name="sessionTerminationReason"> <xs:annotation> <xs:documentation>Reason for session termination</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="End of Journey reached"/> <xs:enumeration value="DAS-TS/ATO-TS Termination Request"/> <xs:enumeration value="Last SP left"/> <xs:enumeration value="Cab inactive"/> <xs:enumeration value="TRN or train length not valid"/> <xs:enumeration value="ETCS-OB in NL"/> <xs:enumeration value="ETCS-OB in SH"/> <xs:enumeration value="ATO-TS Handover cancelled"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **setValueTractionBraking**

type	**xs:string**
used by	element **EnergyData**
annotation	documentation set value by the driver for traction and braking
source	<xs:attribute name="setValueTractionBraking" type="xs:string"> <xs:annotation> <xs:documentation>set value by the driver for traction and braking</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SFERA_version**

type	restriction of **xs:string**
used by	element **MessageHeader**
facets	Kind Value Annotation pattern [0-9]+\.[0-9]+
annotation	documentation SFERA version. The format should be majorVersion.minorVersion (e.g.: 1.43). There should be backwards compatibility for messages with the same major SFERA version.
source	<xs:attribute name="SFERA_version"> <xs:annotation> <xs:documentation>SFERA version. The format should be majorVersion.minorVersion (e.g.: 1.43). There should be backwards compatibility for messages with the same major SFERA version.</xs:documentation> </xs:annotation> <xs:simpleType>

	<xs:restriction base="xs:string"> <xs:pattern value="[0-9]+\.[0-9]+"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **signal_ID_Object**

type	**xs:string**
used by	elements **CancelPreviousSignal** **LastPassedSignal** **Signal_ID**
annotation	documentation Identifier of the signal object
source	<xs:attribute name="signal_ID_Object" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the signal object</xs:documentation> </xs:annotation> </xs:attribute>

attribute **signal_ID_Physical**

type	**xs:string**
used by	elements **CancelPreviousSignal** **LastPassedSignal** **Signal_ID**
annotation	documentation Identifier of the physical signal on the track
source	<xs:attribute name="signal_ID_Physical" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the physical signal on the track</xs:documentation> </xs:annotation> </xs:attribute>

attribute **signalType**

type	**xs:string**
used by	element **SignalPhysicalCharacteristics**
annotation	documentation Type of signal from which the physical characteristics of the signal can be deduced. This will be network specific, as no standardised solution has been identified by the SFERA project. Examples: - CTC (Centralized Traffic Control Signal) - Permissive (Permissive Signal) - Permanent signal - Maximum Sign - Announcement Sign - Disque - Cantonnement BAL 3 aspects
source	<xs:attribute name="signalType" type="xs:string"> <xs:annotation> <xs:documentation>Type of signal from which the physical characteristics of the signal can be deduced. This will be network specific, as no standardised solution has been identified by the SFERA project. Examples: - CTC (Centralized Traffic Control Signal) - Permissive (Permissive Signal) - Permanent signal

	- Maximum Sign - Announcement Sign - Disque - Cantonnement BAL 3 aspects</xs:documentation> </xs:annotation> </xs:attribute>

attribute **slipSlideReported**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Slip/slide reported by TCMS/Train, from Subset-126 (boolean)
source	<xs:attribute name="slipSlideReported" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Slip/slide reported by TCMS/Train, from Subset-126 (boolean)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **sourceCompany**

type	**teltsi_CompanyCode**
facets	Kind Value Annotation minLength 4 maxLength 4 pattern [0-9A-Z]{4}
annotation	documentation ID of the company sending the message. Must be a UIC RICS Code: : https://uic.org/rics
source	<xs:attribute name="sourceCompany" type="teltsi_CompanyCode"> <xs:annotation> <xs:documentation>ID of the company sending the message. Must be a UIC RICS Code: : https://uic.org/rics</xs:documentation> </xs:annotation> </xs:attribute>

attribute **sourceDevice**

type	**deviceType**
used by	element **MessageHeader**
facets	Kind Value Annotation maxLength 128
annotation	documentation ID of the device sending the message. For any given company, this ID must be unique. Maximum 128 characters.
source	<xs:attribute name="sourceDevice" type="deviceType"> <xs:annotation> <xs:documentation>ID of the device sending the message. For any given company, this ID must be unique. Maximum 128 characters.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SP_Altitude**

type	**xs:decimal**
used by	element **SegmentProfile**
annotation	documentation [SUBSET-126] Altitude (in m) at the beginning of the SP. Considering ETRS89 as reference. Starting at -1000m.
source	<xs:attribute name="SP_Altitude" type="xs:decimal"> <xs:annotation> <xs:documentation>[SUBSET-126] Altitude (in m) at the beginning of the SP. Considering ETRS89 as reference. Starting at -1000m.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SP_BeforeOrAfter**

type	restriction of **xs:string**
used by	element **ApplicableIfTrainRunsOnSP**
facets	Kind / Value / Annotation enumeration Before enumeration After
annotation	documentation Indicates whether the mentioned SP is before or after the current SP
source	<xs:attribute name="SP_BeforeOrAfter"> <xs:annotation> <xs:documentation>Indicates whether the mentioned SP is before or after the current SP</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Before"/> <xs:enumeration value="After"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **SP_Direction**

type	restriction of **xs:string**
used by	element **SegmentProfileReference** complexType **SegmentProfileInfo_ComplexType**
facets	Kind / Value / Annotation enumeration Nominal enumeration Reverse
annotation	documentation Travelling direction of the Segment Profile (Nominal or Reverse). If the travelling direction is Reverse, all locations will apply from the end of the segment. Note that Signals and Stopping Points only apply in the nominal direction of the SP.
source	<xs:attribute name="SP_Direction"> <xs:annotation> <xs:documentation>Travelling direction of the Segment Profile (Nominal or Reverse). If the

	travelling direction is Reverse, all locations will apply from the end of the segment. Note that Signals and Stopping Points only apply in the nominal direction of the SP.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Nominal"/> <xs:enumeration value="Reverse"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **SP_Gauge**

type	**xs:int**
used by	element **SegmentProfile**
annotation	documentation Gauge of the segment profile (in mm). Is needed to calculate rolling resistance.
source	<xs:attribute name="SP_Gauge" type="xs:int"> <xs:annotation> <xs:documentation>Gauge of the segment profile (in mm). Is needed to calculate rolling resistance.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SP_ID**

type	restriction of **xs:string**
used by	elements **ApplicableIfTrainRunsOnSP** **EndSignalApplication** **SP_NextSP** **StartSignalApplication** complexType **SegmentProfileIdentification_ComplexType**
facets	Kind Value Annotation maxLength 32
annotation	documentation Segment Profile Identifier
source	<xs:attribute name="SP_ID"> <xs:annotation> <xs:documentation>Segment Profile Identifier</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:maxLength value="32"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **SP_Length**

type	**distanceType**
used by	element **SegmentProfile**
annotation	documentation Length of the section of railway covered by the Segment Profile (m)
source	<xs:attribute name="SP_Length" type="distanceType">

```
  <xs:annotation>
    <xs:documentation>Length of the section of railway covered by the Segment Profile
(m)</xs:documentation>
  </xs:annotation>
</xs:attribute>
```

attribute **SP_Name**

type	**xs:string**
used by	element **SegmentProfile**
annotation	documentation Human-readable name of the Segment Profile. To be used only for debugging purposes.
source	<xs:attribute name="SP_Name" type="xs:string"> <xs:annotation> <xs:documentation>Human-readable name of the Segment Profile. To be used only for debugging purposes.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SP_Status**

type	restriction of **xs:string**
used by	element **SegmentProfile**
facets	Kind / Value / Annotation enumeration Valid enumeration Invalid
annotation	documentation [SUBSET-126] Status of the Segment Profile (Valid or Invalid). From SUBSET-125: 10.1.3.23 The Segment Profile shall indicate its status to the ATO-OB (Segment Profile Status): a) Valid: specifies that the data sent correspond to a SPReq sent; b) Invalid: the requested Segment Profile is not available in the ATO-TS.
source	<xs:attribute name="SP_Status"> <xs:annotation> <xs:documentation>[SUBSET-126] Status of the Segment Profile (Valid or Invalid). From SUBSET-125: 10.1.3.23 The Segment Profile shall indicate its status to the ATO-OB (Segment Profile Status): a) Valid: specifies that the data sent correspond to a SPReq sent; b) Invalid: the requested Segment Profile is not available in the ATO-TS.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Valid"/> <xs:enumeration value="Invalid"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **SP_VersionMajor**

type	**xs:string**
used by	elements **SegmentProfile** **SegmentProfileReference** **SP_Request** complexType **SegmentProfileInfo_ComplexType**

annotation	documentation Identifier of the Segment Profile version, major number.
source	<xs:attribute name="SP_VersionMajor" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the Segment Profile version, major number.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SP_VersionMinor**

type	**xs:string**
used by	elements **SegmentProfile** **SegmentProfileReference** **SP_Request** complexType **SegmentProfileInfo_ComplexType**
annotation	documentation Identifier of the Segment Profile version, minor number.
source	<xs:attribute name="SP_VersionMinor" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the Segment Profile version, minor number.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **specific_SSP_Category**

type	restriction of **xs:string**
used by	complexType **SpecificSSP_ComplexType**
facets	Kind / Value / Annotation enumeration Cant_Deficiency_SSP enumeration Other_SSP_Replaces_CDSSP enumeration Other_SSP_Noreplace_CDSSP
annotation	documentation [SUBSET-126] Indicates the type of specific SSP category.
source	<xs:attribute name="specific_SSP_Category"> <xs:annotation> <xs:documentation>[SUBSET-126] Indicates the type of specific SSP category.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Cant_Deficiency_SSP"/> <xs:enumeration value="Other_SSP_Replaces_CDSSP"/> <xs:enumeration value="Other_SSP_Noreplace_CDSSP"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **speed**

type	**speedType**
used by	elements **AdvisedSpeed** **BrakingForceCurveDataPoint** **MaxSpeed** **MaxSpeedAnnouncement** **PositionSpeed** **TractionForceCurveDataPoint**
annotation	documentation speed (km/h)

source	<xs:attribute name="speed" type="speedType"> <xs:annotation> <xs:documentation>speed (km/h)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SSP_Front**

type	**xs:boolean**
used by	complexType **SSP_ComplexType**
annotation	documentation [SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (no train length delay) or the end of the train (train length delay) has left the element.
source	<xs:attribute name="SSP_Front" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] Qualifier to indicate if a speed limit given for a profile element is to be applied until the front of the train (no train length delay) or the end of the train (train length delay) has left the element.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SSP_NC_CDDIFF**

type	**NC_CDTrainType**
used by	element **CantDeficiencySSP**
facets	Kind / Value / Annotation enumeration 80mm enumeration 100mm enumeration 130mm enumeration 150mm enumeration 165mm enumeration 180mm enumeration 210mm enumeration 225mm enumeration 245mm enumeration 275mm enumeration 300mm
annotation	documentation [SUBSET-126] (If specific_SSP_Category = Cant_Deficiency_SSP) “Cant Deficiency” SSP category for which a different value for the static line speed exists. Used together with V_DIFF to permit certain trains to go faster or lower than the “international basic static speed” given by SSP_Speed.
source	<xs:attribute name="SSP_NC_CDDIFF" type="NC_CDTrainType"> <xs:annotation> <xs:documentation>[SUBSET-126] (If specific_SSP_Category = Cant_Deficiency_SSP) “Cant Deficiency” SSP category for which a different value for the static line speed exists. Used together with V_DIFF to permit certain trains to go faster or lower than the “international basic static speed” given by SSP_Speed.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SSP_NC_DIFF**

type	**NC_TrainType**
used by	element **OtherSpecificSSP**
facets	Kind / Value / Annotation enumeration Freight_Train_P_Position enumeration Freight_Train_G_Position enumeration Passenger_Train
annotation	documentation [SUBSET-126] (If specific_SSP_Category = Other_SSP_*) It is the “other specific” SSP category for which a different value for the static line speed exists. Used together with SS126_V_DIFF to permit trains belonging to the corresponding “other international” train category to go faster or lower than the “international basic static speed” given by SSP_Speed.
source	<xs:attribute name="SSP_NC_DIFF" type="NC_TrainType"> <xs:annotation> <xs:documentation>[SUBSET-126] (If specific_SSP_Category = Other_SSP_*) It is the “other specific” SSP category for which a different value for the static line speed exists. Used together with SS126_V_DIFF to permit trains belonging to the corresponding “other international” train category to go faster or lower than the “international basic static speed” given by SSP_Speed.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **SSP_Speed**

type	**speedType**
used by	complexType **SSP_ComplexType**
annotation	documentation [SUBSET-126] Basic Static Speed Profile speed.
source	<xs:attribute name="SSP_Speed" type="speedType"> <xs:annotation> <xs:documentation>[SUBSET-126] Basic Static Speed Profile speed.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startEndQualifier**

type	restriction of **xs:string**
used by	complexTypes **SegmentProfileInfo_ComplexType** **SP_GenericArea_ComplexType** **TemporaryConstraints_ComplexType**
facets	Kind / Value / Annotation enumeration Starts enumeration Ends enumeration StartsEnds enumeration WholeSP
annotation	documentation [SUBSET-126] Specifies if the element starts, ends, starts and ends or covers the whole concerning Segment Profile
source	<xs:attribute name="startEndQualifier"> <xs:annotation> <xs:documentation>[SUBSET-126] Specifies if the element starts, ends, starts and ends or

	covers the whole concerning Segment Profile</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Starts"/> <xs:enumeration value="Ends"/> <xs:enumeration value="StartsEnds"/> <xs:enumeration value="WholeSP"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **startKm**

type	**xs:double**
used by	element **C_DAS_C_Advice**
annotation	documentation References the kilometre at the beginning of the message, preferably linked with a kilometer sign near the track
source	<xs:attribute name="startKm" type="xs:double"> <xs:annotation> <xs:documentation>References the kilometre at the beginning of the message, preferably linked with a kilometer sign near the track</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startLine**

type	**xs:int**
used by	element **C_DAS_C_Advice**
annotation	documentation References the (national) line number at the beginning of the message
source	<xs:attribute name="startLine" type="xs:int"> <xs:annotation> <xs:documentation>References the (national) line number at the beginning of the message</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startLocation**

type	**distanceType**
used by	elements **ChangeReason** **PlatformChange/PlatformLengthChange** complexTypes **SegmentProfileInfo_ComplexType** **SP_GenericArea_ComplexType** **TemporaryConstraints_ComplexType**
annotation	documentation [SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)
source	<xs:attribute name="startLocation" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Distance of the element's start location from the beginning of the Segment Profile (m)</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **startPointOfDownLineSpeedApplication**

type	**distanceType**
annotation	documentation How far (in m) after the considered signal the speed advice must be respected. 0 = As soon as possible
source	<xs:attribute name="startPointOfDownLineSpeedApplication" type="distanceType"> <xs:annotation> <xs:documentation>How far (in m) after the considered signal the speed advice must be respected. 0 = As soon as possible</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startSignal**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the name of the start signal of the message, if the start of the message is at or near a signal
source	<xs:attribute name="startSignal" type="xs:string"> <xs:annotation> <xs:documentation>References the name of the start signal of the message, if the start of the message is at or near a signal</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startSignalApplicationMustBeClearedBy**

type	**trainPartType**
used by	element **StartSignalApplication**
facets	Kind / Value / Annotation enumeration TrainHead enumeration TrainEnd enumeration TrainLastPantograph
annotation	documentation What part of the train must clear the signal application before applying the signal
source	<xs:attribute name="startSignalApplicationMustBeClearedBy" type="trainPartType"> <xs:annotation> <xs:documentation>What part of the train must clear the signal application before applying the signal</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startStationLong**

type	**xs:string**
used by	element **C_DAS_C_Advice**

annotation	documentation References the start station at the beginning of the message (long form)
source	<xs:attribute name="startStationLong" type="xs:string"> <xs:annotation> <xs:documentation>References the start station at the beginning of the message (long form)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startStationShort**

type	**xs:string**
used by	element **C_DAS_C_Advice**
annotation	documentation References the start station at the beginning of the message (short form)
source	<xs:attribute name="startStationShort" type="xs:string"> <xs:annotation> <xs:documentation>References the start station at the beginning of the message (short form)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startTime**

type	**UTCDateTimeType**
used by	elements **ChangeReason** **ExpectedCurrentData** **TemporarySignalConstraints** complexType **TemporaryConstraints_ComplexType**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Starting date and time (in UTC)
source	<xs:attribute name="startTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Starting date and time (in UTC)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **startValidity**

type	**UTCDateTimeType**
used by	element **C_DAS_C_Advice**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Begin of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.
source	<xs:attribute name="startValidity" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Begin of the advice validity. Either both startValidity and endValidity or none of the two must be specified. Both parameters should refer to a future point in time and endValidity

	must be after startValidity. If either of the two indicates a past time (w.r.t. the clock of the On-Board Unit), which is possible in a situation where Central Unit and OBU are not synchronised, transmission delays etc., then the OBU should treat it as valid but past time indication.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **stationTrack**

type	**xs:string**
used by	element **TimingPoint**
annotation	documentation If the Timing Point is a station: identifier of the track where the train will stop or pass through without stopping at the station. Examples : “1” “14” “2M” “1bis”
source	<xs:attribute name="stationTrack" type="xs:string"> <xs:annotation> <xs:documentation>If the Timing Point is a station: identifier of the track where the train will stop or pass through without stopping at the station. Examples : “1” “14” “2M” “1bis”</xs:documentation> </xs:annotation> </xs:attribute>

attribute **statusReportsEnabled**

type	**xs:boolean**
used by	element **HandshakeRequest**
annotation	documentation Flag notifying whether the DAS is able to send Status Reports to ground.
source	<xs:attribute name="statusReportsEnabled" type="xs:boolean"> <xs:annotation> <xs:documentation>Flag notifying whether the DAS is able to send Status Reports to ground.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **stopLocationTolerance**

type	**distanceType**
used by	element **StoppingPointLocation**
annotation	documentation [SUBSET-126] Required stopping tolerance (m)
source	<xs:attribute name="stopLocationTolerance" type="distanceType"> <xs:annotation> <xs:documentation>[SUBSET-126] Required stopping tolerance (m)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **stoppingPointOffset**

type	**distanceType**
used by	element **StoppingPointInformation**
annotation	documentation Distance (in metres) from the stopping point that the train needs to aim at. Typical use is for coupling trains. - Negative: distance before stopping point - Positive: distance after stopping point
source	<xs:attribute name="stoppingPointOffset" type="distanceType"> <xs:annotation> <xs:documentation>Distance (in metres) from the stopping point that the train needs to aim at. Typical use is for coupling trains. - Negative: distance before stopping point - Positive: distance after stopping point</xs:documentation> </xs:annotation> </xs:attribute>

attribute **stopTime**

type	**xs:duration**
used by	element **StopBeforePassingSignal**
annotation	documentation Stopping time necessary to complete the procedure, only given if StopBeforePassingSignal is “yes”. The time given is the safety procedure time, and is independent from the commercial stop time.
source	<xs:attribute name="stopTime" type="xs:duration"> <xs:annotation> <xs:documentation>Stopping time necessary to complete the procedure, only given if StopBeforePassingSignal is “yes”. The time given is the safety procedure time, and is independent from the commercial stop time.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **superElevationValue**

type	**xs:decimal**
used by	elements **SuperElevationChange** **SuperElevationStart**
annotation	documentation Height difference (in m) between outside and inside rail in a curve. Negative values may be used when the outside rail is lower than the inside rail.
source	<xs:attribute name="superElevationValue" type="xs:decimal"> <xs:annotation> <xs:documentation>Height difference (in m) between outside and inside rail in a curve. Negative values may be used when the outside rail is lower than the inside rail.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TAF_TAP_location_abbreviation**

used by	element **TAF_TAP_Location**
annotation	documentation Abbreviation of TAF/TAP location. To be used in case the infrastructure manager defines a standard abbreviation for the locations within its network.

source	<xs:attribute name="TAF_TAP_location_abbreviation"> <xs:annotation> <xs:documentation>Abbreviation of TAF/TAP location. To be used in case the infrastructure manager defines a standard abbreviation for the locations within its network.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TAF_TAP_location_type**

used by	element **TAF_TAP_Location**
annotation	documentation Type of TAF/TAP location. Examples: Very important station Important station Station Stopping location Junction
source	<xs:attribute name="TAF_TAP_location_type"> <xs:annotation> <xs:documentation>Type of TAF/TAP location. Examples: Very important station Important station Station Stopping location Junction</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TC_ID**

type	**xs:string**
used by	complexType **TC_Identifier_ComplexType**
annotation	documentation Train Characteristics Identifier
source	<xs:attribute name="TC_ID" type="xs:string"> <xs:annotation> <xs:documentation>Train Characteristics Identifier</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TC_VersionMajor**

type	**xs:string**
used by	elements **TC_Request** **TrainCharacteristics** **TrainCharacteristicsRef**
annotation	documentation Identifier of the Train Characteristics version, major number.
source	<xs:attribute name="TC_VersionMajor" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the Train Characteristics version, major number.</xs:documentation> </xs:annotation>

	</xs:attribute>

attribute **TC_VersionMinor**

type	**xs:string**
used by	elements **TC_Request** **TrainCharacteristics** **TrainCharacteristicsRef**
annotation	documentation Identifier of the Train Characteristics version, minor number.
source	<xs:attribute name="TC_VersionMinor" type="xs:string"> <xs:annotation> <xs:documentation>Identifier of the Train Characteristics version, minor number.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **teltsi_LocationSubsidiaryTypeCode**

type	restriction of **xs:token**
used by	element **teltsi_LocationSubsidiaryCode**
facets	Kind / Value / Annotation enumeration 0 documentation: Not Defined documentation: not used enumeration 1 documentation: Track documentation: The track is a uniquely defined part of location enumeration 2 documentation: Private Siding documentation: Tracks are not for open access enumeration 3 documentation: Border Point Code documentation: Special code for the Border Points are allocated at the country border and the points between different IM networks. Location of these points sometimes are not geographically same with the station or yard. Therefore these points are "logical point" enumeration 4 documentation: Sorting Code documentation: Destination station of the wagon has a code in order to provide shunting technology. enumeration 5 documentation: Vehicle Parking Points documentation: All points (tracks) enumeration 6 documentation: Public Loading Places documentation: Is a type of physical location on the open access network where consignor or consignee can load or unload wagons enumeration 7 documentation: Private Loading Places documentation: Is a type of physical location outside the open access network where consignor or consignee can load or unload wagons enumeration 8 documentation: IM Path Tariff Point documentation: Price Segment change between two IM Networks. enumeration 9 documentation

			Depot / Maintenance workshop. Place for overhaul or maintenance of the rolling stock.
	enumeration	10	documentation Switch/turnout documentation The location where two tracks meet or diverge.
	enumeration	11	documentation Grade Crossing documentation The location where two tracks on the same level cross each other.
	enumeration	12	documentation Section of the track documentation Section is inside of the location considered part of a track.
	enumeration	13	documentation Twin track point documentation The spot where is end or start of the twinned track section.
	enumeration	14	documentation Retarder (rail brake) documentation Trackside equipment to control the speed of the wagons running from the shunting hump.
	enumeration	15	documentation Platform documentation The area next to the track which has been raised to make access to railway vehicles easier.
	enumeration	16	documentation Railing documentation barrier Safety equipment used to prevent access to the track by people and animals.
	enumeration	17	documentation Movable scotch block documentation Safety equipment across the track avoiding any unnecessary moving beyond that point.
	enumeration	18	documentation Derailing stop / Trap points / Catch points documentation Safety equipment is on one rail avoiding any unnecessary moving beyond that point.
	enumeration	19	documentation Loading equipment documentation Special equipment to facilitate the loading and unloading on the Public Loading Places.
	enumeration	20	documentation Weighbridge documentation Special equipment is to facilitate the measure of the weight of the wagon.
	enumeration	21	documentation Building documentation Those buildings where IM placed his staff for direct communication with RU staff or the IM buildings serve RU activities as well.
	enumeration	22	documentation Level crossing documentation Place where rail and road crossing in level. on the same level (grade)
	enumeration	23	documentation Bridge documentation Special built structure is over the road
	enumeration	24	documentation Tunnel documentation Structure to to allow a railway line to pass under the surface.
	enumeration	25	documentation Underpass documentation Undercrossing or underground passage under the railway track. (Not used by trains)

	enumeration	26	documentation Block section documentation Block section outside of the location with primary code. In other words: a section on the open track between stations defined by signalling system.
	enumeration	27	documentation Signal documentation A signal is a mechanical or electrical device erected beside a railway line to pass information relating to the state of the line ahead to train drivers/engineers.
	enumeration	28	documentation Sign and board documentation Equipment to inform the board staff for train traffic and shunting.
	enumeration	29	documentation Phase break documentation Border of the power supply systems (catenary).
	enumeration	30	documentation Leap in kilometer documentation The section has deviation in length i. e. the section more or less than called.
	enumeration	32	documentation Balise documentation A balise is an electronic beacon or transponder placed between the rails of a railway as part of an Automatic Train Protection (ATP) system.
	enumeration	33	documentation Hot spot detector documentation Trackside equipment which detects hot wheels or axle-box on passing trains.
	enumeration	34	documentation Flat wheel detector documentation Trackside equipment which detects flat spots on wheels on passing trains.
	enumeration	35	documentation Dynamic wheel load documentation detector Special equipment is in trackside for inspect of the overloaded wagons.
	enumeration	36	documentation Freight yard documentation A freight yard is commercial usage of a physical location which can be used as a sending or a destination station in freight orders of rail freight transports. The freight yard can have his own codification
	enumeration	37	documentation Loading point documentation A loading point is a commercial usage of a physical location. Each loading point is assigned to a yard.
	enumeration	38	documentation IM Network link documentation It allows to link two locations from different IM Networks
	enumeration	39	documentation Reservation code
	enumeration	40	documentation Metastation documentation To mark a meta location that forms the link between different stations that are considered as equal (for the traveller)
	enumeration	41	documentation CompanySpecificIdentifier documentation Company specific identifier of the primary location
	enumeration	42	documentation DIUM stations - Places of acceptance/delivery Station open into international traffic of goods (tariff point included in DIUM) – consignment acceptance/delivery station (loading points are excluded and covered by

			TypeCode 37).		
	enumeration	43	documentation		
				Passengers cars public loading	Is a type of physical location on the open access network where passengers can put their car on a carrying train
	enumeration	44	documentation		
				Passengers cars private loading	Is a type of physical location outside the open access network where passengers can put their car on a carrying train
	enumeration	45	documentation		
				Sewage dump	Place for cleaning purposes - disposal of the waste
	enumeration	46	documentation		
				Refuelling Point	Location where refuelling takes place
	enumeration	47	documentation		
				Mains Supply	Location where energy supply can be provided for the rolling stock e.g. preheating
	enumeration	48	documentation		
				Water Supply	Location where water supply can be provided for the rolling stock
	enumeration	49	documentation		
				Compressed plant	Train on a track with motion stabled with external air supply for braking systems
	enumeration	50	documentation		
				Indoor cleaning platform	Cleaning point -interior
	enumeration	51	documentation		
				Car-wash plant	Cleaning point -outdoor
	enumeration	52	documentation		
				Short dry-cleaning track	Cleaning point
	enumeration	53	documentation		
				Pollution protective plate	Track where floor that avoids pollution of the earth below
	enumeration	54	documentation		
				Sand-filling station	Location where sand is filled
	enumeration	55	documentation		
				Repair track	Location where a

			train/wagon/engine can be repaired
	enumeration	56	documentation Signal box The location of a building containing signalling equipment
	enumeration	57	documentation Intermodal Terminal documentation Intermodal Terminal is a location which provides the space, equipment and operational environment under which the transfer of loading units (freight containers, swap bodies, semi-trailers or trailers) takes place
	enumeration	58	documentation OSJD system based location documentation Location code used within OSJD
	enumeration	59	documentation Train Service Substitute Stop documentation Place outside of railway station or railway stop, where passengers board or leave bus or any other transport mean as substitution of train service. Physical part of Primary Location.
	enumeration	60	documentation Multifunctional rail terminal documentation Facilities for conventional and/or intermodal rail/road transshipment principally open for public use and for all types of cargo. This kind of facility does not only provide transshipment, but also additional services like storage, consignment or road pre/end haulage. Physical part of Primary Location.
	enumeration	61	documentation Relief facility documentation Facilities providing equipment and infrastructure used to overcome a disruption (derailment, collision or other accidents). Physical part of Primary Location.
	enumeration	66	documentation Location ENEE Code documentation Legacy ENEE code of the parent primary location. Different coding of primary location.
	enumeration	70	documentation Network Border documentation Network border between two neighboring IM's; first or last Primary Location on a network. Attribute of primary location.
	enumeration	71	documentation State border documentation Political border between two member states. Attribute of primary location.
	enumeration	72	documentation Administrative border documentation Border point inside a member state to define federal structures or administrative districts or local areas. Attribute of primary location.
	enumeration	74	documentation Operational handover documentation Location where the responsibility for operation changes or can change between two involved IMs. Attribute of primary location.
	enumeration	75	documentation Planning handover documentation Location where the responsibility for timetable planning and path allocation changes or can change between two involved IMs. Attribute of primary location.
	enumeration	76	documentation Other technical facility documentation All technical installations and services that are not included in other facility types. E.g. Pre heating, de icing, air conditioning, Washing/cleaning of rolling stock, Disinfection of rolling stock, Sewage removal and Stationary brake test facilities. Physical part of Primary Location.
	enumeration	90	documentation

	enumeration 99 Test Loc documentation Relation to Station documentation An indicator used to show that this location is a subsidiary of another location.
source	<xs:attribute name="teltsi_LocationSubsidiaryTypeCode"> <xs:annotation> <xs:documentation/> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:token"> <xs:enumeration value="0"> <xs:annotation> <xs:documentation>Not Defined</xs:documentation> <xs:documentation>not used</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="1"> <xs:annotation> <xs:documentation>Track</xs:documentation> <xs:documentation>The track is a uniquely defined part of location</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="2"> <xs:annotation> <xs:documentation>Private Siding</xs:documentation> <xs:documentation>Tracks are not for open access</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="3"> <xs:annotation> <xs:documentation>Border Point Code</xs:documentation> <xs:documentation>Special code for the Border Points are allocated at the country border and the points between different IM networks. Location of these points sometimes are not geographically same with the station or yard. Therefore these points are "logical point"</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="4"> <xs:annotation> <xs:documentation>Sorting Code</xs:documentation> <xs:documentation>Destination station of the wagon has a code in order to provide shunting technology.</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="5"> <xs:annotation> <xs:documentation>Vehicle Parking Points</xs:documentation> <xs:documentation>All points (tracks)</xs:documentation> </xs:annotation> </xs:enumeration> <xs:enumeration value="6"> <xs:annotation> <xs:documentation>Public Loading Places</xs:documentation> <xs:documentation>Is a type of physical location on the open access network where consignor or consignee can load or unload wagons</xs:documentation>

```
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="7">
      <xs:annotation>
        <xs:documentation>Private Loading Places</xs:documentation>
        <xs:documentation>Is a type of physical location outside the open access network where consignor or consignee can load or unload wagons</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="8">
      <xs:annotation>
        <xs:documentation>IM Path Tariff Point</xs:documentation>
        <xs:documentation>Price Segment change between two IM Networks.</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="9">
      <xs:annotation>
        <xs:documentation>Depot / Maintenance workshop.
          Place for overhaul or maintenance of the rolling stock.

                                                    </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="10">
      <xs:annotation>
        <xs:documentation>Switch/turnout</xs:documentation>
        <xs:documentation>The location where two tracks meet or diverge.</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="11">
      <xs:annotation>
        <xs:documentation>Grade Crossing</xs:documentation>
        <xs:documentation>The location where two tracks on the same level cross each other.</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="12">
      <xs:annotation>
        <xs:documentation>Section of the track</xs:documentation>
        <xs:documentation>Section is inside of the location considered part of a track.</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="13">
      <xs:annotation>
        <xs:documentation>Twin track point</xs:documentation>
        <xs:documentation>The spot where is end or start of the twinned track section.</xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="14">
      <xs:annotation>
        <xs:documentation>Retarder (rail brake)</xs:documentation>
        <xs:documentation>Trackside equipment to control the speed of the wagons running from the shunting hump.</xs:documentation>
      </xs:annotation>
```

```
      </xs:enumeration>
      <xs:enumeration value="15">
        <xs:annotation>
          <xs:documentation>Platform</xs:documentation>
          <xs:documentation>The area next to the track which has been raised to make access to
railway vehicles easier.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="16">
        <xs:annotation>
          <xs:documentation>Railing</xs:documentation>
          <xs:documentation>barrier Safety equipment used to prevent access to the track by people
and animals.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="17">
        <xs:annotation>
          <xs:documentation>Movable scotch block</xs:documentation>
          <xs:documentation>Safety equipment across the track avoiding any unnecessary moving
beyond that point.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="18">
        <xs:annotation>
          <xs:documentation>Derailing stop / Trap points / Catch points</xs:documentation>
          <xs:documentation>Safety equipment is on one rail avoiding any unnecessary moving
beyond that point.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="19">
        <xs:annotation>
          <xs:documentation>Loading equipment</xs:documentation>
          <xs:documentation>Special equipment to facilitate the loading and unloading on the Public
Loading Places.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="20">
        <xs:annotation>
          <xs:documentation>Weighbridge</xs:documentation>
          <xs:documentation>Special equipment is to facilitate the measure of the weight of the
wagon.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="21">
        <xs:annotation>
          <xs:documentation>Building</xs:documentation>
          <xs:documentation>Those buildings where IM placed his staff for direct communication with
RU staff or the IM buildings serve RU activities as well.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="22">
        <xs:annotation>
          <xs:documentation>Level crossing</xs:documentation>
          <xs:documentation>Place where rail and road crossing in level. on the same level
(grade)</xs:documentation>
        </xs:annotation>
```

```
      </xs:enumeration>
      <xs:enumeration value="23">
        <xs:annotation>
          <xs:documentation>Bridge</xs:documentation>
          <xs:documentation>Special built structure is over the road</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="24">
        <xs:annotation>
          <xs:documentation>Tunnel</xs:documentation>
          <xs:documentation>Structure to to allow a railway line to pass under the
surface.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="25">
        <xs:annotation>
          <xs:documentation>Underpass</xs:documentation>
          <xs:documentation>Undercrossing or underground passage under the railway track. (Not
used by trains)</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="26">
        <xs:annotation>
          <xs:documentation>Block section</xs:documentation>
          <xs:documentation>Block section outside of the location with primary code. In other words: a
section on the open track between stations defined by signalling system.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="27">
        <xs:annotation>
          <xs:documentation>Signal</xs:documentation>
          <xs:documentation>A signal is a mechanical or electrical device erected beside a railway line
to pass information relating to the state of the line ahead to train
drivers/engineers.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="28">
        <xs:annotation>
          <xs:documentation>Sign and board</xs:documentation>
          <xs:documentation>Equipment to inform the board staff for train traffic and
shunting.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="29">
        <xs:annotation>
          <xs:documentation>Phase break</xs:documentation>
          <xs:documentation>Border of the power supply systems (catenary).</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="30">
        <xs:annotation>
          <xs:documentation>Leap in kilometer</xs:documentation>
          <xs:documentation>The section has deviation in length i. e. the section more or less than
called.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
```

```
      <xs:enumeration value="32">
        <xs:annotation>
          <xs:documentation>Balise</xs:documentation>
          <xs:documentation>A balise is an electronic beacon or transponder placed between the rails of a railway as part of an Automatic Train Protection (ATP) system.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="33">
        <xs:annotation>
          <xs:documentation>Hot spot detector</xs:documentation>
          <xs:documentation>Trackside equipment which detects hot wheels or axle-box on passing trains.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="34">
        <xs:annotation>
          <xs:documentation>Flat wheel detector</xs:documentation>
          <xs:documentation>Trackside equipment which detects flat spots on wheels on passing trains.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="35">
        <xs:annotation>
          <xs:documentation>Dynamic wheel load</xs:documentation>
          <xs:documentation>detector Special equipment is in trackside for inspect of the overloaded wagons.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="36">
        <xs:annotation>
          <xs:documentation>Freight yard</xs:documentation>
          <xs:documentation>A freight yard is commercial usage of a physical location which can be used as a sending or a destination station in freight orders of rail freight transports. The freight yard can have his own codification</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="37">
        <xs:annotation>
          <xs:documentation>Loading point</xs:documentation>
          <xs:documentation>A loading point is a commercial usage of a physical location. Each loading point is assigned to a yard.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="38">
        <xs:annotation>
          <xs:documentation>IM Network link</xs:documentation>
          <xs:documentation>It allows to link two locations from different IM Networks</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="39">
        <xs:annotation>
          <xs:documentation>Reservation code</xs:documentation>
          <xs:documentation/>
        </xs:annotation>
      </xs:enumeration>
```

```
      <xs:enumeration value="40">
        <xs:annotation>
          <xs:documentation>Metastation</xs:documentation>
          <xs:documentation>To mark a meta location that forms the link between different stations
that are considered as equal (for the traveller)</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="41">
        <xs:annotation>
          <xs:documentation>CompanySpecificIdentifier</xs:documentation>
          <xs:documentation>Company specific identifier of the primary location</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="42">
        <xs:annotation>
          <xs:documentation>
                                    DIUM stations - Places of acceptance/delivery    Station open into
international traffic of goods (tariff point included in DIUM) – consignment acceptance/delivery
station (loading points are excluded and covered by TypeCode 37).
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="43">
        <xs:annotation>
          <xs:documentation>
                                    Passengers cars public loading  Is a type of physical location on
the open access network where passengers can put their car on a carrying train
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="44">
        <xs:annotation>
          <xs:documentation>
                                    Passengers cars private loading Is a type of physical location
outside the open access network where passengers can put their car on a carrying train
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="45">
        <xs:annotation>
          <xs:documentation>
                                    Sewage dump   Place for cleaning purposes - disposal of the
waste
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="46">
        <xs:annotation>
          <xs:documentation>
                                    Refuelling Point Location where refuelling takes place
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="47">
        <xs:annotation>
          <xs:documentation>
```

```
                                Mains Supply    Location where energy supply can be provided for the rolling stock e.g. preheating
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="48">
      <xs:annotation>
        <xs:documentation>
                                Water Supply    Location where water supply can be provided for the rolling stock
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="49">
      <xs:annotation>
        <xs:documentation>
                                Compressed plant        Train on a track with motion stabled with external air supply for braking systems
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="50">
      <xs:annotation>
        <xs:documentation>
                                Indoor cleaning platform Cleaning point -interior
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="51">
      <xs:annotation>
        <xs:documentation>
                                Car-wash plant  Cleaning point -outdoor
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="52">
      <xs:annotation>
        <xs:documentation>
                                Short dry-cleaning track Cleaning point
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="53">
      <xs:annotation>
        <xs:documentation>
                                Pollution protective plate      Track where floor that avoids pollution of the earth below
                                </xs:documentation>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="54">
      <xs:annotation>
        <xs:documentation>
                                Sand-filling station    Location where sand is filled
                                </xs:documentation>
      </xs:annotation>
```

```
      </xs:enumeration>
      <xs:enumeration value="55">
        <xs:annotation>
          <xs:documentation>
                                    Repair track      Location where a train/wagon/engine can be
repaired
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="56">
        <xs:annotation>
          <xs:documentation>
                                    Signal box        The location of a building containing signalling
equipment
                                    </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="57">
        <xs:annotation>
          <xs:documentation>Intermodal Terminal</xs:documentation>
          <xs:documentation> Intermodal Terminal is a location which provides the space, equipment
and operational environment under which the transfer of loading units (freight containers, swap
bodies, semi-trailers or trailers) takes place </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="58">
        <xs:annotation>
          <xs:documentation>OSJD system based location</xs:documentation>
          <xs:documentation>Location code used within OSJD</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="59">
        <xs:annotation>
          <xs:documentation>Train Service Substitute Stop</xs:documentation>
          <xs:documentation>Place outside of railway station or railway stop, where passengers board
or leave bus or any other transport mean as substitution of train service. Physical part of Primary
Location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="60">
        <xs:annotation>
          <xs:documentation>Multifunctional rail terminal</xs:documentation>
          <xs:documentation>Facilities for conventional and/or intermodal rail/road transshipment
principally open for public use and for all types of cargo. This kind of facility does not only provide
transshipment, but also additional services like storage, consignment or road pre/end haulage.
Physical part of Primary Location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="61">
        <xs:annotation>
          <xs:documentation>Relief facility</xs:documentation>
          <xs:documentation>Facilities providing equipment and infrastructure used to overcome a
disruption (derailment, collision or other accidents). Physical part of Primary
Location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
```

```
      <xs:enumeration value="66">
        <xs:annotation>
          <xs:documentation>Location ENEE Code</xs:documentation>
          <xs:documentation>Legacy ENEE code of the parent primary location. Different coding of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="70">
        <xs:annotation>
          <xs:documentation>Network Border</xs:documentation>
          <xs:documentation>Network border between two neighboring IM's; first or last Primary Location on a network. Attribute of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="71">
        <xs:annotation>
          <xs:documentation>State border</xs:documentation>
          <xs:documentation>Political border between two member states. Attribute of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="72">
        <xs:annotation>
          <xs:documentation>Administrative border</xs:documentation>
          <xs:documentation>Border point inside a member state to define federal structures or administrative districts or local areas. Attribute of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="74">
        <xs:annotation>
          <xs:documentation>Operational handover</xs:documentation>
          <xs:documentation>Location where the responsibility for operation changes or can change between two involved IMs. Attribute of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="75">
        <xs:annotation>
          <xs:documentation>Planning handover</xs:documentation>
          <xs:documentation>Location where the responsibility for timetable planning and path allocation changes or can change between two involved IMs. Attribute of primary location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="76">
        <xs:annotation>
          <xs:documentation>Other technical facility</xs:documentation>
          <xs:documentation>All technical installations and services that are not included in other facility types. E.g. Pre heating, de icing, air conditioning, Washing/cleaning of rolling stock, Disinfection of rolling stock, Sewage removal and Stationary brake test facilities. Physical part of Primary Location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="90">
        <xs:annotation>
          <xs:documentation>Test Loc</xs:documentation>
          <xs:documentation/>
```

```
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="99">
        <xs:annotation>
          <xs:documentation>Relation to Station</xs:documentation>
          <xs:documentation>An indicator used to show that this location is a subsidiary of another location.</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **temporaryConstraintReference**

type	**xs:string**
used by	element **TemporaryConstraintReason**
annotation	documentation Human-readable identifier of document informing driver of temporary constraint
source	<xs:attribute name="temporaryConstraintReference" type="xs:string"> <xs:annotation> <xs:documentation>Human-readable identifier of document informing driver of temporary constraint</xs:documentation> </xs:annotation> </xs:attribute>

attribute **temporaryConstraintType**

type	restriction of **xs:string**
used by	element **TemporaryConstraints**
facets	Kind / Value / Annotation enumeration ASR enumeration Low_Adhesion enumeration TractionTotalCurrent enumeration RegenerationTotalCurrent enumeration PowerAdvice enumeration EstimatedVoltage enumeration Wind enumeration Unavailable_DAS_OperatingModes enumeration AdvisedSpeed enumeration NetworkSpecificConstraint enumeration PlatformChange enumeration NoAdvicePresentedToDriver
annotation	documentation Reason for the temporary constraint. It is possible to choose one reason among the enumerations; the detailed data has to be put under the related element in the TemporaryConstraint choice. Exceptions are NoAdvicePresentedToDriver, ATO_Inhibition_Zone, DAS_Inhibition_Zone and IM_SDAS_Only, where no additional data is needed.
source	<xs:attribute name="temporaryConstraintType"> <xs:annotation> <xs:documentation>Reason for the temporary constraint. It is possible to choose one reason

	among the enumerations; the detailed data has to be put under the related element in the TemporaryConstraint choice. Exceptions are NoAdvicePresentedToDriver, ATO_Inhibition_Zone, DAS_Inhibition_Zone and IM_SDAS_Only, where no additional data is needed.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="ASR"/> <xs:enumeration value="Low_Adhesion"/> <xs:enumeration value="TractionTotalCurrent"/> <xs:enumeration value="RegenerationTotalCurrent"/> <xs:enumeration value="PowerAdvice"/> <xs:enumeration value="EstimatedVoltage"/> <xs:enumeration value="Wind"/> <xs:enumeration value="Unavailable_DAS_OperatingModes"/> <xs:enumeration value="AdvisedSpeed"/> <xs:enumeration value="NetworkSpecificConstraint"/> <xs:enumeration value="PlatformChange"/> <xs:enumeration value="NoAdvicePresentedToDriver"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **temporarySignalConstraintType**

type	restriction of **xs:string**
used by	element **TemporarySignalConstraints**
facets	Kind / Value / Annotation enumeration TemporarySignal enumeration ExistingSignalAspectModification enumeration CancelledSignal
annotation	documentation Type of temporary signal constraints: Temporary Signal (the identifier must then be new, not used by another element), Existing Signal Aspect Modification (the identifier must point to a Signal described in SP / SP_Points if it is present) or Cancelled Signal (the identifier must point to a Signal described in SP / SP_Points).
source	<xs:attribute name="temporarySignalConstraintType"> <xs:annotation> <xs:documentation>Type of temporary signal constraints: Temporary Signal (the identifier must then be new, not used by another element), Existing Signal Aspect Modification (the identifier must point to a Signal described in SP / SP_Points if it is present) or Cancelled Signal (the identifier must point to a Signal described in SP / SP_Points).</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="TemporarySignal"/> <xs:enumeration value="ExistingSignalAspectModification"/> <xs:enumeration value="CancelledSignal"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **text**

type	**xs:string**
used by	complexType **MultilingualText_ComplexType**
annotation	documentation String containing the text
source	<xs:attribute name="text" type="xs:string"> <xs:annotation> <xs:documentation>String containing the text</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TFC_estimatedVoltage**

type	**voltageType**
used by	complexType **TractionForceCurve_ComplexType**
annotation	documentation This TractionForceCurve is to be used if the estimated voltage of the considered portion of infrastructure is inferior to this value and superior to a lower estimatedVoltage of another TractionForceCurve (if it exists).
source	<xs:attribute name="TFC_estimatedVoltage" type="voltageType"> <xs:annotation> <xs:documentation>This TractionForceCurve is to be used if the estimated voltage of the considered portion of infrastructure is inferior to this value and superior to a lower estimatedVoltage of another TractionForceCurve (if it exists).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TFC_maxCurrentValue**

type	**currentType**
used by	complexType **TractionForceCurve_ComplexType**
annotation	documentation This TractionForceCurve is to be used if the maximum current value of the considered portion of infrastructure is inferior to this value and superior to a lower maxCurrentValue of another TractionForceCurve (if it exists).
source	<xs:attribute name="TFC_maxCurrentValue" type="currentType"> <xs:annotation> <xs:documentation>This TractionForceCurve is to be used if the maximum current value of the considered portion of infrastructure is inferior to this value and superior to a lower maxCurrentValue of another TractionForceCurve (if it exists).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **timeout_ATOTS_Response**

type	**xs:duration**
used by	element **HandshakeAcknowledgement**
annotation	documentation [SUBSET-126] Minimum time the ATO-OB shall wait before repeating a request if there was no response
source	<xs:attribute name="timeout_ATOTS_Response" type="xs:duration"> <xs:annotation> <xs:documentation>[SUBSET-126] Minimum time the ATO-OB shall wait before repeating a

	request if there was no response</xs:documentation> </xs:annotation> </xs:attribute>

attribute **timestamp**

type	**UTCDateTimeType**
used by	elements **MessageHeader** **PassengerMovement** **PlaintextMessage**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Timestamp. Must be in UTC.
source	<xs:attribute name="timestamp" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Timestamp. Must be in UTC.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TP_abbreviation**

type	**xs:string**
used by	elements **NextTimingPointTrainInformation** **PreviousTimingPointTrainInformation** **TimingPoint**
annotation	documentation Abbreviation of the Timing Point name
source	<xs:attribute name="TP_abbreviation" type="xs:string"> <xs:annotation> <xs:documentation>Abbreviation of the Timing Point name</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TP_Accurate_Stopping**

type	restriction of **xs:string**
used by	element **PreviousTimingPoint**
facets	Kind Value Annotation enumeration Undershoot enumeration Accurate enumeration Overshoot
annotation	documentation [SUBSET-126] (If TP_Pass_Stop_Depart = Stopped) This qualifier specifies if the train has stopped accurately or not at the Operational Stopping Point.
source	<xs:attribute name="TP_Accurate_Stopping"> <xs:annotation> <xs:documentation>[SUBSET-126] (If TP_Pass_Stop_Depart = Stopped) This qualifier specifies if the train has stopped accurately or not at the Operational Stopping Point. </xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string">

```
      <xs:enumeration value="Undershoot"/>
      <xs:enumeration value="Accurate"/>
      <xs:enumeration value="Overshoot"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
```

attribute **TP_alignment**

type	restriction of **xs:string**
used by	element **TimingPointConstraints**
facets	Kind / Value / Annotation enumeration Front enumeration Middle enumeration Rear
annotation	documentation [SUBSET-126] Defines if the TP location is applicable from the front, middle or rear of the train
source	<xs:attribute name="TP_alignment"> <xs:annotation> <xs:documentation>[SUBSET-126] Defines if the TP location is applicable from the front, middle or rear of the train</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Front"/> <xs:enumeration value="Middle"/> <xs:enumeration value="Rear"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **TP_ID**

type	**xs:string**
used by	elements **NextTimingPointTrainInformation** **PreviousTimingPointTrainInformation** **TimingPoint** **TimingPointEstimation** complexType **TimingPointIdentification_ComplexType**
annotation	documentation Timing Point Identifier
source	<xs:attribute name="TP_ID" type="xs:string"> <xs:annotation> <xs:documentation>Timing Point Identifier</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TP_Information**

type	restriction of **xs:string**
used by	element **TimingPointConstraints**
facets	Kind / Value / Annotation enumeration None

	enumeration End_of_Journey
annotation	documentation [SUBSET-126] Specifies some information specific for the TP
source	<xs:attribute name="TP_Information"> <xs:annotation> <xs:documentation>[SUBSET-126] Specifies some information specific for the TP</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="None"/> <xs:enumeration value="End_of_Journey"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **TP_latestArrivalTime**

type	**UTCDateTimeType**
used by	element **TimingPointConstraints**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation Time of arrival at Timing Point expressed in UTC. Has to be specified always except at departure stations.
source	<xs:attribute name="TP_latestArrivalTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>Time of arrival at Timing Point expressed in UTC. Has to be specified always except at departure stations.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TP_Pass_Stop_Depart**

type	restriction of **xs:string**
used by	element **PreviousTimingPoint**
facets	Kind Value Annotation enumeration Passed enumeration Stopped enumeration Departed
annotation	documentation [SUBSET-126] Qualifier to indicate if train has stopped at the TP, has departed from the TP or has passed the TP.
source	<xs:attribute name="TP_Pass_Stop_Depart"> <xs:annotation> <xs:documentation>[SUBSET-126] Qualifier to indicate if train has stopped at the TP, has departed from the TP or has passed the TP.</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Passed"/> <xs:enumeration value="Stopped"/> <xs:enumeration value="Departed"/> </xs:restriction>

	</xs:simpleType> </xs:attribute>

attribute **TP_PlannedLatestArrivalTime**

type	**UTCDateTimeType**
used by	element **TimingPointConstraints**
facets	Kind Value Annotation pattern \d{4}\-\d{2}\-\d{2}T\d{2}:\d{2}:\d{2}Z
annotation	documentation The original planned latest arrival time at the Timing Point expressed in UTC. Useful for driver information. Not to be used for calculations.
source	<xs:attribute name="TP_PlannedLatestArrivalTime" type="UTCDateTimeType"> <xs:annotation> <xs:documentation>The original planned latest arrival time at the Timing Point expressed in UTC. Useful for driver information. Not to be used for calculations.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **TP_priority**

type	restriction of **xs:unsignedByte**
used by	element **TimingPoint**
facets	Kind Value Annotation minInclusive 1 maxInclusive 5
annotation	documentation TP_Priority is used to indicate the relative significance of the Timing Point in a Journey from the driver's point of view. This information is intended to be used when choosing the information to display. It does not specify a priority in respecting the timing points when the DAS calculates the speed profile (the attributes TP_latestArrivalTime and arrivalWindow in the Journey Profile are the absolute time constraints on timing points). It has 5 possible values : 5: very important station, junction or network border 4: important station, junction or shunting area 3: station, junction or regional border 2: other stopping locations 1: all other timing points
source	<xs:attribute name="TP_priority"> <xs:annotation> <xs:documentation>TP_Priority is used to indicate the relative significance of the Timing Point in a Journey from the driver's point of view. This information is intended to be used when choosing the information to display. It does not specify a priority in respecting the timing points when the DAS calculates the speed profile (the attributes TP_latestArrivalTime and arrivalWindow in the Journey Profile are the absolute time constraints on timing points). It has 5 possible values : 5: very important station, junction or network border 4: important station, junction or shunting area 3: station, junction or regional border 2: other stopping locations 1: all other timing points</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:unsignedByte"> <xs:minInclusive value="1"/> <xs:maxInclusive value="5"/> </xs:restriction>

```
  </xs:simpleType>
</xs:attribute>
```

attribute **TP_StopSkipPass**

type	restriction of **xs:string**
used by	elements **NextTimingPointTrainInformation** **PreviousTimingPointTrainInformation** **TimingPointConstraints**
facets	Kind / Value / Annotation enumeration Stopping_Point enumeration Skipped_Stopping_Point enumeration Passing_Point
annotation	documentation [SUBSET-126] Specifies if the Timing Point is a Passing Point, an operational Stopping Point or a skipped Stopping Point
source	<xs:attribute name="TP_StopSkipPass"> <xs:annotation> <xs:documentation>[SUBSET-126] Specifies if the Timing Point is a Passing Point, an operational Stopping Point or a skipped Stopping Point</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="Stopping_Point"/> <xs:enumeration value="Skipped_Stopping_Point"/> <xs:enumeration value="Passing_Point"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **TP_type**

type	**xs:string**
used by	element **TimingPoint**
annotation	documentation Type of Timing Point. Examples: Very important station Important station Station Stopping location Junction Regional border Connexion Network border Signal with detection Grid or shunting area Service installation Service stopping location Movable bridge Other
source	<xs:attribute name="TP_type" type="xs:string"> <xs:annotation> <xs:documentation>Type of Timing Point. Examples: Very important station Important station Station Stopping location Junction

	Regional border Connexion Network border Signal with detection Grid or shunting area Service installation Service stopping location Movable bridge Other</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trackIdentifier**

used by	complexType **LineAndTrackIdentifier_ComplexType**
annotation	documentation Identification of track on a line (ex : “V1” or “Voie 1” or “Line 1”)
source	<xs:attribute name="trackIdentifier"> <xs:annotation> <xs:documentation>Identification of track on a line (ex : “V1” or “Voie 1” or “Line 1”)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **tractionForce**

type	**forceType**
used by	element **TractionForceCurveDataPoint**
annotation	documentation Traction Force (in N)
source	<xs:attribute name="tractionForce" type="forceType"> <xs:annotation> <xs:documentation>Traction Force (in N)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **tractionReactionTime**

type	**xs:int**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Delay between traction command and start of tractioning. Expressed in s
source	<xs:attribute name="tractionReactionTime" type="xs:int"> <xs:annotation> <xs:documentation>Delay between traction command and start of tractioning. Expressed in s</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trainCategoryCode**

type	restriction of **xs:string**
used by	elements **OtherSpecificSSP** **RT_TrainInformation** **SignalTrainCategory/Signal_OtherTrainCategory** complexType **TC_Features_ComplexType**
facets	Kind Value Annotation maxLength 10
annotation	documentation Code for the train category (freeform text, limited to 10 characters)
source	<xs:attribute name="trainCategoryCode"> <xs:annotation> <xs:documentation>Code for the train category (freeform text, limited to 10 characters)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:maxLength value="10"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **trainHold**

type	**xs:boolean**
used by	element **StoppingPointDepartureDetails**
annotation	documentation [SUBSET-126] The variable defines if the train is requested to be held at the Stopping Point (1) or not (0)
source	<xs:attribute name="trainHold" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] The variable defines if the train is requested to be held at the Stopping Point (1) or not (0)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trainLength**

type	**xs:integer**
used by	elements **RT_TrainInformation** **TrainCharacteristicsRef** complexType **TC_Features_ComplexType**
annotation	documentation The total length of a train. Expressed in m
source	<xs:attribute name="trainLength" type="xs:integer"> <xs:annotation> <xs:documentation>The total length of a train. Expressed in m</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trainMaxSpeed**

type	**xs:integer**
used by	element **TrainCharacteristicsRef** complexType **TC_Features_ComplexType**
annotation	documentation

	The maximum possible speed of the train in km/h
source	<xs:attribute name="trainMaxSpeed" type="xs:integer"> <xs:annotation> <xs:documentation>The maximum possible speed of the train in km/h</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trainMoving**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Train is moving, from Subset-126 (boolean)
source	<xs:attribute name="trainMoving" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Train is moving, from Subset-126 (boolean)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trainWeight**

type	**xs:int**
used by	element **TrainCharacteristicsRef** complexType **TC_Features_ComplexType**
annotation	documentation The sum of all weights of wagons and traction units. Expressed in tonnes (t)
source	<xs:attribute name="trainWeight" type="xs:int"> <xs:annotation> <xs:documentation>The sum of all weights of wagons and traction units. Expressed in tonnes (t)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trp_ae**

type	**xs:decimal**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
source	<xs:attribute name="trp_ae" type="xs:decimal"> <xs:annotation> <xs:documentation>Tunnel Resistance Prediction coefficient a_e (for engines). Unit: kg.m^(2b_e-1). Typical value 1140. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trp_aw**

type	**xs:decimal**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
source	<xs:attribute name="trp_aw" type="xs:decimal"> <xs:annotation> <xs:documentation>Tunnel Resistance Prediction coefficient a_w (for wagons). Unit: kg.m^(2b_w-1). Typical value 662. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trp_be**

type	**xs:decimal**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
source	<xs:attribute name="trp_be" type="xs:decimal"> <xs:annotation> <xs:documentation>Tunnel Resistance Prediction coefficient b_e (for engines). No unit. Typical value 1.48. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **trp_bw**

type	**xs:decimal**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
source	<xs:attribute name="trp_bw" type="xs:decimal"> <xs:annotation> <xs:documentation>Tunnel Resistance Prediction coefficient b_w (for wagons). No unit. Typical value 1.75. Reference: https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **tunnelCategory**

type	restriction of **xs:string**
used by	element **Tunnel**
facets	Kind: enumeration; Value: SingleTrack; Annotation:

	enumeration DoubleTrack enumeration Wide-crossSection
annotation	documentation [SUBSET-126] Category of the Tunnel (single or double track)
source	<xs:attribute name="tunnelCategory"> <xs:annotation> <xs:documentation>[SUBSET-126] Category of the Tunnel (single or double track)</xs:documentation> </xs:annotation> <xs:simpleType> <xs:restriction base="xs:string"> <xs:enumeration value="SingleTrack"/> <xs:enumeration value="DoubleTrack"/> <xs:enumeration value="Wide-crossSection"/> </xs:restriction> </xs:simpleType> </xs:attribute>

attribute **tunnelFactor**

type	**xs:double**
used by	element **Tunnel**
annotation	documentation Aerodynamic cross section of the tunnel in expressed in m^2. Used in predictive tunnel resistance equation developed by Z. Filipovic. See https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)
source	<xs:attribute name="tunnelFactor" type="xs:double"> <xs:annotation> <xs:documentation>Aerodynamic cross section of the tunnel in expressed in m^2. Used in predictive tunnel resistance equation developed by Z. Filipovic. See https://www.sciencedirect.com/science/article/pii/S2210970616300415 (5)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **unableToStop**

type	**xs:boolean**
used by	element **B2G_StatusReport**
annotation	documentation [SUBSET-126] ATO: Unable to stop at the next Stopping Point, from Subset-126 (boolean)
source	<xs:attribute name="unableToStop" type="xs:boolean"> <xs:annotation> <xs:documentation>[SUBSET-126] ATO: Unable to stop at the next Stopping Point, from Subset-126 (boolean)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **UTC_Offset**

type	**xs:unsignedByte**
used by	element **SegmentProfile**

annotation	documentation [SUBSET-126] Offset to add to the UTC time in order to calculate the local time. Unsigned value Resolution: 15 min 0 = UTC - 14:00 56 = UTC ± 0 112 = UTC + 14:00 113 - 127 = Spare
source	<xs:attribute name="UTC_Offset" type="xs:unsignedByte"> <xs:annotation> <xs:documentation>[SUBSET-126] Offset to add to the UTC time in order to calculate the local time. Unsigned value Resolution: 15 min 0 = UTC - 14:00 56 = UTC ± 0 112 = UTC + 14:00 113 - 127 = Spare </xs:documentation> </xs:annotation> </xs:attribute>

attribute **V_DIFF**

type	**speedType**
used by	complexType **SpecificSSP_ComplexType**
annotation	documentation [SUBSET-126] Absolute Positive Speed associated to a train category
source	<xs:attribute name="V_DIFF" type="speedType"> <xs:annotation> <xs:documentation>[SUBSET-126] Absolute Positive Speed associated to a train category</xs:documentation> </xs:annotation> </xs:attribute>

attribute **value**

type	**xs:string**
used by	element **NetworkSpecificParameter**
annotation	documentation Value of the element
source	<xs:attribute name="value" type="xs:string"> <xs:annotation> <xs:documentation>Value of the element</xs:documentation> </xs:annotation> </xs:attribute>

attribute **verticalOffset**

type	**distanceType**
used by	complexType **PhysicalCharacteristics_ComplexType**
annotation	documentation Vertical offset (in m) from the top of the track to the middle of the object.

source	<xs:attribute name="verticalOffset" type="distanceType"> <xs:annotation> <xs:documentation>Vertical offset (in m) from the top of the track to the middle of the object.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **virtualMarker**

type	**xs:boolean**
used by	element **KM_Reference**
annotation	documentation To be used for km reference points defined in reference documents (as opposed to physical markers on the side of the track).
source	<xs:attribute name="virtualMarker" type="xs:boolean"> <xs:annotation> <xs:documentation>To be used for km reference points defined in reference documents (as opposed to physical markers on the side of the track).</xs:documentation> </xs:annotation> </xs:attribute>

attribute **visualIdentifier**

type	**xs:string**
used by	element **SignalPhysicalCharacteristics**
annotation	documentation Visual identifier of the signal. Typically a sign placed over or under the signal itself facilitates the visual identification by the driver.
source	<xs:attribute name="visualIdentifier" type="xs:string"> <xs:annotation> <xs:documentation>Visual identifier of the signal. Typically a sign placed over or under the signal itself facilitates the visual identification by the driver.</xs:documentation> </xs:annotation> </xs:attribute>

attribute **voltageValue**

type	**voltageType**
used by	element **EstimatedVoltageConstraint** complexType **Voltage_ComplexType**
annotation	documentation Voltage value (V)
source	<xs:attribute name="voltageValue" type="voltageType"> <xs:annotation> <xs:documentation>Voltage value (V)</xs:documentation> </xs:annotation> </xs:attribute>

attribute **wagons**

type	**xs:int**
used by	complexType **TC_Features_ComplexType**
annotation	documentation Number of wagons
source	<xs:attribute name="wagons" type="xs:int"> <xs:annotation> <xs:documentation>Number of wagons</xs:documentation> </xs:annotation> </xs:attribute>

attribute **windHeading**

type	**xs:unsignedShort**
used by	element **Wind**
annotation	documentation Wind heading, in degrees from North
source	<xs:attribute name="windHeading" type="xs:unsignedShort"> <xs:annotation> <xs:documentation>Wind heading, in degrees from North</xs:documentation> </xs:annotation> </xs:attribute>

attribute **windSpeed**

type	**speedType**
used by	element **Wind**
annotation	documentation Wind speed
source	<xs:attribute name="windSpeed" type="speedType"> <xs:annotation> <xs:documentation>Wind speed</xs:documentation> </xs:annotation> </xs:attribute>

attribute **XPath**

type	**xs:string**
used by	complexType **Error_ComplexType**
annotation	documentation XML Path to identify the location of the error. See syntax in Appendix E.
source	<xs:attribute name="XPath" type="xs:string"> <xs:annotation> <xs:documentation>XML Path to identify the location of the error. See syntax in Appendix E.</xs:documentation> </xs:annotation> </xs:attribute>

XML Schema documentation generated by **XMLSpy** Schema Editor **http://www.altova.com/xmlspy**